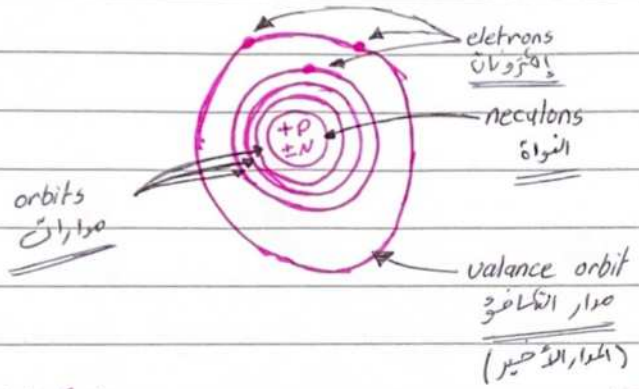
Electronics 1 , Eng Khawla 2023

بسم الله الرحمن الرحيم ..

Transistor Consists (يحتوي) From Two Diods ..

Materials Consists From ATOMS (ذرات) ..

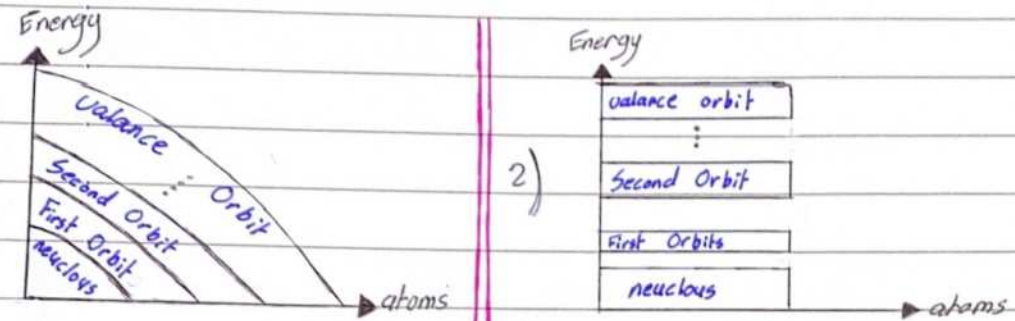


Valance Electrons (إلكترونات التكافؤ) : Number (عدد) of electrons in valance orbit (مدار التكافؤ) ..

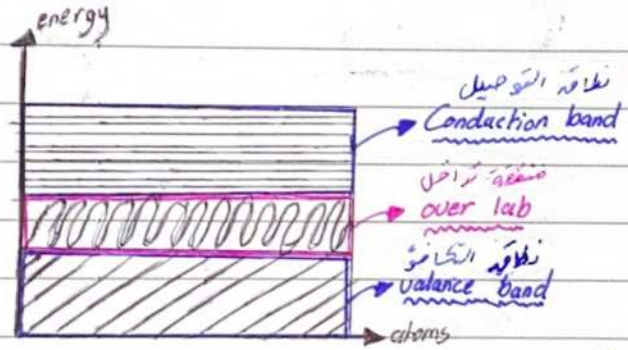
1) Conductors (الموصلات)	2) Semi Conductor (أشباه الموصلات)	3) Isolators (العوازل)
Groups (المجموعات) : (1), 2, 3 (الأفضل)	Groups : 4	Groups : 5, 6, (7) (الأفضل)
Ex : Copper (نحاس), Gold (الذهب)	Ex : Si, Ge (عناصر) GaAs (مركب) (جرمانيوم سيليكون)	Ex : Wood (خشب)
	• بطبيعتها عازلة تعتاج لطاقة بسيطة حتى تتحول لموصلات	• تعتاج لطاقة هائلة حتى تتحول لموصلات

ليث، عمرو

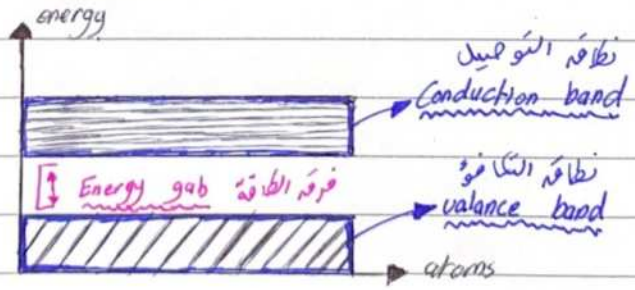
Representation of ATOMS (تمثيل الذرات), one of the following :-



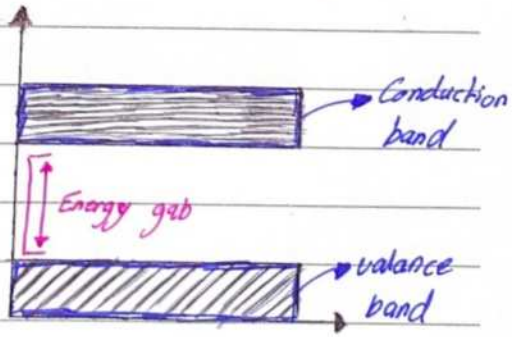
(a.) Conductors :-



(b.) Semi Conductors :-



(c.) Isolators :- (طاقة تصنع لـ cast)



Energy gab (Semi Conductor) < (smaller) Energy gab (Isolator)

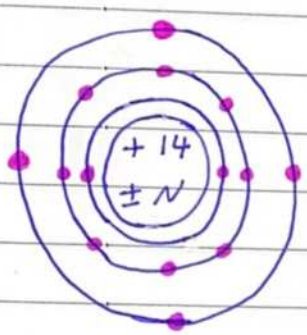
ليث، عمرو

Switch (المفتاح) Consists (يحتوي) From Transistor ; And The Transistor Consists From Two Diode ..

Semi Conductor ..

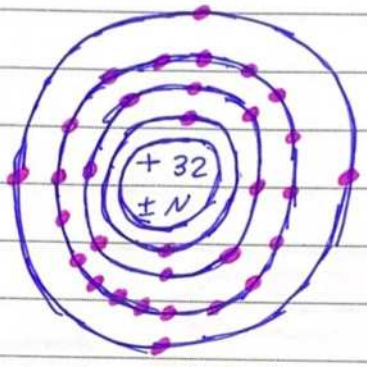
● → electron

Rep. [Si] 14



● → 2, 8, 4

Rep [Ge] 32

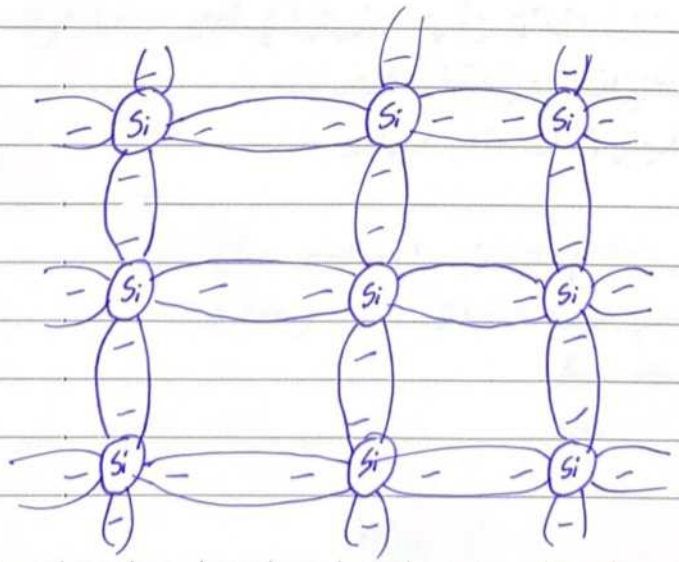


● → 2, 8, 18, 4

⊖ أيهم الأكثر إستقراراً : Si ← لأن المدار الأخير أقرب للنواة بالتالي طاقة الربط أكبر .

Covalent Bounds : (رابطة تساهمية)

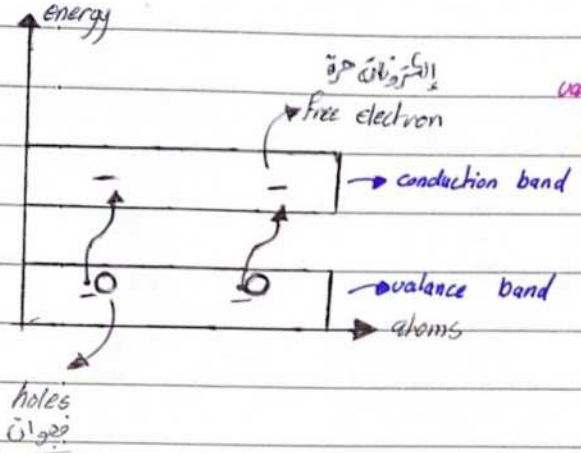
↳ • تشارك الأيونات دون فقدان أو كسب



• Piece of pure Crystal
بلورة نقية خالية من الشوائب
(رابطة قوية جداً)

ليث، عمرو

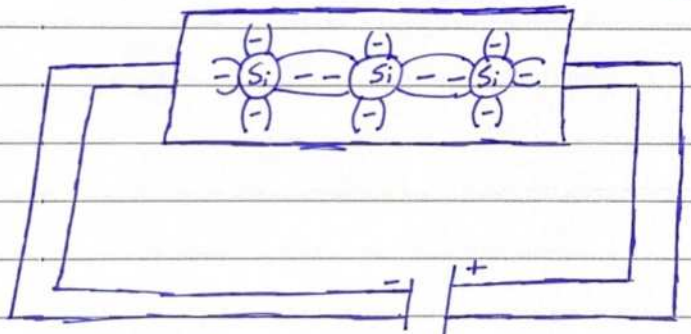
holes and free electron :-



١) تنتقل الإلكترونات من الـ valance band إلى الـ conduction band فتترك مكانها فجوة تسمى الـ holes ..

٢) يأتي إلكترون آخر ويستقر في الـ holes ..

• الذرة بطبيعتها تبحث عن الإستقرار



* حتى تتطرد الإلكترونات يجب أن تكون الطاقة (قوة الشوئية) أعلى من قوة الربط بالنواة وبالتالي تنجذب الإلكترونات نحو القطب الموجب ..
* تنتقل الإلكترونات من اليسار لليمين وتترك مكانها فجوات ..
* الأقرب للبطارية ينتقل أولاً
* ينشأ تيارين :: ① تيار من حركة الإلكترونات (free electrons)
② تيار معاكس نتيجة الفجوات (holes)
* حاملات التيار :: ① free electrons (current carries)
② holes

ليث، عمرو

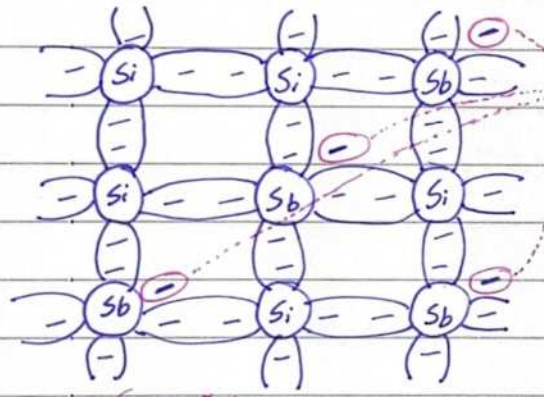
Doping (التطعيم) :: Adding impurity (غير نقية) atoms to pure silicon (Si) or Germanium (Ge)..

إضافة ذرات غير نقية للسيليكون النقي أو الجرمانيوم النقي..

Doping (أقسام) ::

*n-type :: Adding impurity atoms that five negative [Free electron] 5-th Group on valance orbit to pure silicon or Germamium..

5th Group Examples : Sb , As , P , N , Bi

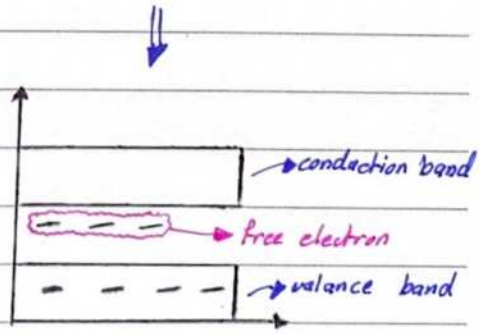


Sb :: Donor Atoms (مانحة للإلكترون)

Free electron :: Majority current carriers.. (حاملات التيار الأساسية)

holes :: Minority current carriers.. (حاملات التيار الثانوية)

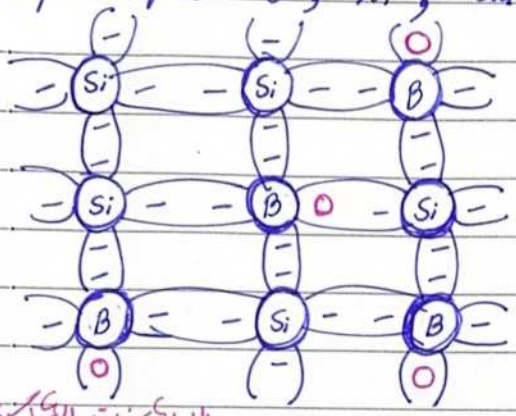
* الـ Free electron دائماً بال energy gab إذا تم توجيهه وتزويده بالطاقة يذهب لـ Conduction band



ليث، عمرو

* p-type : Adding impurity atoms that have 3 valance electrons to the pure Silicon or Germanium .. 3th Group
(Si) (Ge)

3th Group Examples: B , Al , Ga , In , Tl



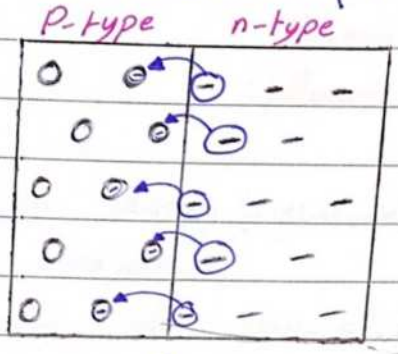
O : holes

B : acceptor atoms
الإلكترونات المكتسبة

* عند تزويده بمصدر جهد الـ holes يصبح عددها أكبر بالتالي : holes : Majority current carriers
Free electron : Minority current carriers

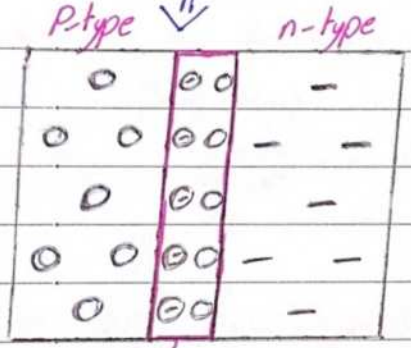
ليث، عمرو

Diode Formation :: (.. تشكيل الدايود ..)



* لا تنتقل جميع الإلكترونات لأن الإلكترونات في P-type تتنافر مع الإلكترونات القادمة من n-type عنصرها و أي أن

قوة التنافر > قوة الجذب
أقوى



* ينشأ مجال كهربائي ينتقل من الموجب إلى السالب وينشأ عنه قوة دافعة كهربائية (فولتية) عاكسة على المجال الكهربائي ..

Depletion Region
(إستنزاف الإلكترونات)

Si ⇒ $V_p = 0.7$ V

Ge ⇒ $V_p = 0.3$ V

GeAs ⇒ $V_p = 1.2$ V

ليث، عمرو

* تشغيل الـ Diode ..

1) Forward Biasing
2) Reverse Biasing

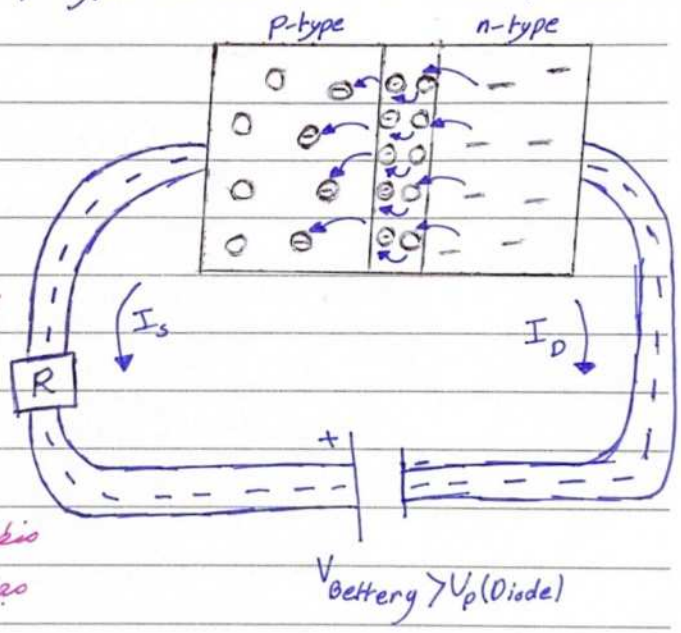
Forward Biasing (إنحياز أمامي) :

Two Conditions so

A Connect the n-type of the diode with the negative side of the Bettery & Conneat p-type of the diode with the positive side of the Bettery.

B $U_{Bettery} > U_{p\,(Diode)}$

$U_{Bettery}$ جهد البطارية . $U_{p\,(Diode)}$ جهد الدايود



* Depletion Region : تقل وتصغر لكن لاتختفي

Resistence المقاومة

لابد أن يكون هناك إستقطاب لحظي

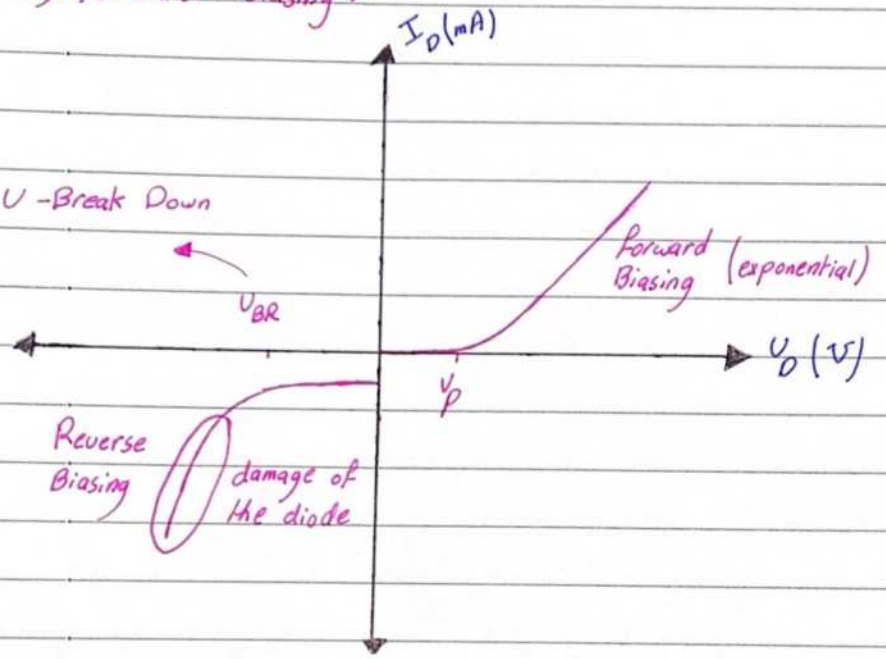
* Diode not ideal { منطقة الإستنزاف / مجال قوة داخلية

* آلية العمل : تنشأ قوة التنافر بين free electron الموجودة في n-type مع القطب السالب للبطارية ولازم قوة التنافر تكون أعلى من V_p حتى تنتقل الإلكترونات تتم عملية إختراقه للحاجز ، في قوة جذب مع الـ Bettery positive side وتتكرر العملية فتصبح هناك حركة الإلكترونات في مسار مغلق بالتالي ينشأ تيار كهربائي (closed loop) - Switch - (إذا كان عبارة عن Dide)

ideal → $V_p = 0$

ليث، عمرو

Cont, Forward Biasing :-



((Characteristive Curve of the diode))

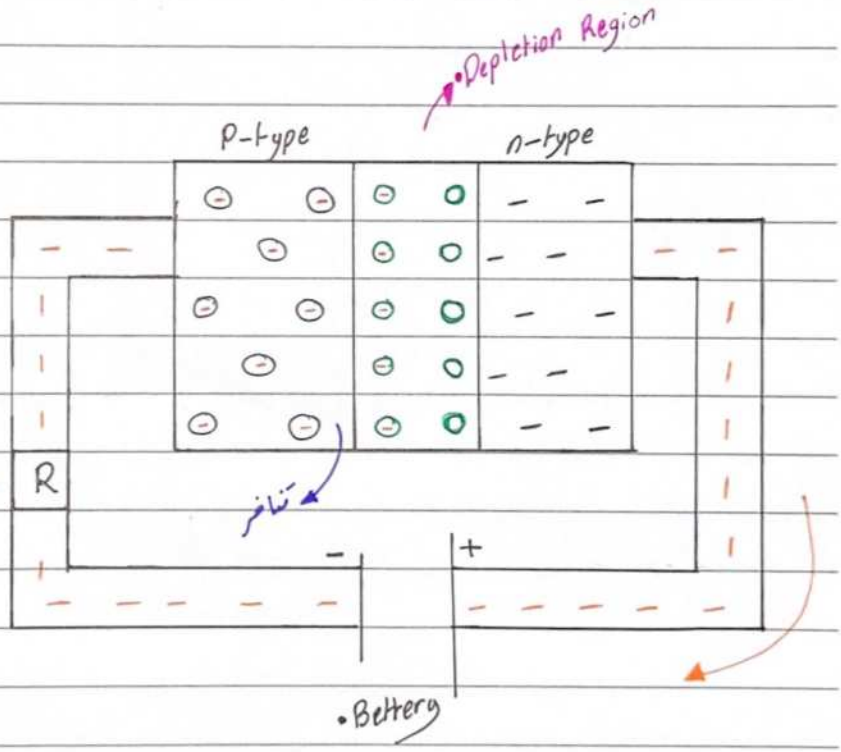
ليث، عمرو

Reverse Biasing :

One Condition :

A Connect the n-type of the Diode with positive side of the Battery

B connect p-type of the Diode with negative side of the Battery.



* آلية العمل : الـ Free electron تنجذب نحو القطب الموجب للبطارية بعض و تستمر بالفجوات (الموجودة بالـ p-type) ، و في p-type و تتنافر مع بعض و تمنعها من الدخول لـ p-type و بالتالي ينقطع التيار الكهربائي ..

* Depletion Region will increase → open ← Switch عبارة عن Diode لو كان الـ

* التيار قريب من الصفر و لكن لا يساوي صفر لأن diode ليس ideal مثالي

* إذا زدنا قيمة الـ Battery قيمة التنافر تزداد و تتفرق الإلكترونات الـ Depletion Region بالتالي الـ Diode يتضرر .. The Diode will be damage

ليث، عمرو

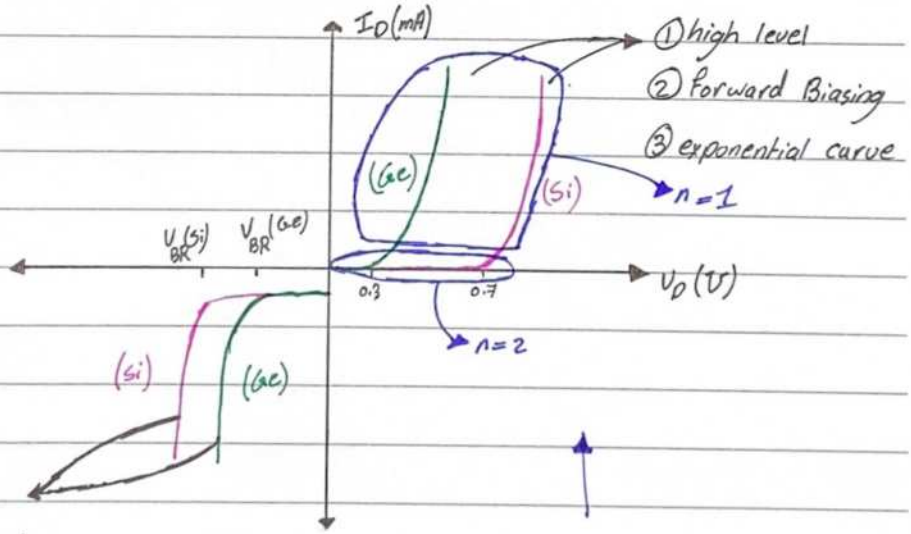
Sympol of Diode :- (رمز الدايود)

*أي رمز كهربائي له صفين بالتركيب ..

① + Anode — Cathode −

② + P n −

Characterstic Curve of the Curve :-



n: integer number

Shockley's equation :- $I_D = I_s\left(e^{(V_D/n.V_T)} - 1\right)$

I_D : Forward Biasing

I_s : Reverse Biasing

V_D : voltage Diode

$V_T = \frac{K \cdot T_K}{q}$; $q = 1.6 * 10^{-19}$; $K = 1.38 * 10^{-23}$

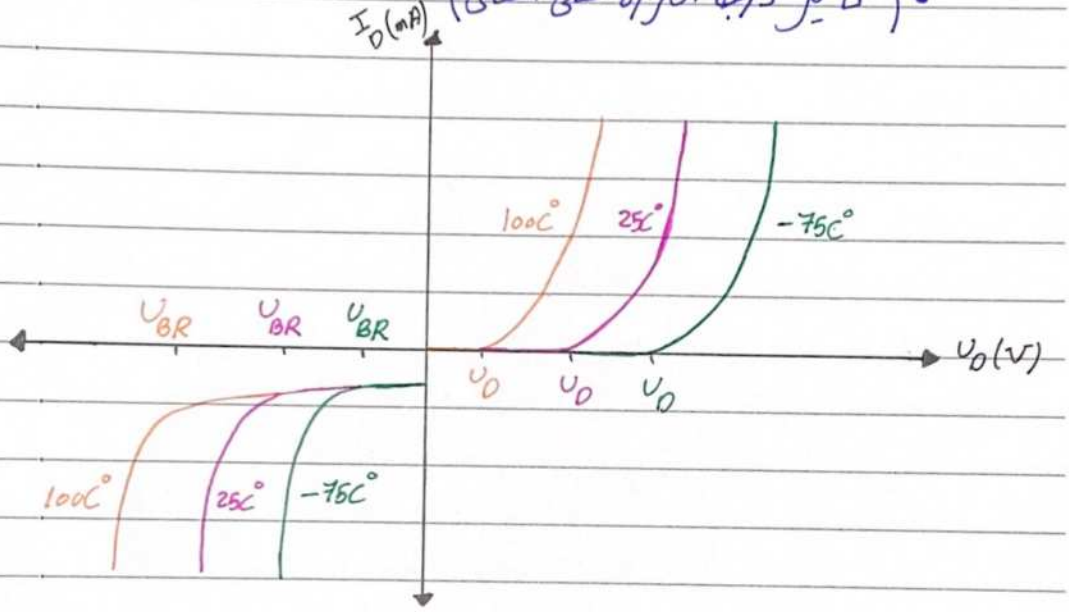
T_K : Temperature in Kelvin

$T_K = T_C^\circ + 273^\circ$

ليث، عمرو

Effect of tempreature on the Characterestic Curve:.

• (تأثير درجة الحرارة على المنحنى)



* إذا زادت درجة الحرارة في حالة Forward :

- حركة الإلكترونات أسرع (التيار) بالتالي Depletion Region تصغر ولكن لا تختفي ، الـ V_D و الـ V_{BR} إزاحة لليسار ..

* إذا قلّت درجة الحرارة :

- حركة الإلكترونات سرعتها تقل (التيار) بالتالي Depletion Region تزداد و الـ V_D والـ V_{BR} إزاحة لليمين ..

ليث، عمرو

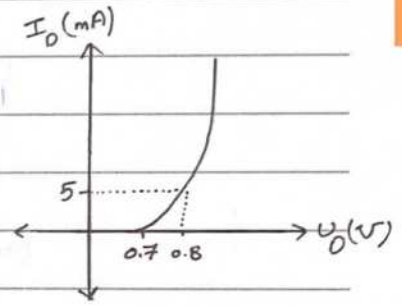
Diod Two Types :: 1) not ideal → هنالك مستويات للمقاومة بكن حسابها

2) ideal → $U_D = IR$

$R = \frac{U}{I}$

Resistance levels
مستويات المقاومة

1) DC or Static Resistance (R_D) :

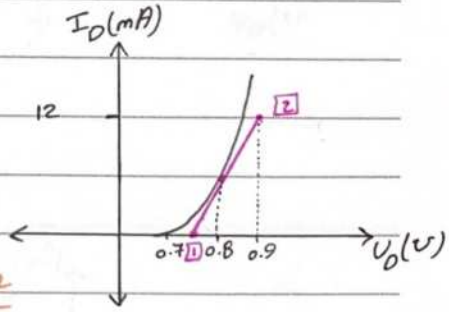


Suppose $U_D = 0.8$ V Find R_D.

$R_D = \frac{U}{I} = \frac{0.8}{5*10^{-3}} = 160\ \Omega$

* في حالة DC نأخذ نقطة وحيدة U_D و I_D ونقابلها ع منحنى يعطينا I_D أو U_D وأنا أوجد R_D ..

2) AC or Dynamic Resistance (r_d) :



Suppose $V_D = 0.8$ V Find r_d.

$r_d = \frac{\Delta V}{\Delta I} = \frac{U_2 - U_1}{I_2 - I_1} = \frac{0.9 - 0.75}{12*10^{-3} - 0} = 12.5\ \Omega$

$r_d = \frac{26\ mV}{I_D}$

* في حالة AC نأخذ نقطة ومماس عند هذه النقطة ثم نأخذ نقطتين من أطراف المماس ونقسم فرق الجهد على فرق التيار ..

ليث، عمرو

3) Average Resistance (r_{avg})

Suppose $V_1 = 0.8\ V$, $I_1 = 5\ mA$

$V_2 = 0.75\ V$, $I_2 = 4\ mA$

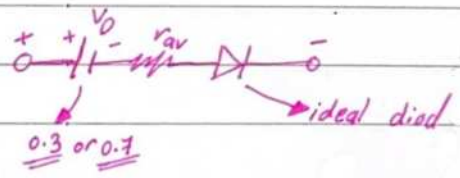
Find r_{avg} :-

$$r_{avg} = \frac{V_2 - V_1}{I_2 - I_1} = \frac{0.75 - 0.8}{(4-5)*10^{-3}} = 50\ \Omega$$

[Equivelent Circuits of the Diods]

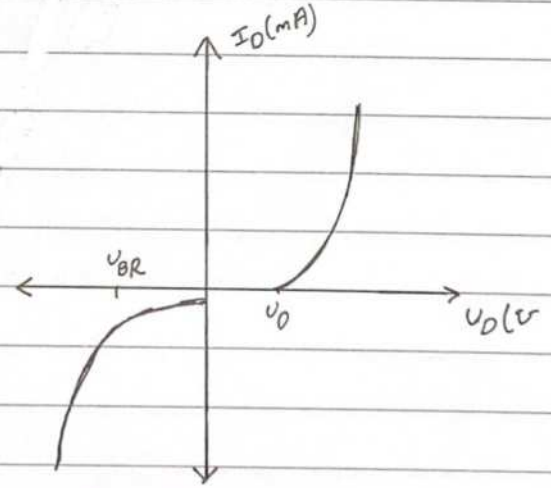
(■ الدوائر المكافئة للديود ..)

[1] Piece-wise linear equation :-

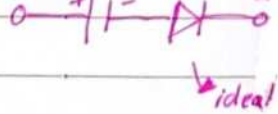


0.3 or 0.7

V_D : value not fixed



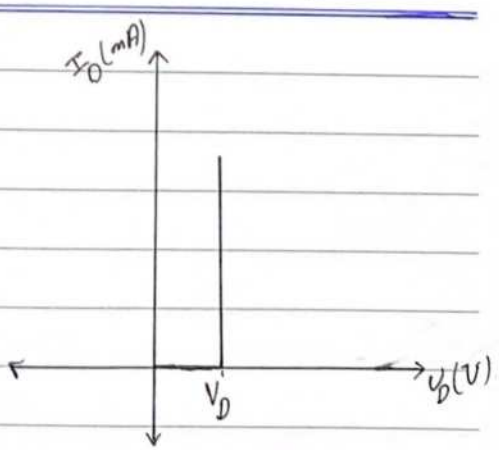
[2] Simplified :-



$r_{av} = 0$

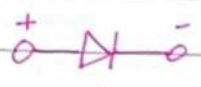
V_f : Fixed value

$r_{av} = \frac{V_2 - V_1}{I_2 - I_1} = 0 \rightarrow V_2 = V_1$



ليث، عمرو

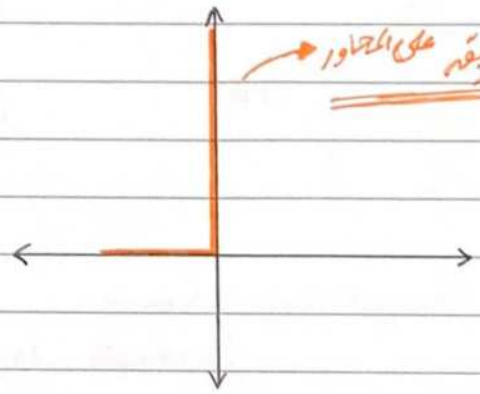
3 ideal :-



ideal diod

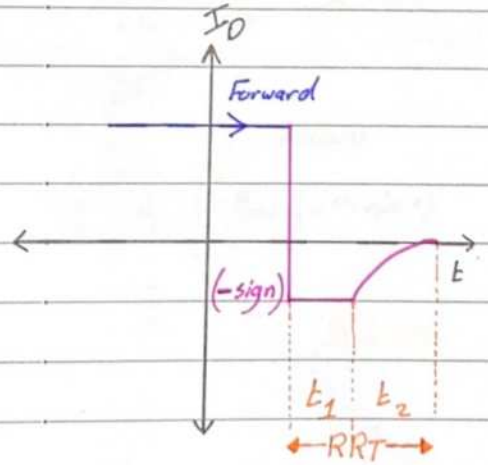
$r_{av} = 0$

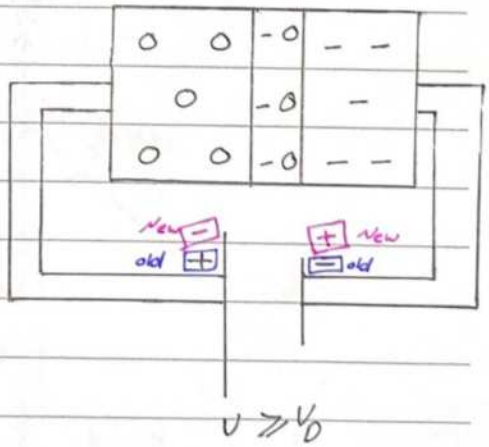
$V_D = 0$



• مُنطبِقة على المحاور

Reverse Recovery Time (RRT)





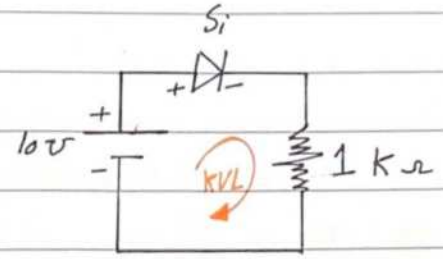
$RRT = t_1 + t_2 \approx 10^{-12}$ (very small)

* نعكس الجهد فتصبح العلبة .. Reverse

ليث، عمرو

Ch.2 :: Diods Applications

#Load Line (خط الحمل) ::



$-10 + V_D + I_D R = 0$

$-10 + 0.7 + I_D(1000) = 0$

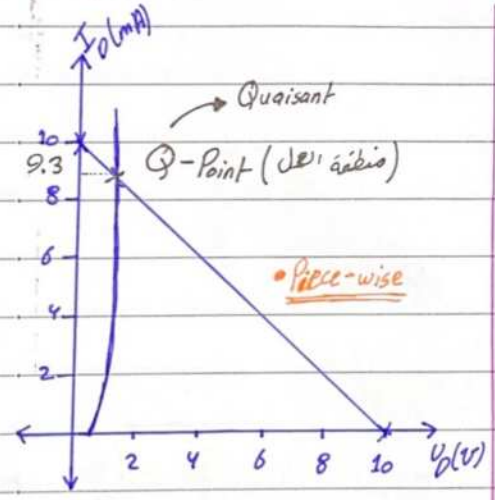
$I_D = 9.3\ mA$

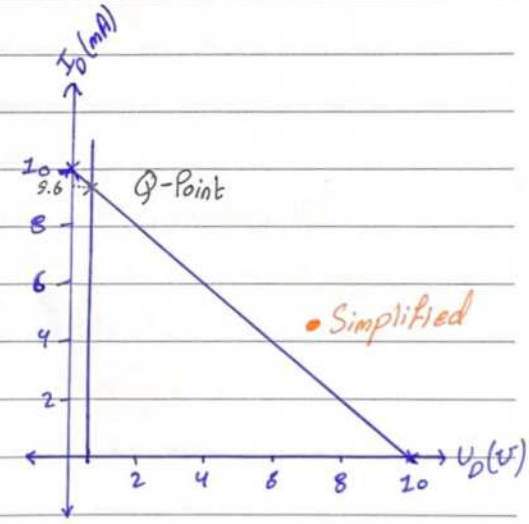
* قبل الحل :: ١ طريقة التوصيل : Forward ① Reverse ②

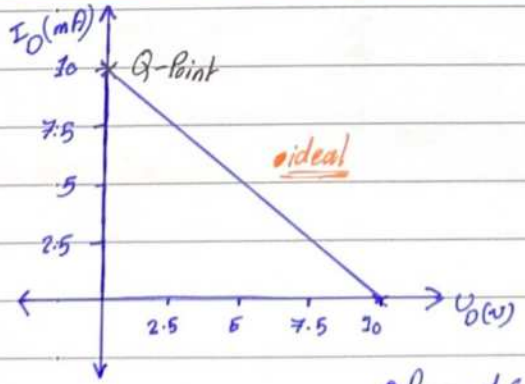
• when: $I_D = 0 \rightarrow V_D = 10\ V$ (10,0)

$V_D = 0 \rightarrow I_D = 10\ mA$ (0,10)

(لإيجاد النقطتين اللي مر فيها الـ Load Line)







□ في غرفة بعيت I :

• Piece-wise $I_D = 9.3\ mA$

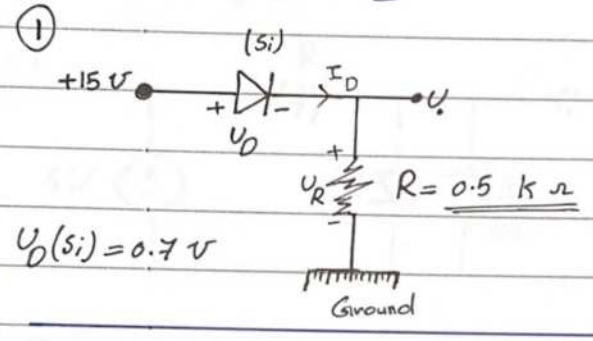
• Simplified $I_D = 9.6\ mA$

• ideal $I_D = 10\ mA$

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Examples: [series]

(1)



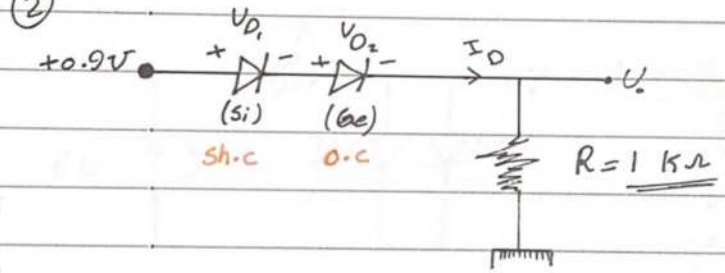
$-15 + V_D + V_R = 0$

$-15 + 0.7 + I_D(0.5\,k) = 0$

$I_D = \frac{14.3}{0.5\,k} = 28.6\ mA$

$V_o = I_D(R) = 14.3\ V$

(2)



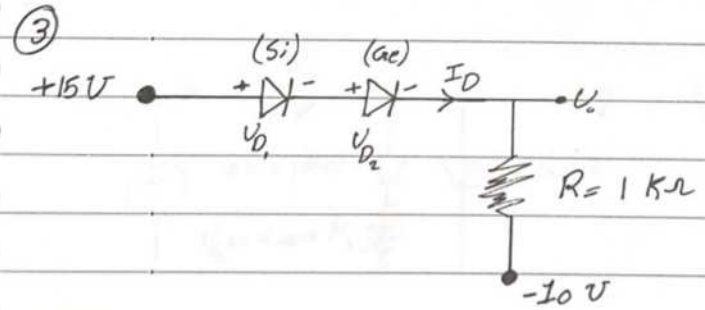
$I_D = 0$

$V_{D_1} = 0$ (short circuit)

$V_{D_2} = 0.9$ (open circuit)

$V_o = 0$

(3)



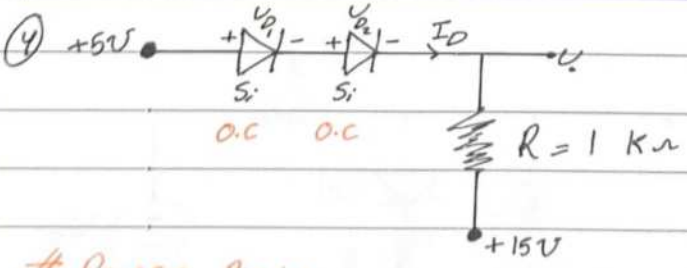
$-15 + V_{D_1} + V_{D_2} + I_D(R) - 10 = 0$

$I_D = \frac{24}{1\,K} = 24\ mA$

$V_o = I_D R - 10$

$V_o = 14\ V$

(4)



* إذا طلع عنّا مصدرين وأكثر بنشوف إتجاه الأكبر..

Reverse Biasing

$I_D = 0$

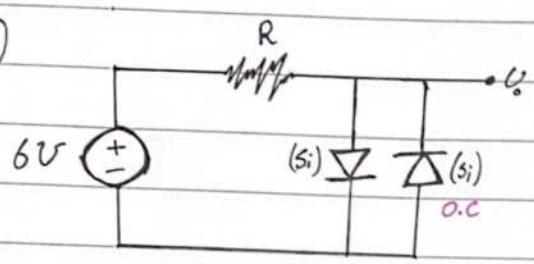
$V_o = 15\ V$

$15 - 5 + V_{D_1} + V_{D_2} = 0$

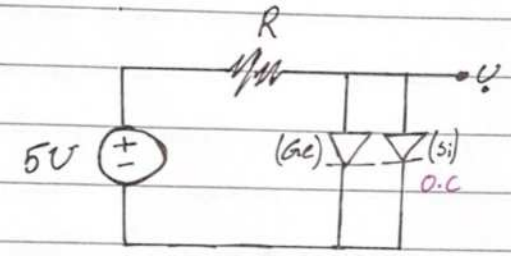
$V_{D_1} = V_{D_2} = -5\ V$

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#Examples [parallel]

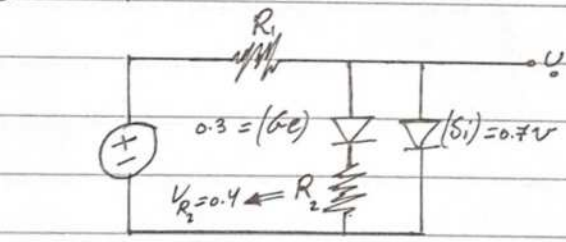
① 

$V_o = 0.7\ V$

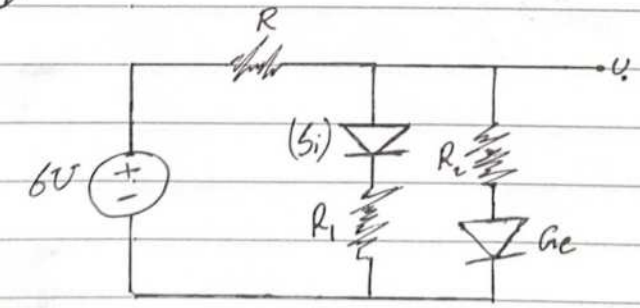
② 

$V_o = 0.3\ V$ ← الجهد الأقل

* V_o تكون نفس جهد الأقل مروراً والثاني open circuit ولذلك لا يمكن أن يكون الجهد مختلف بالتوازي

③ 

$V_o = 0.7\ V$

④ 

حالتين

Two Cases:

1) $R_1 = R_2 \Rightarrow V_o = 0.3\ V$

2) $R_1 \neq R_2 \Rightarrow$ We must apply KVL to Solve

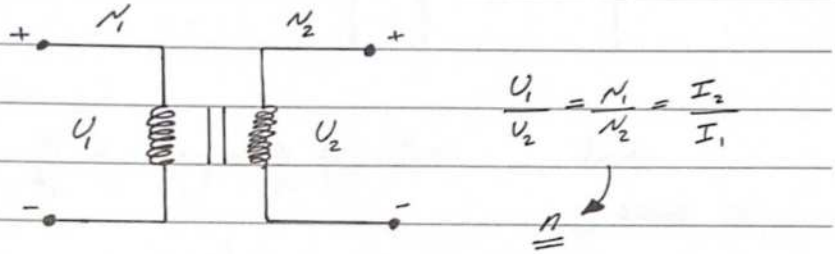
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AC Application (Diod) 8-

1. المحول

1 Transformer : " Convert from AC Signal to AC Signal with Changing second Signal .. "

* جهاز يقوم بتحويل من جهود منخفضة إلى جهود عالية أو العكس ..



$$\frac{V_{rms}}{} = \frac{V_p}{\sqrt{2}}$$

* آلية العمل : " عند مرور تيار كهربائي في الملف الإبتدائي يتولد مجال مغناطيسي ينتقل عبر القلب الحديدي (core) إلى الملف الثانوي مما ينجم عنه نشوء تيار في هذه الملفات الثانوية " ..

UP-voltage : $V_2 > V_1$

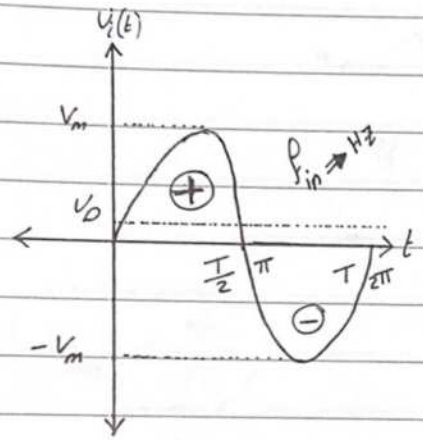
down-voltage : $V_1 > V_2$

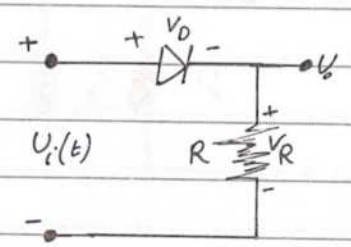
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#AC Application (Diod) 8-

2 Rectifier : "convert the AC Signal to DC Signal"

A. Half wave rectifier : (H.W.R)





• Find :
1] Sketch $U_o(t)$
2] V_{DC}
3] PIV
4] f_{out}

* positive half cycle : $(U_R = U_o)$

$$U_D + U_R - U_i(t) = 0$$
$$U_R = U_i(t) - U_D$$
$$U_R = U_m - U_D$$
$$U_o = U_m - U_D$$

* negative half cycle : $(U_R = U_o)$

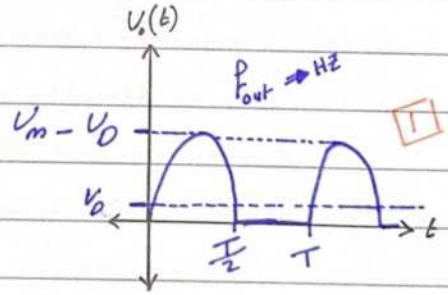
$I_D = 0$; Diode open circuit
$U_R = 0$.
$U_o = 0$.

* V_{DC} :

$$V_{DC} = 0.318\,(U_m - U_D)$$

$\frac{1}{\pi}$

2



1

* PIV : Peak Inverse Voltage

$PIV \geq V_{max}$ ← عادةً نبوخذ المساواة

$PIV = V_{max}$

3

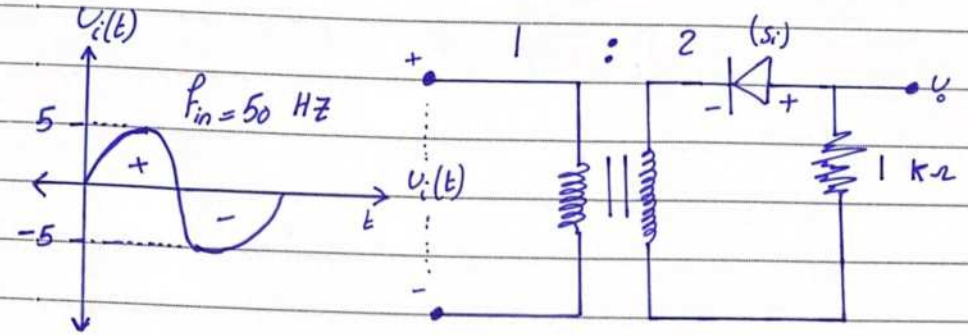
* f_{out} : Out Frequency

$f_{out} = f_{in}$

4

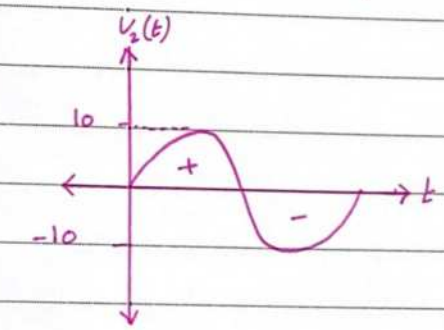
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Examples:



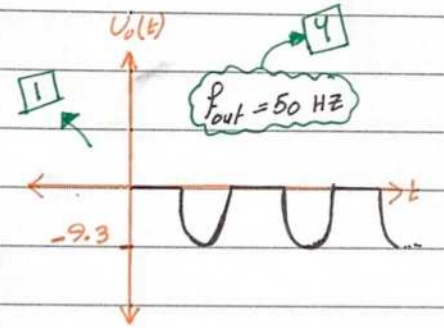
$$* \frac{V_1}{V_2} = \frac{N_1}{N_2} \Rightarrow V_2 = \frac{N_2}{N_1} V_1 \Rightarrow V_2 = \frac{2}{1}(5) = 10\ V$$

Find:
1] Sketch $v_o(t)$
2] V_{OC}
3] PIV
4] f_{out}



*Positive half cycle:
open circuit
$I_D = 0 \Rightarrow V_R = V_o = 0$

*negative half cycle:
$v_2(t) - V_D + V_R = 0$
$10 - 0.7 + V_R = 0$
$V_o = V_R = -9.3\ V$



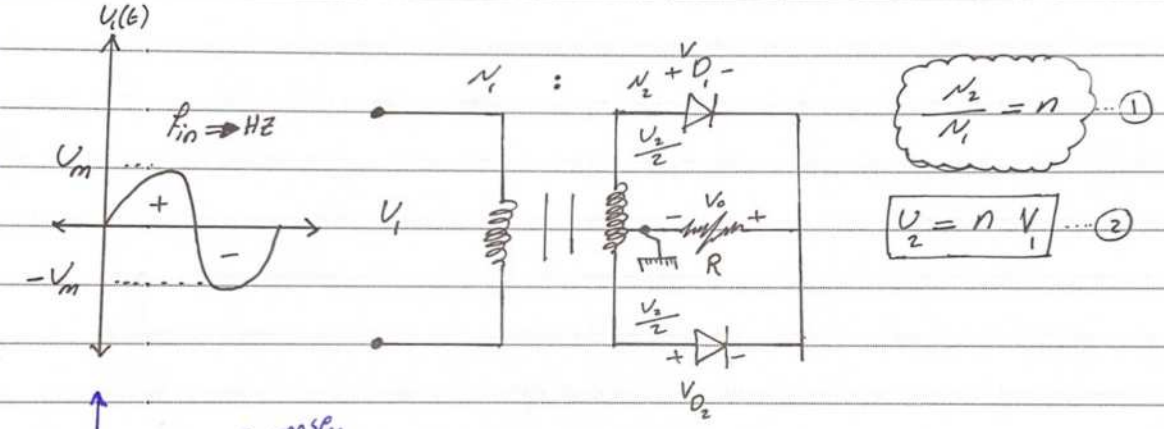
$$* V_{OC} = 0.318(-10 + 0.7) = -2.96\ V \rightarrow [2]$$

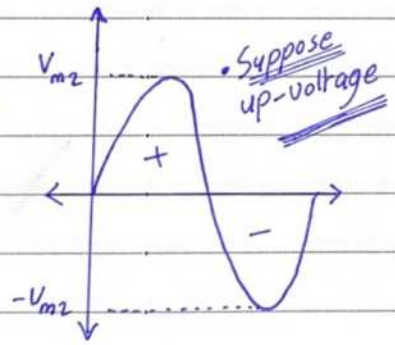
$$* PIV = 10\ V \rightarrow [3]$$

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B. Full wave rectifier: (F.W.R) → Center-tapped (F.W.R)
→ Bridge (F.W.R)

1) Center-tapped: ("Ground the center of secondry coil")





• Suppose up-voltage

*Positive half cycle:

$$V_{D_1} + V_o - \frac{V_2}{2} = 0$$

$$V_o = \frac{V_2}{2} - V_{D_1}$$

*negative half cycle:

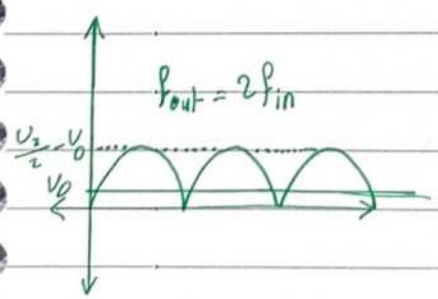
$$V_{D_2} + V_o - \frac{V_2}{2} = 0$$

$$V_o = \frac{V_2}{2} - V_{D_2}$$

$$*V_{DC} = 0.636 \left(\frac{V_{m2}}{2} - V_D \right)$$

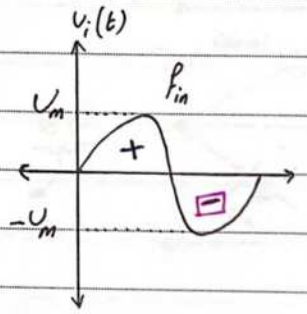
$\frac{2}{\pi}$

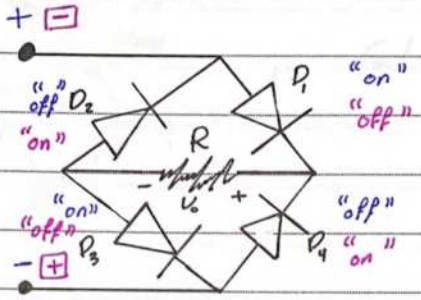
$$*PIV = V_{m2}$$



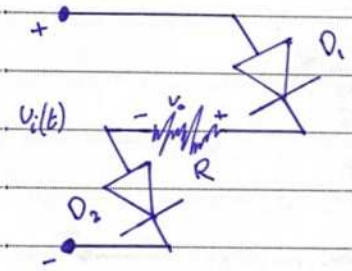
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2) Bridge : "النصف العلوي والنصف السفلي متساويات"





* Positive half cycle :-

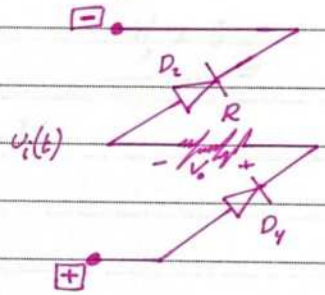


$$-v_i(t) + 2V_D + V_o = 0$$

$$V_o = v_i(t) - 2V_D$$

$$\boxed{V_o = V_{max} - 2V_D}$$

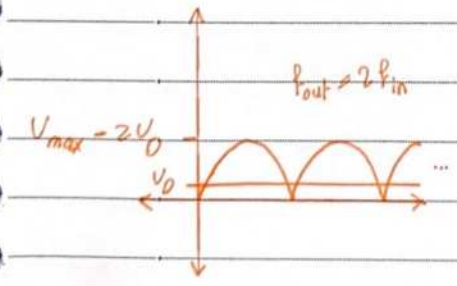
* Negative half cycle :-



$$v_i(t) - 2V_D - V_o = 0$$

$$V_o = v_i(t) - 2V_D$$

$$\boxed{V_o = V_{max} - 2V_D}$$



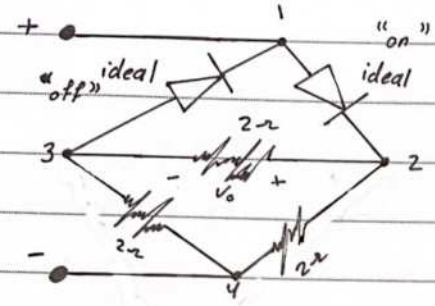
- $V_{DC} = 0.636\,(V_{max} - 2V_D)$

V_p (rectifier)

- $PIV = V_{max}$

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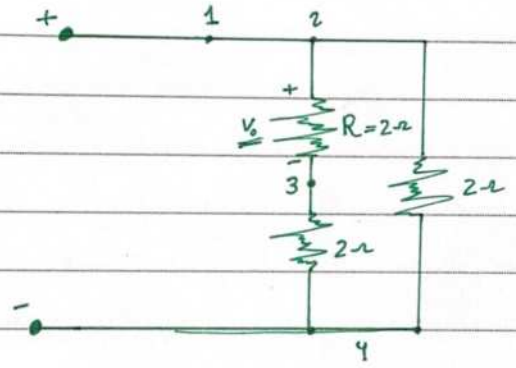
Example:



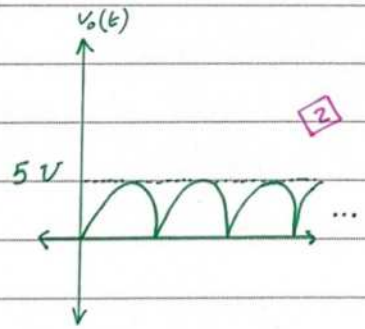
Find:- 1) $V_o(t)$

2) Sketch $V_o(t)$

V_{rms}



$V_o = 5\,V$ [1]

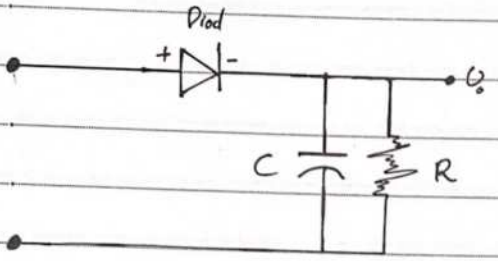


[2]

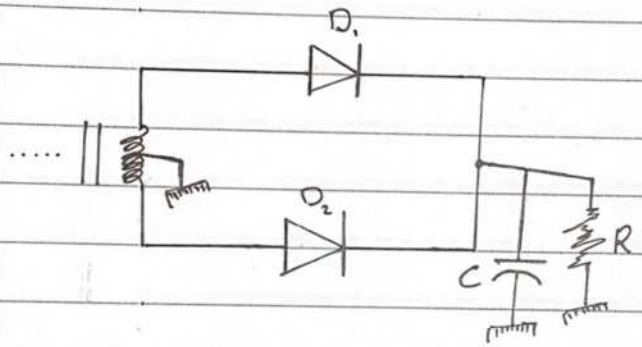
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AC Application (Diod) :-

3 Filter: "دائرة قادرة على تمرير (أو تضعيف) ترددات معينة وتضعيف أخرى"

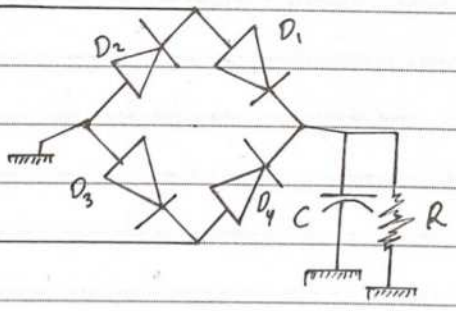


• Rectifier + Filter



• Rectifier + Filter

(center-tapped)

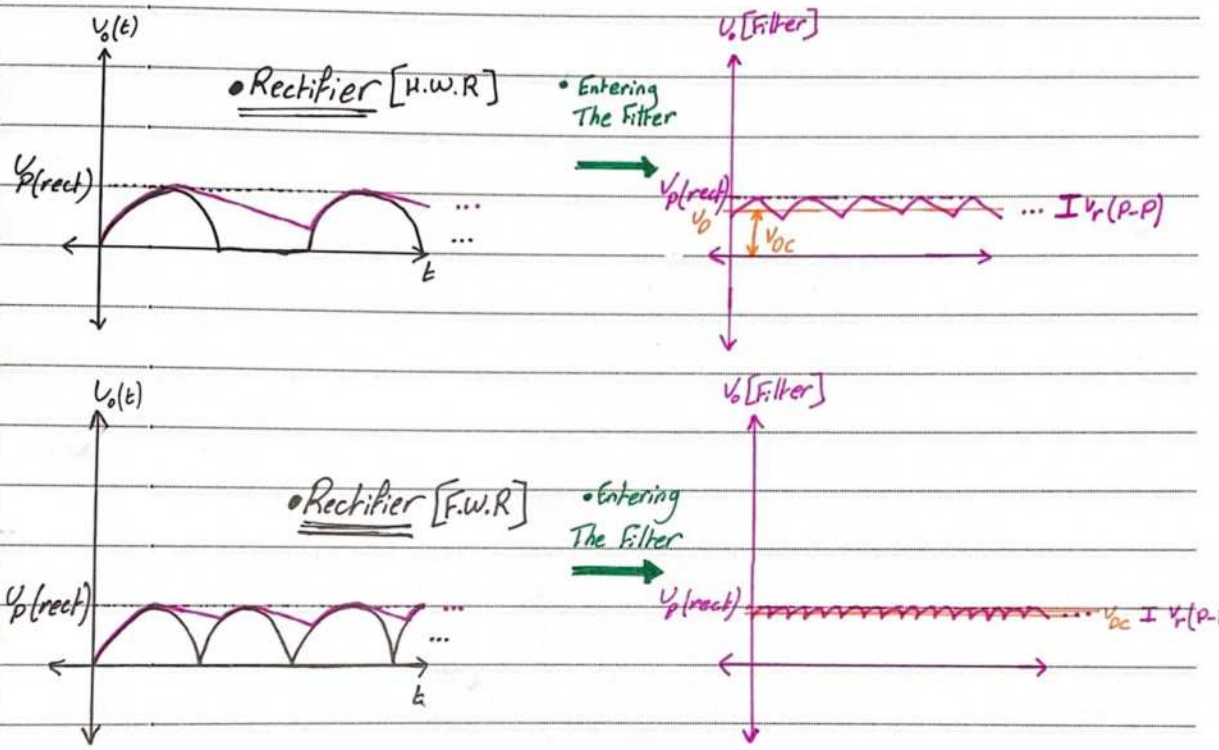


• Rectifier + Filter

(Bridge)

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Filter Graphically :

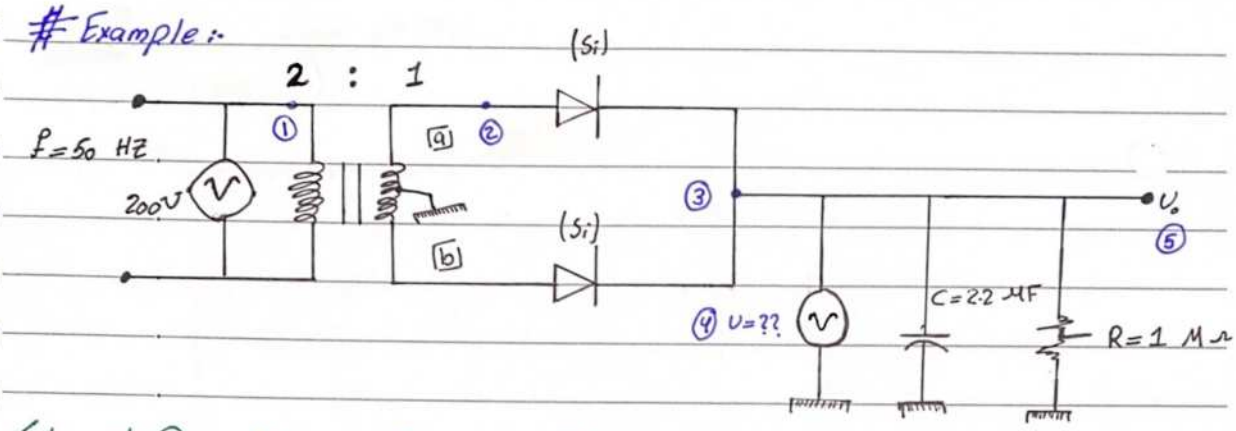


* Ripple Factor :: (معامل التموّج) • Ripple : "التموّج الذي حدث على موجة DC"

- $r = \frac{V_{r\,(P-P)}}{V_{dc\,(ripple)}} * 100\%$
- $V_{r(P-P)} = \frac{1}{f_{out} * RC} * [V_{p(rect)}]$
- $V_{dc\,(ripple)} = \left[1 - \frac{1}{2 f_{out} RC}\right] * [V_{p(rect)}]$

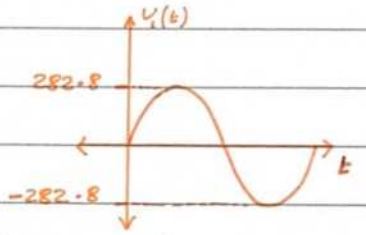
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Example :



Sol:: at ① : $V_{rms} = 200\ V \Rightarrow V_p = \sqrt{2} * 200 = 282.8\ V$

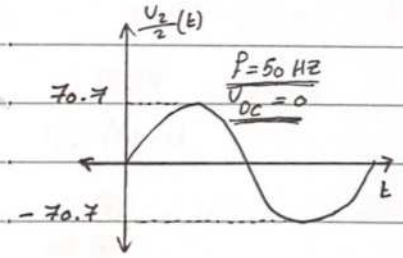
$f = 50\ Hz$, $V_{DC} = 0$



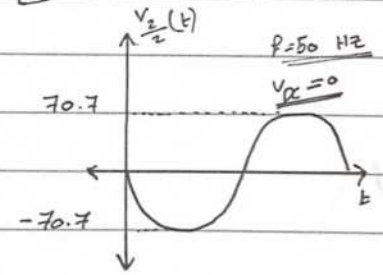
at ② : $\frac{V_2}{V_1} = \frac{N_2}{N_1} \Rightarrow V_2 = \frac{1}{2}(282.8) \Rightarrow V_2 = 141.4\ V$

$f = 50\ Hz$, $V_{DC} = 0$

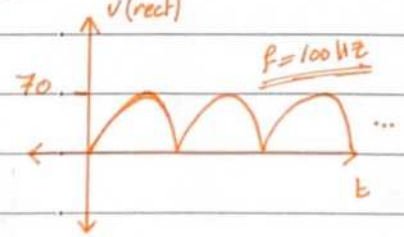
a



b



at ③ : The output of rectifier :



• $V_{DC} = 0.636[70] = 44.52\ V$

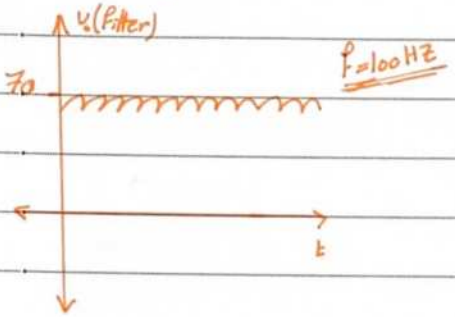
at ④ : The value of voltage :

$V_{DC} = 44.52\ V$

• Page 27

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at ⑤: The output of Filter:



- $V_r(P\text{-}P) = \frac{1}{f_o RC}\left[V_p(rect)\right]$

$$V_r(P\text{-}P) = \frac{1}{(100)(1*10^6)(2.2*10^{-6})}\left[70\right] = 0.32\ V$$

- $V_{DC} = \left[1 - \frac{1}{2f_o RC}\right]\left[V_p(rect)\right]$

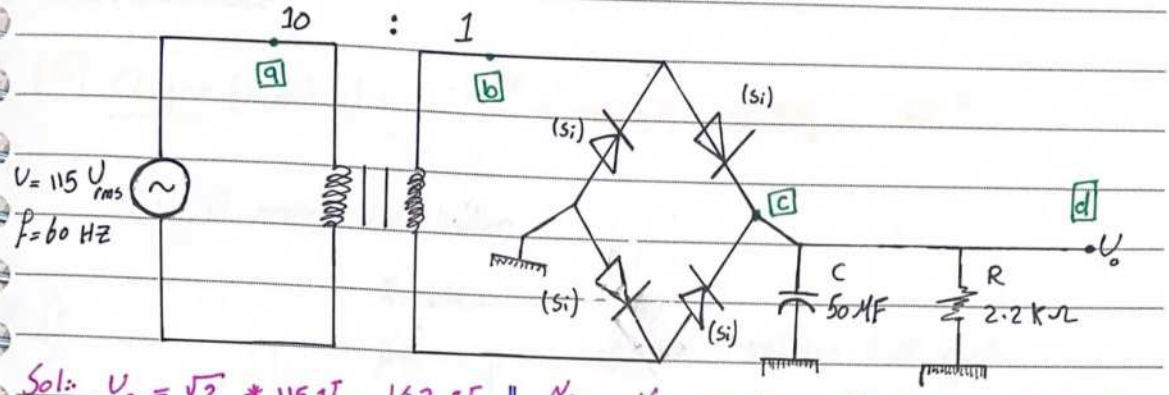
$$V_{DC} = \left[1 - \frac{1}{2(100)(10^6)(2.2*10^{-6})}\right]\left[70\right] = 69.84\ V$$

- $r\% = \frac{V_r(P\text{-}P)}{V_{DC}(Ripple)} * 100\%$

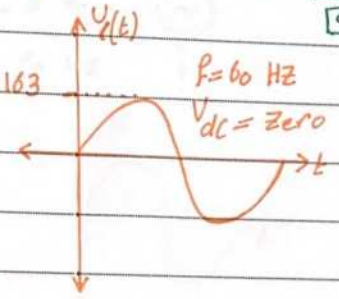
$$r\% = \frac{0.32}{69.84} * 100\% = 0.46\%$$

ليث، عمرو

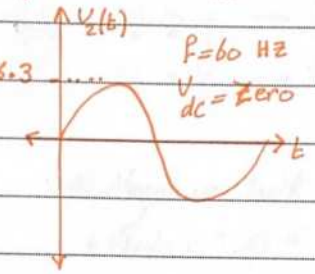
• Extra Example :-



Sol:- $V_p = \sqrt{2} * 115\ V = 163\ V$ [a]



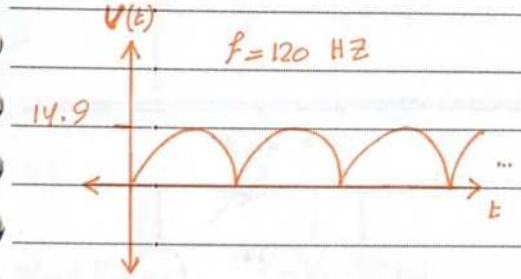
$\frac{N_2}{N_1} = \frac{V_2}{V_1} \Rightarrow V_2 = \frac{1}{10}(163) = 16.3\ V$ [b]



• After Rectifier (Bridge) :

$V_{p(rect)} = V_{m_2} - 2V_D \Rightarrow V_{p(rect)} = 14.9\ V$ [c]

($V_{m_2} = 16.3$, $2V_D = 0.7$)



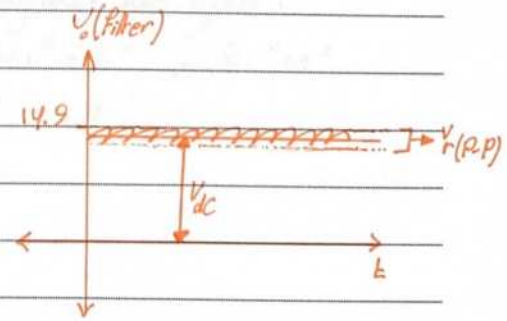
$V_{dc} = 0.636(14.9) = 9.48\ V$

• After Filter :

$V_{r(P-P)} = \frac{1}{fRC}(V_{p(rect)}) = 1.13\ V$

$V_{dc(Ripple)} = \left[1 - \frac{1}{2fRC}\right] * V_{p(rect)} = 14.3\ V$

$r = \frac{1.13}{14.3} * 100\ \% = 7.90\ \%$



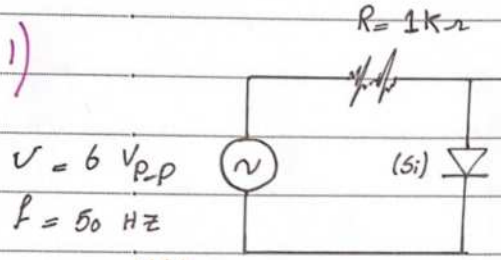
ليث، عمرو

AC Application (Diod) :-

[4] Clipper (Limiter) : "قص الموجة أو تحديدها"

H.W.R → part of limiter

1)



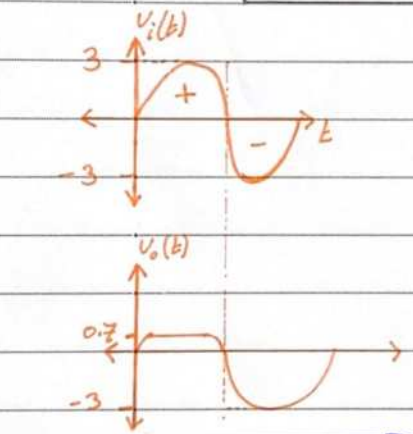
$V_o(t) = V_i(t)$

*Positive half cycle:

$V_i < V_D$ → "off" o.c

$V_i \geq V_D$ → "on"

$V_o = V_D = 0.7\,V$



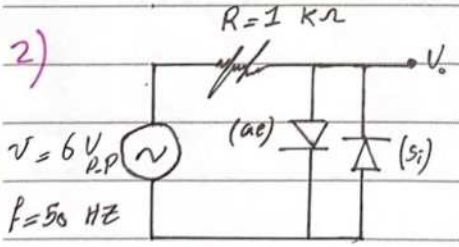
* Negative half cycle:

• Diode : "off"

open circuit (o.c)

$V_o(t) = V_i(t)$

2)



*Positive half cycle:

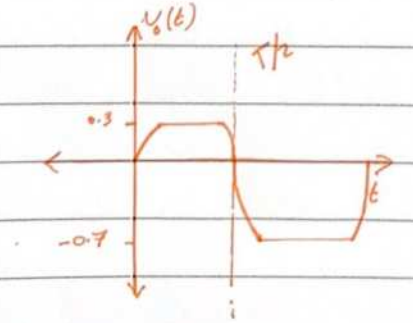
Ge : "on" , Si : "off"

$V_o = 0.3\,V$

* Negative half cycle :

Ge : "off" , Si : "on"

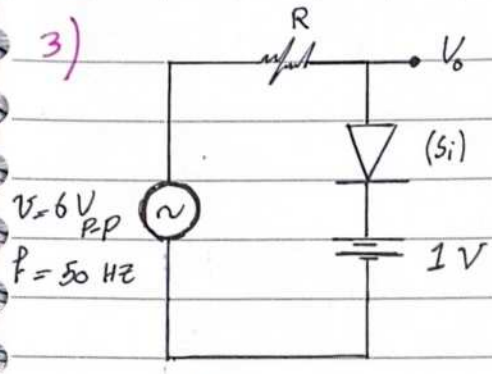
$V_o = -0.7$

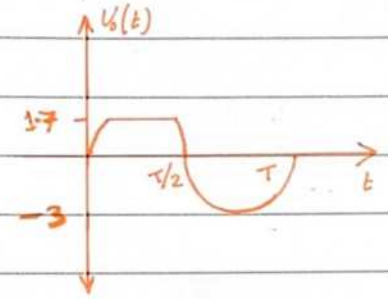


• Page 30

ليث، عمرو

3)





* Positive half cycle:

AC : Forward , DC : Reverse

$V_{DC} > V_{AC}$ ⟹ Reverse , Diode o.c

$V_o = V_i(t)$

$V_{AC} > V_{DC}$ ⟹ Forward , Diode "on"

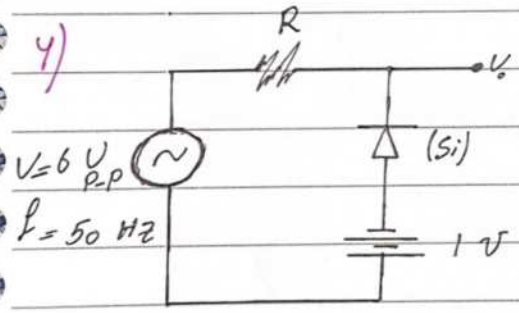
(KVL) → $V_o = 1 + 0.7 = 1.7\ V$

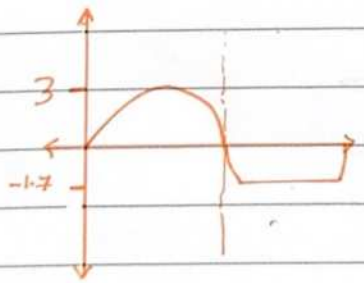
* Negative half cycle:

Diode "off" → open-circuit

$V_o(t) = V_i(t)$

4)





* Positive half cycle:

Diode "off" → open-circuit

$V_o(t) = V_i(t)$

* Negative half cycle:

AC : Forward , DC : Reverse

$V_{DC} > V_{AC}$, Diode "off"

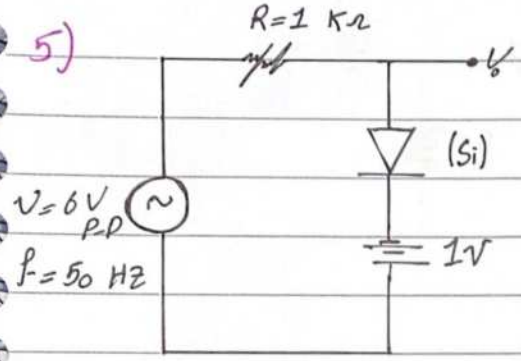
$V_o(t) = V_i(t)$

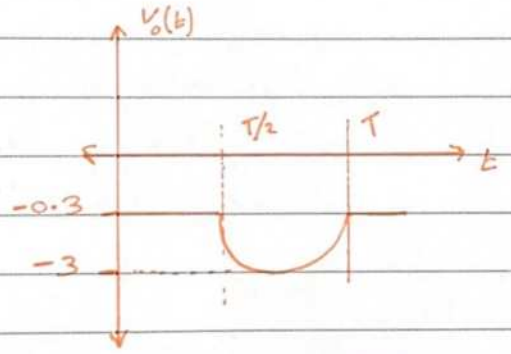
$V_{DC} < V_{AC}$, Diode "on"

(KVL) $V_o = -1 - 0.7 = -1.7$

ليث، عمرو

5)





*Positive half cycle:

DC And AC Forward.

$V_o = -0.3\ V$

*Negative half cycle:

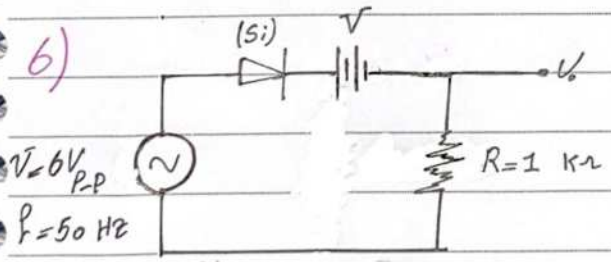
Diode "off" → open-circuit

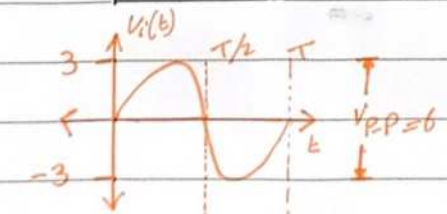
$V_o(t) = V_i(t)$ when $V_{AC} > V_{DC}$

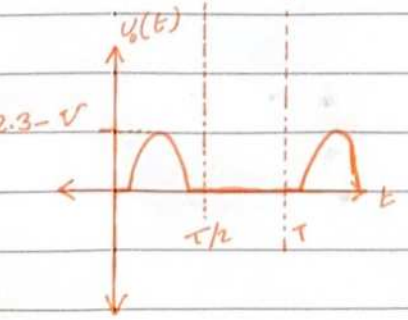
But when $V_{DC} > V_{AC}$ (KVL)

$V_o = -0.3\ V$

6)







*Positive half cycle:

$V_{DC} > V_{AC}$ → Diode "off"

→ open-circuit $V_o = V_R = 0\ V$

$V_{AC} > V_{DC}$ → Diode "on"

$V_o = 3 - V_D - V$

$V_o = 2.3 - V$

*Negative half cycle:

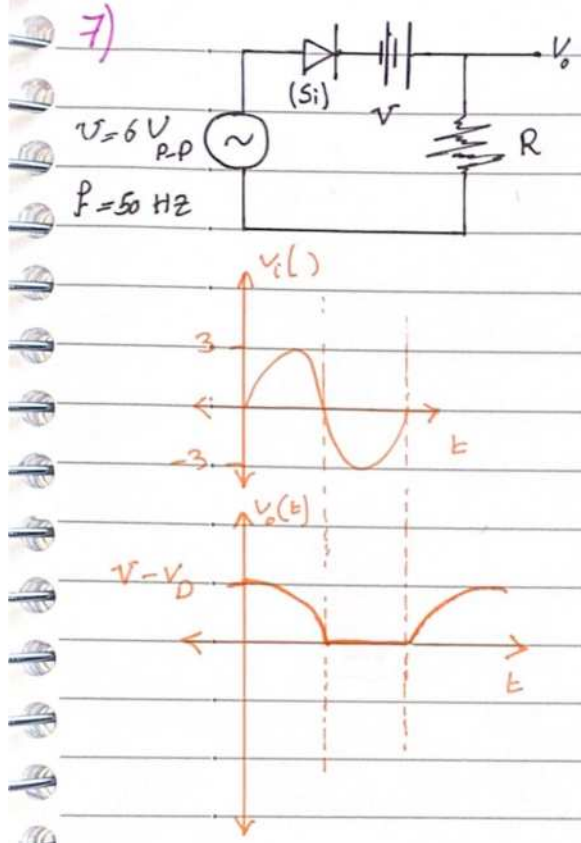
• Diode "off" → open-circuit

$V_o = V_R = 0\ V$

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ليث، عمرو

7)



* Positive half cycle:

V_{DC} : Forward V : Forward

(KVL) $V_o = V_i - V_D + V$ when $v_i = 0$

$V_o = -V_D + V$

* Negative half cycle:

V_{DC} : Forward V_{AC} : Reverse

$V_{DC} > V_{AC}$ → Diode "on"

$V_o \rightarrow V_i + V_D - V + V_o = 0$

$V_o = -V_i - V_D + V$ when $v_i = 0$

$V_o = -V_D + V$

$V_{AC} > V_{DC}$ → Diode "off"

$V_o = V_R = 0\ V$

ليث، عمرو

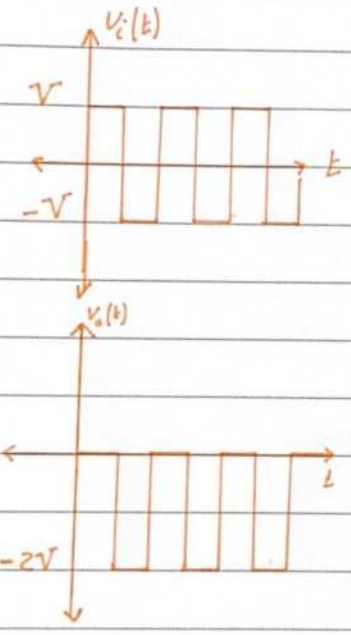
Clamper: "يقفز بالشحنة لأعلى أو لأسفل بمقدار معين (shift) ويحتوي على مكثف"

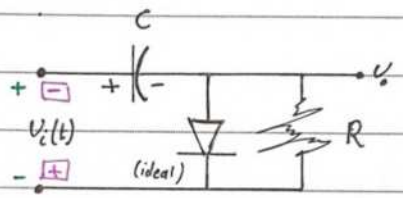
* خطوات الحل: ١) نختار الموجة اللي تعمل على تشغيل الـ Diode
+ Positive half cycle or - Negative half cycle

٢) نحسب V_c

٣) نحسب V_o

Example:





* Positive half cycle: (Diode "on")

$V_c - V = 0$ $\quad V_D(ideal) = 0\ V$

$V_c = V$

• $V_o = 0$ (لأنه توازي مع الـ Diode)

* Negative half cycle: (Diode "off")

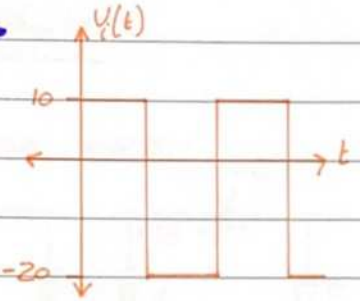
$V_c + V_o + V = 0$

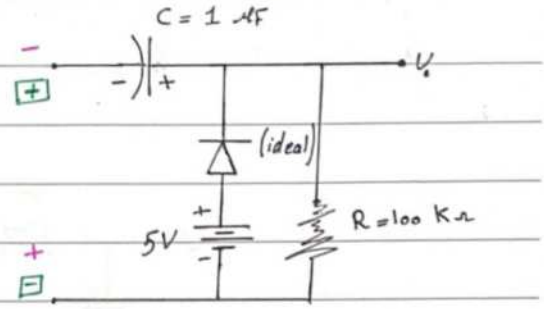
$V_o = -V_c - V$

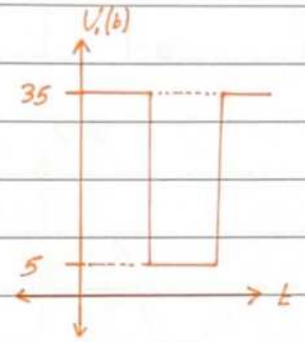
• $V_o = -2V$

ليث، عمرو

Example :-







* Negative half cycle : (Diode "on")

$20 - V_c + 5 = 0$

$V_c = 25\ V$

$V_o = 5\ V$

* Positive half cycle : (Diode "off")

$-10 - V_c + V_o = 0$

$-10 - 25 + V_o = 0$

$V_o = 35\ V$

$\tau = RC$ (زمن التفريغ)

$\tau = 0.1\ s$

* لو كان الـ Diode ليس مثالي $V_D = 0.7$ ← (Si) :

* Negative half cycle : (Diode "on")

$20 - V_c - V_D + 5 = 0$ $\boxed{V_D = 0.7}$

$V_c = 24.3\ V$

$V_o = 4.3\ V$

* Positive half cycle : (Diode "off")

$-10 - V_c + V_o = 0$

$V_o = +34.3\ V$

* ملاحظة : إذا لقيت قيمة V_o ← 0 ← يعني الـ Diode ← ideal

إذا لقيت قيمة V_o ← عدد صحيح يعني الـ Diode ← ideal وفي حالة نظري

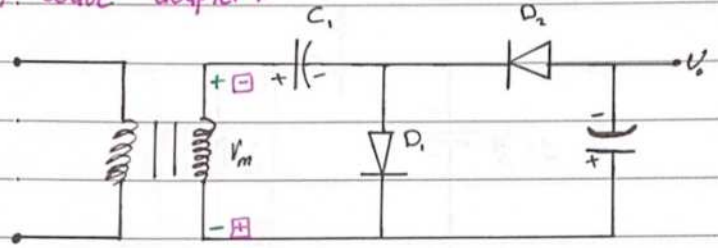
إذا لقيت قيمة V_o ← 0.3 , 0.7 يعني الـ Diode ← not ideal

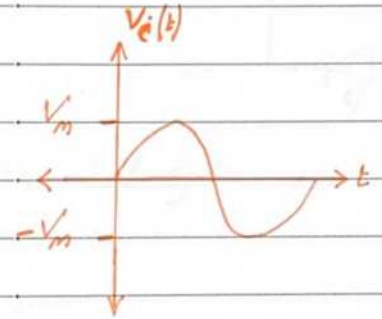
إذا لقيت قيمة V_o ← عدد عشري يعني الـ Diode ← not ideal وفي حالة نظري

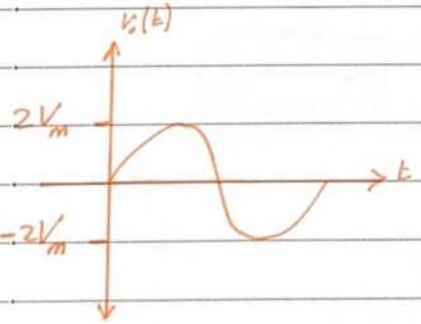
ليث، عمرو

* Voltage Multiplier :-

[a] Half-wave doupler :-







* Positive half cycle :

D_1 : "on" ,, D_2 : "off"

$-V_m + V_{c1} = 0$

$VC_1 = V_m$

* Negative half cycle :

D_1 : "off" ,, D_2 : "on"

$V_m + V_{c1} - V_{c2} = 0$

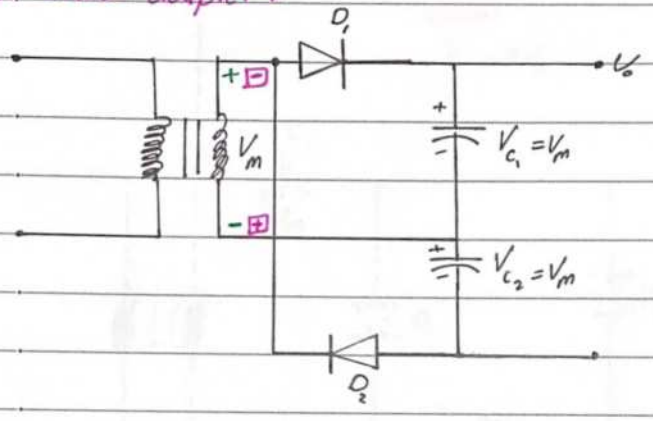
$VC_2 = V_m + V_{c1}$

$VC_2 = 2V_m = V_o$

ليث، عمرو

*Voltage Multiplier :-

(b) Full-wave doupler :-



*Positive half cycle:

$-V_m + V_{c_1} = 0$

$V_{c_1} = V_m$

*Negative half cycle :

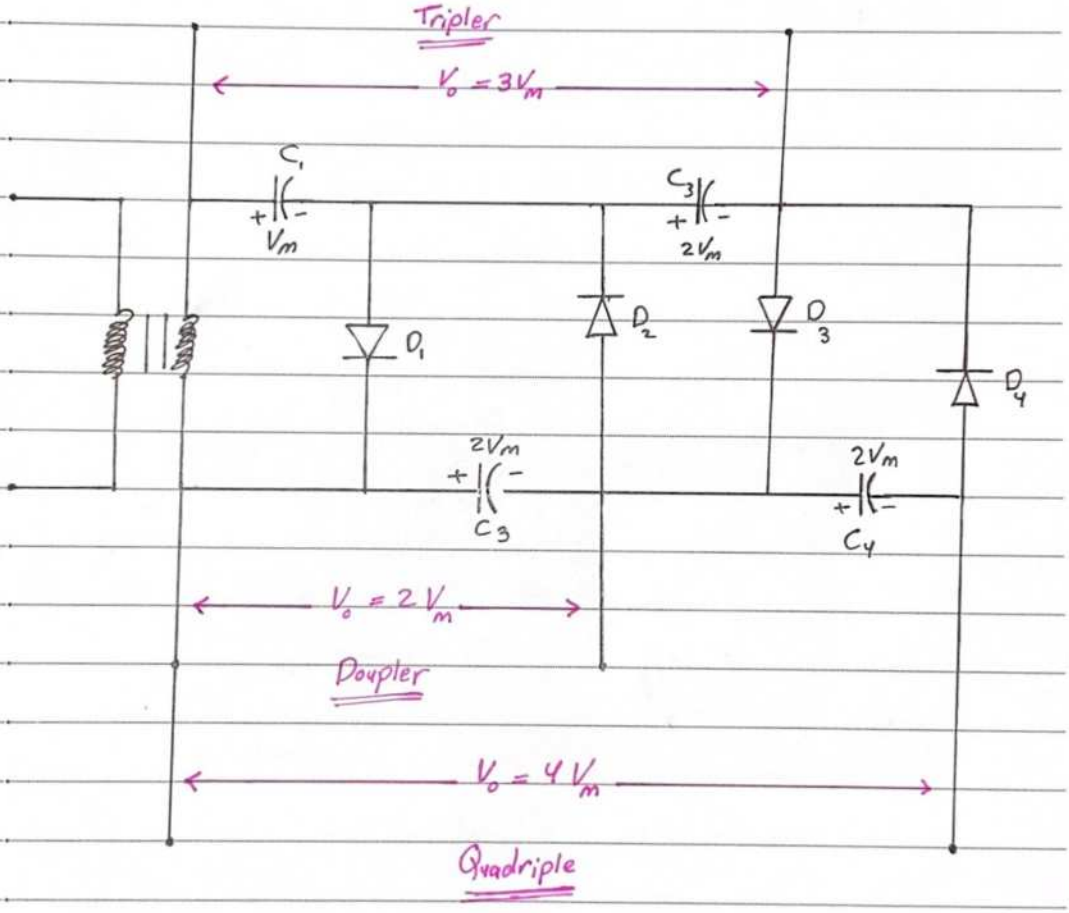
$V_m - V_{c_2} = 0$

$V_{c_2} = V_m$

$$V_o = 2V_m$$

ليث، عمرو

* Voltage Tripler & Quadriple :-

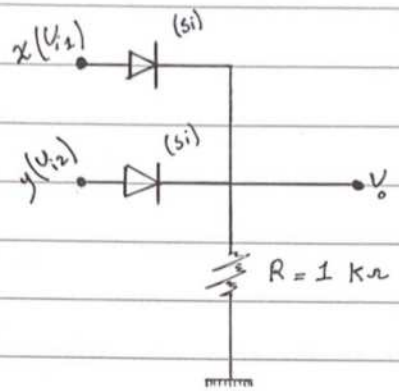


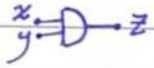
ليث، عمرو

OR / AND Gate :

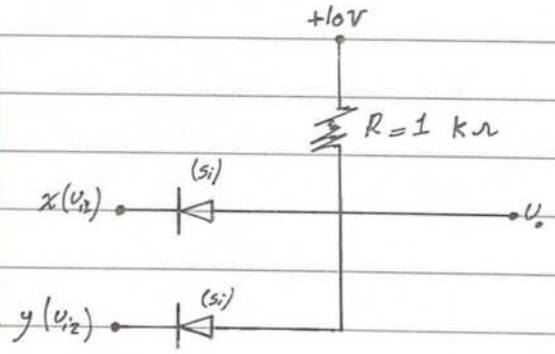
OR: 

$x(V_{i1})$	$y(V_{i2})$	$z(V_o)$
0	0	0
0	1	1
1	0	1
1	1	1



AND: 

$x(V_{i1})$	$y(V_{i2})$	$z(V_o)$
0	0	0
0	1	0
1	0	0
1	1	1



- قيمة المداخل إذا كانت 1 هذا يعني أن له قيمة 5V و 10V و...
وإذا كانت 0 يعني أن ليس له قيمة ..

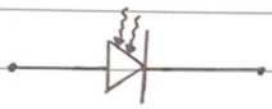
ليث، عمرو

LED Diode : Light Emitting Diode

Symbol: 

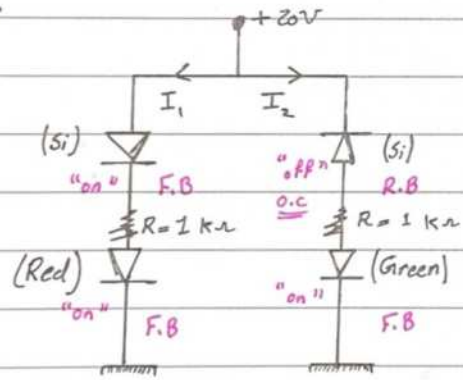
- أشهر إستخداماته : Fiber
- يضيء بحسب نوع الشوائب داخله.
- كل لون له ڤولتية معددة تُعطى (ليست للحفظ)
- يحول الموجات الكهربائية إلى ضوئية

Photo Diode :

Symbol: 

- يحول الموجات الضوئية إلى كهربائية

#Example :



- Green = 2 V
- Red = 1.8 V

Find I_1 , I_2 :

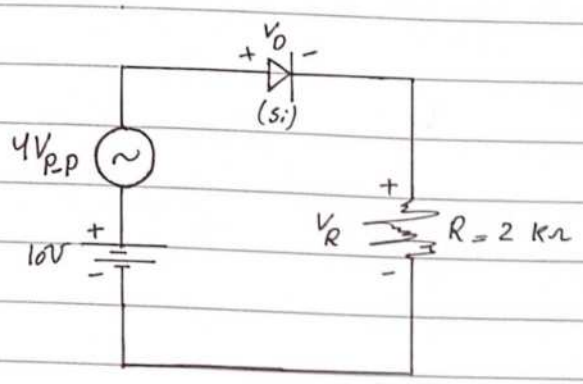
Solution : $-20 + 0.7 + I_1(1\,k\Omega) + 1.8 = 0$

$$I_1 = \frac{20 - 0.7 - 1.8}{1\,k\Omega} = 17.5\ mA$$

ليث، عمرو

* Network with DC & AC Sources:

• نعمل على .. Superposition



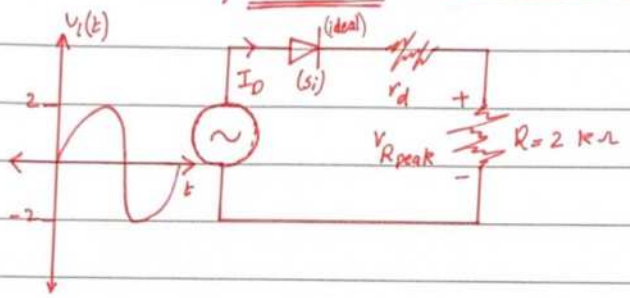
1) DC Source: AC Source "sh.c"

$$-10 + V_D + V_R = 0$$

$$-10 + 0.7 + V_R = 0 \rightarrow V_R = 9.3\ V$$

$$I_D = \frac{V_R}{R} = \frac{9.3}{2\ k} = 4.65\ mA$$

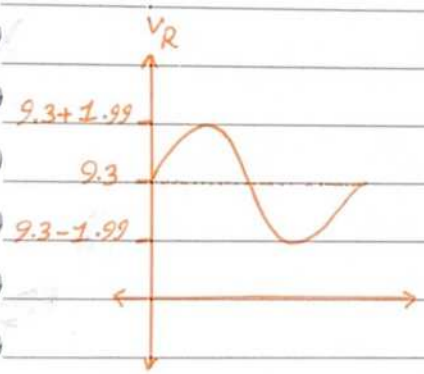
2) AC Source: DC Source "sh.c"



$$r_d = \frac{26\ mV}{4.65\ mA} = 5.59\ \Omega$$

$$V_{R_{peak}} = V_{source} * \left(\frac{R}{R + r_d}\right) = 2 * \frac{2\ k}{2k + 5.59}$$

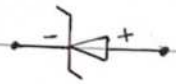
$$V_{R_{peak}} = 1.99\ V$$



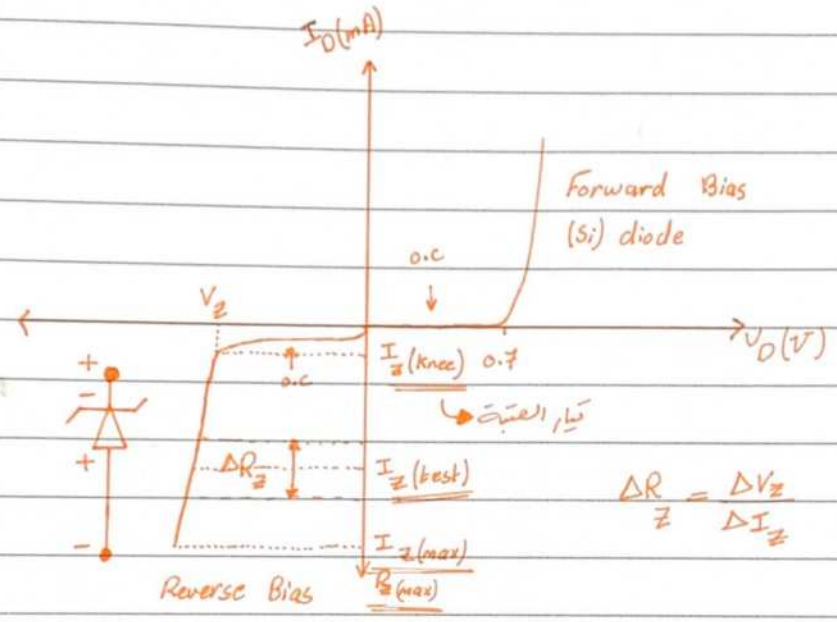
$$V_{D_{peak}} = V_{peak} - V_{R_{peak}} = 2 - 1.99 = 0.01\ mV$$

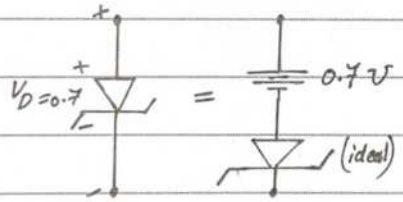
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* Zener Diode :

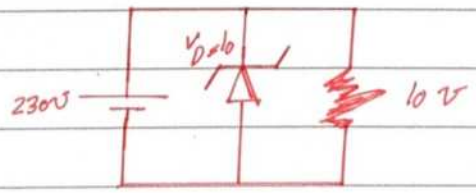
Symbol: 

* Characteristic Curve OF Zener Diode :.





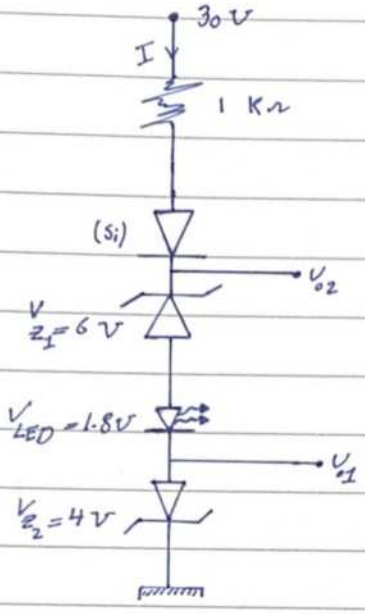
• In Real life :.



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* Zener diode :

• DC Application.



• Find :

1) V_{o1}

2) V_{o2}

3) I

• $V_{o1} = V_{Z_2} = 0.7$ V

• $V_{o2} = V_{o1} + V_{LED} + V_{Z1}$

$V_{o2} = 0.7 + 1.8 + 6$

$V_{o2} = 8.5$ V

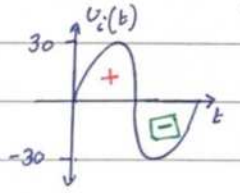
• $-30 + V_R + V_{(Si)} + V_{o2} = 0$

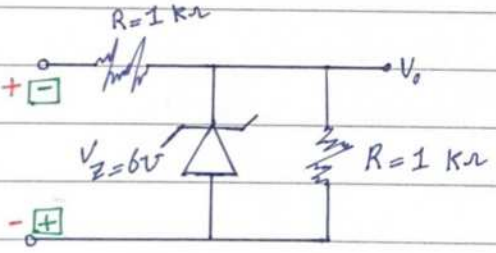
$V_R = 30 - V_{(Si)} - V_{o2}$

$V_R = 20.8$

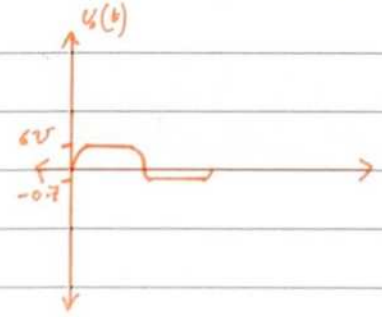
$I = \frac{20.8}{1\text{ K}} = 20.8$ mA

• AC Application.





• Sketch : $V_o(t)$.



+ Positive half cycle:

$V_o = V_{R_L} = V_Z = 6$

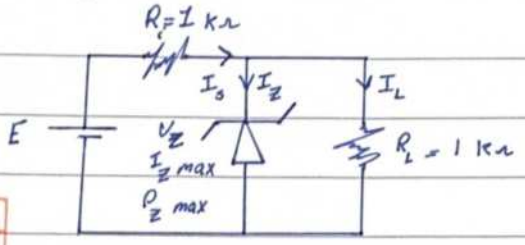
- Negative half cycle:

$V_o = V_{R_L} = V_Z = -0.7$

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* Zener Diode as Voltage regulator:

① Fixed input Fixed Load: R_L



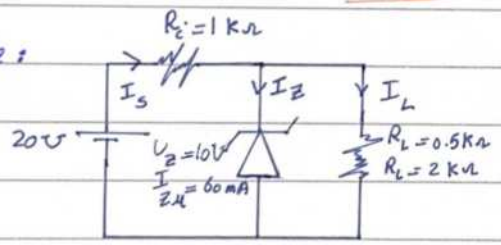
Voltage Divider: $V_{RL} = \left[\frac{V}{S}(E)\right]\left[\frac{R_L}{R_i + R_L}\right]$

• Find $I_S, I_Z, I_L, V_{RL}, V_{R_i}$

• طريقة الحل: " نعتبر أن الـ Zener على أنه open circuit ونجد بقية المقاومة R_L عن طريقة الفولتج ديفايدر " إذا كان $V_{R_L} \geq V_Z$ فإن الـ Zener Diode "on"

وأما إذا كان $V_{R_L} < V_Z$ فإن الـ Zener Diode "off"

Example:



Find: $I_s, I_Z, I_L, P_Z, V_{R_i}, V_{R_L}$

[1] $R_L = 0.5\ K\Omega$ $\quad V_{R_L} < V_Z$

$V_{R_L} = \frac{20 * 0.5}{1.5} = 6.67 < 10V$

Zener Diode open circuit "off"

$I_S = I_L = 13.3\ mA$, $I_Z = 0$

$P_Z = 0$, $V_{R_L} = 6.67\ V$

$V_{R_i} = 20 - 6.67 = 13.3\ V$

[2] $R_L = 2\ K\Omega$ $\quad V_{R_L} > V_Z$

$V_{R_L} = \frac{20 * 2}{3} = 13.3V > 10V$

Zener Diode "on"

$V_{R_L} = 10.V$, $V_{R_i} = 20 - 10 = 10\ V$

$I_S = \frac{10}{1K} = 10\ mA$, $I_L = \frac{10}{2K} = 5\ mA$

$I_Z = I_S - I_L = 5\ mA$

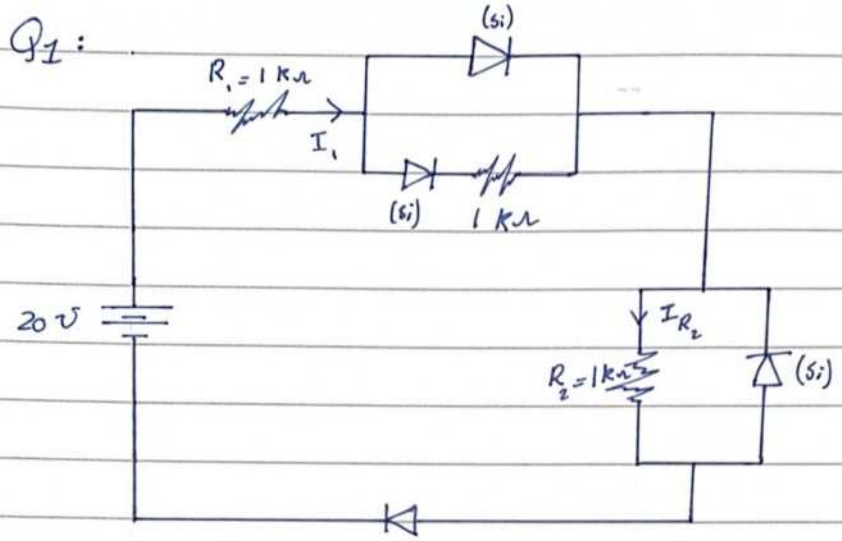
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$P_Z = 5 * 10 = 50\ mWatt$

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Quiz 2: From page (16 - 29)

Q1:



Find:

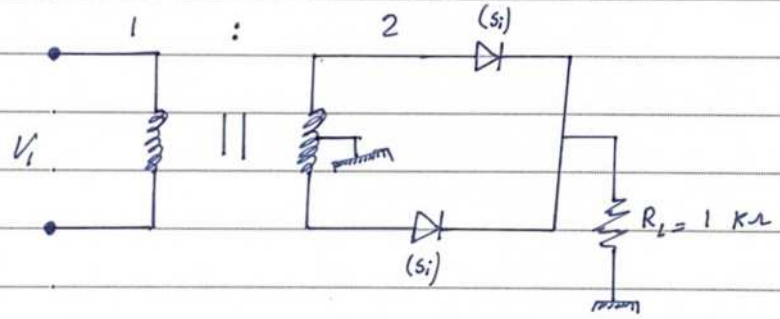
① $I_1 = 9.5\ mA$

② $I_{R_1} = 9.5\ mA$

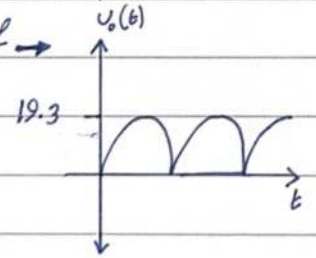
③ $V_{R_1} = 9.5\ V$

④ $V_{R_1} = 9.5\ V$

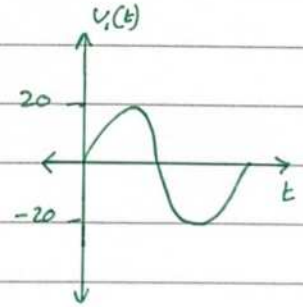
Q2:



if →



Find: $V_i(t)$



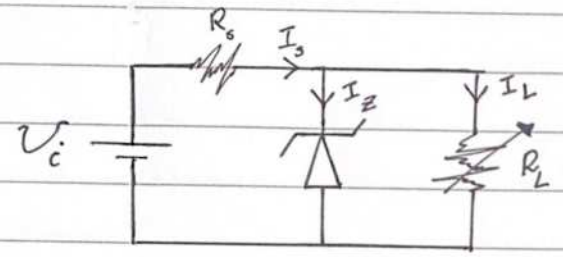
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* Zener Diode as Voltage regulator:

② Fixed input Variable Load:

$V_{L\,min}$ $R_{L\,min}$

$V_{L\,max}$ $R_{L\,max}$ → $P_{Z\,max}$



$$V_L = \frac{R_L}{R_L + R_s} V_i \qquad V_L = V_Z$$

- $R_{L\,min} = \dfrac{R_s V_Z}{V_i - V_Z}$

To find $R_{L\,max}$: - $R_{L\,max} = \dfrac{V_Z}{I_{L\,min}}$

$$I_{L\,min} = \frac{V_Z}{R_{L\,max}}$$

$$I_s = I_Z + I_L \Rightarrow I_L = I_s - I_Z$$

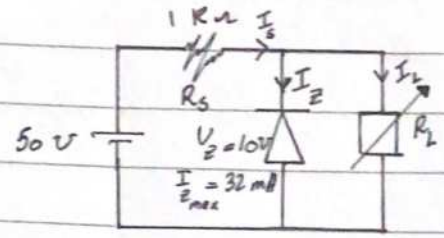
$$I_{L\,min} = \frac{V_i - V_Z}{R_s} - I_{Z\,max}$$

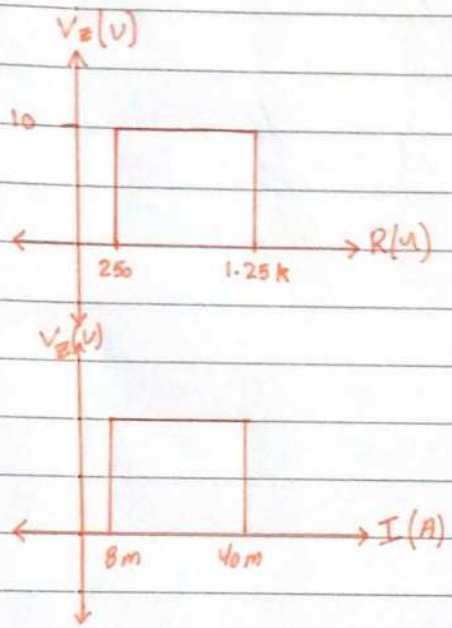
- $R_{L\,max} = \dfrac{V_Z}{I_{L\,min}}$

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*Example :





- $R_{L\,min} = \frac{1\,K * 10}{50 - 10} = 250\,\Omega$

$$I_s = \frac{50 - 10}{1\,K\Omega} = 40\,mA$$

$$I_{L\,min} = I_s - I_{z\,max}$$

$$I_{L\,min} = 40 - 32 = 8\,mA$$

- $R_{L\,max} = \frac{V_z}{I_{L\,min}} = \frac{10\,V}{8\,mA} = 1.25\,K\Omega$

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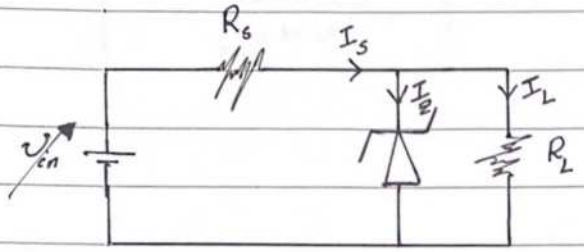
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* Zener Diode as Voltage regulator:

③ Variable Input Fixed Load:

$V_{in\ min}$

$V_{in\ max}$



- V_{in} min :

$$\frac{V_L}{} = \frac{V_Z}{} = \frac{R_L}{R_L + R_S} V_i$$

- $$V_{in\ min} = \frac{(R_L + R_S)\, V_Z}{R_L}$$

- $V_{in\ max}$:

$$V_{in} = V_S + V_Z$$

$$V_{in\ max} = V_{S\ max} + V_Z$$

$$V_{S\ max} \propto I_{S\ max}$$

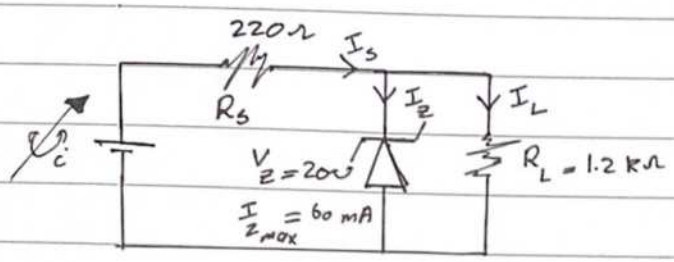
$$I_{S\ max} = I_{Z\ max} + I_L$$

$$I_{S\ max} = I_{Z\ max} + \frac{V_Z}{R_L}$$

- $$V_{in\ max} = I_{S\ max} * R_S + V_Z$$

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*Example:



$$\bullet\ V_{in\,min} = \frac{(R_L + R_s)\,V_z}{R_L}$$

$$V_{in\,min} = \frac{[1200 + 220] * 20}{1200}$$

$$V_{in\,min} = 23.67\ V$$

$$\bullet\ V_{in\,max} = I_{s\,max} * R_s + V_z$$

$$I_L = \frac{20}{1.2k} = 16.67\,mA$$

$$I_{s\,max} = I_{z\,max} + I_L$$

$$I_{s_{max}} = 76.67\,mA$$

$$V_{in\,max} = (76.67m)(220) + 20$$

$$V_{in\,max} = 36.87\ V$$

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• إلى هنا إمتحان اليد

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• Final Material

Ch.3 : BJT

"BJT": Bipolar Junction Transistor

• في ديسمبر من سنة 1947 حدثت ثورة صناعية بسبب إختراع الـ Transistor ، الـ Trasistor الأساسي يسمى (point-contact)

• مميزات الـ Transistor :
1. صغير الحجم وخفيف
2. لا يتطلب حرارة ولا يضيع الحرارة

• في هذا القسم ستكون أول مناقشتنا عن الأجهزة التي تمتلك ثلاث أطراف سنجد أن كلها مكبرات (Amplifier) تكبر الفولتية أو التيار أو الـ power ..
- إحدى الثلاث أطراف يتحكم بالإثنين الآخرين ..

• الـ Transistor هو جهاز يمكن إستعماله كـ (مكبر أو Switch)

• أنواع الـ Transistor التي يتحكم فيها تيار ← "current control current"
1. pnp ← "Positive Negative Positive"
2. npn ← "Negative Positive Negative"

• Transistor = Transfer + Resistor

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أكبر منطقة لأصغر منطقة :

collecter ١

Emitter ٢

Base ٣

• الـ Base صغيرة جداً حتى لا تستقر الإلكترونات بداخلها

C (Collecter)

∴ PnP ١

B (Base)

E (Emitter)

+ Positive

- Negative

- Negative

+ Positive

C (collecter)

∴ NpN ٢

B (Base)

E (Emitter)

- Negative

+ Positive

+ Positive

- Negative

"npn"

لازم الـ I_B يكون شغال بالأول حتى يعطين تيار I_c ، وهذا معنى current control current

$I_c = \beta I_B$

I_c

I_B

V_{BB}

V_{CC}

D.R

D.R

I_E

- إتجاهات التيار تبعاً لأقطاب البطاريات

KCL ∴ $I_B + I_c = I_E$

$V_B, V_C, V_E, V_{BE}, V_{CE}, V_{BC}$

• $V_{BB} \geq V_{BE}$

• طريقة العمل : تحدث قوة تنافر بين أقطاب البطاريات السالبة مع الـ n بالأسفل ، فتنتقل الإلكترونات لأعلى و تخترق الـ D.R و يستقر قليل منها في فجوات الـ P وقليل يذهب في إتجاه الـ Base و النسبة الأكبر نحو الـ Collecter ..

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• ملاحظة: في الـ pnp هناك إختلاف بالرسمة وهي:

1. نعكس أقطاب البطارية
2. المنطقة الوسطى تكون n-type ومحصورة بين منطقتين p-type
3. عكس إتجاه التيارات

• ملاحظة: المشترك بين الـ pnp والـ npn هي معادلة KCL:

$$I_C + I_B = I_E$$

#Note: $V_{CE} = V_C - V_E$

$V_{BE} = V_B - V_E$

$V_{CB} = V_C - V_B$

• There Are three common Config..

1. common-emitter
2. common-Base
3. common-Collecter

Common → تأريض

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3.5 COMMON-EMITTER CONFIGURATION

The most frequently encountered transistor configuration appears in Fig. 3.12 for the *pnp* and *npn* transistors. It is called the *common-emitter configuration* because the emitter is common to both the input and output terminals (in this case common to both the base and collector terminals).

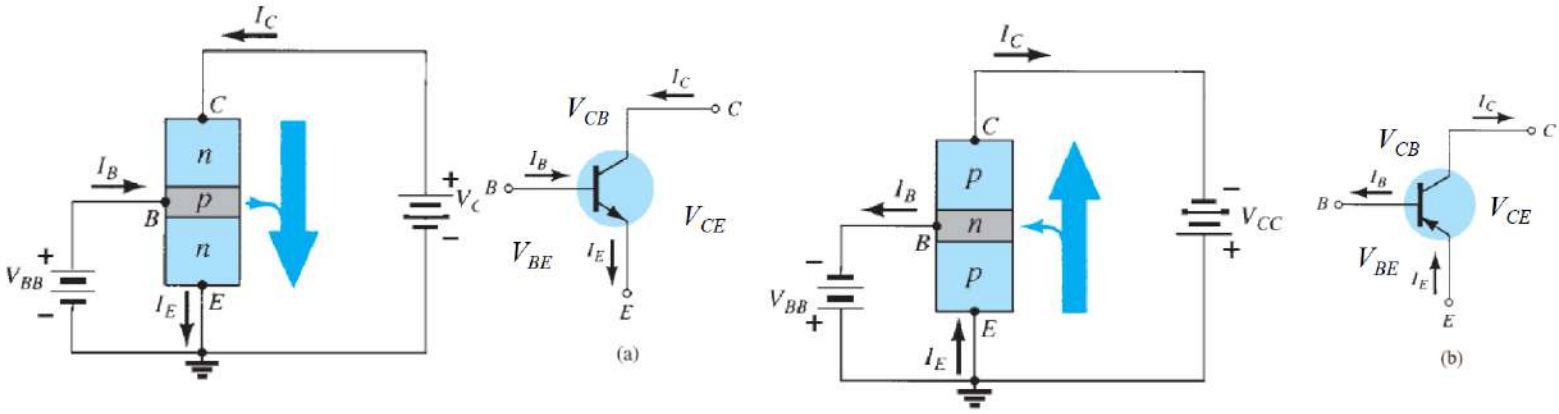


FIG. 3.12
Notation and symbols used with the common-emitter configuration: (a) npn transistor; (b) pnp transistor.

Two sets of characteristics are again necessary to describe fully the behavior of the common-emitter configuration: one for the *input* or *base–emitter* circuit and one for the *output* or *collector–emitter* circuit. Both are shown in Fig. 3.13.

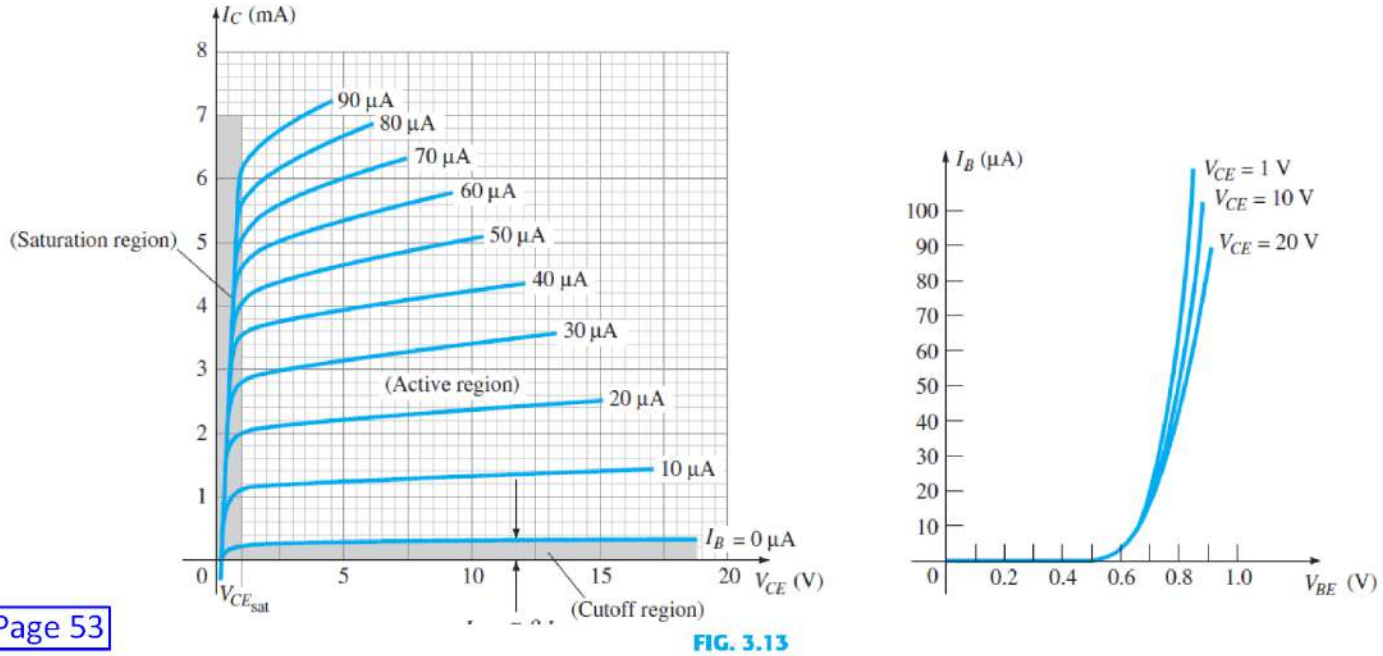


FIG. 3.13
Characteristics of a silicon transistor in the common-emitter configuration: (a) collector characteristics; (b) base characteristics.

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■ من خلال الرسمة السابقة :.

الـ V_{CE} " إذا غيرناه " V_{BE} " يعمل عند فولتية الـ Diode 0.7

■ يوجد لدينا ٣ مناطق :.

١ Active Region ← المدى الفعّال :.

الـ BE ← "Forward Biasing" ، الـ BC ← "Reverse"

$V_{BE}=0.7$

في هذا المنطقة يعمل الـ Transistor كمُكبِّر (Amplifier)

↳ Amplifier Voltage, Current and Power

٢ Cutoff Region ← المدى المقطوع :.

الـ BE والـ BC ← "Reverse Biasing"

في هذه المنطقة يعمل الـ Transistor كـ "Switch" ← "open circuit"

$I_C = 0\ mA$ ، $V_{CE} = V_{CC}$

٣ Saturation Region ← مدى الإشباع :.

الـ BE والـ BC ← "Forward Biasing"

في هذه المنطقة يعمل الـ Transistor كـ "Switch"

$V_{CE} = 0$ نظرياً ، $V_{CE} \approx (0.2 \sim 0.5)$ عملياً

• ملاحظة :. الـ I_C والـ I_E يقاسوا بـ "mA" "milli Ampere" والـ I_B بـ "µA" "MicroAmpere"

• إذا كان $I_B = 0$ فإن الـ $I_C = 0$

• بالنسبة لـ β ← $I_C = \beta I_B$ نأخذها من نصف الرسمة الشمال ، وتكون بالوسط تقريباً لأنه إذا إقتربنا من الإشباع فإن الوجة ستتمدد أو تُقعّر

$\beta(static) = \frac{I_C}{I_B}$ ، $\beta(Dynamic) = \frac{\Delta I_C}{\Delta I_B}$ Page 54

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Faculty of Engineering Technology
Electrical Engineering Department

Chapter 3 Bipolar Junction Transistor (BJT)

Lecture on 1

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CHAPTER OBJECTIVES ●

- Become familiar with the basic construction and operation of the Bipolar Junction Transistor.
- Be able to apply the proper biasing to insure operation in the active region.
- Recognize and be able to explain the characteristics of an *npn* or *pnp* transistor.
- Become familiar with the important parameters that define the response of a transistor.
- Be able to test a transistor and identify the three terminals.

On December 23, 1947, however, the electronics industry was to experience the advent of a completely new direction of interest and development. It was on the afternoon of this day that Dr. S. William Shockley, Walter H. Brattain, and John Bardeen demonstrated the amplifying action of the first transistor at the Bell Telephone Laboratories as shown in Fig. 3.1 . The original transistor (a point-contact transistor) is shown in Fig. 3.2 . The advantages of this three-terminal solid-state device over the tube were immediately obvious:

- It was smaller and lightweight;
- it had no heater requirement or heater loss;
- it had a rugged construction;
- it was more efficient since less power was absorbed by the device itself;
- it was instantly available for use, requiring no warm-up period; and lower operating voltages were possible.

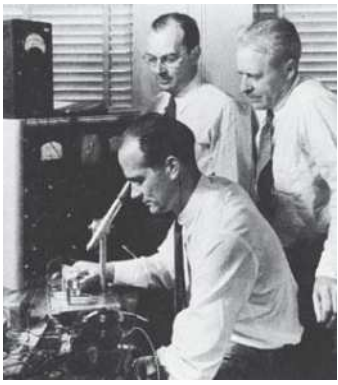


FIG. 3.1
Coinventors of the first transistor at Bell Laboratories.

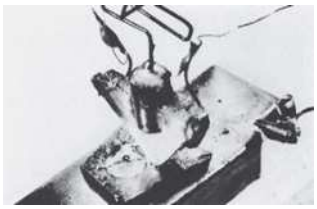
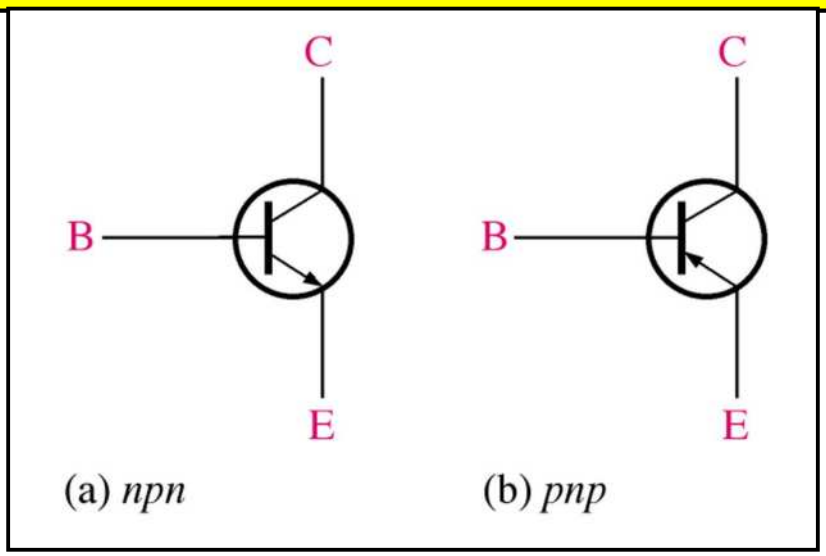


FIG. 3.2
The first transistor. (Courtesy of AT&T Archives and History Center.)

Note that this chapter is our first discussion of devices with three or more terminals. You will find that all amplifiers (devices that increase the voltage, current, or power level) have at least three terminals, with one controlling the flow or potential between the other two.

A **transistor** is a device which can be used as either an **amplifier** or a **switch**. Let's first consider its operation in a more simple view as a current controlling device.



TRANSISTOR CONSTRUCTION

The transistor is a three-layer semiconductor device consisting of either two *n* - and one *p* -type layers of material or two *p* - and one *n* -type layers of material. The former is called an *npn transistor* , and the latter is called a *pnp transistor* . Both are shown in Fig. 3.3 with the proper dc biasing. The emitter layer is heavily doped, with the base and collector only lightly doped. The outer layers have widths much greater than the sandwiched p - or n -type material.

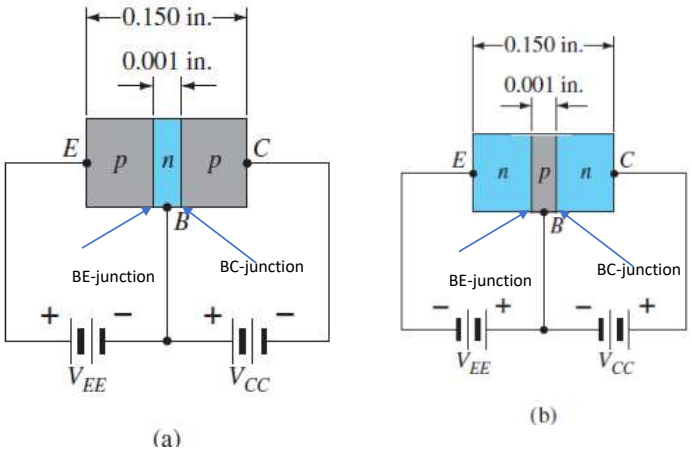


FIG. 3.3
Types of transistors: (a) pnp; (b) npn.

TRANSISTOR OPERATION

The basic operation of the transistor will now be described using the *pnp* transistor of Fig. 3.3a . The operation of the *npn* transistor is exactly the same if the roles played by the electron and hole are interchanged. In Fig. 3.4a the *pnp* transistor has been redrawn without the base-to-collector bias. Note the similarities between this situation and that of the *forward-biased* diode in Chapter 1 . The depletion region has been reduced in width due to the applied bias, resulting in a heavy flow of majority carriers from the *p* - to the *n* -type material.

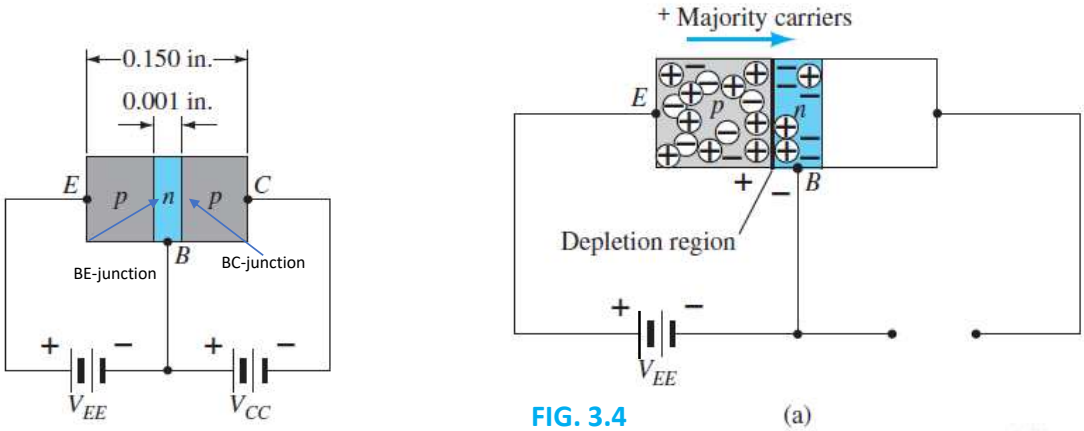


FIG. 3.4
Biasing a transistor: (a) forward-bias; (b) reverse-bias.

TRANSISTOR OPERATION

Let us now remove the base-to-emitter bias of the *pnp* transistor of Fig. 3.3a as shown in Fig. 3.4b . Consider the similarities between this situation and that of the *reverse-biased* diode of Section 1.6 . Recall that the flow of majority carriers is zero, resulting in only a minority-carrier flow, as indicated in Fig. 3.4b. In summary, therefore:
One p–n junction of a transistor is reverse-biased, whereas the other is forward-biased

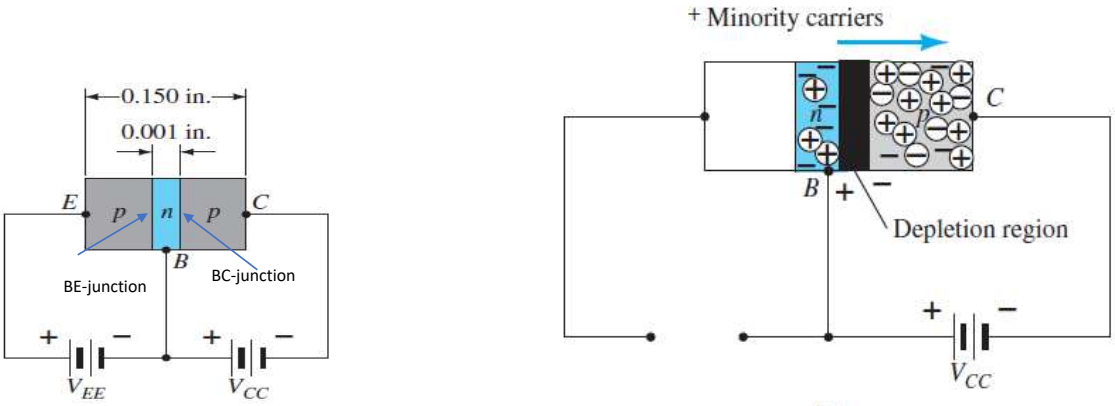
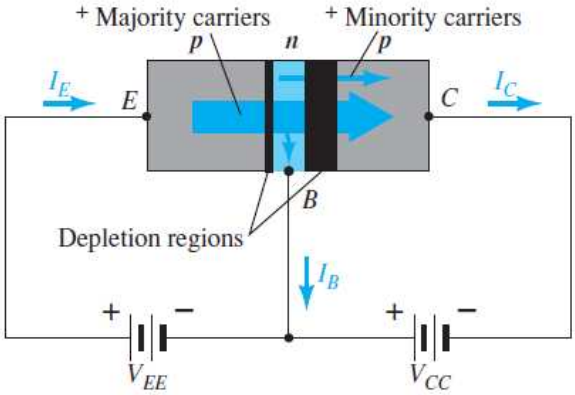


FIG. 3.4
Biasing a transistor: (a) forward-bias; (b) reverse-bias.

In Fig. 3.5 both biasing potentials have been applied to a *pnp* transistor, with the resulting majority- and minority-carrier flows indicated. Note in Fig. 3.5 the widths of the depletion regions, indicating clearly which junction is forward-biased and which is reverse-biased. As indicated in Fig. 3.5 , a large number of majority carriers will diffuse across the forward biased *p–n* junction into the *n* -type material. The question then is whether these carriers will contribute directly to the base current I_B or pass directly into the *p*-type material. Since the sandwiched *n* -type material is very thin and has a low conductivity, a very small number of these carriers will take this path of high resistance to the base terminal. The magnitude of the base current is typically on the order of microamperes, as compared to milliamperes for the emitter and collector currents. The larger number of these majority carriers will diffuse across the reverse-biased junction into the *p* -type material connected to the collector terminal as indicated in Fig. 3.5

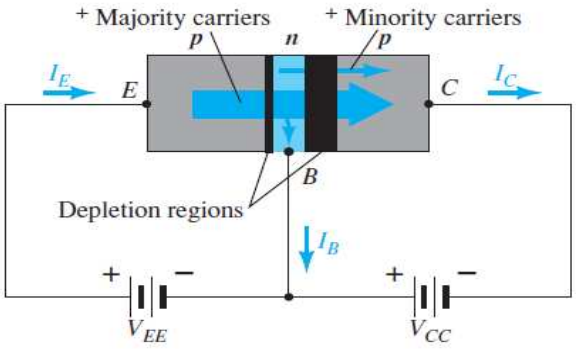


Applying Kirchhoff's current law to the transistor of Fig. 3.5 as if it were a single node, we obtain

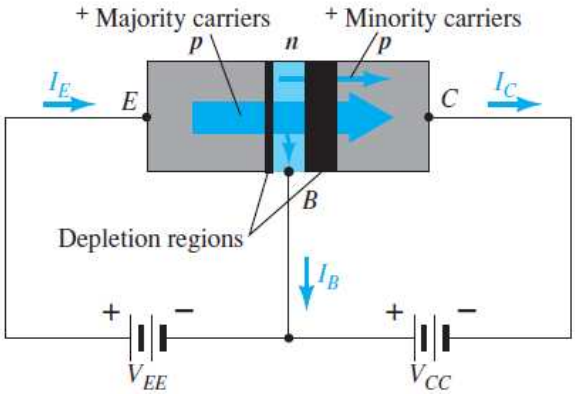
$$I_E = I_C + I_B$$

and find that the emitter current is the sum of the collector and base currents. The collector current, however, comprises two components—the majority and the minority carriers as indicated in Fig. 3.5 . The minority-current component is called the *leakage current* and is given the symbol I_{CO} (I_C current with emitter terminal Open). The collector current, therefore, is determined in total by

$$I_C = I_{C\text{majority}} + I_{CO\text{minority}}$$



For general-purpose transistors, I_C is measured in milliamperes and I_{CO} is measured in microamperes or nanoamperes. I_{CO} , like I_S for a reverse-biased diode, is temperature sensitive and must be examined carefully when applications of wide temperature ranges are considered .



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COMMON-BASE CONFIGURATION

The notation and symbols are indicated in Fig. 3.6 for the common-base configuration with *pnp* and *npn* transistors. The common-base terminology is derived from the fact that the base is common to both the input and output sides of the configuration.

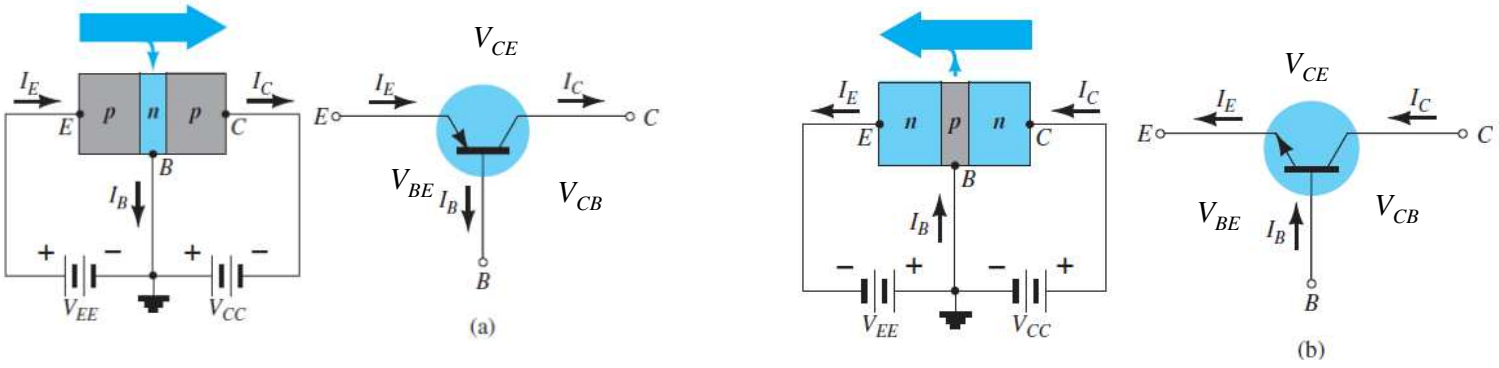


FIG. 3.6 *Notation and symbols used with the common-base configuration: (a) pnp transistor; (b) npn transistor.*

For the transistor:

The arrow in the graphic symbol defines the direction of emitter current (conventional flow) through the device.

All the current directions appearing in Fig. 3.6 are the actual directions as defined by the choice of conventional flow. Note in each case that $I_E = I_C + I_B$. Note also that the applied biasing (voltage sources) are such as to establish current in the direction indicated for each branch. That is, compare the direction of I_E to the polarity of V_{EE} for each configuration and the direction of I_C to the polarity of V_{CC}.

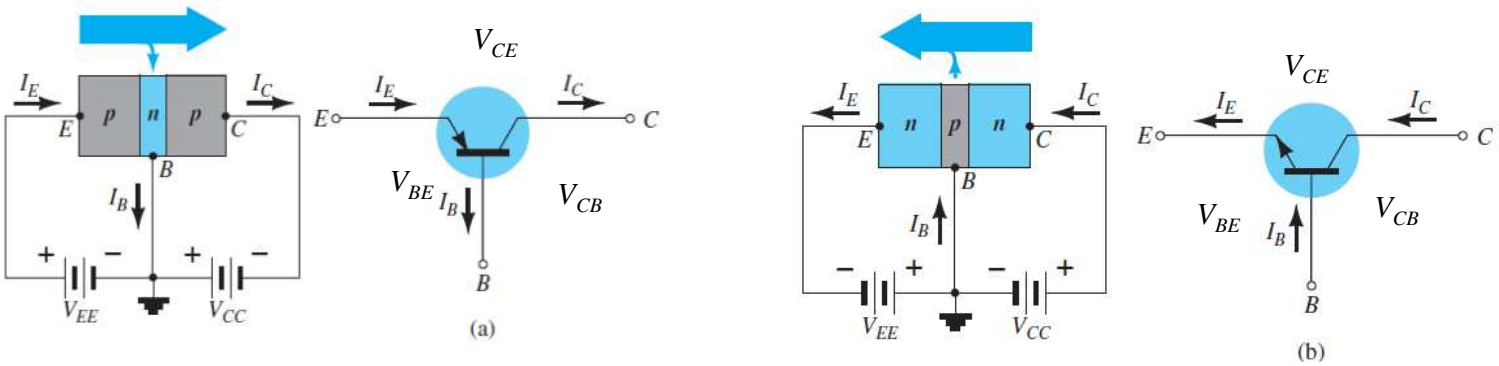
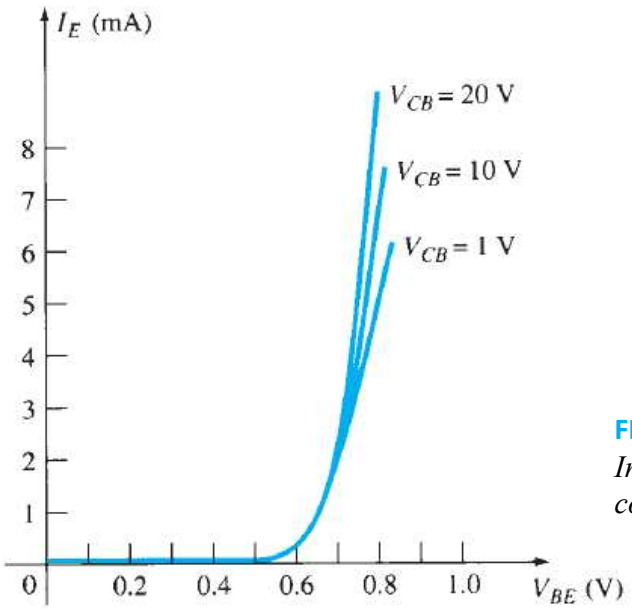


FIG. 3.6 *Notation and symbols used with the common-base configuration: (a) pnp transistor; (b) npn transistor.*

The input set for the common-base amplifier as shown in Fig. 3.7 relates an input current (I_E) to an input voltage (V_{BE}) for various levels of output voltage (V_{CB}).



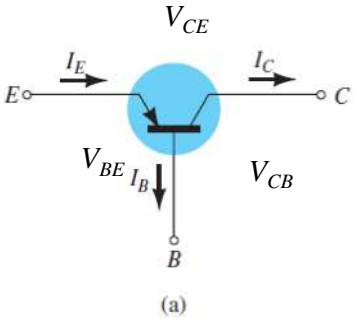
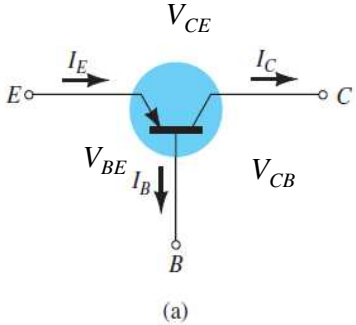


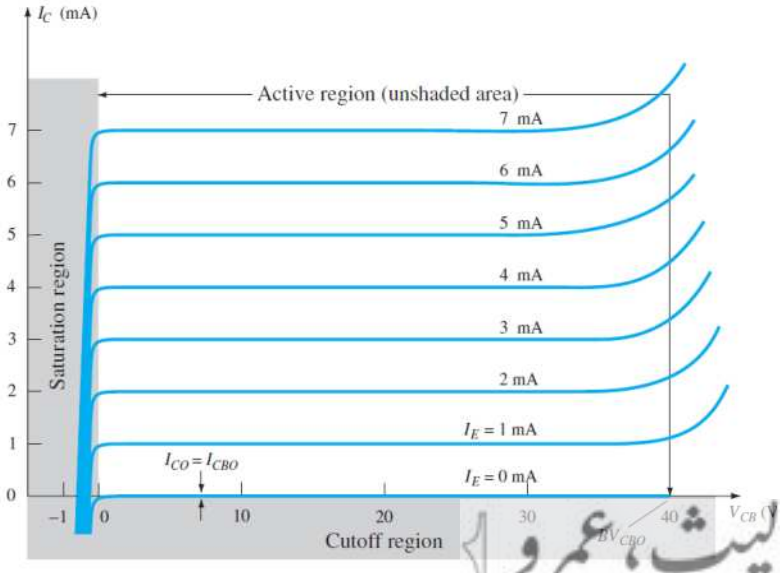
FIG. 3.7
Input or driving point characteristics for a common-base silicon transistor amplifier.

The output set relates an output current (I_C) to an output voltage (V_{CB}) for various levels of input current (I_E) as shown in Fig. 3.8 . The output or *collector* set of characteristics has three basic regions of interest, as indicated in Fig. 3.8 *: the active* , *cutoff* , and *saturation regions.* The active region is the region normally employed for linear (undistorted) amplifiers. In particular:

In the active region the base–emitter junction is forward-biased, whereas the collector–base junction is reverse-biased.

FIG. 3.8
Output or collector characteristics for a common-base transistor amplifier.





The active region is defined by the biasing arrangements of Fig. 3.6 . At the lower end of the active region the emitter current (I_E) is zero, and the collector current is simply that due to the reverse saturation current I_{CO} , as indicated in Fig. 3.9 . The current I_{CO} is so small (microamperes) in magnitude compared to the vertical scale of I_C (milliamperes) that it appears on virtually the same horizontal line as I_C 0. The circuit conditions that exist when I_E 0 for the common-base configuration are shown in Fig. 3.9 . The notation most frequently used for I_{CO} on data and specification sheets

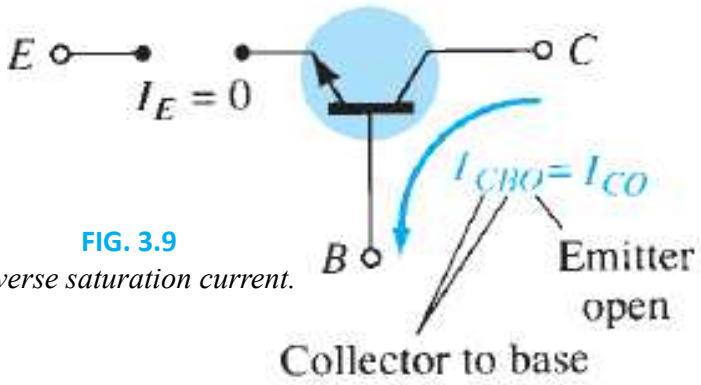
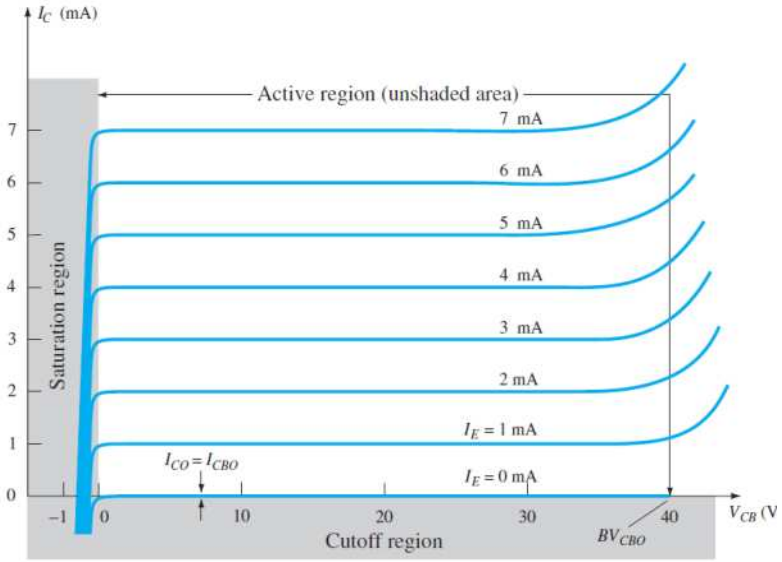


FIG. 3.9
Reverse saturation current.



As indicated in Fig. 3.9 , I_{CBO} (the collector-to-base current with the emitter leg open). its effect can be ignored. However, for higher power units I_{CBO} will still appear in the microampere range. In addition, keep in mind that I_{CBO} , like I_S , for the diode (both reverse leakage currents) is temperature sensitive. At higher temperatures the effect of I_{CBO} may become an important factor since it increases so rapidly with temperature.

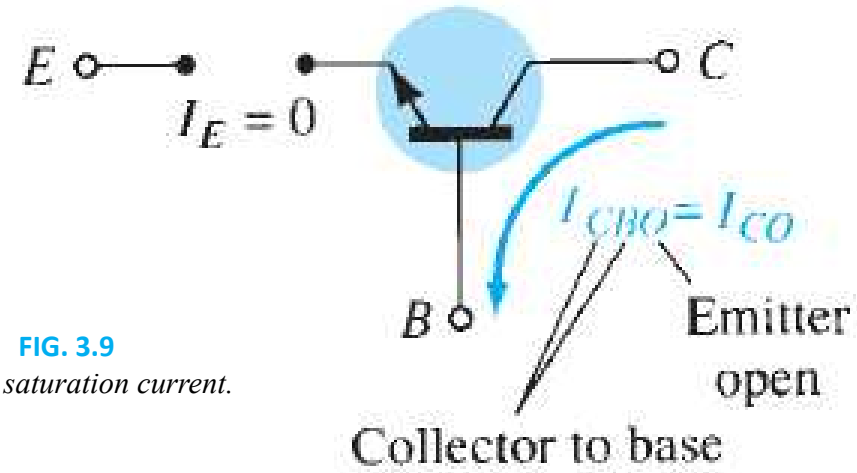


FIG. 3.9
Reverse saturation current.

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Note in Fig. 3.8 that as the emitter current increases above zero, the collector current increases to a magnitude essentially equal to that of the emitter current as determined by the basic transistor-current relations. Note also the almost negligible effect of V_{CB} on the collector current for the active region. The curves clearly indicate that *a first approximation to the relationship between I_E and I_C in the active region is given by*

$$I_C \cong I_E$$

As inferred by its name, the **cutoff region** is defined as that region where the collector current is 0 A, as revealed on Fig. 3.8 .

In addition: In the cutoff region the base–emitter and collector–base junctions of a transistor are both reverse-biased.

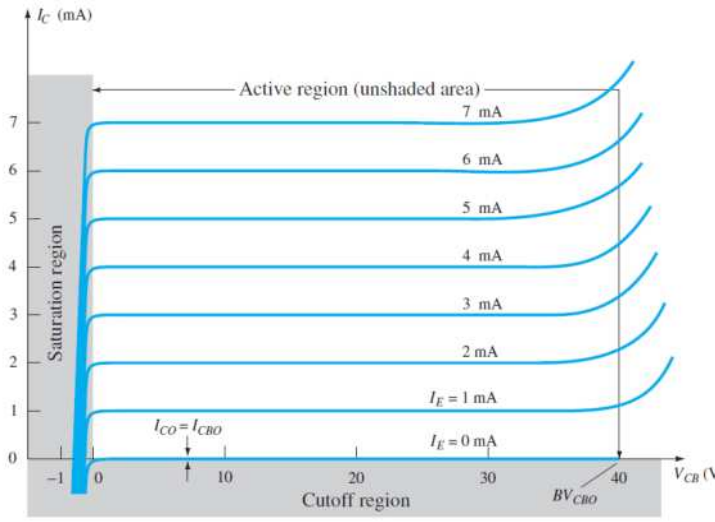


FIG. 3.8
Output or collector characteristics for a common-base transistor amplifier.

The saturation region is defined as that region of the characteristics to the left of V_{CB} 0 V. The horizontal scale in this region was expanded to clearly show the dramatic change in characteristics in this region. Note the exponential increase in collector current as the voltage V_{CB} increases toward 0 V.

In the saturation region the base–emitter and collector–base junctions are forward-biased.

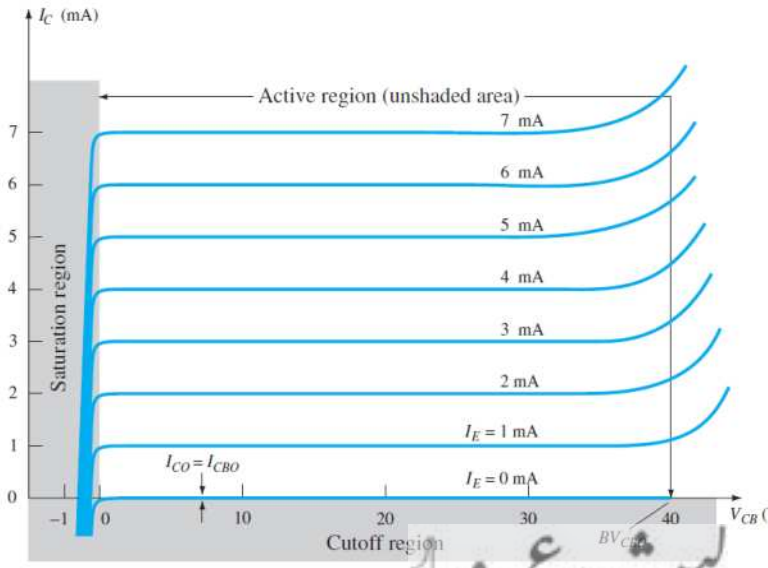


FIG. 3.8
Output or collector characteristics for a common-base transistor amplifier.

The input characteristics of Fig. 3.7 reveal that for fixed values of collector voltage (V_{CB}), as the base-to-emitter voltage increases, the emitter current increases in a manner that closely resembles the diode characteristics. In fact, increasing levels of V_{CB} have such a small effect on the characteristics that as a first approximation the change due to changes in V_{CB} **can be ignored** and the characteristics drawn assumed to be the following:

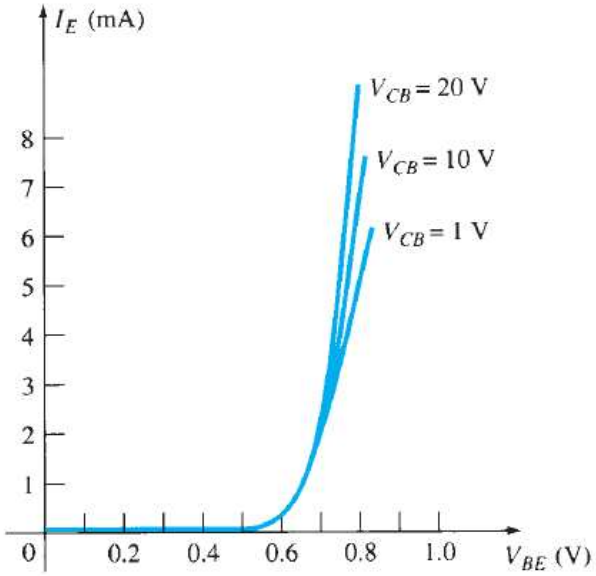


FIG. 3.7
Input or driving point characteristics for a common-base silicon transistor amplifier.

As shown in Fig. 3.10a . If we then apply the piecewise linear approach, the characteristics of Fig. 3.10b result. Taking it a step further and ignoring the slope of the curve and therefore the resistance associated with the forward-biased junction results in the characteristics of Fig. 3.10c . For the analysis to follow in this book the equivalent model of Fig. 3.10c will be employed for all dc analysis of transistor networks. **That is, once a transistor is in the "on" state, the base-to-emitter voltage will be assumed to be the following**: $V_{BE} \cong 0.7$ V

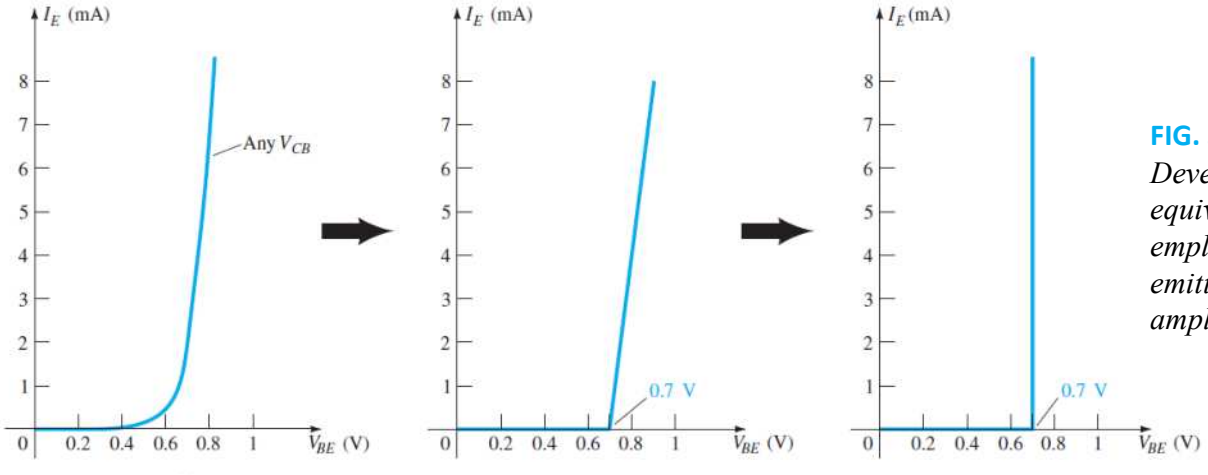


FIG. 3.10
Developing the equivalent model to be employed for the base-to-emitter region of an amplifier in the dc mode.

with the transistor in the "on" or active state the voltage from base to emitter will be 0.7 V at *any* level of emitter current as controlled by the external network

EXAMPLE 3.1

a. Using the characteristics of Fig. 3.8, determine the resulting collector current if $I_E = 3$ mA and $V_{CB} = 10$ V.
b. Using the characteristics of Fig. 3.8, determine the resulting collector current if I_E remains at 3 mA but V_{CB} is reduced to 2 V.
c. Using the characteristics of Figs. 3.7 and 3.8, determine V_{BE} if $I_C = 4$ mA and $V_{CB} = 20$ V.
d. Repeat part (c) using the characteristics of Figs. 3.8 and 3.10c.

Solution:

a. The characteristics clearly indicate that $I_C \cong I_E = \mathbf{3\ mA}$.
b. The effect of changing V_{CB} is negligible and I_C continues to be **3 mA**.
c. From Fig. 3.8, $I_E \cong I_C = 4$ mA. On Fig. 3.7 the resulting level of V_{BE} is about **0.74 V**.
d. Again from Fig. 3.8, $I_E \cong I_C = 4$ mA. However, on Fig. 3.10c, V_{BE} is **0.7 V** for any level of emitter current.

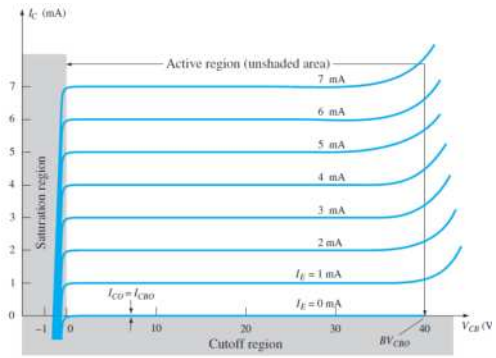


FIG. 3.8

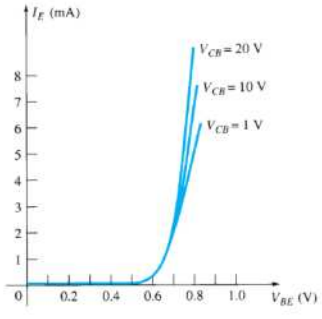
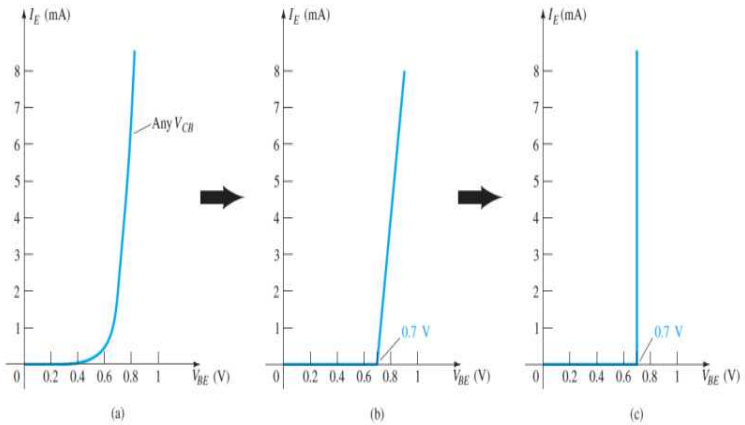


FIG. 3.7



Alpha (A)

DC Mode In the dc mode the levels of I_C and I_E due to the majority carriers are related by a quantity called *alpha* and defined by the following equation:

$$\alpha = \frac{I_C}{I_E}$$

where I_C and I_E are the levels of current at the point of operation. For practical devices alpha typically extends from 0.90 to 0.998, with most values approaching the high end of the range. Since alpha is defined solely for the majority carriers, Eq. (3.2) becomes

$$I_C = \alpha I_E + I_{CBO}$$

For the characteristics of Fig. 3.8 when I_E 0 mA, I_C is therefore equal to I_{CBO} ,. In other words, when I_E 0 mA, I_C also appears to be 0 mAf or the range of V_{CB} values.

AC Mode For ac situations where the point of operation moves on the characteristic curve, an ac alpha is defined by

$$\alpha_{ac} = \left.\frac{\Delta I_C}{\Delta I_E}\right|_{V_{CB}=\text{constant}}$$

Biasing

The proper biasing of the common-base configuration in the active region can be determined quickly using the approximation $I_C \cong I_E$ and assuming for the moment that $I_B \cong 0\ \mu A$. The result is the configuration of Fig. 3.11 for the *pnp* transistor. The arrow of the symbol defines the direction of conventional flow for $I_E \cong I_C$. The dc supplies are then inserted with a polarity that will support the resulting current direction. For the *npn* transistor the polarities will be reversed.

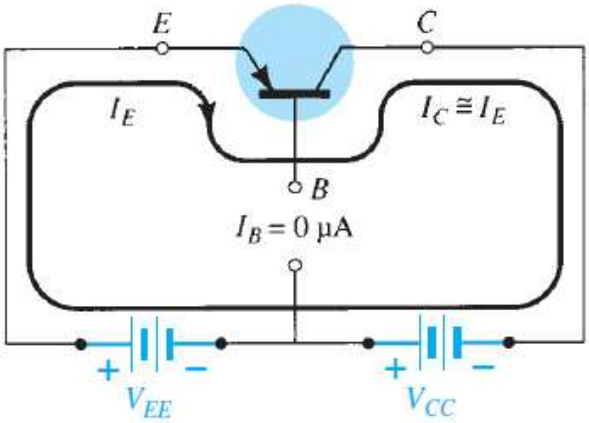


FIG. 3.11
Establishing the proper biasing management for a common-base pnp transistor in the active region.

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Chapter 3 Bipolar Junction Transistor

Lecture 2

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Alpha (A)

DC Mode In the dc mode the levels of I_C and I_E due to the majority carriers are related by a quantity called *alpha* and defined by the following equation:

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where I_C and I_E are the levels of current at the point of operation. For practical devices alpha typically extends from 0.90 to 0.998, with most values approaching the high end of the range. Since alpha is defined solely for the majority carriers, Eq. (3.2) becomes

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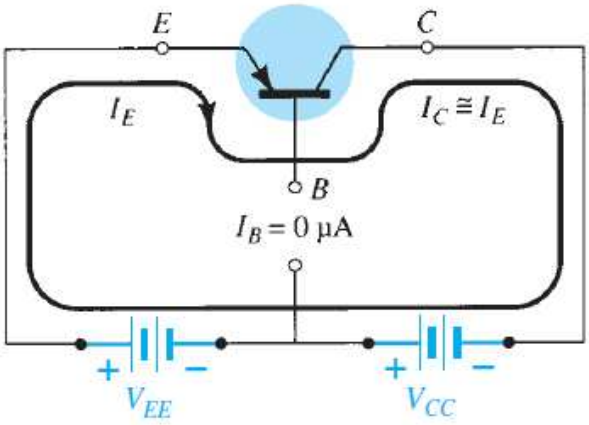


FIG. 3.11
Establishing the proper biasing management for a common-base pnp transistor in the active region.

Breakdown Region

As the applied voltage V_{CB} increases there is a point where the curves take a dramatic upswing in Fig. 3.8. This is due primarily to an avalanche effect similar to that described for the diode in Chapter 1 when the reverse-bias voltage reached the breakdown region.

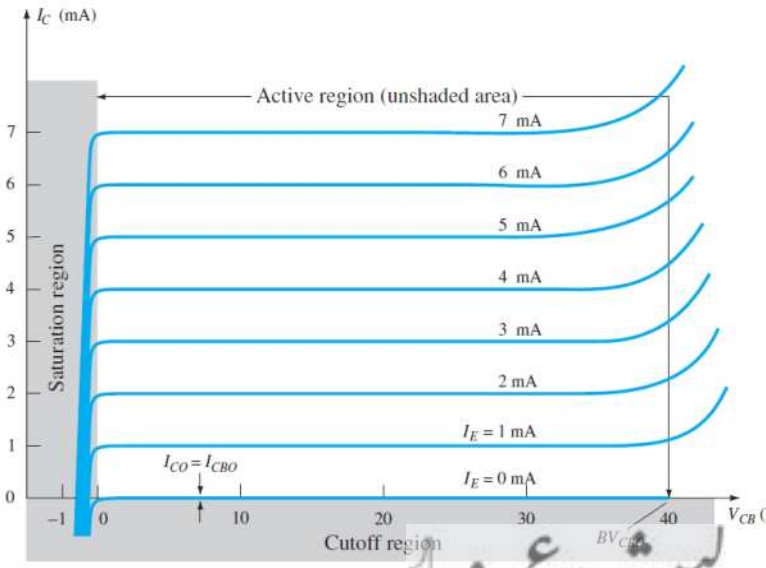


FIG. 3.8
Output or collector characteristics for a common-base transistor amplifier.

As stated earlier the base-to-collector junction is reversed biased in the active region, but there is a point where too large a reverse-bias voltage will lead to the avalanche effect. The result is a large increase in current for small increases in the base-to-collector voltage. The largest permissible base-to-collector voltage is labeled BV_{CBO} as shown in Fig. 3.8. It is also referred to as $V_{(BR)CBO}$ as shown on the characteristics of Fig. 3.23 to be discussed later.

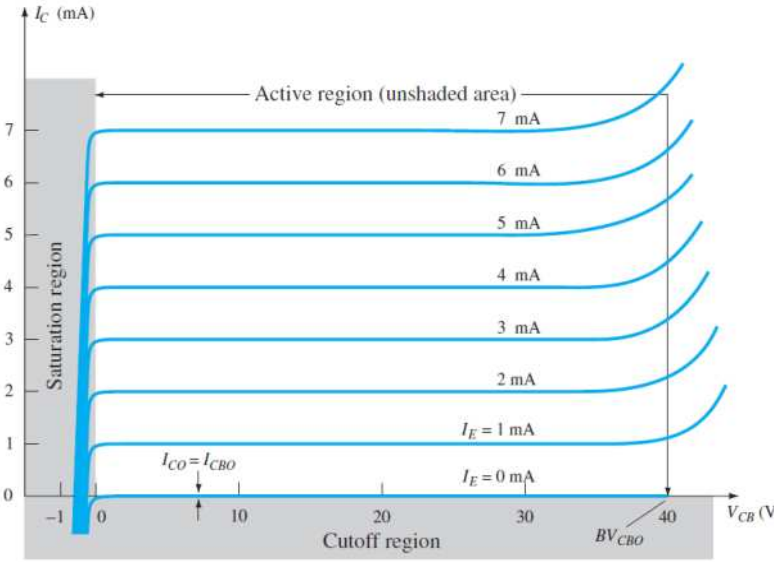
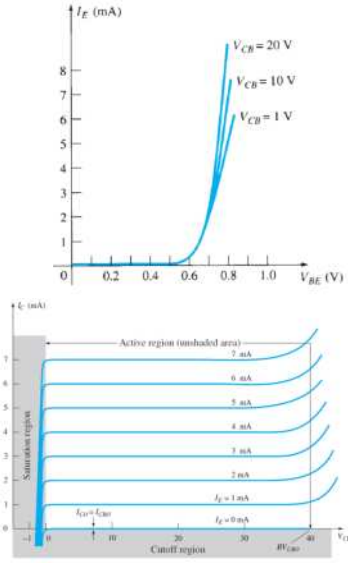


FIG. 3.8
Output or collector characteristics for a common-base transistor amplifier.

TRANSISTOR AMPLIFYING ACTION



Now that the relationship between I_C and I_E has been established in Section 3.4, the basic amplifying action of the transistor can be introduced on a surface level using the network of Fig. 3.12. The dc biasing does not appear in the figure since our interest will be limited to the ac response. For the common-base configuration the ac input resistance determined by the characteristics of Fig. 3.7 is quite small and typically varies from 10 to 100 Ω. The output resistance as determined by the curves of Fig. 3.8 is quite high (the more horizontal the curves the higher the resistance) and typically varies from 50 kΩ to 1 MΩ (100 kΩ for the transistor of Fig. 3.12). The difference in resistance is due to the forward-biased junction at the input (base to emitter) and the reverse-biased junction at the output (base to collector). Using a common value of 20 Ω for the input resistance, we find that

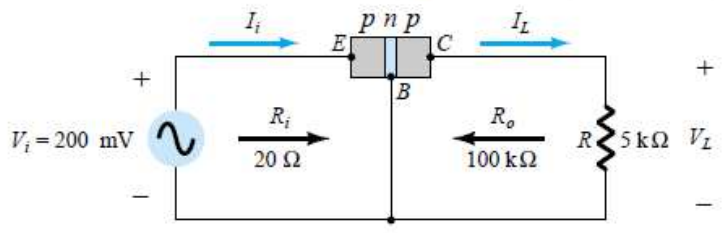
$$I_i = \frac{V_i}{R_i} = \frac{200\text{ mV}}{20\ \Omega} = \mathbf{10\ mA}$$

If we assume for the moment that $\alpha_{ac} = 1$ ($I_c = I_e$),

$$I_L = I_i = 10\text{ mA}$$

and

$$V_L = I_L R$$
$$= (10\text{ mA})(5\text{ k}\Omega)$$
$$= \mathbf{50\ V}$$



Basic voltage amplification action of the common-base configuration.

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The voltage amplification is

$$A_v = \frac{V_L}{V_i} = \frac{50\text{ V}}{200\text{ mV}} = \mathbf{250}$$

Typical values of voltage amplification for the common-base configuration vary from 50 to 300. The current amplification (I_C/I_E) is always less than 1 for the common-base configuration. This latter characteristic should be obvious since $I_C = \alpha I_E$ and α is always less than 1.

The basic amplifying action was produced by transferring a current I from a low- to a high-*resistance* circuit. The combination of the two terms in italics results in the label *transistor;* that is,

*trans*fer + re*sistor* → *transistor*

3.5 COMMON-EMITTER CONFIGURATION

The most frequently encountered transistor configuration appears in Fig. 3.12 for the *pnp* and *npn* transistors. It is called the *common-emitter configuration* because the emitter is common to both the input and output terminals (in this case common to both the base and collector terminals).

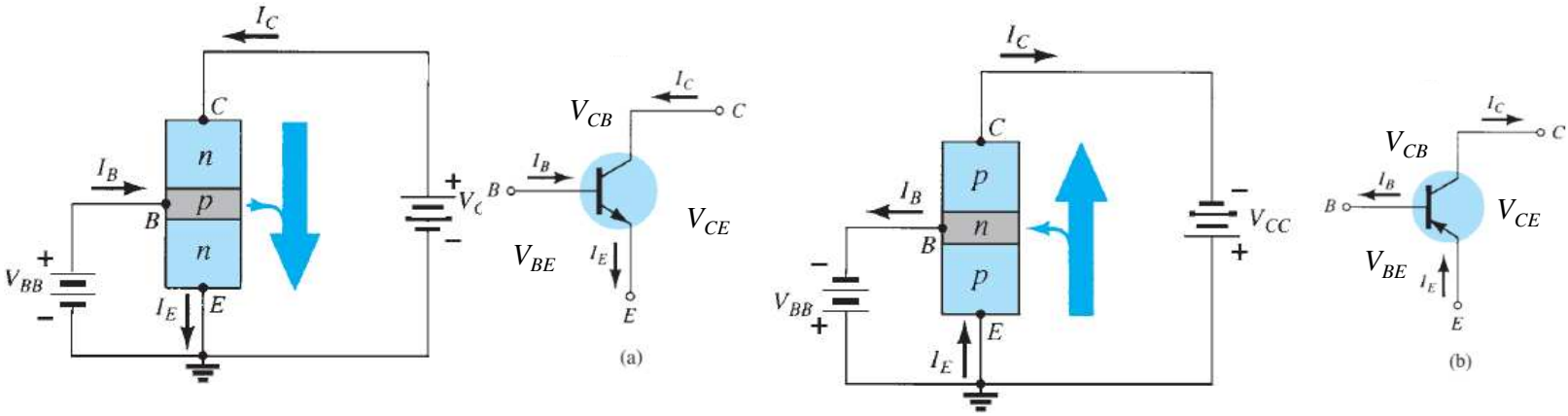


FIG. 3.12
Notation and symbols used with the common-emitter configuration: (a) npn transistor; (b) pnp transistor.

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Two sets of characteristics are again necessary to describe fully the behavior of the common-emitter configuration: one for the *input* or *base–emitter* circuit and one for the *output* or *collector–emitter* circuit. Both are shown in Fig. 3.13.

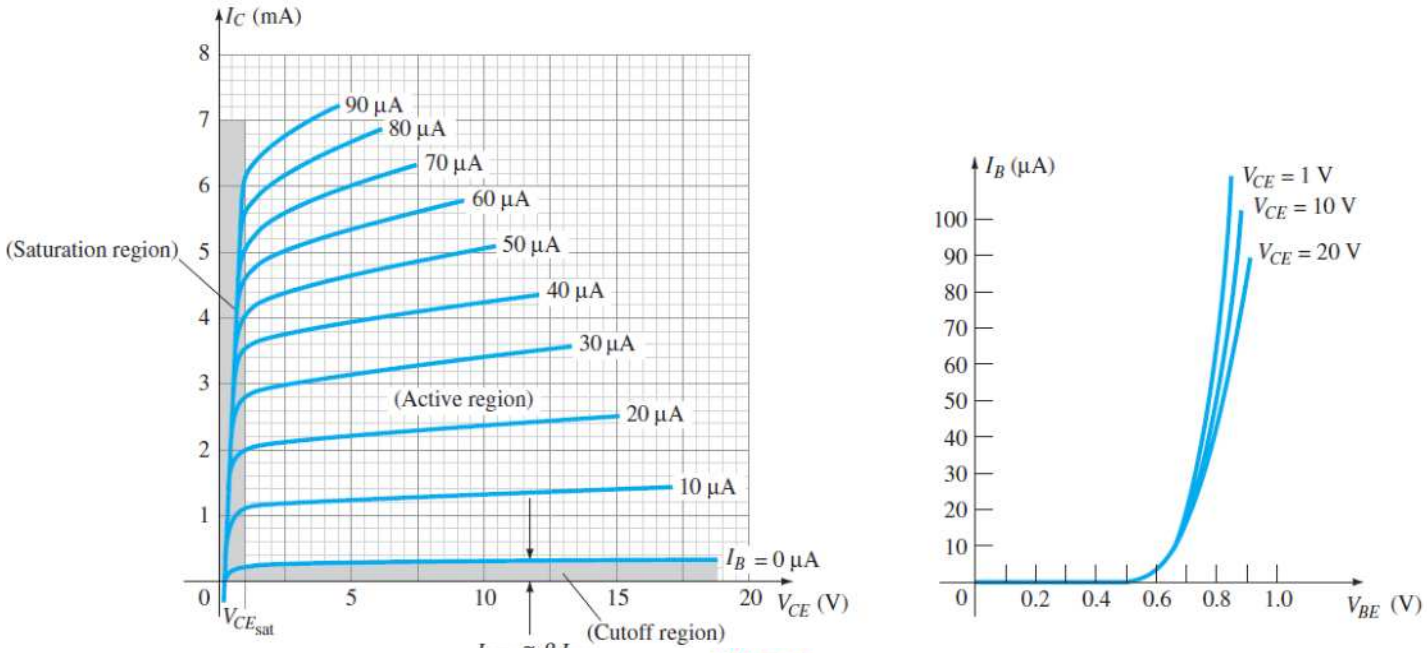
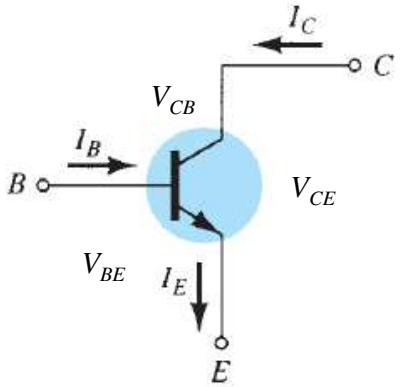


FIG. 3.13

Characteristics of a silicon transistor in the common-emitter configuration: (a) collector characteristics; (b) base characteristics.

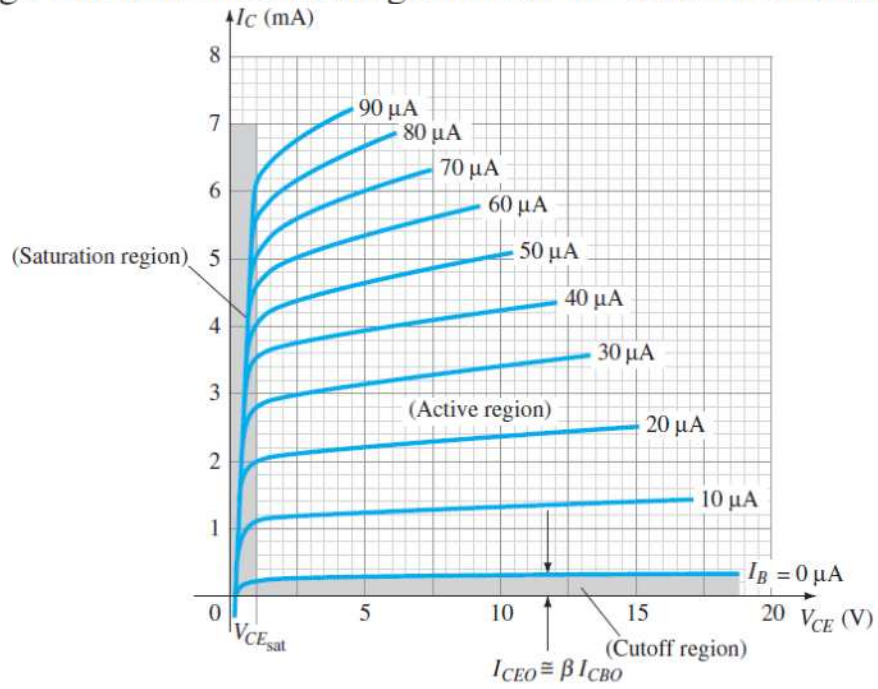
The emitter, collector, and base currents are shown in their actual conventional current direction. Even though the transistor configuration has changed, the current relations developed earlier for the common-base configuration are still applicable. That is, $I_E = I_C + I_B$ and $I_C = \alpha I_E$.

For the common-emitter configuration the output characteristics are a plot of the output current (I_C) versus output voltage (V_{CE}) for a range of values of input current (I_B). The input characteristics are a plot of the input current (I_B) versus the input voltage (V_{BE}) for a range of values of output voltage (V_{CE}).

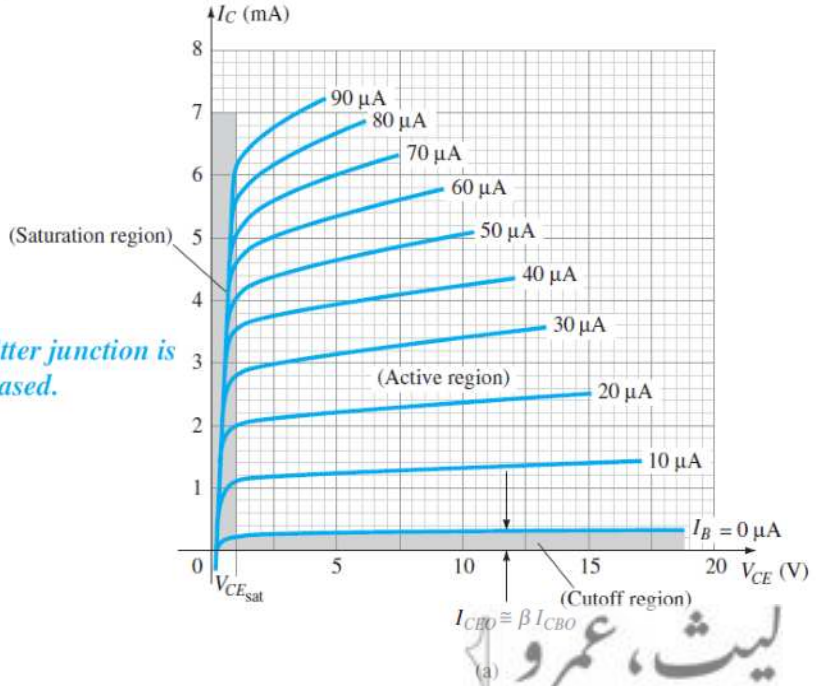


(a)

Note that on the characteristics of Fig. the magnitude of I_B is in microamperes, compared to milliamperes of I_C. Consider also that the curves of I_B are not as horizontal as those obtained for I_E in the common-base configuration, indicating that the collector-to-emitter voltage will influence the magnitude of the collector current.



The active region for the common-emitter configuration is that portion of the upper-right quadrant that has the greatest linearity, that is, that region in which the curves for I_B are nearly straight and equally spaced. In Fig. this region exists to the right of the vertical dashed line at $V_{CE_{sat}}$ and above the curve for I_B equal to zero. The region to the left of $V_{CE_{sat}}$ is called the saturation region.



In the active region of a common-emitter amplifier, the base–emitter junction is forward-biased, whereas the collector–base junction is reverse-biased.

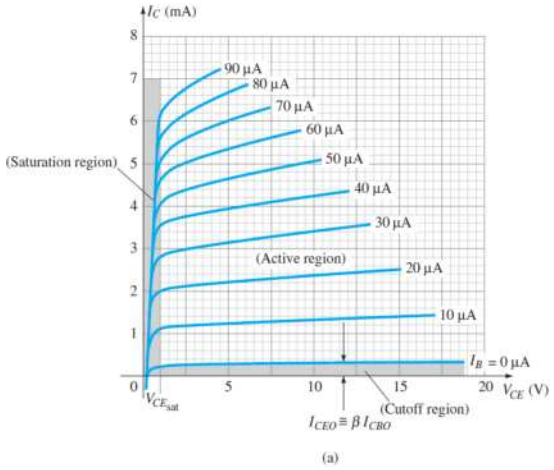
The cutoff region for the common-emitter configuration is not as well defined as for the common-base configuration. Note on the collector characteristics of Fig. 3.14 that I_C is not equal to zero when I_B is zero. For the common-base configuration, when the input current I_E was equal to zero, the collector current was equal only to the reverse saturation current I_{CO}, so that the curve $I_E = 0$ and the voltage axis were, for all practical purposes, one.

The reason for this difference in collector characteristics can be derived through the proper manipulation of Eqs. (3.3) and (3.6). That is,

Eq. (3.6): $$I_C = \alpha I_E + I_{CBO}$$

Substitution gives Eq. (3.3): $$I_C = \alpha(I_C + I_B) + I_{CBO}$$

Rearranging yields $$I_C = \frac{\alpha I_B}{1 - \alpha} + \frac{I_{CBO}}{1 - \alpha}$$



If we consider the case discussed above, where $I_B = 0$ A, and substitute a typical value of α such as 0.996, the resulting collector current is the following:

$$I_C = \frac{\alpha(0\text{ A})}{1 - \alpha} + \frac{I_{CBO}}{1 - 0.996}$$

$$= \frac{I_{CBO}}{0.004} = 250 I_{CBO}$$

If I_{CBO} were 1 μA, the resulting collector current with $I_B = 0$ A would be 250(1 μA) = 0.25 mA, as reflected in the characteristics of Fig. 3.14.

For future reference, the collector current defined by the condition $I_B = 0$ μA will be assigned the notation indicated by the following equation:

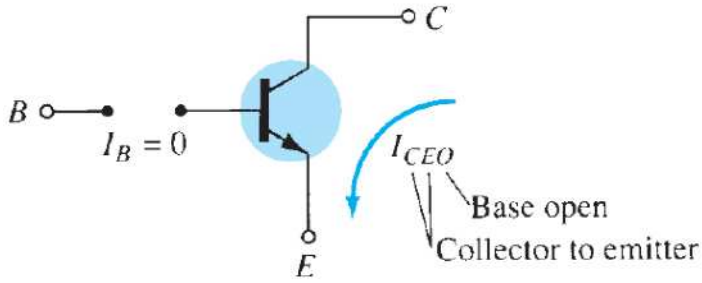
$$I_{CEO} = \left.\frac{I_{CBO}}{1 - \alpha}\right|_{I_B = 0\,\mu\text{A}} \tag{3.9}$$

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For linear (least distortion) amplification purposes, cutoff for the common-emitter configuration will be defined by $I_C = I_{CEO}$.

In other words, the region below $I_B = 0\ \mu A$ is to be avoided if an undistorted output signal is required.

When employed as a switch in the logic circuitry of a computer, a transistor will have two points of operation of interest: one in the cutoff and one in the saturation region. The cutoff condition should ideally be $I_C = 0$ mA for the chosen V_{CE} voltage. Since I_{CEO} is typically low in magnitude for silicon materials, *cutoff will exist for switching purposes when $I_B = 0\ \mu A$ or $I_C = I_{CEO}$ for silicon transistors only. For germanium transistors, however, cutoff for switching purposes will be defined as those conditions that exist when $I_C = I_{CBO}$.* This condition can normally be obtained for germanium transistors by reverse-biasing the base-to-emitter junction a few tenths of a volt.



Recall for the common-base configuration that the input set of characteristics was approximated by a straight-line equivalent that resulted in $V_{BE} = 0.7$ V for any level of I_E greater than 0 mA. For the common-emitter configuration the same approach can be taken, resulting in the approximate equivalent of Fig. 3.15. The result supports our earlier conclusion that for a transistor in the "on" or active region the base-to-emitter voltage is 0.7 V. In this case the voltage is fixed for any level of base current.

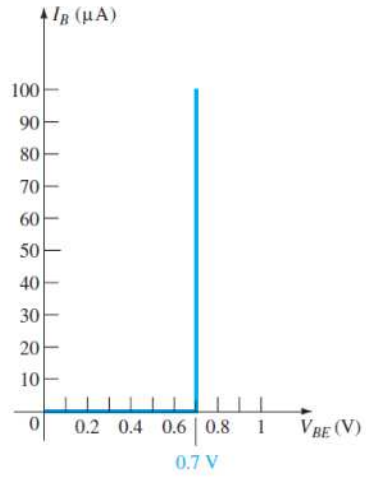


FIG. 3.15
Piecewise-linear equivalent for the diode characteristics of Fig. 3.13b.

Beta (β)

DC Mode In the dc mode the levels of I_C and I_B are related by a quantity called *beta* and defined by the following equation:

$$\beta_{dc} = \frac{I_C}{I_B} \tag{3.10}$$

where I_C and I_B are determined at a particular operating point on the characteristics. For practical devices the level of β typically ranges from about 50 to over 400, with most in the midrange. As for α, the parameter β reveals the relative magnitude of one current with respect to the other. For a device with a β of 200, the collector current is 200 times the magnitude of the base current.

On specification sheets β_{dc} is usually included as h_{FE} with the italic letter h derived from an ac *h*ybrid equivalent circuit to be introduced in Chapter 5. The subscript FE is derived from *f*orward-current amplification and common-*e*mitter configuration, respectively.

AC Mode For ac situations an ac beta is defined as follows:

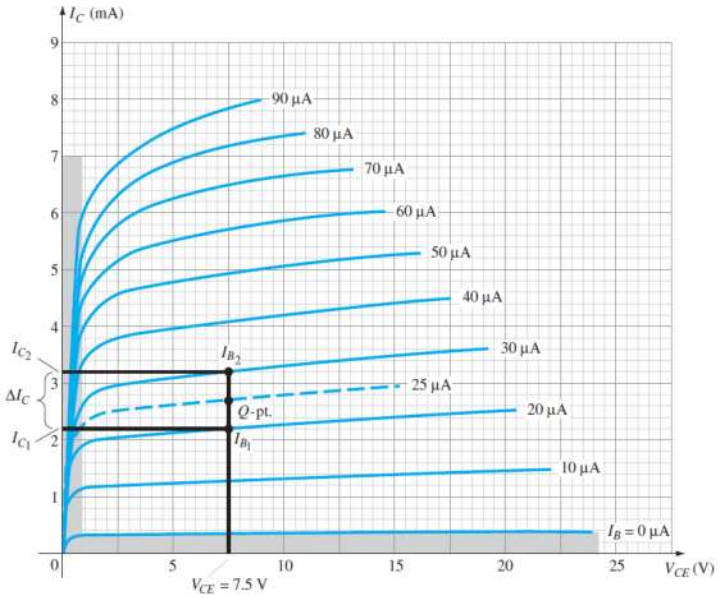
$$\beta_{ac} = \left.\frac{\Delta I_C}{\Delta I_B}\right|_{V_{CE}=\text{constant}} \tag{3.11}$$

The formal name for β_{ac} is *common-emitter, forward-current, amplification factor*. Since the collector current is usually the output current for a common-emitter configuration and the base current is the input current, the term *amplification* is included in the nomenclature above.

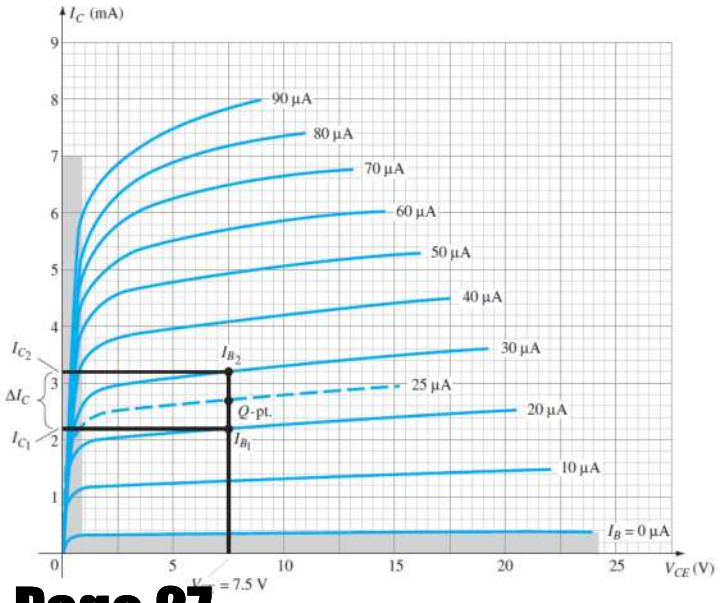
On specification sheets β_{ac} is normally referred to as h_{fe}. Note that the only difference between the notation used for the dc beta, specifically, $\beta_{dc} = h_{FE}$, is the type of lettering for each subscript quantity.

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The use of Eq. (3.11) is best described by a numerical example using an actual set of characteristics such as appearing in Fig. 3.13a and repeated in Fig. 3.17. Let us determine β_{ac} for a region of the characteristics defined by an operating point of $I_B = 25\ \mu A$ and $V_{CE} = 7.5$ V as indicated on Fig. 3.16. The restriction of $V_{CE} =$ constant requires that a vertical



The restriction of $V_{CE} =$ constant requires that a vertical line be drawn through the operating point at $V_{CE} = 7.5$ V. At any location on this vertical line the voltage V_{CE} is 7.5 V, a constant. The change in $I_B(\Delta I_B)$ as appearing in Eq. (3.11) is then defined by choosing two points on either side of the Q-point along the vertical axis of about equal distances to either side of the Q-point. For this situation the $I_B = 20\ \mu A$ and $30\ \mu A$ curves meet the requirement without extending too far from the Q-point. They also



$$\beta_{ac} = \left.\frac{\Delta I_C}{\Delta I_B}\right|_{V_{CE}=\text{constant}} = \frac{I_{C_2} - I_{C_1}}{I_{B_2} - I_{B_1}}$$
$$= \frac{3.2\ \text{mA} - 2.2\ \text{mA}}{30\ \mu A - 20\ \mu A} = \frac{1\ \text{mA}}{10\ \mu A}$$
$$= \mathbf{100}$$

If we determine the dc beta at the Q-point, we obtain

$$\beta_{dc} = \frac{I_C}{I_B} = \frac{2.7\ \text{mA}}{25\ \mu A} = \mathbf{108}$$

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If the characteristics of a transistor are approximated by those appearing in Fig. 3.17, the level of β_{ac} would be the same in every region of the characteristics. Note that the step in I_B is fixed at 10 μA and the vertical spacing between curves is the same at every point in the characteristics—namely, 2 mA. Calculating the β_{ac} at the Q-point indicated results in

$$\beta_{ac} = \left.\frac{\Delta I_C}{\Delta I_B}\right|_{V_{CE}=\text{constant}} = \frac{9\text{ mA} - 7\text{ mA}}{45\ \mu\text{A} - 35\ \mu\text{A}} = \frac{2\text{ mA}}{10\ \mu\text{A}} = \mathbf{200}$$

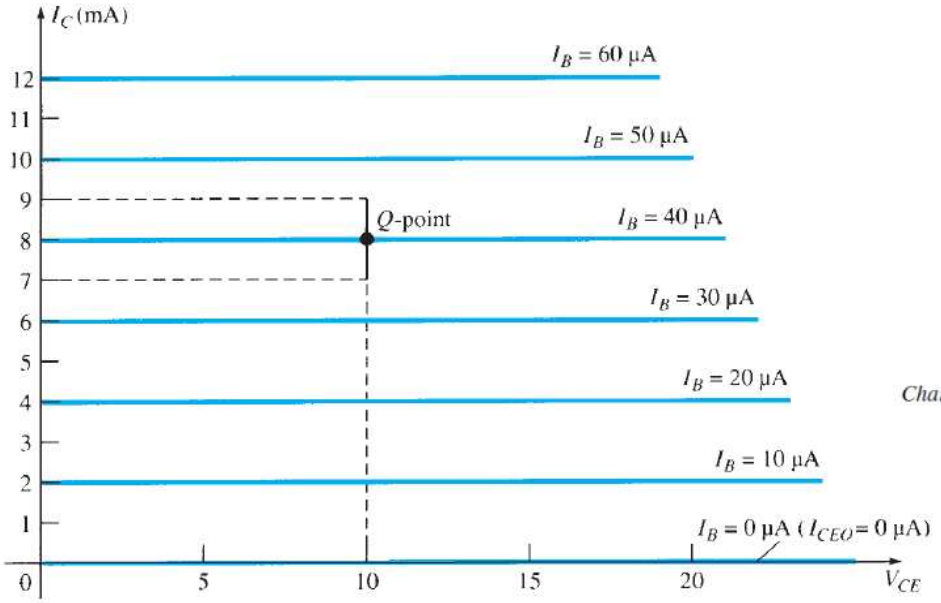
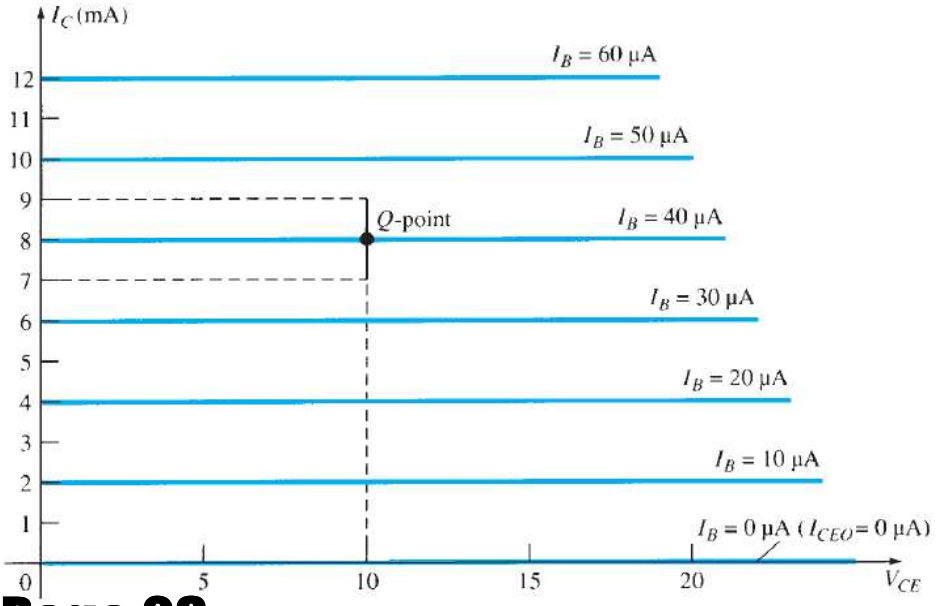


FIG. 3.17
Characteristics in which β_{ac} is the same everywhere and $\beta_{ac} = \beta_{dc}$.

Determining the dc beta at the same Q-point results in

$$\beta_{dc} = \frac{I_C}{I_B} = \frac{8\text{ mA}}{40\ \mu\text{A}} = \mathbf{200}$$

revealing that if the characteristics have the appearance of Fig. 3.17, the magnitudes of β_{ac} and β_{dc} *will be the same* at every point on the characteristics. In particular, note that $I_{CEO} = 0\ \mu$A.



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A relationship can be developed between β and α using the basic relationships introduced thus far. Using $\beta = I_C/I_B$, we have $I_B = I_C/\beta$, and from $\alpha = I_C/I_E$ we have $I_E = I_C/\alpha$. Substituting into

$$I_E = I_C + I_B$$

we have

$$\frac{I_C}{\alpha} = I_C + \frac{I_C}{\beta}$$

and dividing both sides of the equation by I_C results in

$$\frac{1}{\alpha} = 1 + \frac{1}{\beta}$$

or

$$\beta = \alpha\beta + \alpha = (\beta + 1)\alpha$$

so that

$$\boxed{\alpha = \frac{\beta}{\beta + 1}} \qquad \textbf{(3.12)}$$

or

$$\boxed{\beta = \frac{\alpha}{1 - \alpha}} \qquad \textbf{(3.13)}$$

In addition, recall that

$$I_{CEO} = \frac{I_{CBO}}{1 - \alpha}$$

but using an equivalence of

$$\frac{1}{1 - \alpha} = \beta + 1$$

derived from the above, we find that

$$I_{CEO} = (\beta + 1)I_{CBO}$$

or

$$\boxed{I_{CEO} \cong \beta I_{CBO}} \qquad \textbf{(3.14)}$$

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as indicated on Fig. 3.13a. Beta is a particularly important parameter because it provides a direct link between current levels of the input and output circuits for a common-emitter configuration. That is,

$$\boxed{I_C = \beta I_B} \tag{3.15}$$

and since

$$\begin{aligned} I_E &= I_C + I_B \\ &= \beta I_B + I_B \end{aligned}$$

we have

$$\boxed{I_E = (\beta + 1) I_B} \tag{3.16}$$

Both of the equations above play a major role in the analysis in Chapter 4.

If $\beta = 120$, and $I_C = 10$ mA.
Find I_E, I_B, and α

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Chapter 3 Bipolar Junction Transistor

Lecture 3

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Biasing (common base)

The proper biasing of the common-base configuration in the active region can be determined quickly using the approximation $I_C \cong I_E$ and assuming for the moment that $I_B \cong 0\ \mu A$. The result is the configuration of Fig. 3.11 for the *pnp* transistor. The arrow of the symbol defines the direction of conventional flow for $I_E \cong I_C$. The dc supplies are then inserted with a polarity that will support the resulting current direction. For the *npn* transistor the polarities will be reversed.

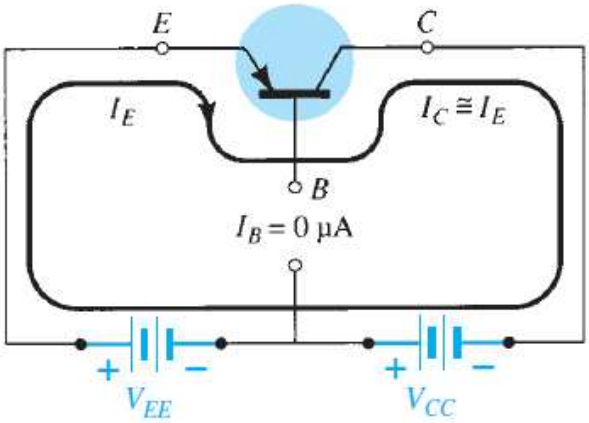
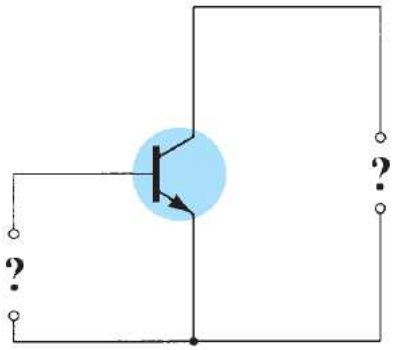


FIG. 3.11
Establishing the proper biasing management for a common-base pnp transistor in the active region.

Biasing

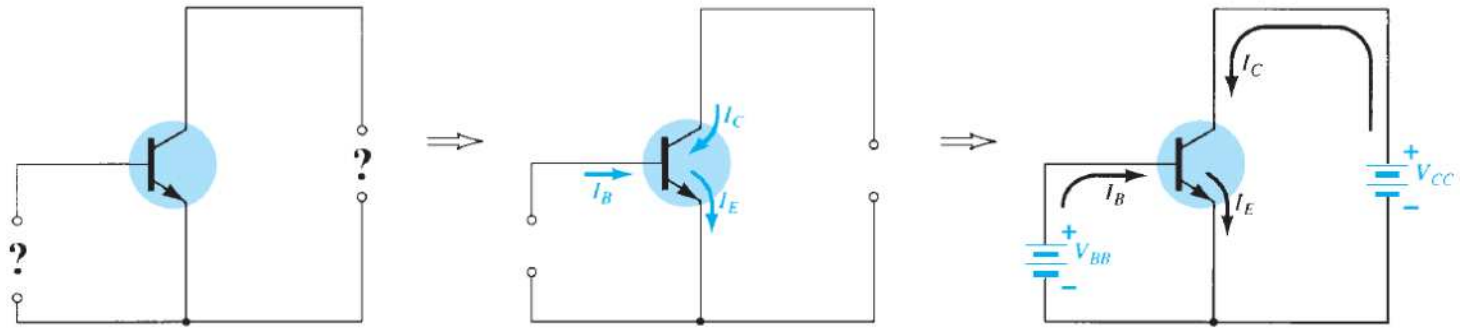
The proper biasing of a common-emitter amplifier can be determined in a manner similar to that introduced for the common-base configuration. Let us assume that we are presented with an *npn* transistor such as shown in Fig. 3.18a and asked to apply the proper biasing to place the device in the active region.



(a)

FIG. 3.18
Determining the proper biasing arrangement for a common-emitter npn transistor configuration.

The first step is to indicate the direction of I_E as established by the arrow in the transistor symbol as shown in Fig. 3.18b. Next, the other currents are introduced as shown, keeping in mind Kirchhoff's current law relationship: $I_C + I_B = I_E$. That is, I_E is the sum of I_C and I_B and both I_C and I_B must enter the transistor structure. Finally, the supplies are introduced with polarities that will support the resulting directions of I_B and I_C as shown in Fig. 3.18c to complete the picture. The same approach can be applied to *pnp* transistors. If the transistor of Fig. 3.18 was a *pnp* transistor, all the currents and polarities of Fig. 3.18c would be reversed.



(a) (b) (c)

FIG. 3.18
Determining the proper biasing arrangement for a common-emitter npn transistor configuration.

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Breakdown Region

As with the common-base configuration, there is a maximum collector-emitter voltage that can be applied and still remain in the active stable region of operation. In Fig. 3.19 the characteristics of Fig. 3.8 have been extended to demonstrate the impact on the characteristics at high levels of V_{CE}. At high levels of base current the currents almost climb vertically, whereas at lower levels a region develops that seems to back up on itself. This region is particularly noteworthy because an increase in current is resulting in a drop in voltage—totally different from that of any resistive element where an increase in current results in an increase in potential drop across the resistor. Regions of this nature are said to have a

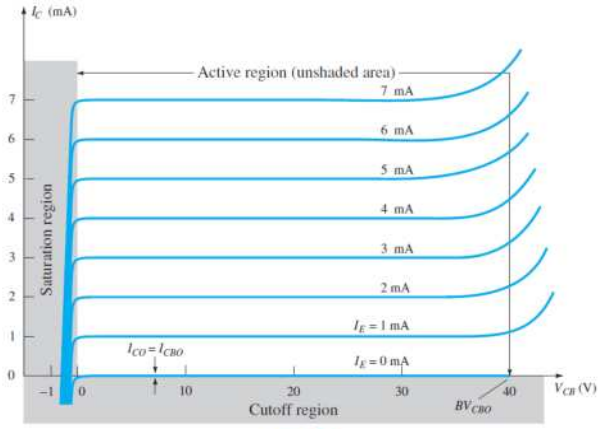
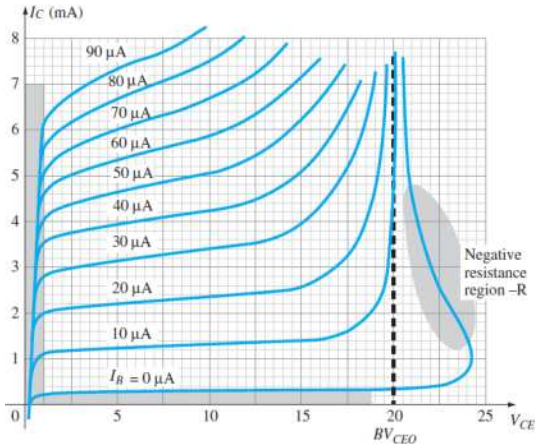


FIG. 3.8



negative-resistance characteristic. Although the concept of a negative resistance may seem strange at this point, this text will introduce devices and systems that rely on this type of characteristic to perform their desired task.

The recommended maximum value for a transistor under normal operating conditions is labeled BV_{CEO} as shown in Fig. 3.19 or $V_{(BR)CEO}$ as shown in Fig. 3.23. It is less than BV_{CBO}

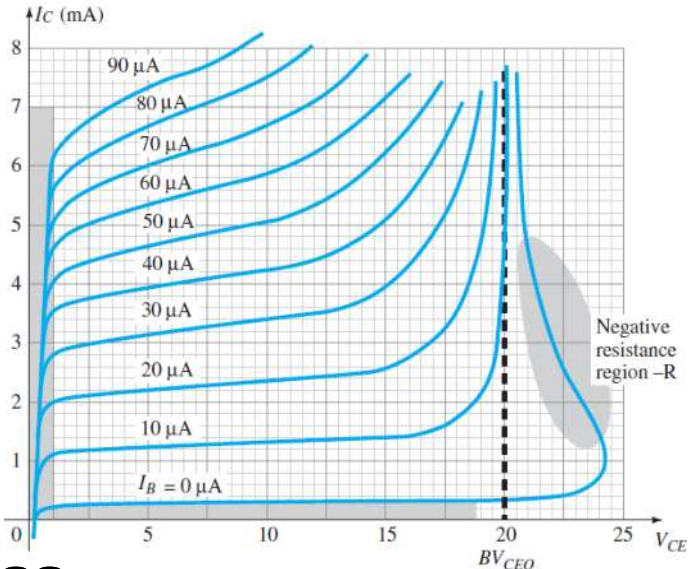


FIG. 3.19
Examining the breakdown region of a transistor in the common-emitter configuration.

3.6 COMMON-COLLECTOR CONFIGURATION

The third and final transistor configuration is the *common-collector configuration*, shown in Fig. 3.20 with the proper current directions and voltage notation. The common-collector configuration is used primarily for impedance-matching purposes since it has a high input impedance and low output impedance, opposite to that of the common-base and common-emitter configurations.

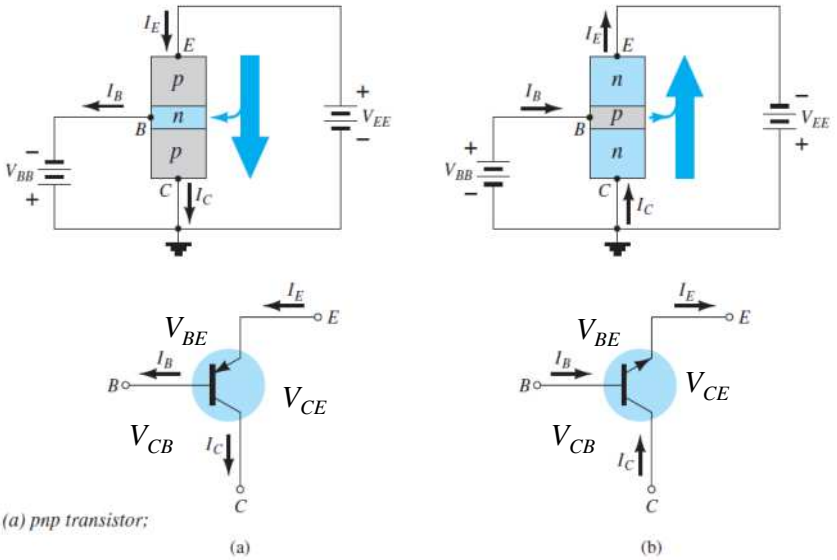


FIG. 3.20
Notation and symbols used with the common-collector configuration: (a) pnp transistor; (b) npn transistor.

A common-collector circuit configuration is provided in Fig. 3.21 with the load resistor connected from emitter to ground. Note that the collector is tied to ground even though the transistor is connected in a manner similar to the common-emitter configuration.

From a design viewpoint, there is no need for a set of common-collector characteristics to choose the parameters of the circuit of Fig. 3.21 .
It can be designed using the common-emitter characteristics of Section 3.5 .

For all practical purposes,

- The output characteristics of the common-collector configuration are the same as for the common-emitter configuration.
- For common-collector configuration, the output characteristics are a plot of I_E versus V_{CE} for a range of values of I_B .

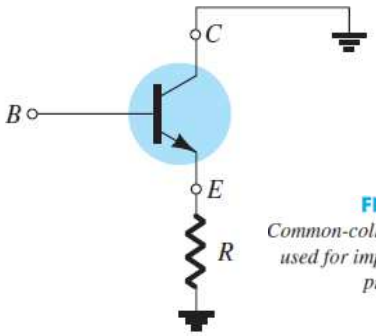
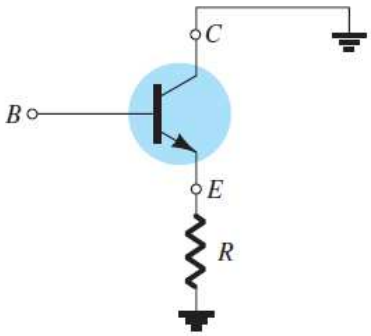


FIG. 3.21
Common-collector configuration used for impedance-matching purposes.

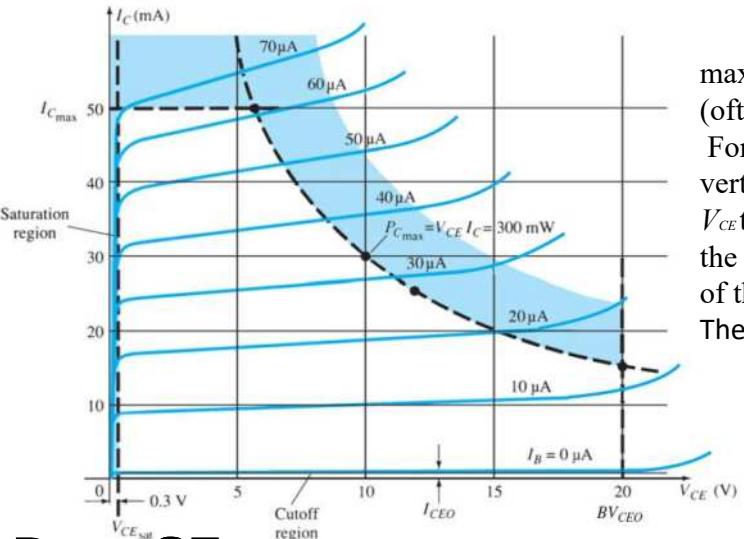
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- The input current, therefore, is the same for both the common-emitter and common-collector characteristics.
- The horizontal voltage axis for the common-collector configuration is obtained by simply changing the sign of the collector-to-emitter voltage of the common-emitter characteristics.
- Finally, there is an almost unnoticeable change in the vertical scale of I_C of the common-emitter characteristics if I_C is replaced by I_E for the common-collector characteristics (since $\alpha \cong 1$). For the input circuit of the common-collector configuration the common-emitter base characteristics are sufficient for obtaining the required information.



3.7 LIMITS OF OPERATION

For each transistor there is a region of operation on the characteristics that will ensure that the maximum ratings are not being exceeded and the output signal exhibits minimum distortion. Such a region has been defined for the transistor characteristics of Fig. 3.22. All of the limits of operation are defined on a typical transistor specification sheet described in Section 3.8.



maximum collector current and maximum collector-to-emitter voltage (often abbreviated as BV_{CEO} or $V_{(BR)CEO}$ on the specification sheet).

For the transistor given, I_{Cmax} was specified as 50 mA, BV_{CEO} as 20 V. The vertical line on the characteristics defined as V_{CEsat} specifies the minimum V_{CE} that can be applied without falling into the nonlinear region labelled the *saturation* region. The level of V_{CEsat} is typically in the neighbourhood of the 0.3 V specified for this transistor.

The maximum dissipation level is defined by the following equation:

$$P_{C_{max}} = V_{CE} I_C \qquad (3.17)$$

FIG. 3.22
Defining the linear (undistorted) region of operation for a transistor.

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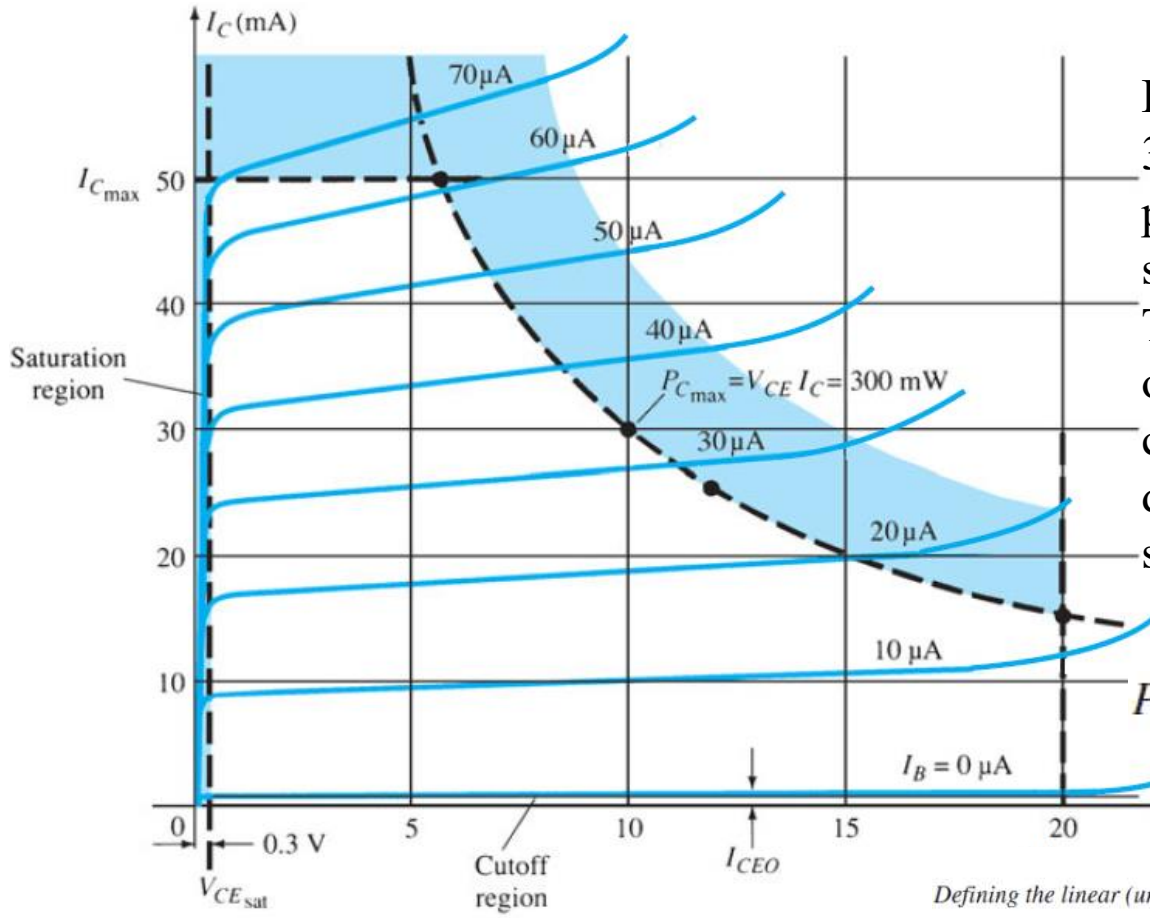
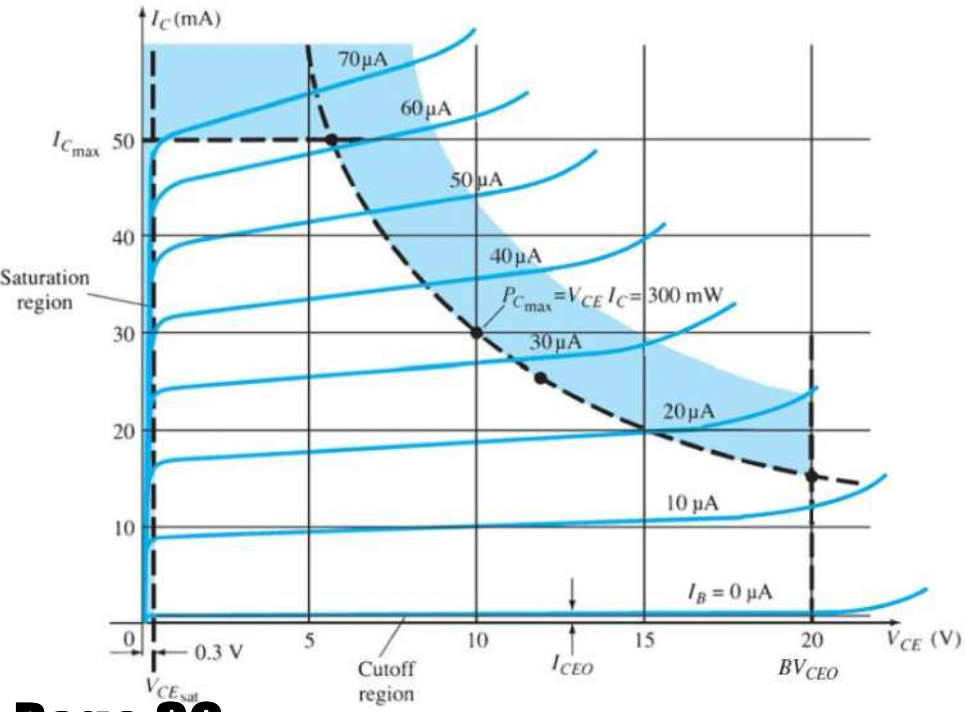


FIG. 3.22
Defining the linear (undistorted) region of operation for a transistor.

For the device of Fig. 3.22 , the collector power dissipation was specified as 300 mW. The question then arises of how to plot the collector power dissipation curve specified by the fact that

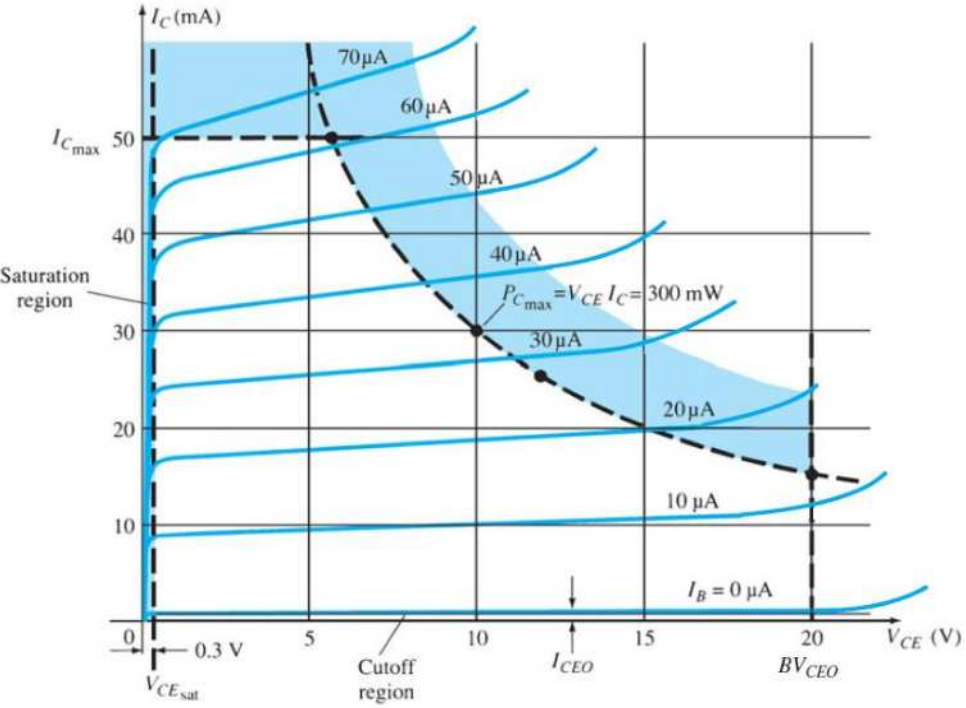
$$P_{C_{max}} = V_{CE} I_C = 300 \text{ mW}$$

At $I_{C_{max}}$ At any point on the characteristics the product of V_{CE} and I_C must be equal to 300 mW. If we choose I_C to be the maximum value of 50 mA and substitute into the relationship above, we obtain



$$V_{CE} I_C = 300 \text{ mW}$$
$$V_{CE}(50 \text{ mA}) = 300 \text{ mW}$$
$$V_{CE} = \frac{300 \text{ mW}}{50 \text{ mA}} = \mathbf{6\ V}$$

At $V_{CE_{max}}$ As a result we find that if $I_C = 50$ mA, then $V_{CE} = 6$ V on the power dissipation curve as indicated in Fig. 3.22. If we now choose V_{CE} to be its maximum value of 20 V, the level of I_C is the following:

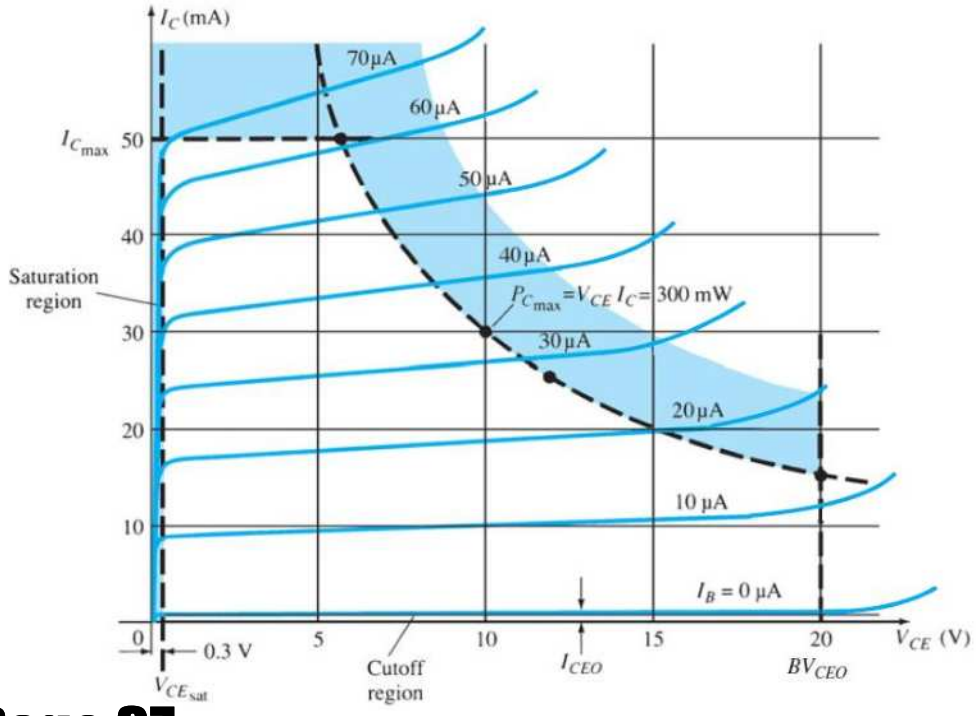


$$(20\text{ V})I_C = 300\text{ mW}$$

$$I_C = \frac{300\text{ mW}}{20\text{ V}} = \mathbf{15\text{ mA}}$$

defining a second point on the power curve.

At $I_C = \frac{1}{2}I_{C_{max}}$ If we now choose a level of I_C in the midrange such as 25 mA and solve for the resulting level of V_{CE}, we obtain



$$V_{CE}(25\text{ mA}) = 300\text{ mW}$$

$$V_{CE} = \frac{300\text{ mW}}{25\text{ mA}} = \mathbf{12\text{ V}}$$

If the characteristic curves are unavailable or do not appear on the specification sheet (as is often the case), one must simply be sure that I_C, V_{CE}, and their product $V_{CE}I_C$ fall into the following range:

$$\begin{aligned} I_{CEO} &\leqq I_C \leqq I_{C_{max}} \\ V_{CE_{sat}} &\leqq V_{CE} \leqq V_{CE_{max}} \\ V_{CE}I_C &\leqq P_{C_{max}} \end{aligned} \tag{3.18}$$

For the common-base characteristics the maximum power curve is defined by the following product of output quantities:

$$P_{C_{max}} = V_{CB}I_C \tag{3.19}$$

3.8 TRANSISTOR SPECIFICATION SHEET

Since the specification sheet is the communication link between the manufacturer and user, it is particularly important that the information provided be recognized and correctly understood. Although all the parameters have not been introduced, a broad number will now be familiar. The remaining parameters will be introduced in the chapters that follow. Reference will then be made to this specification sheet to review the manner in which the parameter is presented.

The information provided as Fig. 3.23 is provided by the Fairchild Semiconductor Corporation. The 2N4123 is a general-purpose *npn* transistor with the casing and terminal

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MAXIMUM RATINGS

Rating	Symbol	2N4123	Unit
Collector-Emitter Voltage	V_{CEO}	30	Vdc
Collector-Base Voltage	V_{CBO}	40	Vdc
Emitter-Base Voltage	V_{EBO}	5.0	Vdc
Collector Current – Continuous	I_C	200	mAdc
Total Device Dissipation @ $T_A = 25°C$ Derate above 25°C	P_D	625 5.0	mW mW°C
Operating and Storage Junction Temperature Range	T_j, T_{stg}	–55 to +150	°C

THERMAL CHARACTERISTICS

Characteristic	Symbol	Max	Unit
Thermal Resistance, Junction to Case	$R_{\theta JC}$	83.3	°C W
Thermal Resistance, Junction to Ambient	$R_{\theta JA}$	200	°C W

FAIRCHILD
SEMICONDUCTOR ™

2N4123



TO-92

General Purpose Transistor NPN Silicon

For more information
Textbook 146-152

Characteristic	Symbol	Min	Max	Unit
OFF CHARACTERISTICS				
Collector-Emitter Breakdown Voltage (1) ($I_C = 1.0$ mAdc, $I_E = 0$)	$V_{(BR)CEO}$	30		Vdc
Collector-Base Breakdown Voltage ($I_C = 10$ µAdc, $I_E = 0$)	$V_{(BR)CBO}$	40		Vdc
Emitter-Base Breakdown Voltage ($I_E = 10$ µAdc, $I_C = 0$)	$V_{(BR)EBO}$	5.0	–	Vdc
Collector Cutoff Current ($V_{CB} = 20$ Vdc, $I_E = 0$)	I_{CBO}	–	50	nAdc
Emitter Cutoff Current ($V_{BE} = 3.0$ Vdc, $I_C = 0$)	I_{EBO}	–	50	nAdc
ON CHARACTERISTICS				
DC Current Gain(1) ($I_C = 2.0$ mAdc, $V_{CE} = 1.0$ Vdc) ($I_C = 50$ mAdc, $V_{CE} = 1.0$ Vdc)	h_{FE}	50 25	150 –	–
Collector-Emitter Saturation Voltage(1) ($I_C = 50$ mAdc, $I_B = 5.0$ mAdc)	$V_{CE(sat)}$	–	0.3	Vdc
Base-Emitter Saturation Voltage(1) ($I_C = 50$ mAdc, $I_B = 5.0$ mAdc)	$V_{BE(sat)}$	–	0.95	Vdc
SMALL-SIGNAL CHARACTERISTICS				
Current-Gain – Bandwidth Product ($I_C = 10$ mAdc, $V_{CE} = 20$ Vdc, f = 100 MHz)	f_T	250		MHz
Output Capacitance ($V_{CB} = 5.0$ Vdc, $I_E = 0$, f = 100 MHz)	C_{obo}	–	4.0	pF
Input Capacitance ($V_{BE} = 0.5$ Vdc, $I_C = 0$, f = 100 kHz)	C_{ibo}	–	8.0	pF
Collector-Base Capacitance ($I_E = 0$, $V_{CB} = 5.0$ V, f = 100 kHz)	C_{cb}	–	4.0	pF
Small-Signal Current Gain ($I_C = 2.0$ mAdc, $V_{CE} = 10$ Vdc, f = 1.0 kHz)	h_{fe}	50	200	–
Current Gain – High Frequency ($I_C = 10$ mAdc, $V_{CE} = 20$ Vdc, f = 100 MHz) ($I_C = 2.0$ mAdc, $V_{CE} = 10$ V, f = 1.0 kHz)	h_{fe}	2.5 50	– 200	–

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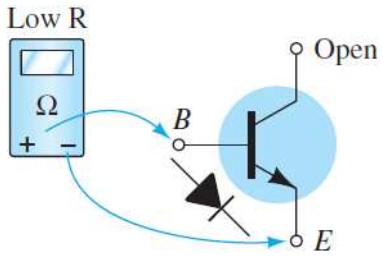


FIG. 3.27
Checking the forward-biased base-to-emitter junction of an npn transistor.

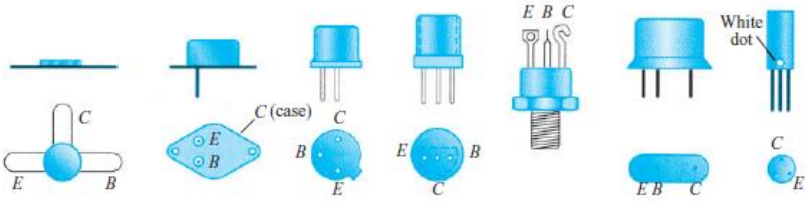
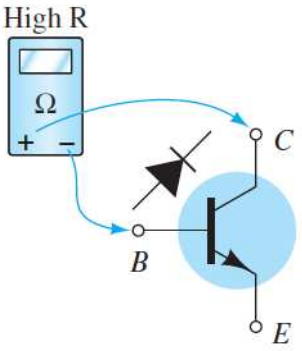


FIG. 3.30
Transistor terminal identification.



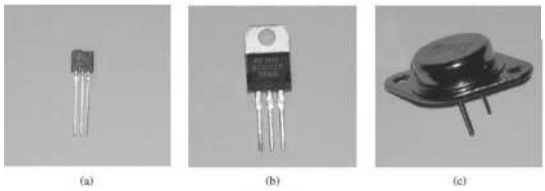


FIG. 3.29
Various types of general-purpose or switching transistors: (a) low power; (b) medium power; (c) medium to high power.

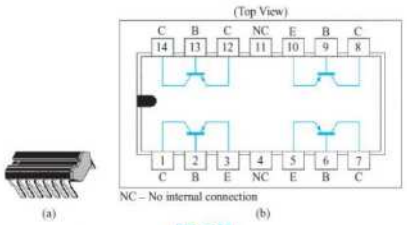


FIG. 3.32
Type Q2T2905 Texas Instruments quad pnp silicon transistor: (a) appearance; (b) pin connections.

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Faculty of Engineering Technology
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Chapter 4 DC Biasing—BJTs

Lecture 1

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CHAPTER OBJECTIVES ●

- Be able to determine the dc levels for the variety of important BJT configurations.
- Understand how to measure the important voltage levels of a BJT transistor configuration and use them to determine whether the network is operating properly.
- Become aware of the saturation and cutoff conditions of a BJT network and the expected voltage and current levels established by each condition.
- Be able to perform a load-line analysis of the most common BJT configurations.
- Become acquainted with the design process for BJT amplifiers.
- Understand the basic operation of transistor switching networks.
- Begin to understand the troubleshooting process as applied to BJT configurations.
- Develop a sense for the stability factors of a BJT configuration and how they affect its operation due to changes in specific characteristics and environmental changes.

The analysis or design of a transistor amplifier requires a knowledge of both the DC and the AC response of the system. Too often it is assumed that the transistor is a magical device that can raise the level of the applied ac input without the assistance of an external energy source. In actuality, ***any increase in ac voltage, current, or power is the result of a transfer of energy from the applied dc supplies.***

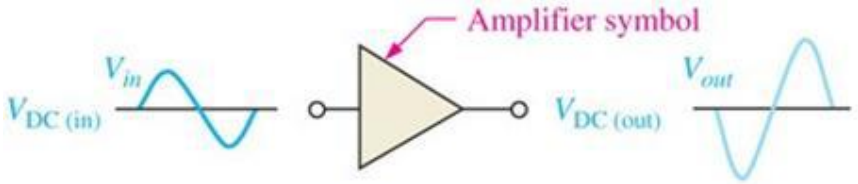
The analysis or design of any electronic amplifier therefore has two components: a dc and an ac portion. Fortunately, the superposition theorem is applicable, and the investigation of the dc conditions can be totally separated from the ac response. However, one must keep in mind that during the design or synthesis stage the choice of parameters for the required dc levels will affect the ac response, and vice versa.

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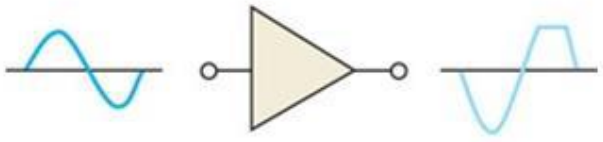
The dc level of operation of a transistor is controlled by a number of factors, including the range of possible operating points on the device characteristics A transistor must be properly biased with a dc voltage in order to operate as a linear amplifier.

A dc operating point must be set so that signal variations at the input terminal are amplified and accurately reproduced at the output terminal.

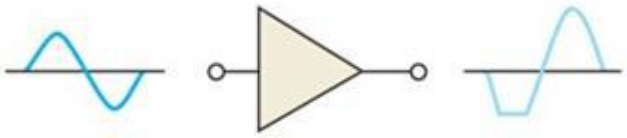
- If an amplifier is not biased with the correct DC voltages, it can go into saturation and cutoff.
- Figure 1(a) shows the correct Linear operation with amplified output.
- Figure 1(b) shows nonlinear operation where the amplifier is in cutoff. The clipping in the positive cycle is always due to cutoff.
- Figure 1(c) shows nonlinear operation where the amplifier is in saturation. The clipping in the negative cycle is always due to saturation.



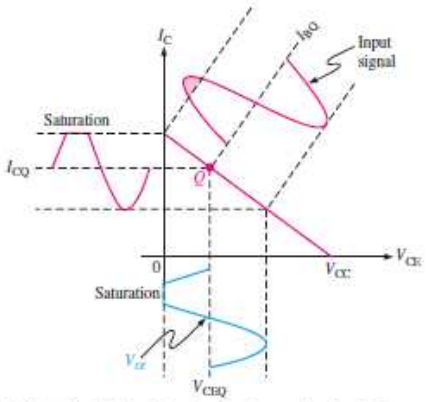
(a) Linear operation: larger output has same shape as input except that it is inverted



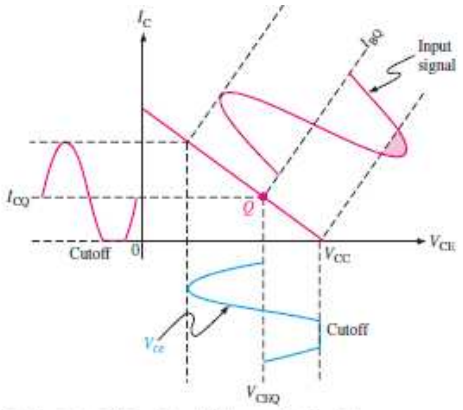
(b) Nonlinear operation: output voltage limited (clipped) by cutoff



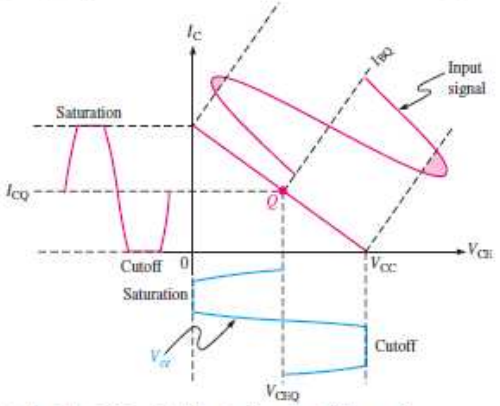
(c) Nonlinear operation: output voltage limited (clipped) by saturation



(a) Transistor is driven into saturation because the Q-point is too close to saturation for the given input signal.



(b) Transistor is driven into cutoff because the Q-point is too close to cutoff for the given input signal.



(c) Transistor is driven into both saturation and cutoff because the input signal is too large.

Although a number of networks are analyzed in this chapter, there is an underlying similarity in the analysis of each configuration due to the recurring use of the following important basic relationships for a transistor:

$$V_{BE} \cong 0.7 \text{ V} \tag{4.1}$$

$$I_E = (\beta + 1)I_B \cong I_C \tag{4.2}$$

$$I_C = \beta I_B \tag{4.3}$$

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The term *biasing* appearing in the title of this chapter is an all-inclusive term for the application of dc voltages to establish a fixed level of current and voltage. For transistor amplifiers the resulting dc current and voltage establish an *operating point* on the characteristics that define the region that will be employed for amplification of the applied signal. Because the operating point is a fixed point on the characteristics, it is also called the *quiescent point* (abbreviated *Q* -point). By definition, *quiescent* means quiet, still, inactive. Figure.1 shows a general output device characteristic with four operating points indicated.

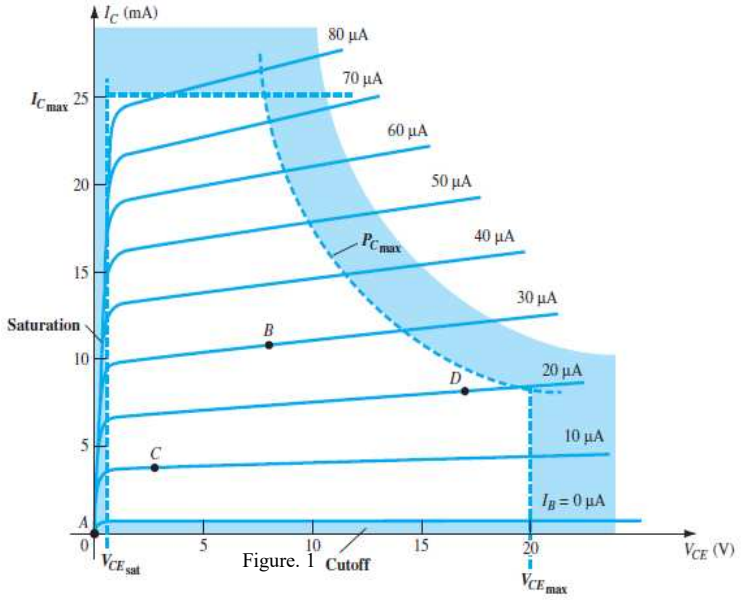
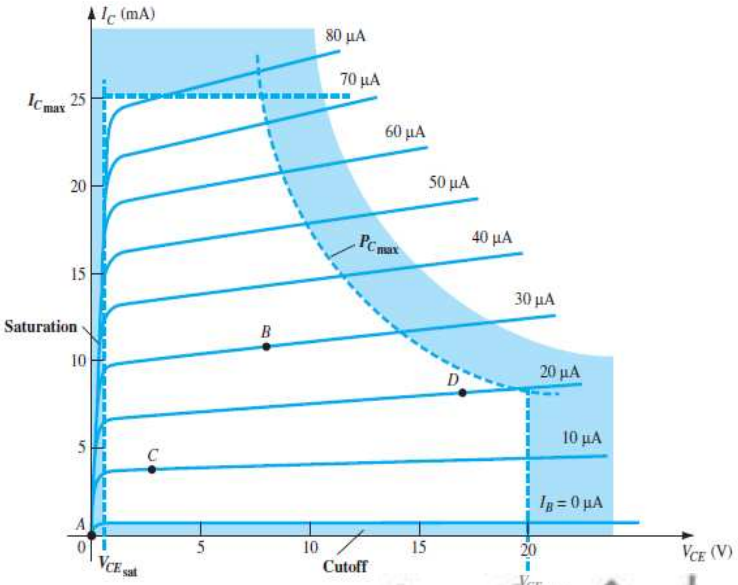


Figure. 1 **Various operating points within the limits of operation of a transistor**

The biasing circuit can be designed to set the device operation at any of these points or others within the *active region.* The maximum power constraint is defined by the curve P_{Cmax} in the same figure. At the lower end of the scales are the *cutoff region* , defined by $I_B \leq 0$ mA, and the *saturation region,* defined by $V_{CE} \leq V_{CE\,sat}$

Confining ourselves to the *active* region, we can select many different operating areas or points. The chosen *Q* -point often depends on the intended use of the circuit.



The chosen Q -point often depends on the intended use of the circuit.
For the BJT to be biased in its linear or active operating region the following must be true:

1. ***The base–emitter junction must be forward-biased (p-region voltage more positive), with a resulting forward-bias voltage of about 0.6 V to 0.7 V.***

2. ***The base–collector junction must be reverse-biased (n-region more positive), with the reverse-bias voltage being any value within the maximum limits of the device.***

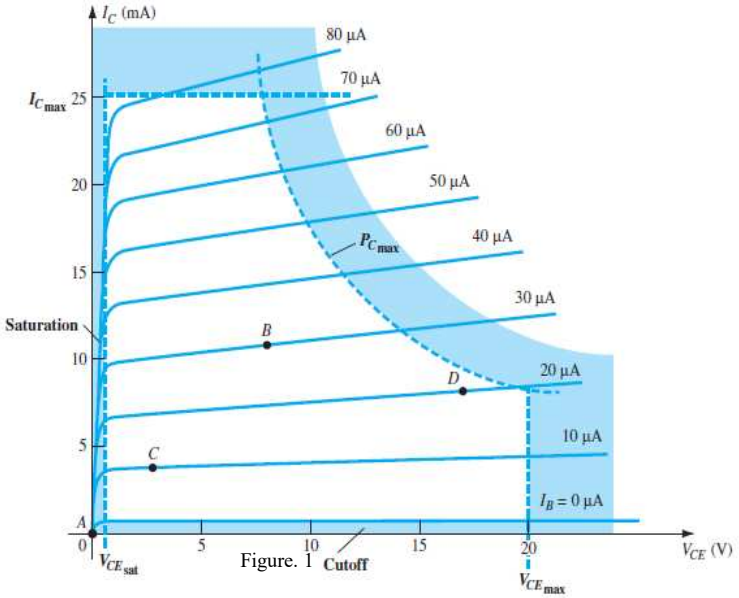


Figure. 1 **Various operating points within the limits of operation of a transistor**

Operation in the cutoff, saturation, and linear regions of the BJT characteristic are provided
as follows:

1. *Linear-region operation:*

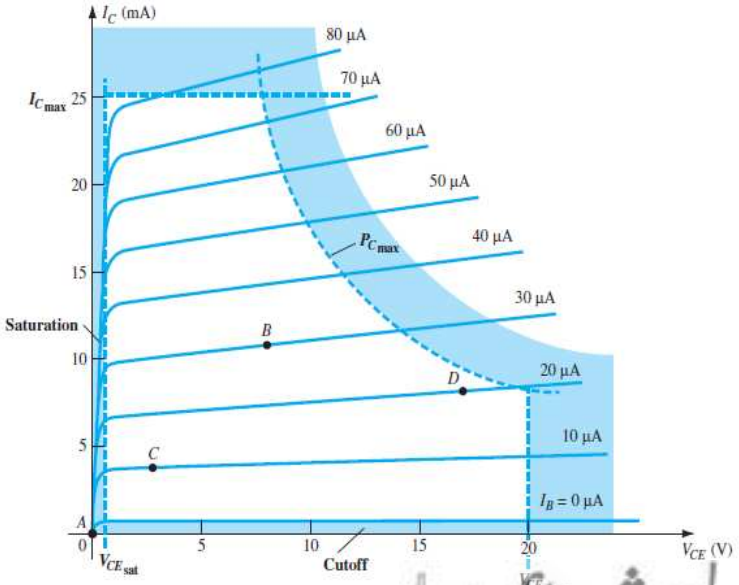
Base–emitter junction forward-biased
Base–collector junction reverse-biased

2. *Cutoff-region operation:*

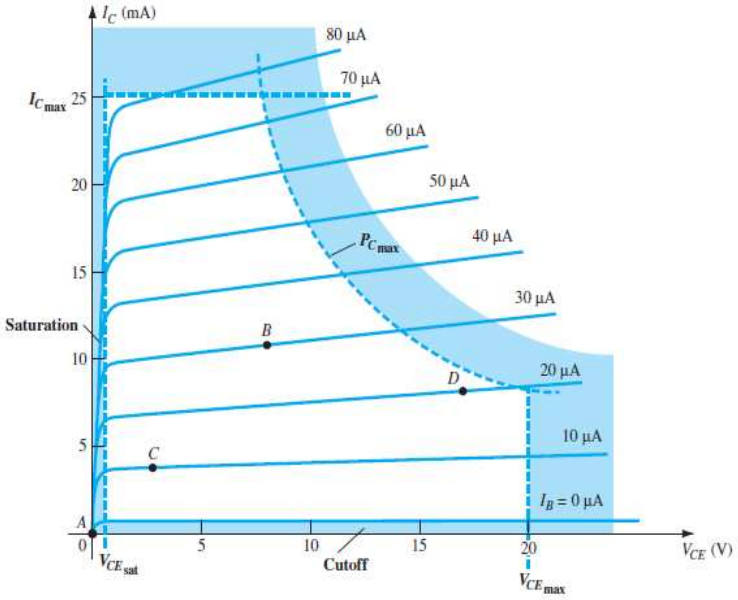
Base–emitter junction reverse-biased
Base–collector junction reverse-biased

3. *Saturation-region operation:*

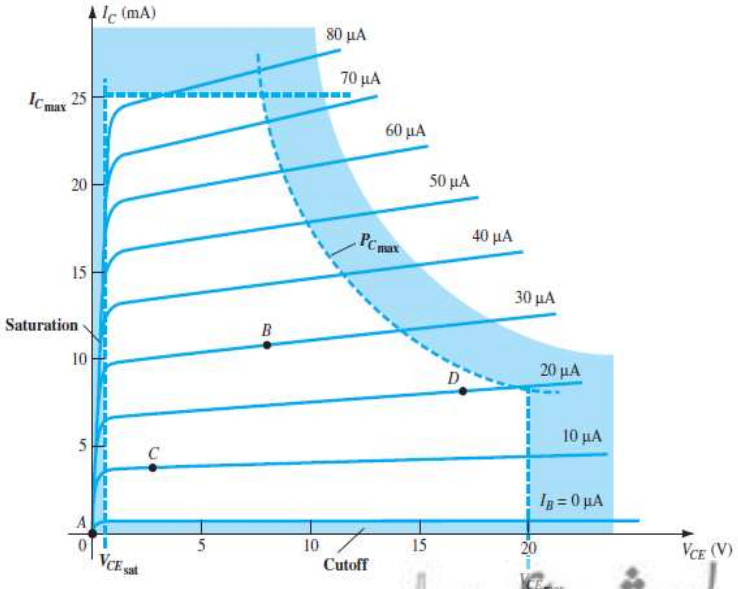
Base–emitter junction forward-biased
Base–collector junction forward-biased



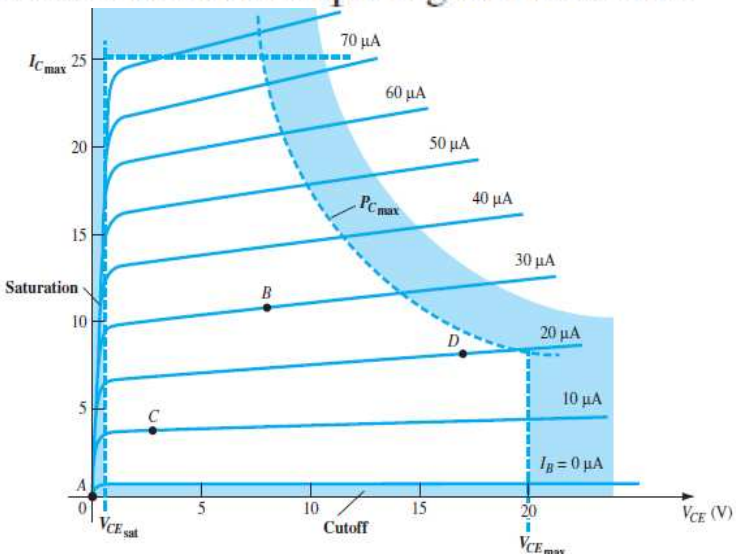
If no bias were used, the device would initially be completely off, resulting in a *Q*-point at *A*—namely, zero current through the device (and zero voltage across it). Because it is necessary to bias a device so that it can respond to the entire range of an input signal, point *A* would not be suitable.



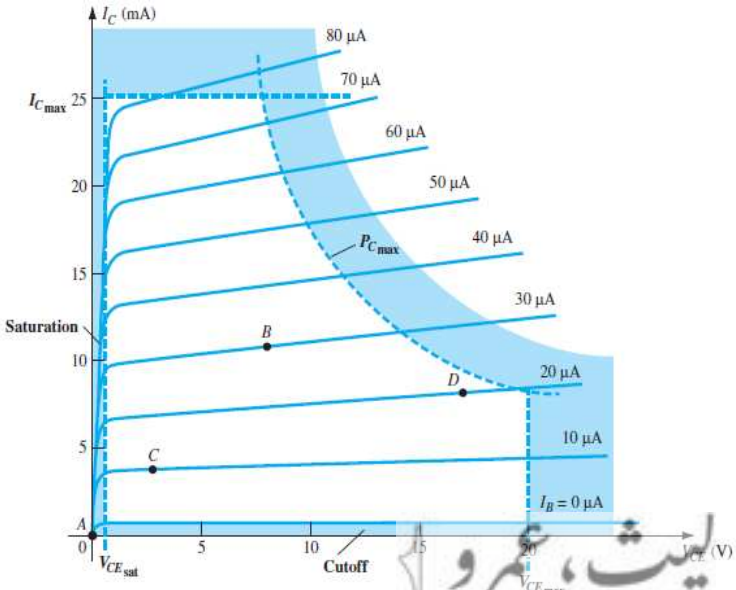
For point *B*, if a signal is applied to the circuit, the device will vary in current and voltage from the operating point, allowing the device to react to (and possibly amplify) both the positive and negative excursions of the input signal. If the input signal is properly chosen, the voltage and current of the device will vary, but not enough to drive the device into *cutoff* or *saturation*.



Point C would allow some positive and negative variation of the output signal, but the peak-to-peak value would be limited by the proximity of $V_{CE} = 0$ V and $I_C = 0$ mA. Operating at point C also raises some concern about the nonlinearities introduced by the fact that the spacing between I_B curves is rapidly changing in this region. In general, it is preferable to operate where the gain of the device is fairly constant (or linear) to ensure that the amplification over the entire swing of input signal is the same. Point B is a region of more linear spacing and therefore more linear operation, as shown in Fig. 4.1.

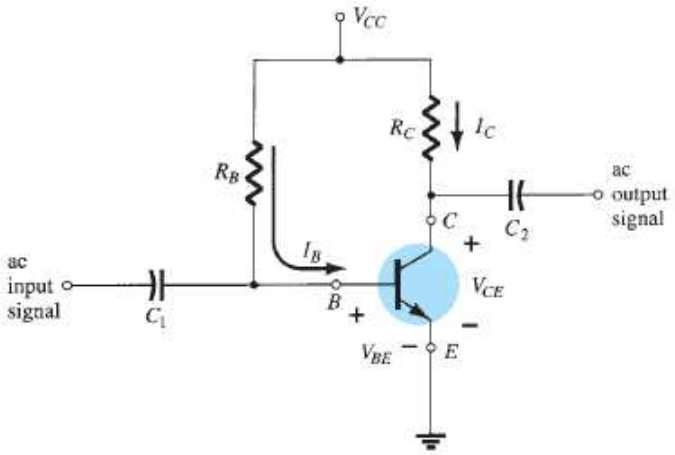


Point D sets the device operating point near the maximum voltage and power level. The output voltage swing in the positive direction is thus limited if the maximum voltage is not to be exceeded. Point B therefore seems the best operating point in terms of linear gain and largest possible voltage and current swing.

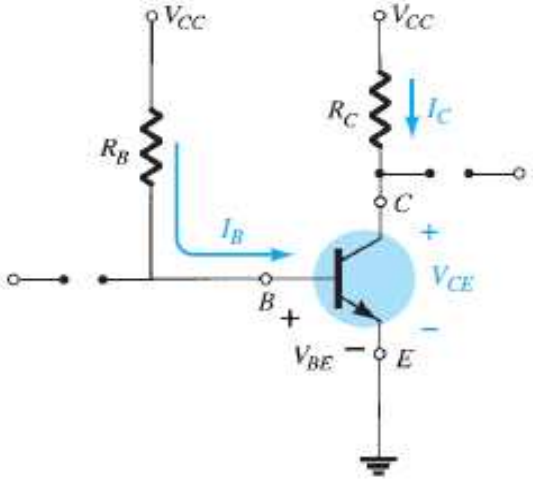


FIXED-BIAS CONFIGURATION

The network employs an *npn* transistor, the equations and calculations apply equally well to a *pnp* transistor configuration merely by changing all current directions and voltage polarities.

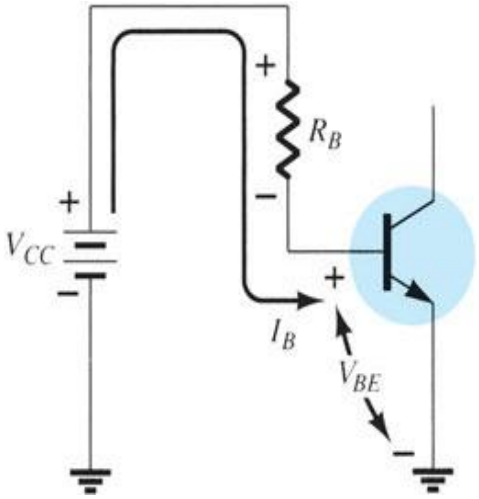


Fixed-bias circuit.



DC equivalent

Forward Bias of Base–Emitter



Writing Kirchhoff's voltage equation , in the clockwise direction for the loop, we obtain

$$+V_{CC} - I_B R_B - V_{BE} = 0$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

Collector–Emitter Loop

The collector–emitter section of the network appears in Fig. 4.5 with the indicated direction of current I_C and the resulting polarity across R_C. The magnitude of the collector current is related directly to I_B through

$$I_C = \beta I_B \tag{4.5}$$

Applying Kirchhoff's voltage law in the clockwise direction around the indicated closed loop of Fig. 4.5 results in the following:

$$V_{CE} + I_C R_C - V_{CC} = 0$$

and

$$V_{CE} = V_{CC} - I_C R_C \tag{4.6}$$

which states that the voltage across the collector–emitter region of a transistor in the fixed-bias configuration is the supply voltage less the drop across R_C.

As a brief review of single- and double-subscript notation recall that

$$V_{CE} = V_C - V_E \tag{4.7}$$

where V_{CE} is the voltage from collector to emitter and V_C and V_E are the voltages from collector and emitter to ground, respectively. *In this case,* since $V_E = 0$ V, we have

$$V_{CE} = V_C \tag{4.8}$$

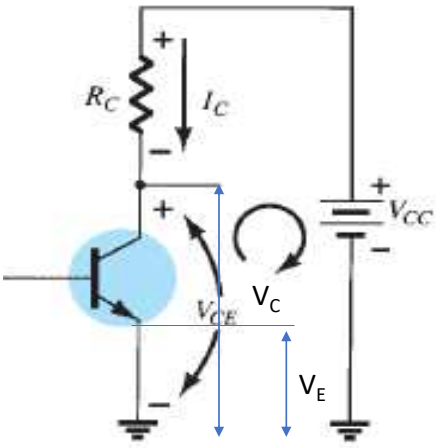


Fig. 4.5

In addition, because

$$V_{BE} = V_B - V_E$$

and $V_E = 0$ V, then

$$V_{BE} = V_B$$

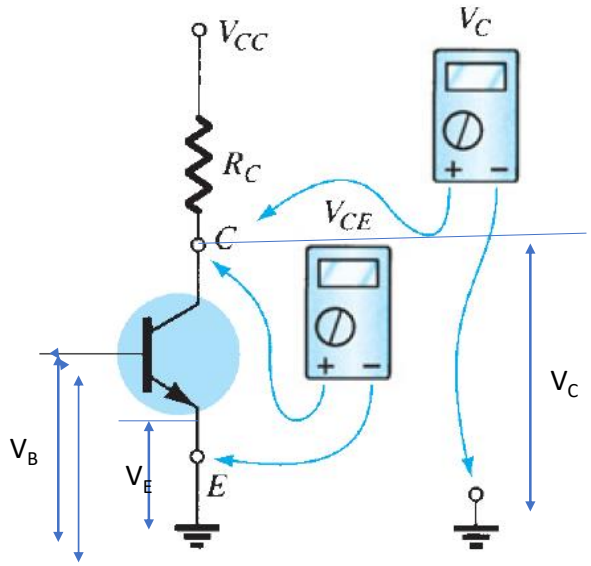
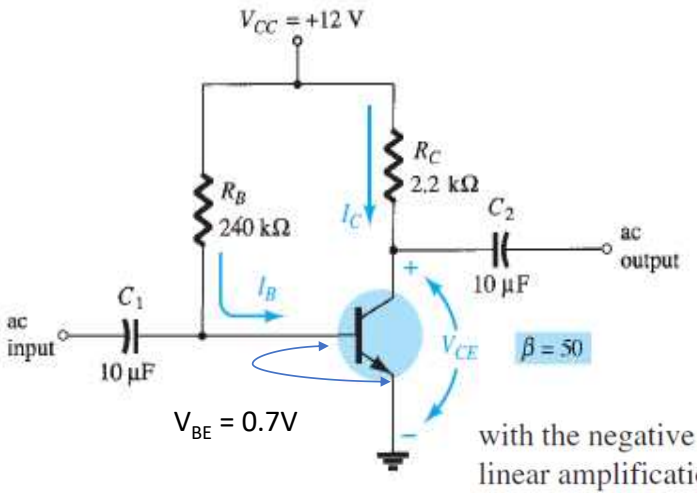


FIG. 4.6
Measuring V_{CE} and V_C.

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EXAMPLE 1 Determine the following for the fixed-bias configuration of following Figure:

a. I_{BQ} and I_{CQ}.
b. V_{CEQ}.
c. V_B and V_C.
d. V_{BC}.



Solution:

a. Eq. (4.4): $$I_{B_Q} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12\text{ V} - 0.7\text{ V}}{240\text{ k}\Omega} = \mathbf{47.08\ \mu A}$$

Eq. (4.5): $$I_{C_Q} = \beta I_{BQ} = (50)(47.08\ \mu\text{A}) = \mathbf{2.35\ mA}$$

b. Eq. (4.6): $$V_{CE_Q} = V_{CC} - I_C R_C$$
$$= 12\text{ V} - (2.35\text{ mA})(2.2\text{ k}\Omega)$$
$$= \mathbf{6.83\ V}$$

c. $V_B = V_{BE} = \mathbf{0.7\ V}$
$V_C = V_{CE} = \mathbf{6.83\ V}$

d. Using double-subscript notation yields
$$V_{BC} = V_B - V_C = 0.7\text{ V} - 6.83\text{ V}$$
$$= \mathbf{-6.13\ V}$$

with the negative sign revealing that the junction is reversed-biased, as it should be for linear amplification.

Transistor Saturation

The term *saturation* is applied to any system where levels have reached their maximum values. A saturated sponge is one that cannot hold another drop of water. For a transistor operating in the saturation region, the current is a maximum value *for the particular design*. Change the design and the corresponding saturation level may rise or drop. Of course, the highest saturation level is defined by the maximum collector current as provided by the specification sheet.

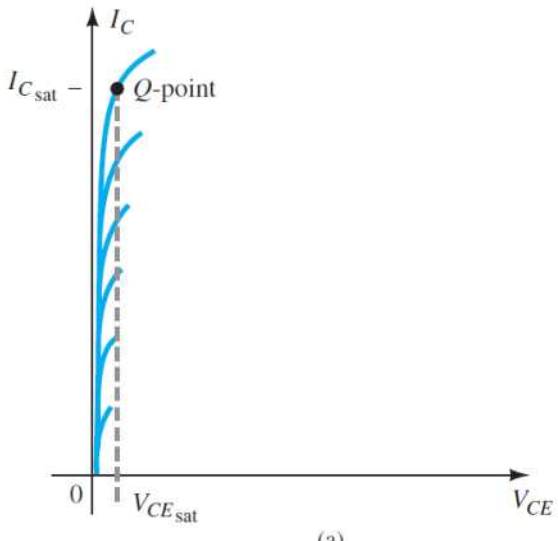


FIG. 4.8
Saturation regions: (a) actual; (b) approximate.

If we approximate the curves of Fig. 4.8a by those appearing in Fig. 4.8b, a quick, direct method for determining the saturation level becomes apparent. In Fig. 4.8b, the current is relatively high, and the voltage V_{CE} is assumed to be 0 V. Applying Ohm's law, we can determine the resistance between collector and emitter terminals as follows:

$$R_{CE} = \frac{V_{CE}}{I_C} = \frac{0\text{ V}}{I_{C_{sat}}} = 0\ \Omega$$

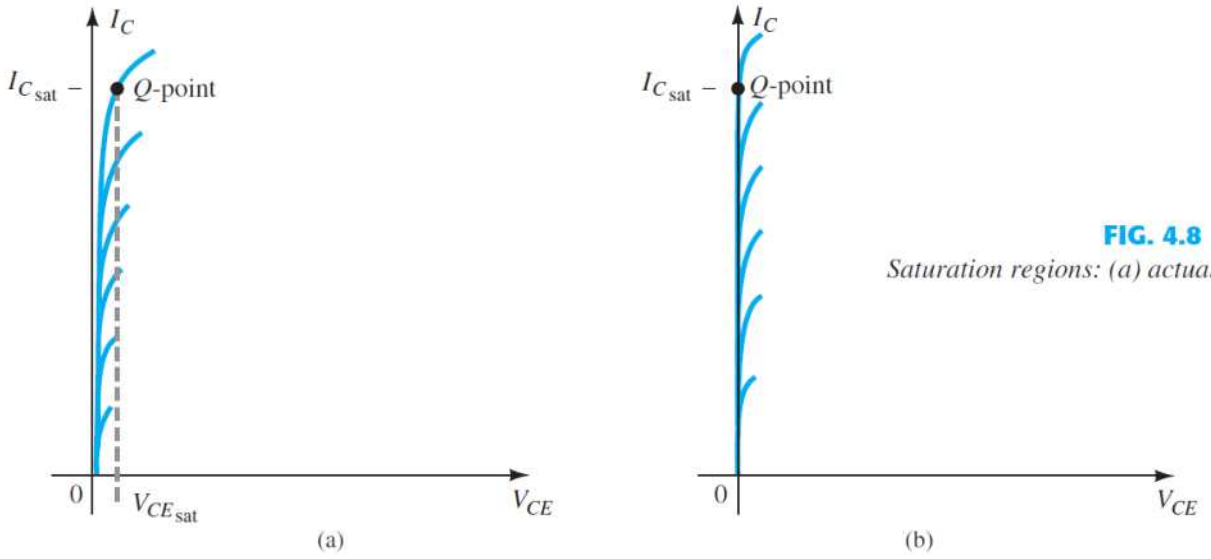
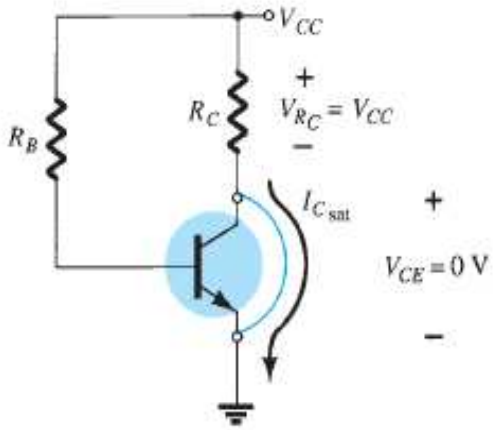


FIG. 4.8
Saturation regions: (a) actual; (b) approximate.

The resulting saturation current for the fixed-bias configuration is

$$I_{C_{sat}} = \frac{V_{CC}}{R_C}$$



Once $I_{C_{sat}}$ is known, we have some idea of the maximum possible collector current for the chosen design and the level to stay below if we expect linear amplification.

لیث، عمرو

EXAMPLE 4.2 Determine the saturation level for the network of Fig. 4.7.

Solution:

$$I_{C_{sat}} = \frac{V_{CC}}{R_C} = \frac{12\text{ V}}{2.2\text{ k}\Omega} = \mathbf{5.45\text{ mA}}$$

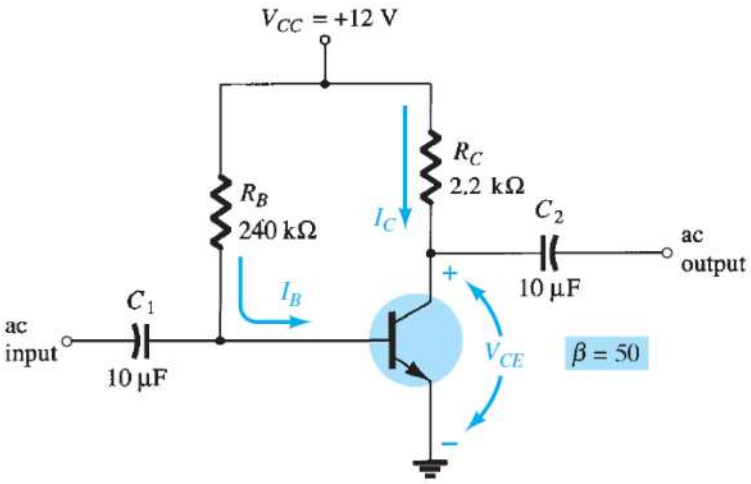


FIG. 4.7

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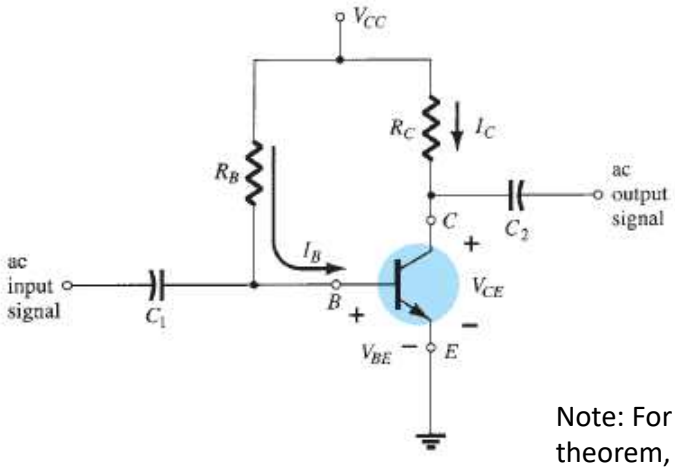
Chapter 4: DC Biasing—BJTs

Lecture 2

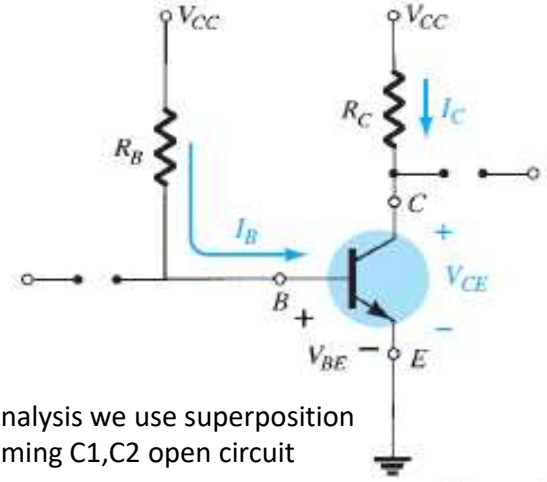
Dr . Majed Dwairi
majeddw@bau.edu.jo
majeddw@gmail.com
Mob. 0797146959

FIXED-BIAS CONFIGURATION

The network employs an *npn* transistor, the equations and calculations apply equally well to a *pnp* transistor configuration merely by changing all current directions and voltage polarities.



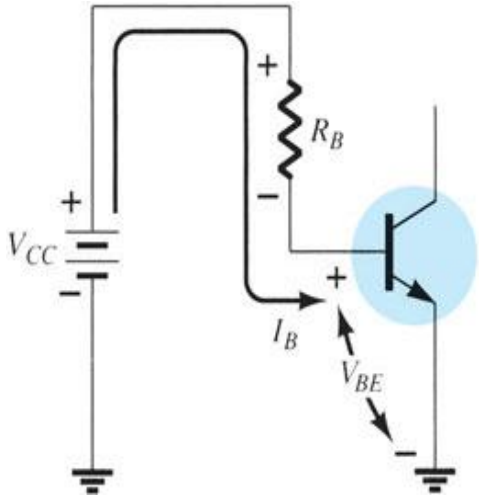
Fixed-bias circuit.



Note: For Dc analysis we use superposition theorem, assuming C1,C2 open circuit

DC equivalent

Forward Bias of Base–Emitter



Writing Kirchhoff's voltage equation , in the clockwise direction for the loop, we obtain

$$+V_{CC} - I_B R_B - V_{BE} = 0$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

Collector–Emitter Loop

The collector–emitter section of the network appears in Fig. 4.5 with the indicated direction of current I_C and the resulting polarity across R_C. The magnitude of the collector current is related directly to I_B through

$$I_C = \beta I_B \tag{4.5}$$

Applying Kirchhoff's voltage law in the clockwise direction around the indicated closed loop of Fig. 4.5 results in the following:

$$V_{CE} + I_C R_C - V_{CC} = 0$$

and

$$V_{CE} = V_{CC} - I_C R_C \tag{4.6}$$

which states that the voltage across the collector–emitter region of a transistor in the fixed-bias configuration is the supply voltage less the drop across R_C.

As a brief review of single- and double-subscript notation recall that

$$V_{CE} = V_C - V_E \tag{4.7}$$

where V_{CE} is the voltage from collector to emitter and V_C and V_E are the voltages from collector and emitter to ground, respectively. *In this case,* since $V_E = 0$ V, we have

$$V_{CE} = V_C \tag{4.8}$$

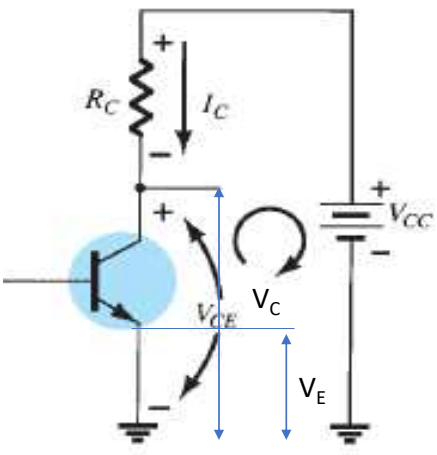


Fig. 4.5

In addition, because

$$V_{BE} = V_B - V_E$$

and $V_E = 0$ V, then

$$V_{BE} = V_B$$

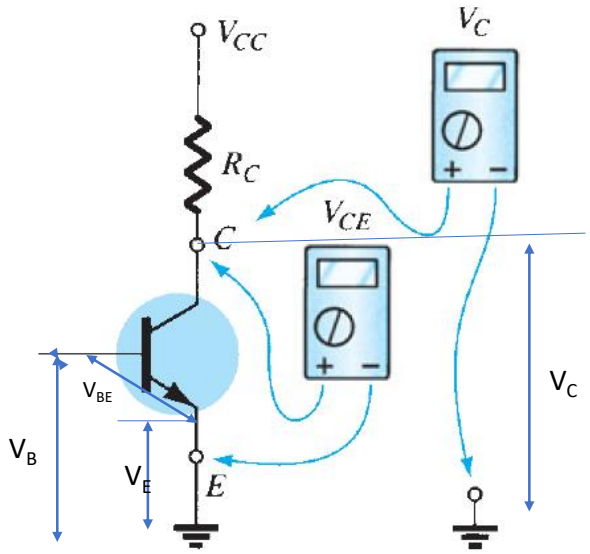
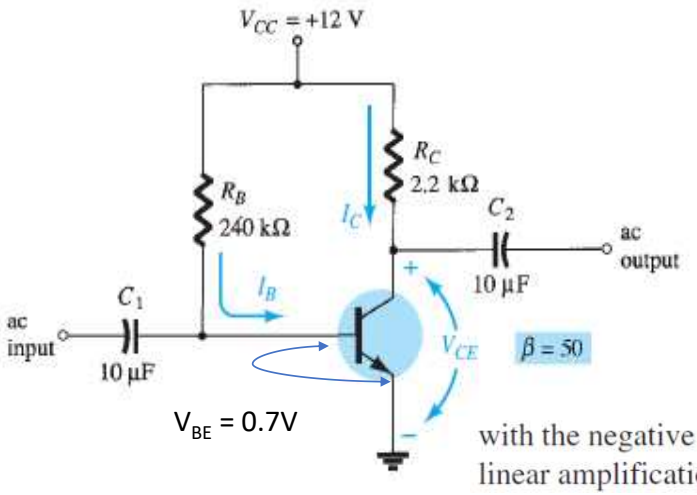


FIG. 4.6
Measuring V_{CE} and V_C.

EXAMPLE.1 Determine the following for the fixed-bias configuration of following Figure:

a. I_{BQ} and I_{CQ}.
b. V_{CEQ}.
c. V_B and V_C.
d. V_{BC}.



Solution:

a. Eq. (4.4): $$I_{B_Q} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12\text{ V} - 0.7\text{ V}}{240\text{ k}\Omega} = \mathbf{47.08\ \mu A}$$

Eq. (4.5): $$I_{C_Q} = \beta I_{BQ} = (50)(47.08\ \mu\text{A}) = \mathbf{2.35\ mA}$$

b. Eq. (4.6): $$V_{CE_Q} = V_{CC} - I_C R_C$$
$$= 12\text{ V} - (2.35\text{ mA})(2.2\text{ k}\Omega)$$
$$= \mathbf{6.83\ V}$$

c. $V_B = V_{BE} = \mathbf{0.7\ V}$
$V_C = V_{CE} = \mathbf{6.83\ V}$

d. Using double-subscript notation yields

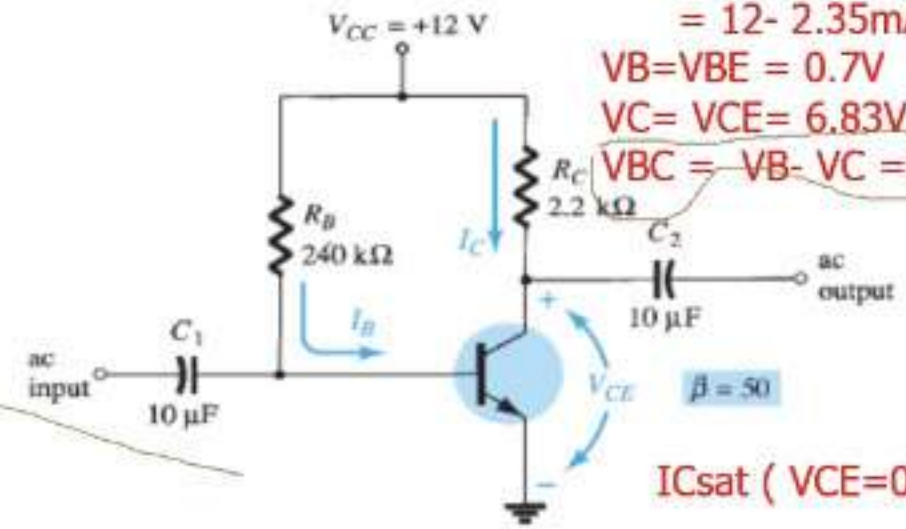
$$V_{BC} = V_B - V_C = 0.7\text{ V} - 6.83\text{ V}$$
$$= \mathbf{-6.13\ V}$$

with the negative sign revealing that the junction is reversed-biased, as it should be for linear amplification.

EXAMPLE.1 Determine the following for the fixed-bias configuration of following Figure:

a. I_{BQ} and I_{CQ}
b. V_{CEQ}
c. V_B and V_C.
d. V_{BC}.

$I_{C\,sat}$



IB=(VCC-VBE)/RB
=(12-0.7)/240K = 47 Micro A
IC=beta * IB = 50 * 47 Micro A =2.35mA

VCE=VCC- IC*RC
= 12- 2.35mA * 2.2K = 6.83 V
VB=VBE = 0.7V
VC= VCE= 6.83V
VBC = VB- VC =0.7-6.83 = -6.13 V

ICsat (VCE=0) =VCC/RC = 12/2.2K =5.45mA

Transistor Saturation

The term *saturation* is applied to any system where levels have reached their maximum values. A saturated sponge is one that cannot hold another drop of water. For a transistor operating in the saturation region, the current is a maximum value *for the particular design*. Change the design and the corresponding saturation level may rise or drop. Of course, the highest saturation level is defined by the maximum collector current as provided by the specification sheet.

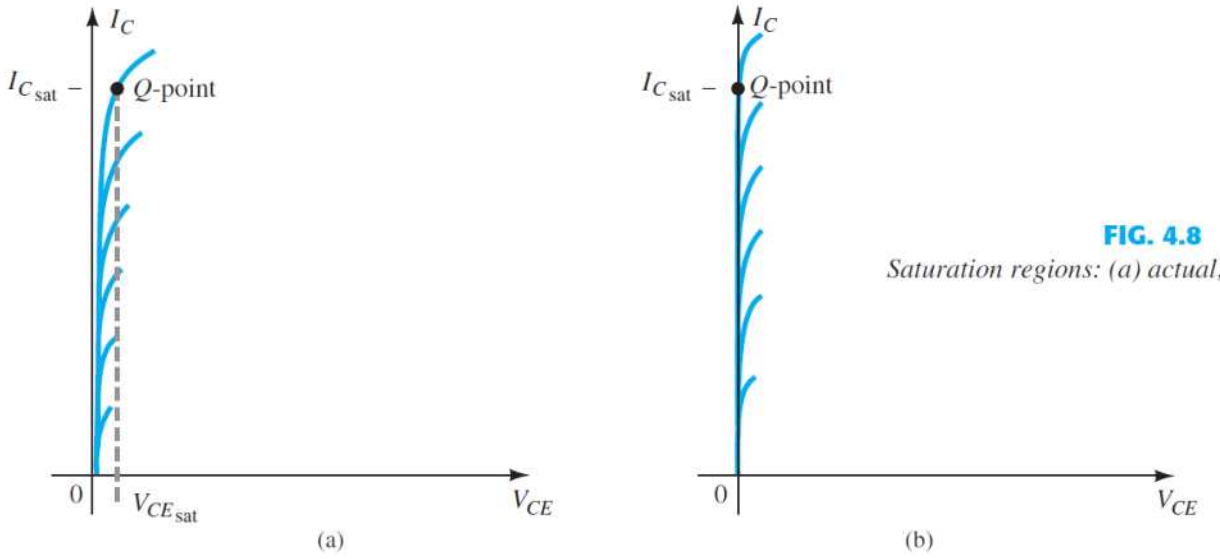


FIG. 4.8
Saturation regions: (a) actual; (b) approximate.

If we approximate the curves of Fig. 4.8a by those appearing in Fig. 4.8b, a quick, direct method for determining the saturation level becomes apparent. In Fig. 4.8b, the current is relatively high, and the voltage V_{CE} is assumed to be 0 V. Applying Ohm's law, we can determine the resistance between collector and emitter terminals as follows:

$$R_{CE} = \frac{V_{CE}}{I_C} = \frac{0\text{ V}}{I_{C_{sat}}} = 0\ \Omega$$

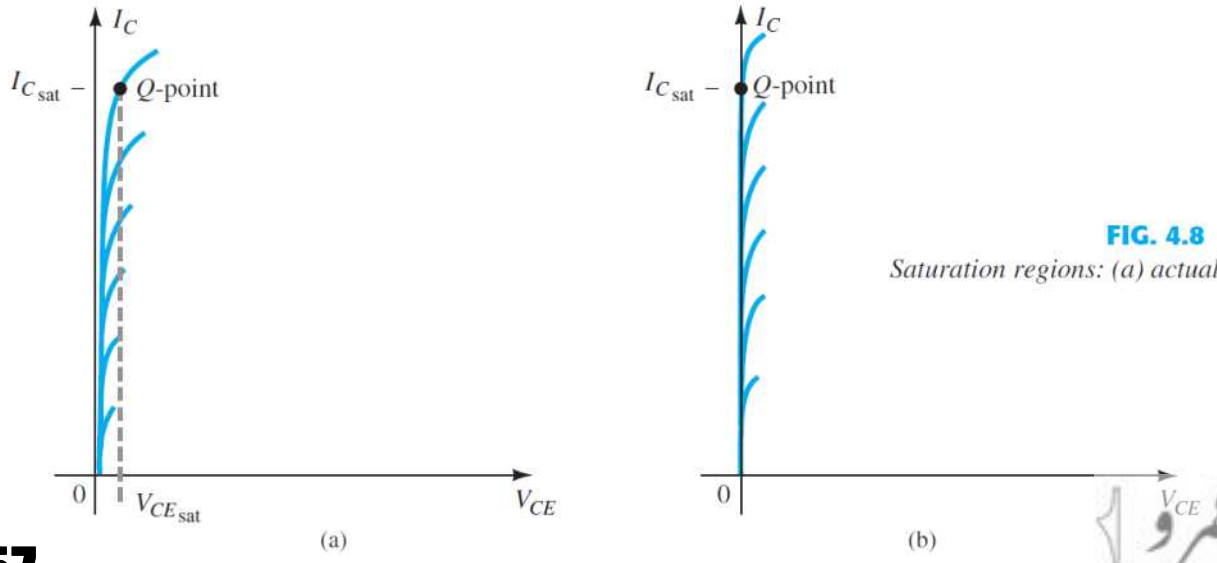
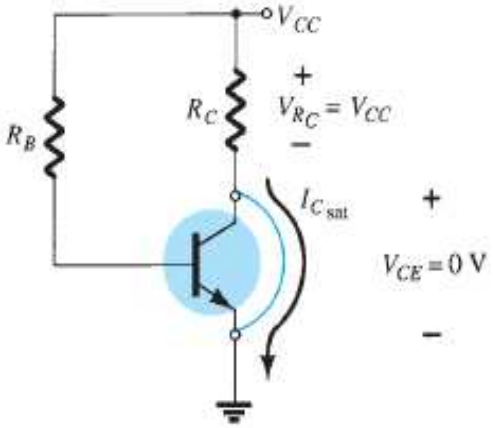


FIG. 4.8
Saturation regions: (a) actual; (b) approximate.

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The resulting saturation current for the fixed-bias configuration is

$$I_{C_{sat}} = \frac{V_{CC}}{R_C}$$



Once $I_{C_{sat}}$ is known, we have some idea of the maximum possible collector current for the chosen design and the level to stay below if we expect linear amplification.

EXAMPLE 4.2 Determine the saturation level for the network of Fig. 4.7.

Solution:

$$I_{C_{sat}} = \frac{V_{CC}}{R_C} = \frac{12\text{ V}}{2.2\text{ k}\Omega} = \mathbf{5.45\text{ mA}}$$

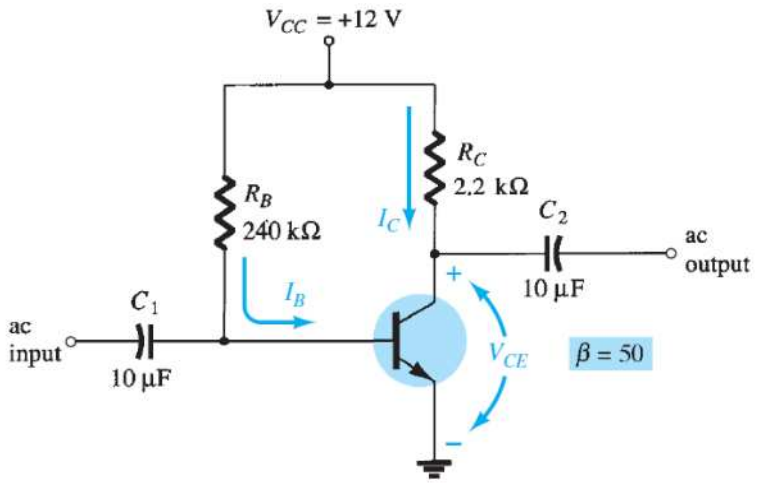
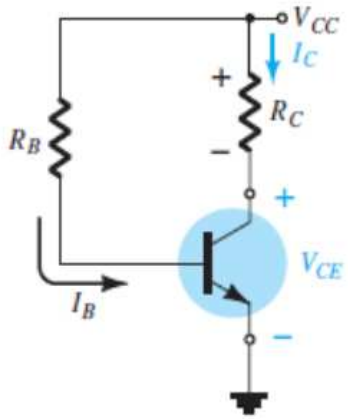


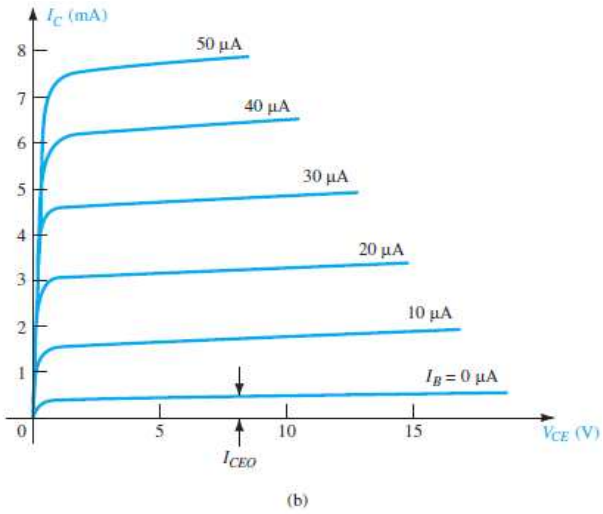
FIG. 4.7

Load-Line Analysis

Recall that the load-line solution for a diode network was found by superimposing the actual diode characteristics of the diode on a plot of the network equation involving the same network variables. The intersection of the two plots defined the actual operating conditions for the network.

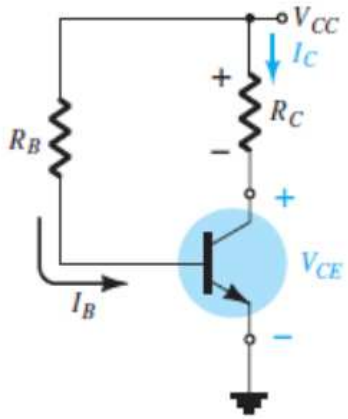
It is referred to as load-line analysis because the load (network resistors) of the network defined the slope of the straight line connecting the points defined by the network parameters.

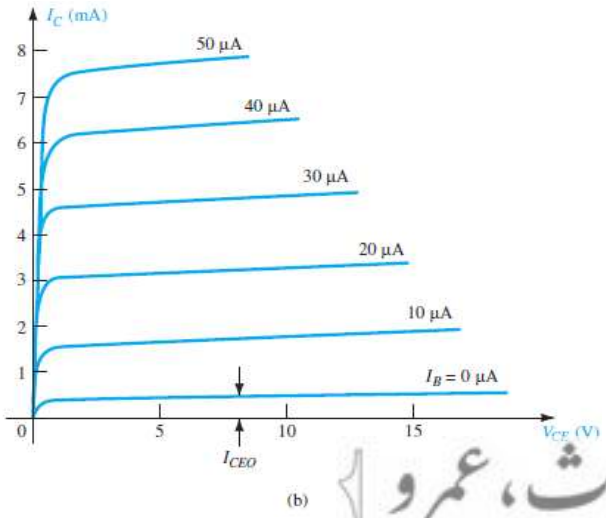




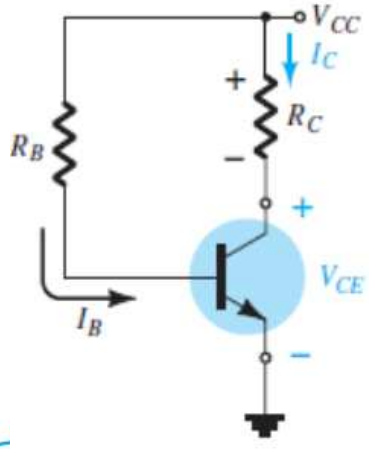
Load–Line Analysis

The same approach can be applied to BJT networks. The characteristics of the BJT are superimposed on a plot of the network equation defined by the same axis parameters. The load resistor R_C for the fixed-bias configuration will define the slope of the network equation and the resulting intersection between the two plots. The smaller the load resistance, the steeper the slope of the network load line





The network of Fig. shown establishes an output equation that relates the variables I_C and V_{CE} in the following manner:



$$V_{CE} = V_{CC} - I_C R_C$$

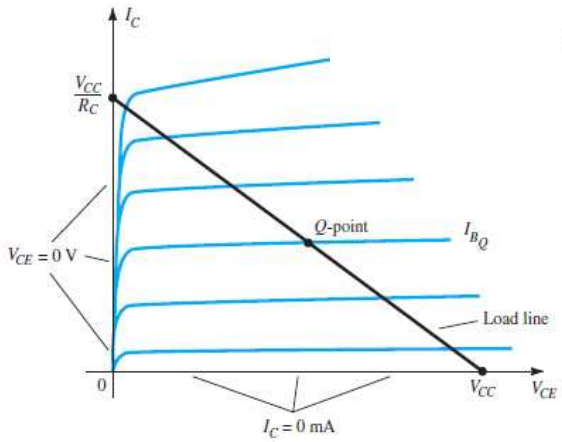
The output characteristics of the transistor also relate the same two variables I_C and V_{CE} as shown in Fig. We must now superimpose the straight line defined by the above equation on the characteristics.

$$V_{CE} = V_{CC} - (0)R_C$$

$$V_{CE} = V_{CC}|_{I_C = 0\text{ mA}}$$

$$0 = V_{CC} - I_C R_C$$

$$I_C = \frac{V_{CC}}{R_C}\bigg|_{V_{CE} = 0\text{ V}}$$



If the level of I_B is changed by varying the value of R_B, the Q-point moves up or down the load line as shown in Fig. 4.13 for increasing values of I_B. If V_{CC} is held fixed and R_C increased, the load line will shift as shown in Fig. 4.14. If I_B is held fixed, the Q-point will move as shown in the same figure. If R_C is fixed and V_{CC} decreased, the load line shifts as shown in Fig. 4.15.

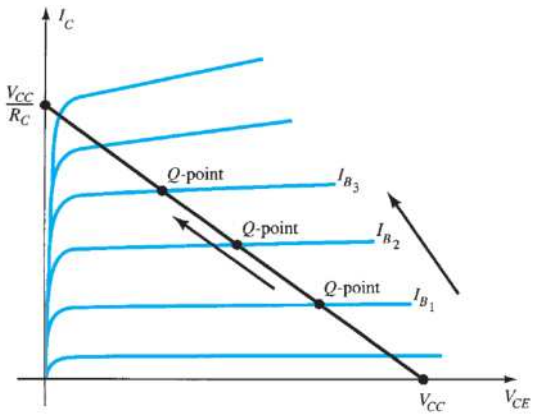


FIG. 4.13
Movement of the Q-point with increasing level of I_B.

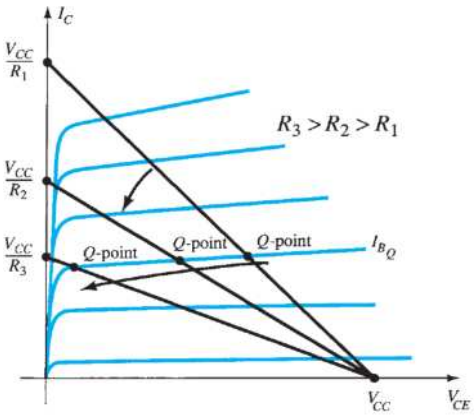


FIG. 4.14
Effect of an increasing level of R_C on the load line and the Q-point.

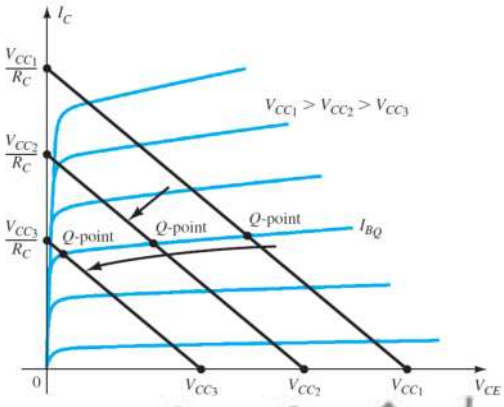
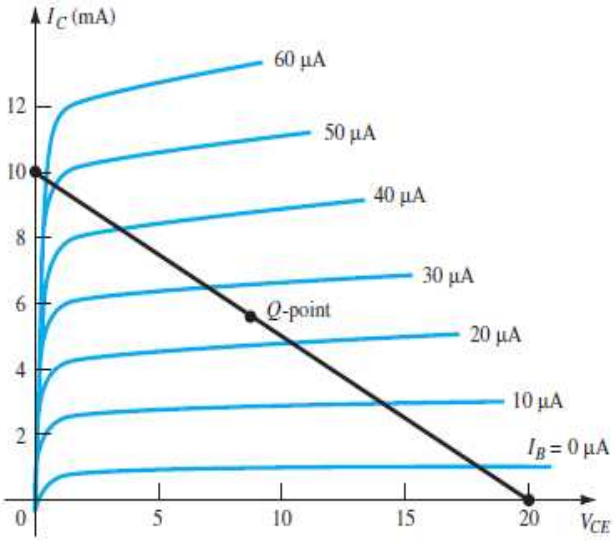


FIG. 4.15
Effect of lower values of V_{CC} on the load line and the Q-point.

EXAMPLE 4.3 Given the load line of the following characteristics and the defined Q-point, determine the required values of V_{CC}, R_C, and R_B for a fixed-bias configuration.



Solution:

$$V_{CE} = V_{CC} = \mathbf{20\ V} \text{ at } I_C = 0\ \text{mA}$$

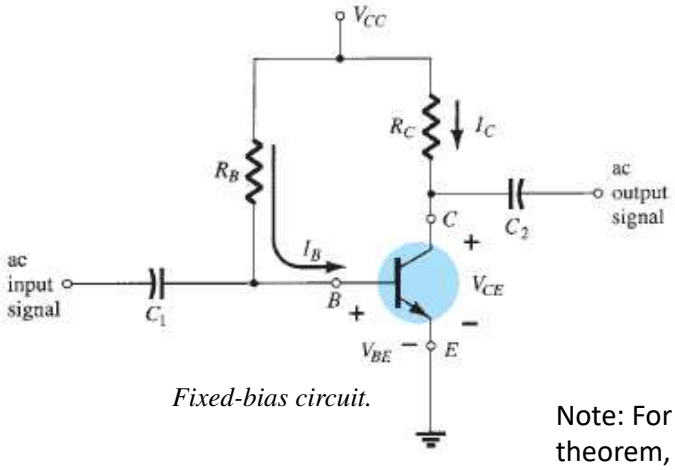
$$I_C = \frac{V_{CC}}{R_C} \text{ at } V_{CE} = 0\ \text{V}$$

$$R_C = \frac{V_{CC}}{I_C} = \frac{20\ \text{V}}{10\ \text{mA}} = \mathbf{2\ k\Omega}$$

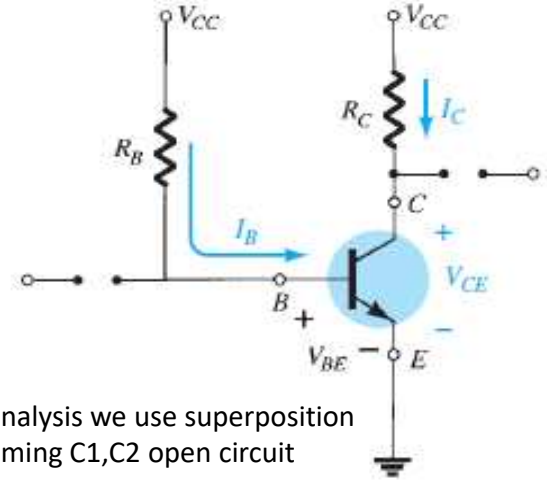
$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$R_B = \frac{V_{CC} - V_{BE}}{I_B} = \frac{20\ \text{V} - 0.7\ \text{V}}{25\ \mu\text{A}} = \mathbf{772\ k\Omega}$$

FIXED-BIAS CONFIGURATION



Fixed-bias circuit.



Note: For Dc analysis we use superposition theorem, assuming C1,C2 open circuit

DC equivalent

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$I_C = \beta I_B \qquad \textbf{(4.5)}$$

$$V_{CE} = V_{CC} - I_C R_C \qquad \textbf{(4.6)}$$

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EMITTER-BIAS CONFIGURATION

The dc bias network of the following Figure 4.17 contains an emitter resistor to improve the stability level over that of the fixed-bias configuration. The more stable a configuration, the less its response will change due to undesirable changes in temperature and parameter variations.

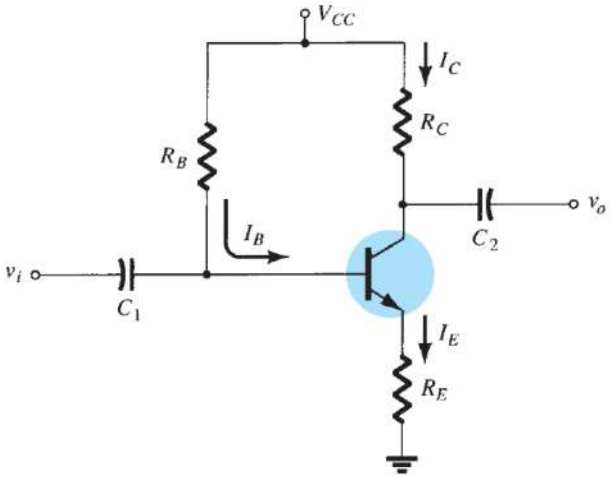


FIG. 4.17

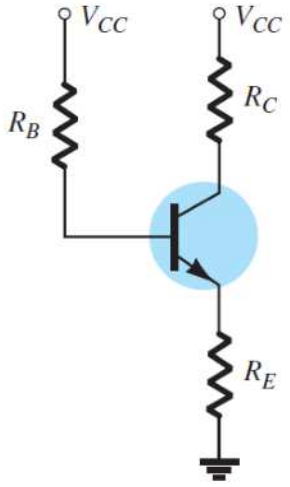


FIG. 4.18
DC equivalent of Fig. 4.17.

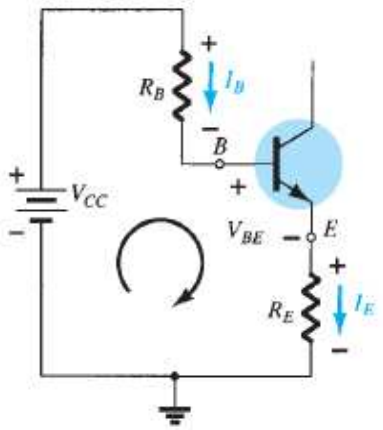
Base–Emitter Loop:

The base–emitter loop of the network of Fig. 4.18 can be redrawn as shown in Fig. 4.19. Writing Kirchhoff's voltage law around the indicated loop in the clockwise direction results in the following equation:

$$+V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0 \qquad \textbf{(4.15)}$$

Recall from Chapter 3 that

$$I_E = (\beta + 1) I_B$$



Base–Emitter Loop:

Substituting for I_E in Eq. (4.15) results in

$$V_{CC} - I_B R_B - V_{BE} - (\beta + I) I_B R_E = 0$$

Grouping terms then provides the following:

$$-I_B(R_B + (\beta + 1)R_E) + V_{CC} - V_{BE} = 0$$

Multiplying through by (-1), we have

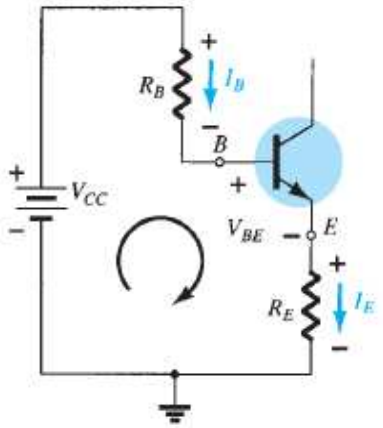
$$I_B(R_B + (\beta + 1)R_E) - V_{CC} + V_{BE} = 0$$

with

$$I_B(R_B + (\beta + 1)R_E) = V_{CC} - V_{BE}$$

and solving for I_B gives

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}$$



Note that the only difference between this equation for I_B and that obtained for the fixed-bias configuration is the term $(\beta + 1)R_E$.

There is an interesting result that can be derived from Eq. (4.17) if the equation is used to sketch a series network that would result in the same equation. Such is the case for the network of Fig. 4.20. Solving for the current I_B results in the same equation as obtained above. Note that aside from the base-to-emitter voltage V_{BE}, the resistor R_E is *reflected* back to the input base circuit by a factor $(\beta + 1)$. In other words, the emitter resistor, which is part of the collector–emitter loop, "appears as" $(\beta + 1)R_E$ in the base–emitter loop. Because β is typically 50 or more, the emitter resistor appears to be a great deal larger in the base circuit. In general, therefore, for the configuration of Fig. 4.21,

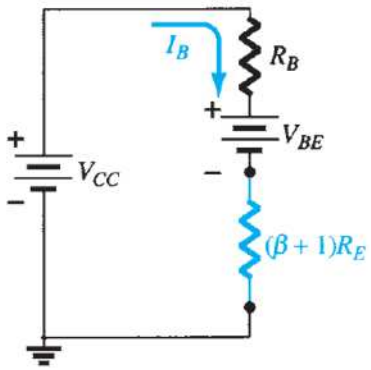


FIG. 4.20
Network derived from Eq. (4.17).

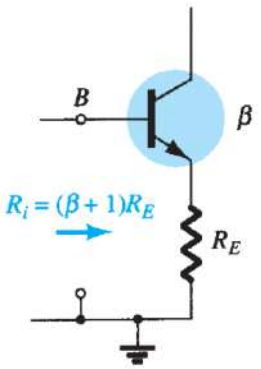


FIG. 4.21
Reflected impedance level of R_E.

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} \quad \textbf{(4.17)}$$

$$R_i = (\beta + 1)R_E \quad \textbf{(4.18)}$$

Equation (4.18) will prove useful in the analysis to follow. In fact, it provides a fairly easy way to remember Eq. (4.17). Using Ohm's law, we know that the current through a system is the voltage divided by the resistance of the circuit. For the base–emitter circuit the net voltage is $V_{CC} - V_{BE}$. The resistance levels are R_B plus R_E reflected by $(\beta + 1)$. The result is Eq. (4.17).

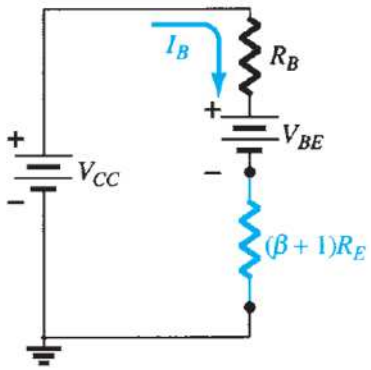


FIG. 4.20
Network derived from Eq. (4.17).

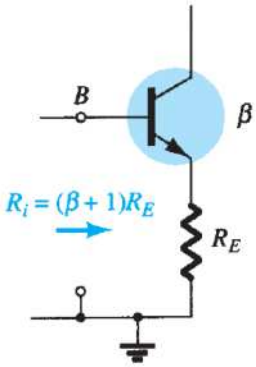


FIG. 4.21
Reflected impedance level of R_E.

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} \qquad \textbf{(4.17)}$$

$$R_i = (\beta + 1)R_E \qquad \textbf{(4.18)}$$

Collector–Emitter Loop

The collector–emitter loop appears in Fig. 4.22. Writing Kirchhoff's voltage law for the indicated loop in the clockwise direction results in

$$+I_E R_E + V_{CE} + I_C R_C - V_{CC} = 0$$

Substituting $I_E \cong I_C$ and grouping terms gives

$$V_{CE} - V_{CC} + I_C(R_C + R_E) = 0$$

and

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

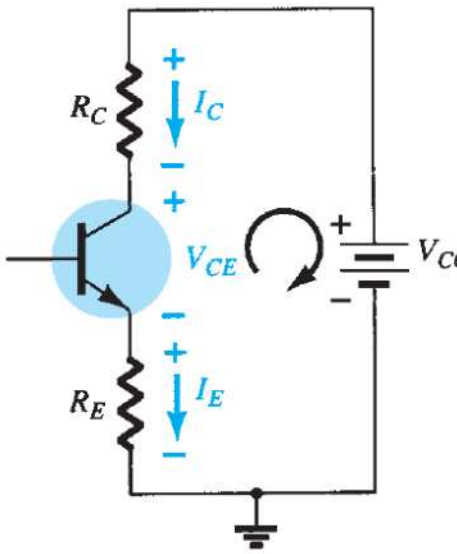


FIG. 4.22
Collector–emitter loop.

The single-subscript voltage V_E is the voltage from emitter to ground and is determined by

$$V_E = I_E R_E$$

whereas the voltage from collector to ground can be determined from

$$V_{CE} = V_C - V_E$$

and

$$V_C = V_{CE} + V_E$$

or

$$V_C = V_{CC} - I_C R_C$$

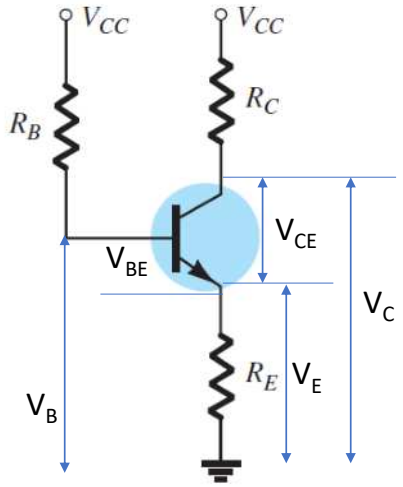


FIG. 4.18

The voltage at the base with respect to ground can be determined using Fig. 4.18

$$V_B = V_{CC} - I_B R_B$$

or

$$V_B = V_{BE} + V_E$$

EXAMPLE 4.4

For the emitter-bias network of Fig. 4.23, determine:

a. I_B.
b. I_C.
c. V_{CE}.
d. V_C.
e. V_E.
f. V_B.
g. V_{BC}.

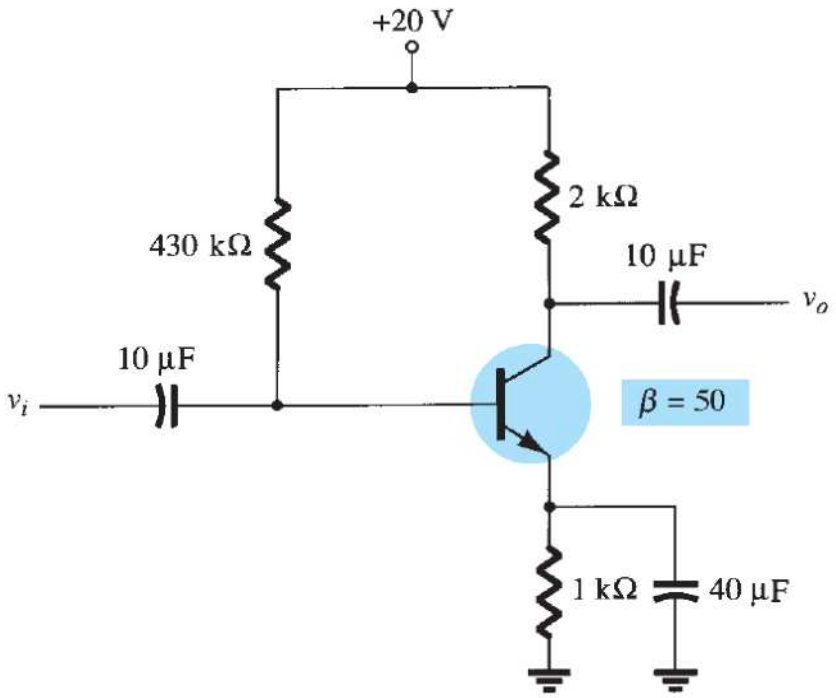


FIG. 4.23
Emitter-stabilized bias circuit for Example 4.4.

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FIG. 4.22
Collector–emitter loop.

$$V_B = V_{CC} - I_B R_B \quad (4.23)$$

$$V_B = V_{BE} + V_E \quad (4.24)$$

VCE = VC - VE
VC or VC -0
0: ground

VBC=VB-VC=-13.3 V

ICsat=VCC/(RC+RE)
=20/(2+1) = 6.67mA

EXAMPLE 4.4 For the emitter-bias network of Fig. 4.23, determine:

a. I_B.
b. I_C.
c. V_{CE}.
d. V_C.
e. V_E.
f. V_B.
g. V_{BC}.

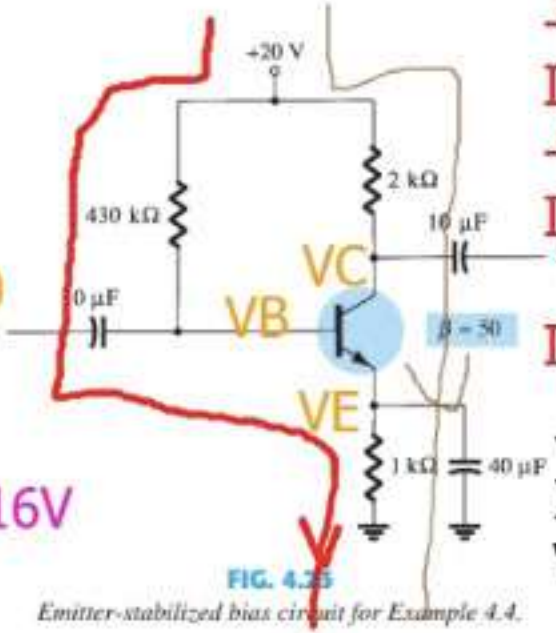


FIG. 4.23
Emitter-stabilized bias circuit for Example 4.4.

-VCC + IC*RC + VC =0

-VE+ IE * RE = 0
VE= IE*RE=2*1=2V
VC= VCE+VE= 14+2=16V

VB=VCC- IB *RB
or
VB= VBE+ VE=2.7V

-VCC+IB*RB+VBE+IE*RE=0
IE=(beta +1) * IB
-VCC + IB*RB + VBE + (beta+1) *IB*RE =0
IB= (VCC-VBE)/ RB + (beta+1)*RE
=20-0.7/430+51*1= 40 micro A
IC = beta * IB = 50*40= 2mA

-VCC +IC * RC +VCE + IE*RE = 0
IC=IE
VCE=VCC-IC(RC+RE)
= 20-2(2+1) =14V
VC=VCC - IC*RC
=20-2*2= 16V

Solution:

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Solution:

a. Eq. (4.17): $$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20\text{ V} - 0.7\text{ V}}{430\text{ k}\Omega + (51)(1\text{ k}\Omega)}$$

$$= \frac{19.3\text{ V}}{481\text{ k}\Omega} = \mathbf{40.1\ \mu A}$$

b. $I_C = \beta I_B$

$= (50)(40.1\ \mu\text{A})$

$\cong$ **2.01 mA**

c. Eq. (4.19): $V_{CE} = V_{CC} - I_C(R_C + R_E)$

$= 20\text{ V} - (2.01\text{ mA})(2\text{ k}\Omega + 1\text{ k}\Omega) = 20\text{ V} - 6.03\text{ V}$

$=$ **13.97 V**

d. $V_C = V_{CC} - I_C R_C$

$= 20\text{ V} - (2.01\text{ mA})(2\text{ k}\Omega) = 20\text{ V} - 4.02\text{ V}$

$=$ **15.98 V**

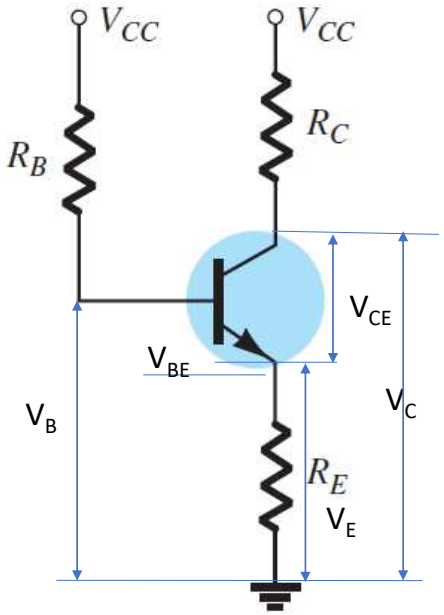


FIG. 4.18

e. $V_E = V_C - V_{CE}$

$= 15.98\text{ V} - 13.97\text{ V}$

$=$ **2.01 V**

or $V_E = I_E R_E \cong I_C R_E$

$= (2.01\text{ mA})(1\text{ k}\Omega)$

$=$ **2.01 V**

f. $V_B = V_{BE} + V_E$

$= 0.7\text{ V} + 2.01\text{ V}$

$=$ **2.71 V**

g. $V_{BC} = V_B - V_C$

$= 2.71\text{ V} - 15.98\text{ V}$

$=$ **−13.27 V** (reverse-biased as required)

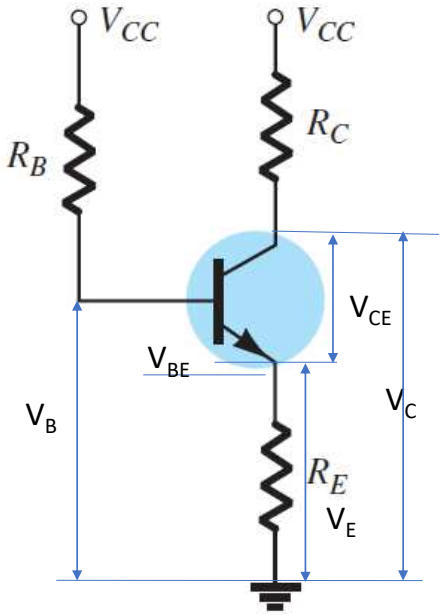


FIG. 4.19

Improved Bias Stability

The addition of the emitter resistor to the dc bias of the BJT provides improved stability, that is, the dc bias currents and voltages remain closer to where they were set by the circuit when outside conditions, such as temperature and transistor beta, change. Although a mathematical analysis is provided in Section 4.12, some comparison of the improvement can be obtained as demonstrated by Example 4.5.

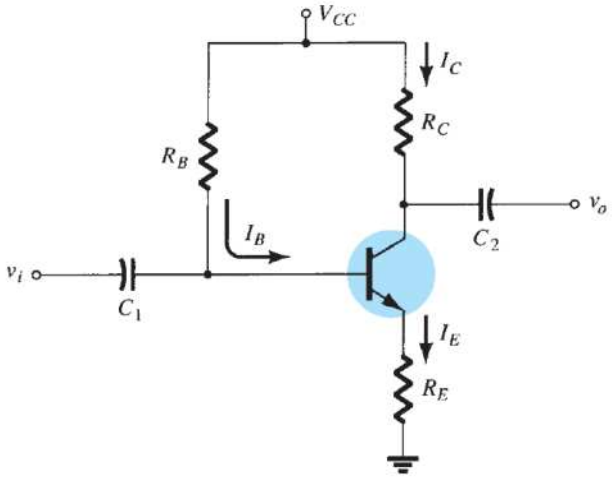


FIG. 4.17

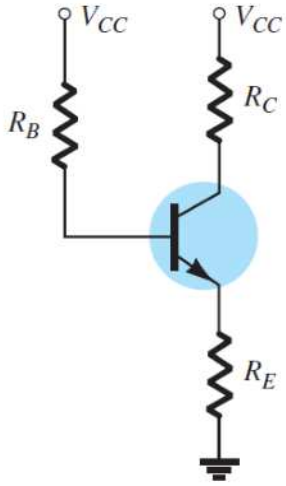


FIG. 4.18
DC equivalent of Fig. 4.17.

EXAMPLE 4.5 Prepare a table and compare the bias voltage and currents of the circuits of Fig. 4.7 and Fig. 4.23 for the given value of $\beta = 50$ and for a new value of $\beta = 100$. Compare the changes in I_C and V_{CE} for the same increase in β.

Solution: Using the results calculated in Example 4.1 and then repeating for a value of $\beta = 100$ yields the following:

Effect of β variation on the response of the fixed-bias configuration of Fig. 4.7.

β	I_B (μA)	I_C (mA)	V_{CE} (V)
50	47.08	2.35	6.83
100	47.08	4.71	1.64

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

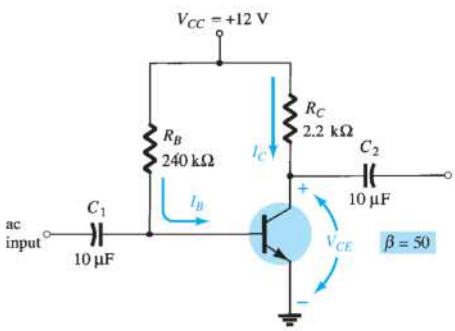


FIG. 4.7

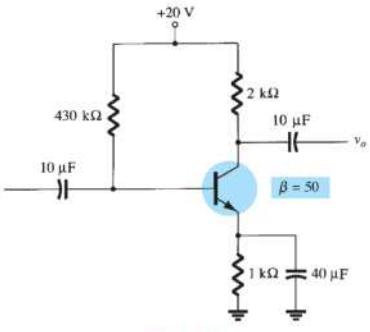


FIG. 4.23

The BJT collector current is seen to change by 100% due to the 100% change in the value of β. The value of I_B is the same, and V_{CE} decreased by 76%.

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Using the results calculated in Example 4.4 and then repeating for a value of $\beta = 100$, we have the following:

Effect of β variation on the response of the emitter-bias configuration of Fig. 4.23.

β	I_B (μA)	I_C (mA)	V_{CE} (V)
50	40.1	2.01	13.97
100	36.3	3.63	9.11

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}$$

Now the BJT collector current increases by about 81% due to the 100% increase in β. Notice that I_B decreased, helping maintain the value of I_C—or at least reducing the overall change in I_C due to the change in β. The change in V_{CE} has dropped to about 35%. The network of Fig. 4.23 is therefore more stable than that of Fig. 4.7 for the same change in β.

Saturation Level

The collector saturation level or maximum collector current for an emitter-bias design can be determined using the same approach applied to the fixed-bias configuration: Apply a short circuit between the collector–emitter terminals as shown in Fig. 4.24 and calculate the resulting collector current. For Fig. 4.24

$$I_{C_{sat}} = \frac{V_{CC}}{R_C + R_E} \qquad (4.25)$$

The addition of the emitter resistor reduces the collector saturation level below t obtained with a fixed-bias configuration using the same collector resistor.

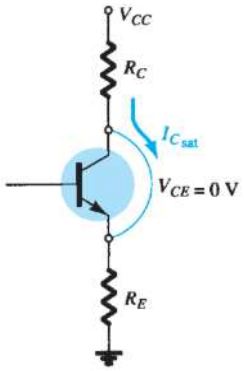


FIG. 4.24
Determining $I_{C_{sat}}$ for the emitt stabilized bias circuit.

EXAMPLE 4.6 Determine the saturation current for the network of Example 4.4.

Solution:

$$I_{C_{sat}} = \frac{V_{CC}}{R_C + R_E}$$

$$= \frac{20\text{ V}}{2\text{ k}\Omega + 1\text{ k}\Omega} = \frac{20\text{ V}}{3\text{ k}\Omega}$$

$$= \mathbf{6.67\text{ mA}}$$

which is about three times the level of I_{C_Q} for Example 4.4.

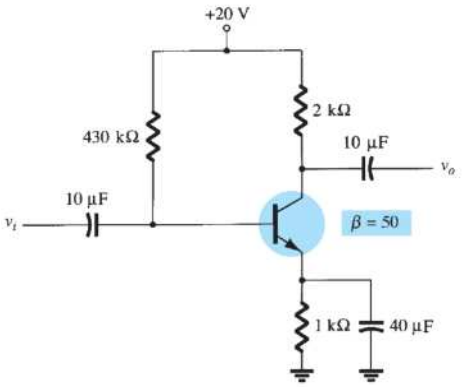


FIG. 4.23
Emitter-stabilized bias circuit for Example 4.4.

Load-Line Analysis

The load-line analysis of the emitter-bias network is only slightly different from that encountered for the fixed-bias configuration. The level of I_B as determined by Eq. (4.17) defines the level of I_B on the characteristics of Fig. 4.25 (denoted I_{B_Q}).

The collector–emitter loop equation that defines the load line is

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

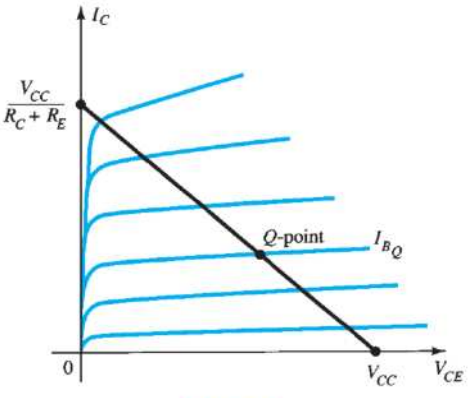


FIG. 4.25
Load line for the emitter-bias configuration.

Choosing $I_C = 0$ mA gives

$$V_{CE} = V_{CC}\big|_{I_C=0\text{ mA}} \tag{4.26}$$

as obtained for the fixed-bias configuration. Choosing $V_{CE} = 0$ V gives

$$I_C = \frac{V_{CC}}{R_C + R_E}\bigg|_{V_{CE}=0\text{ V}} \tag{4.27}$$

as shown in Fig. 4.25. Different levels of I_{B_Q} will, of course, move the Q-point up or down the load line.

EXAMPLE 4.7

a. Draw the load line for the network of Fig. 4.26a on the characteristics for the transistor appearing in Fig. 4.26b.
b. For a Q-point at the intersection of the load line with a base current of 15 μA, find the values of I_{C_Q} and V_{CE_Q}.
c. Determine the dc beta at the Q-point.
d. Using the beta for the network determined in part c, calculate the required value of R_B and suggest a possible standard value.

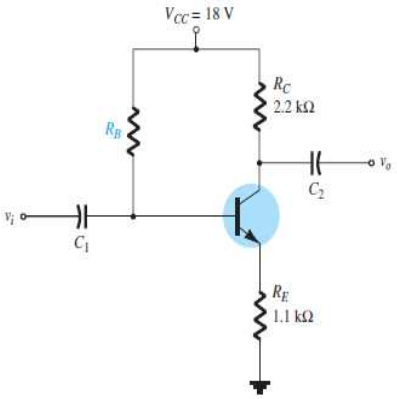


FIG. 4.26a

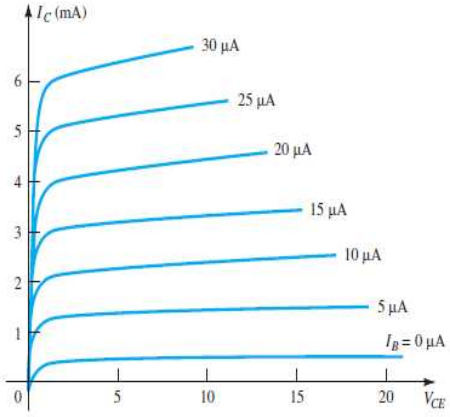


FIG. 4.26b

Solution:

a. Two points on the characteristics are required to draw the load line.

At $V_{CE} = 0$ V: $I_C = \dfrac{V_{CC}}{R_C + R_E} = \dfrac{18\text{ V}}{2.2\text{ k}\Omega + 1.1\text{ k}\Omega} = \dfrac{18\text{ V}}{3.3\text{ k}\Omega} = 5.45\text{ mA}$

At $I_C = 0$ mA: $V_{CE} = V_{CC} = 18$ V

The resulting load line appears in Fig. 4.27.

b. From the characteristics of Fig. 4.27 we find

$V_{CE_Q} \cong 7.5\text{ V}, I_{C_Q} \cong 3.3\text{ mA}$

c. The resulting dc beta is:

$$\beta = \frac{I_{C_Q}}{I_{B_Q}} = \frac{3.3\text{ mA}}{15\ \mu\text{A}} = 220$$

d. Applying Eq. 4.17:

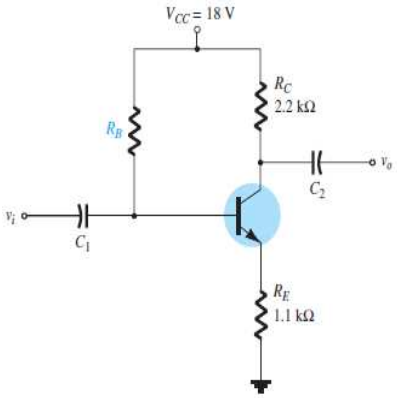
$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{18\text{ V} - 0.7\text{ V}}{R_B + (220 + 1)(1.1\text{ k}\Omega)}$$

and $15\ \mu\text{A} = \dfrac{17.3\text{ V}}{R_B + (221)(1.1\text{ k}\Omega)} = \dfrac{17.3\text{ V}}{R_B + 243.1\text{ k}\Omega}$

so that $(15\ \mu\text{A})(R_B) + (15\ \mu\text{A})(243.1\text{ k}\Omega) = 17.3$ V

and $(15\ \mu\text{A})(R_B) = 17.3\text{ V} - 3.65\text{ V} = 13.65\text{ V}$

resulting in $R_B + \dfrac{13.65\text{ V}}{15\ \mu\text{A}} = 910\text{ k}\Omega$



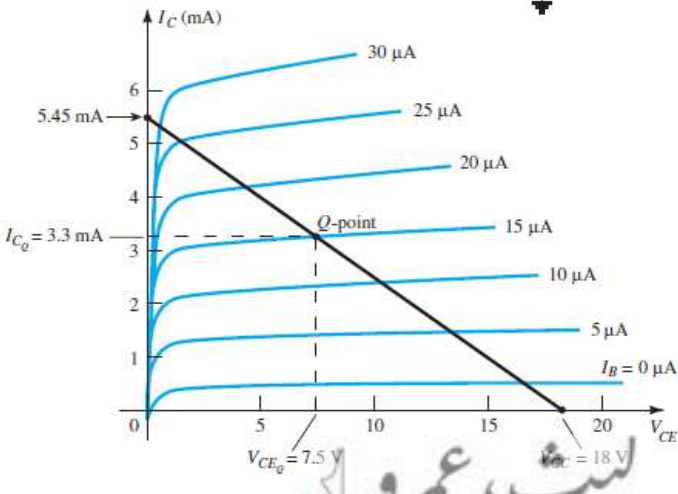


FIG. 4.27

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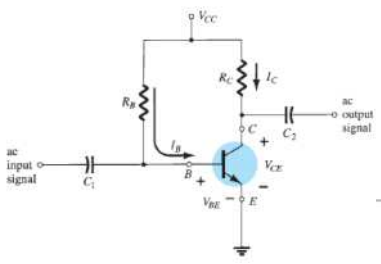
Chapter 4: DC Biasing—BJTs

Lecture 3

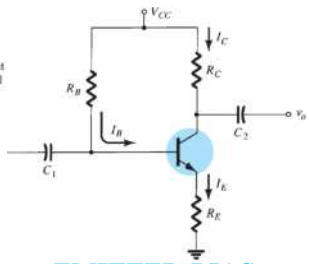
Dr . Faeik T Al Rabee
faeik.alrabee@bau.edu.jo

4.5 VOLTAGE-DIVIDER BIAS CONFIGURATION

Fig. 4.28 is such a network. If analyzed on an exact basis, the sensitivity to changes in beta is quite small. If the circuit parameters are properly chosen, the resulting levels of I_{C_Q} and V_{CE_Q} can be almost totally independent of beta. Recall from previous discussions that a Q-point is defined by a fixed level of I_{C_Q} and V_{CE_Q} as shown in Fig. 4.29. The level of I_{B_Q} will change with the change in beta, but the operating point on the characteristics defined by I_{C_Q} and V_{CE_Q} can remain fixed if the proper circuit parameters are employed.



FIXED-BIAS CONFIGURATION



EMITTER-BIAS CONFIGURATION

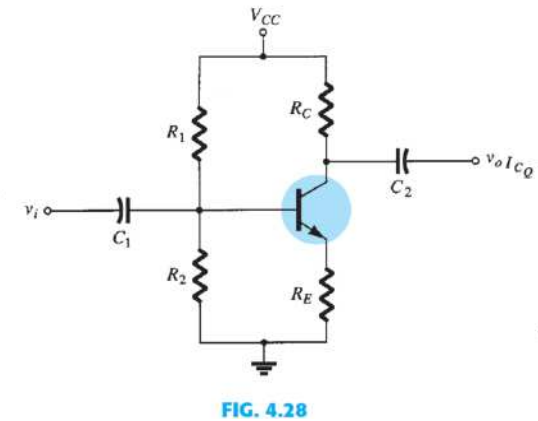


FIG. 4.28

As noted earlier, there are two methods that can be applied to analyze the voltage-divider configuration. The reason for the choice of names for this configuration will become obvious in the analysis to follow. The first to be demonstrated is the *exact method*, which can be applied to *any* voltage-divider configuration. The second is referred to as the *approximate method* and can be applied only if specific conditions are satisfied. The approximate approach permits a more direct analysis with a savings in time and energy. It is also particularly helpful in the design mode to be described in a later section. All in all, the approximate approach can be applied to the majority of situations and therefore should be examined with the same interest as the exact method.

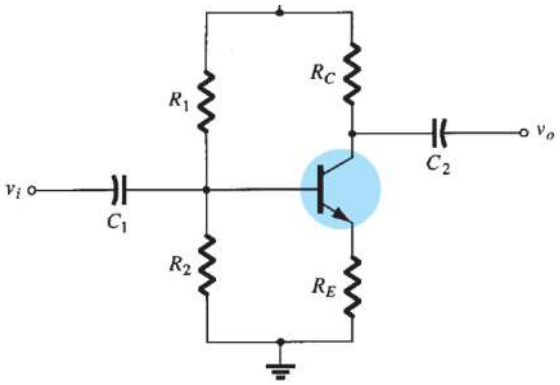


FIG. 4.28

Exact Analysis

For the dc analysis the network of Fig. 4.28 can be redrawn as shown in Fig. 4.30. The input side of the network can then be redrawn as shown in Fig. 4.31 for the dc analysis. The Thévenin equivalent network for the network to the left of the base terminal can then be found in the following manner:

R_{Th} The voltage source is replaced by a short-circuit equivalent as shown in Fig. 4.32:

$$R_{Th} = R_1 \| R_2 \qquad \text{(4.2}$$

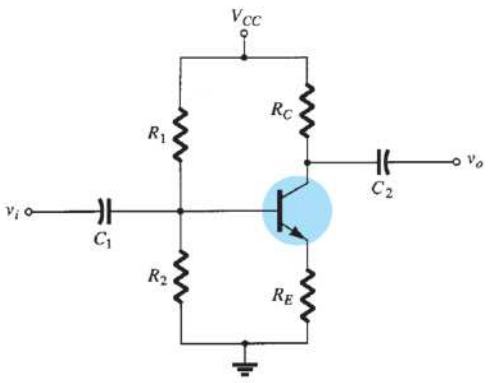


FIG. 4.28

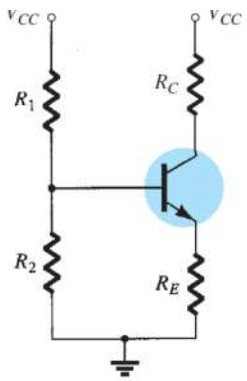


FIG. 4.30

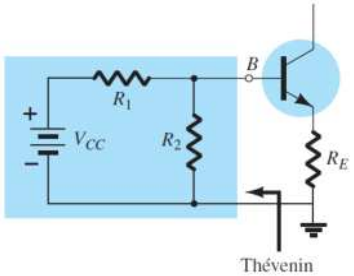


FIG. 4.31

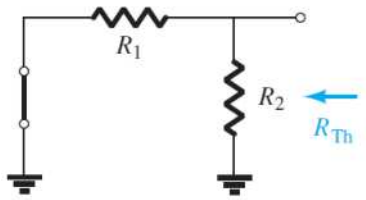


FIG. 4.32
Determining R_{Th}

E_{Th} The voltage source V_{CC} is returned to the network and the open-circuit Thévenin voltage of Fig. 4.33 determined as follows:

Applying the voltage-divider rule gives

$$E_{\text{Th}} = V_{R_2} = \frac{R_2 V_{CC}}{R_1 + R_2} \tag{4.29}$$

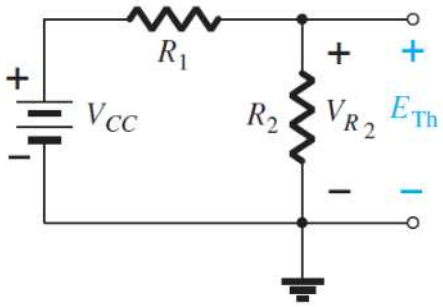


FIG. 4.33

The Thévenin network is then redrawn as shown in Fig. 4.34, and I_{B_Q} can be determined by first applying Kirchhoff's voltage law in the clockwise direction for the loop indicated:

$$E_{\text{Th}} - I_B R_{\text{Th}} - V_{BE} - I_E R_E = 0$$

Substituting $I_E = (\beta + 1)I_B$ and solving for I_B yields

$$I_B = \frac{E_{\text{Th}} - V_{BE}}{R_{\text{Th}} + (\beta + 1)R_E} \tag{4.30}$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} \tag{4.17}$$

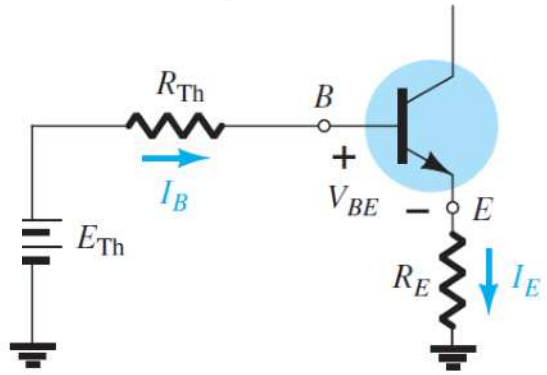


FIG. 4.34
Inserting the Thévenin equivalent circuit.

Although Eq. (4.30) initially appears to be different from those developed earlier, note that the numerator is again a difference of two voltage levels and the denominator is the base resistance plus the emitter resistor reflected by $(\beta + 1)$—certainly very similar to Eq. (4.17).

لیث، عمرو

Once I_B is known, the remaining quantities of the network can be found in the same manner as developed for the emitter-bias configuration. That is,

$$V_{CE} = V_{CC} - I_C(R_C + R_E) \tag{4.31}$$

which is exactly the same as Eq. (4.19). The remaining equations for V_E, V_C, and V_B are also the same as obtained for the emitter-bias configuration.

EMITTER-BIAS CONFIGURATION

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} \tag{4.17}$$

VOLTAGE-DIVIDER BIAS CONFIGURATION

$$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} \tag{4.30}$$

EXAMPLE 4.8 Determine the dc bias voltage V_{CE} and the current I_C for the voltage-divider configuration of Fig. 4.35

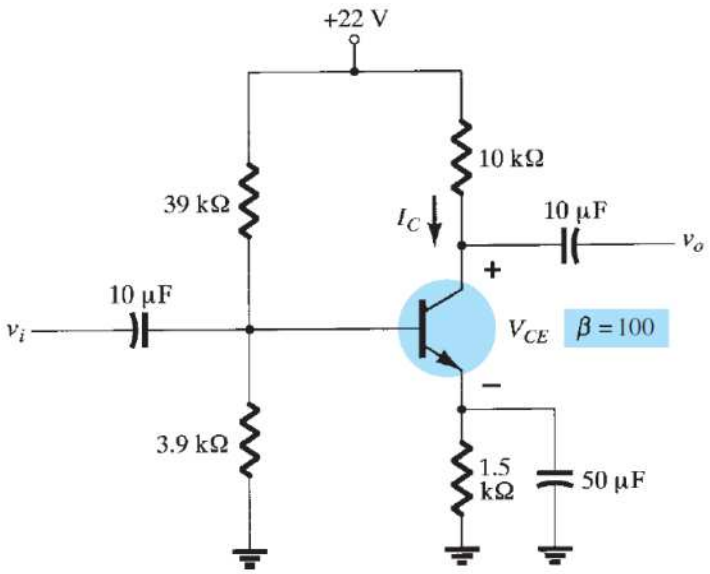


FIG. 4.35
Beta-stabilized circuit for Example 4.8.

لیث، عمرو

Solution:

$$R_{Th} = R_1 \| R_2$$

$$= \frac{(39\ \text{k}\Omega)(3.9\ \text{k}\Omega)}{39\ \text{k}\Omega + 3.9\ \text{k}\Omega} = 3.55\ \text{k}\Omega$$

$$E_{Th} = \frac{R_2 V_{CC}}{R_1 + R_2}$$

$$= \frac{(3.9\ \text{k}\Omega)(22\ \text{V})}{39\ \text{k}\Omega + 3.9\ \text{k}\Omega} = 2\ \text{V}$$

$$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E}$$

$$= \frac{2\ \text{V} - 0.7\ \text{V}}{3.55\ \text{k}\Omega + (101)(1.5\ \text{k}\Omega)} = \frac{1.3\ \text{V}}{3.55\ \text{k}\Omega + 151.5\ \text{k}\Omega}$$

$$= 8.38\ \mu\text{A}$$

$$I_C = \beta I_B$$

$$= (100)(8.38\ \mu\text{A})$$

$$= \mathbf{0.84\ mA}$$

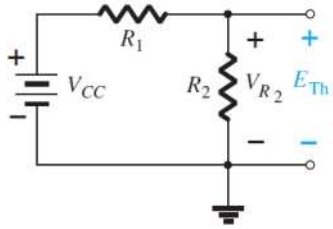


FIG. 4.33
Determining E_{Th}.

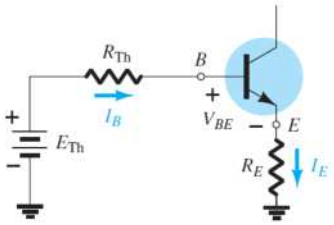


FIG. 4.34
Inserting the Thévenin equivalent circuit.

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

$$= 22\ \text{V} - (0.84\ \text{mA})(10\ \text{k}\Omega + 1.5\ \text{k}\Omega)$$

$$= 22\ \text{V} - 9.66\ \text{V}$$

$$= \mathbf{12.34\ V}$$

Approximate Analysis

The input section of the voltage-divider configuration can be represented by the network of Fig. 4.36. The resistance R_i is the equivalent resistance between base and ground for the transistor with an emitter resistor R_E. Recall from Section 4.4 [Eq. (4.18)] that the reflected resistance between base and emitter is defined by $R_i = (\beta + 1)R_E$. If R_i is much larger than the resistance R_2, the current I_B will be much smaller than I_2 (current always seeks the path of least resistance) and I_2 will be approximately equal to I_1. If we accept the approximation that I_B is essentially 0 A compared to I_1 or I_2, then $I_1 = I_2$, and R_1 and R_2 can be considered series elements. The voltage across R_2, which is actually the base voltage, can be

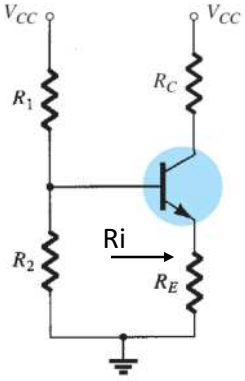


FIG. 4.30

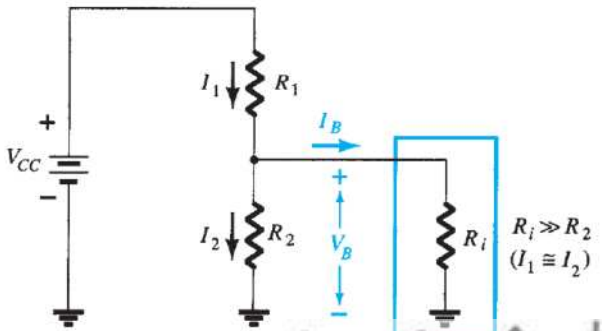


FIG. 4.36

determined using the voltage-divider rule (hence the name for the configuration). That is,

$$V_B = \frac{R_2 V_{CC}}{R_1 + R_2} \tag{4.32}$$

Because $R_i = (\beta + 1)R_E \cong \beta R_E$ the condition that will define whether the approximate approach can be applied is

$$\beta R_E \geq 10R_2 \tag{4.33}$$

In other words, if β times the value of R_E is at least 10 times the value of R_2, the approximate approach can be applied with a high degree of accuracy.

Once V_B is determined, the level of V_E can be calculated from

$$V_E = V_B - V_{BE}$$

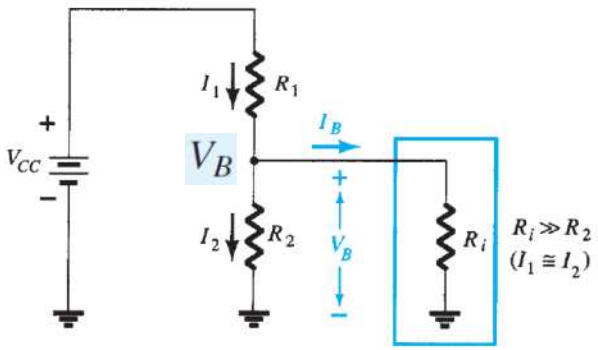


FIG. 4.36

and the emitter current can be determined from

$$I_E = \frac{V_E}{R_E}$$

and

$$I_{C_Q} \cong I_E$$

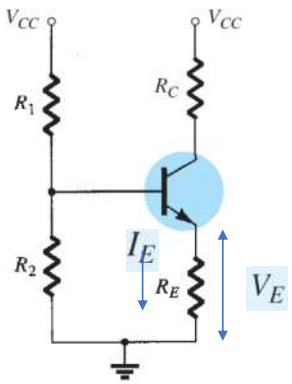


FIG. 4.30

The collector-to-emitter voltage is determined by

$$V_{CE} = V_{CC} - I_C R_C - I_E R_E$$

but because $I_E \cong I_C$,

$$V_{CE_Q} = V_{CC} - I_C(R_C + R_E) \tag{4.37}$$

Note in the sequence of calculations from Eq. (4.33) through Eq. (4.37) that β does not appear and I_B was not calculated. The Q-point (as determined by I_{C_Q} and V_{CE_Q}) is therefore independent of the value of β.

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EXAMPLE 4.9 Repeat the analysis of Fig. 4.35 using the approximate technique, and compare solutions for I_{C_Q} and V_{CE_Q}.

Solution: Testing:

$$\beta R_E \geq 10R_2$$
$$(100)(1.5\ \text{k}\Omega) \geq 10(3.9\ \text{k}\Omega)$$
$$150\ \text{k}\Omega \geq 39\ \text{k}\Omega\ (\textit{satisfied})$$

$$V_B = \frac{R_2 V_{CC}}{R_1 + R_2}$$
$$= \frac{(3.9\ \text{k}\Omega)(22\ \text{V})}{39\ \text{k}\Omega + 3.9\ \text{k}\Omega}$$
$$= 2\ \text{V}$$

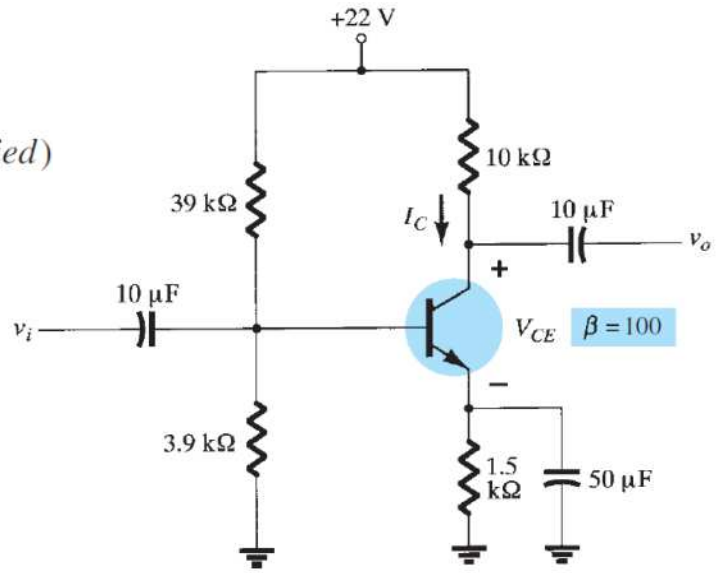


FIG. 4.35
Beta-stabilized circuit for Example 4.8.

Note that the level of V_B is the same as E_{Th} determined in Example 4.7. Essentially, therefore, the primary difference between the exact and approximate techniques is the effect of R_{Th} in the exact analysis that separates E_{Th} and V_B.

Eq. (4.34): $$V_E = V_B - V_{BE}$$
$$= 2\ \text{V} - 0.7\ \text{V}$$
$$= 1.3\ \text{V}$$

$$I_{CQ} \cong I_E = \frac{V_E}{R_E} = \frac{1.3\ \text{V}}{1.5\ \text{k}\Omega} = \mathbf{0.867\ mA}$$

compared to 0.84 mA with the exact analysis. Finally,

$$V_{CE_Q} = V_{CC} - I_C(R_C + R_E)$$
$$= 22\ \text{V} - (0.867\ \text{mA})(10\ \text{kV} + 1.5\ \text{k}\Omega)$$
$$= 22\ \text{V} - 9.97\ \text{V}$$
$$= \mathbf{12.03\ V}$$

versus 12.34 V obtained in Example 4.8.

لیث، عمرو

The results for I_{C_Q} and V_{CE_Q} are certainly close, and considering the actual variation in parameter values, one can certainly be considered as accurate as the other. The larger the level of R_i compared to R_2, the closer is the approximate to the exact solution. Example 4.11 will compare solutions at a level well below the condition established by Eq. (4.33).

EXAMPLE 4.10 Repeat the exact analysis of Example 4.8 if β is reduced to 50, and compare solutions for I_{C_Q} and V_{CE_Q}.

Solution: This example is not a comparison of exact versus approximate methods, but a testing of how much the Q-point will move if the level of β is cut in half. R_{Th} and E_{Th} are the same:

$$R_{\text{Th}} = 3.55\ \text{k}\Omega, \qquad E_{\text{Th}} = 2\ \text{V}$$

$$I_B = \frac{E_{\text{Th}} - V_{BE}}{R_{\text{Th}} + (\beta + 1)R_E}$$

$$= \frac{2\ \text{V} - 0.7\ \text{V}}{3.55\ \text{k}\Omega + (51)(1.5\ \text{k}\Omega)} = \frac{1.3\ \text{V}}{3.55\ \text{k}\Omega + 76.5\ \text{k}\Omega}$$

$$= 16.24\ \mu\text{A}$$

$$I_{C_Q} = \beta I_B$$

$$= (50)(16.24\ \mu\text{A})$$

$$= \mathbf{0.81\ mA}$$

$$V_{CE_Q} = V_{CC} - I_C(R_C + R_E)$$

$$= 22\ \text{V} - (0.81\ \text{mA})(10\ \text{k}\Omega + 1.5\ \text{k}\Omega)$$

$$= \mathbf{12.69\ V}$$

لیث، عمرو

Tabulating the results, we have:

Effect of β variation on the response of the voltage-divider configuration of Fig. 4.35.

β	I_{C_Q} (mA)	V_{CE_Q} (V)
100	0.84 mA	12.34 V
50	0.81 mA	12.69 V

The results clearly show the relative insensitivity of the circuit to the change in β. Even though β is drastically cut in half, from 100 to 50, the levels of I_{C_Q} and V_{CE_Q} are essentially the same.

Important Note: Looking back on the results for the fixed-bias configuration, we find the current decreased from 4.71 mA to 2.35 mA when beta dropped from 100 to 50. For the voltage-divider configuration, the same change in beta only resulted in a change in current from 0.84 mA to 0.81 mA. Even more noticeable is the change in V_{CE_Q} for the fixed-bias configuration. Dropping beta from 100 to 50 resulted in an increase in voltage from 1.64 to 6.83 V (a change of over 300%). For the voltage-divider configuration, the increase in voltage was only from 12.34 V to 12.69 V, which is a change of less than 3%. In summary, therefore, changing beta by 50% resulted in a change in an important network parameter of over 300% for the fixed-bias configuration and less than 3% for the voltage-divider configuration—a significant difference.

So, for linear amplification, voltage divider Bias Configuration is the best circuit as it is stable essentially that β does not appear in the equations

لیث، عمرو

EXAMPLE 4.11 Determine the levels of I_{C_Q} and V_{CE_Q} for the voltage-divider configuration of Fig. 4.37 using the exact and approximate techniques and compare solutions. In this case, the conditions of Eq. (4.33) *will not be satisfied* and the results will reveal the difference in solution if the criterion of Eq. (4.33) is ignored.

Eq. (4.33):

$$\beta R_E \geq 10R_2$$
$$(50)(1.2\,\text{k}\Omega) \geq 10(22\,\text{k}\Omega)$$
$$60\,\text{k}\Omega \not\geq 220\,\text{k}\Omega \text{ (not satisfied)}$$

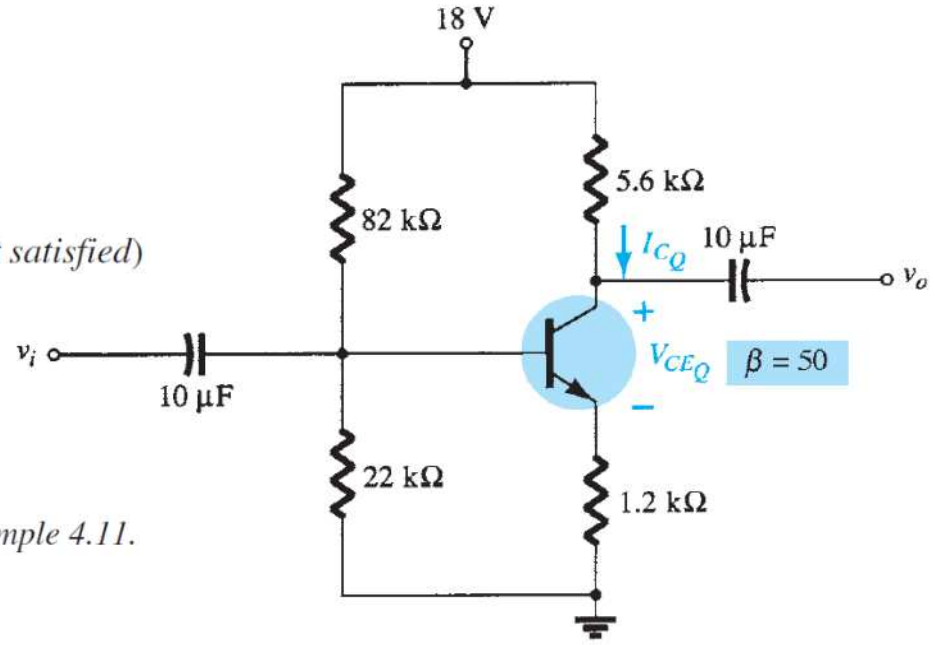


FIG. 4.37
Voltage-divider configuration for Example 4.11.

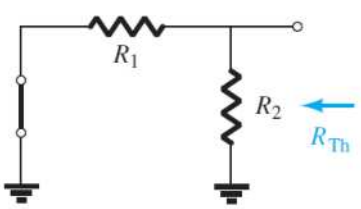
$$R_{\text{Th}} = R_1 \| R_2 = 82\,\text{k}\Omega \| 22\,\text{k}\Omega = 17.35\,\text{k}\Omega$$

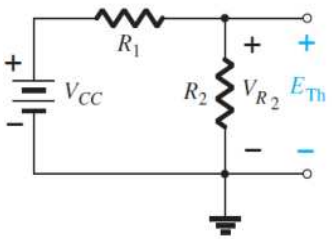
$$E_{\text{Th}} = \frac{R_2 V_{CC}}{R_1 + R_2} = \frac{22\,\text{k}\Omega(18\text{ V})}{82\,\text{k}\Omega + 22\,\text{k}\Omega} = 3.81\text{ V}$$

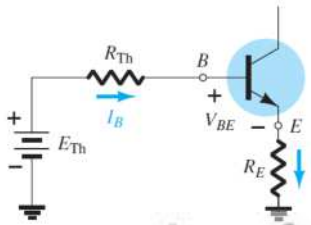
$$I_B = \frac{E_{\text{Th}} - V_{BE}}{R_{\text{Th}} + (\beta + 1)R_E} = \frac{3.81\text{ V} - 0.7\text{ V}}{17.35\,\text{k}\Omega + (51)(1.2\,\text{k}\Omega)} = \frac{3.11\text{ V}}{78.55\,\text{k}\Omega} = 39.6\,\mu\text{A}$$

$$I_{C_Q} = \beta I_B = (50)(39.6\,\mu\text{A}) = \mathbf{1.98\text{ mA}}$$

$$V_{CE_Q} = V_{CC} - I_C(R_C + R_E)$$
$$= 18\text{ V} - (1.98\text{ mA})(5.6\,\text{k}\Omega + 1.2\,\text{k}\Omega)$$
$$= \mathbf{4.54\text{ V}}$$







Approximate analysis:

$$V_B = \frac{R_2 V_{CC}}{R_1 + R_2}$$

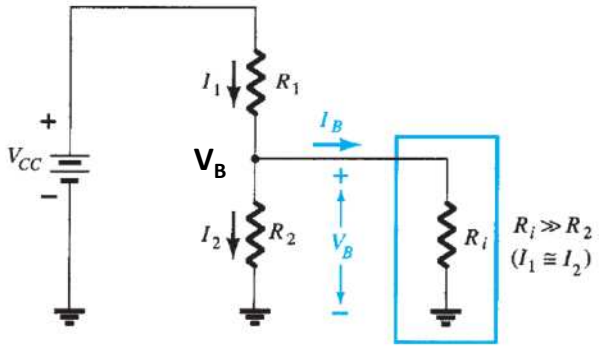


FIG. 4.36

$$V_B = E_{Th} = 3.81\ \text{V}$$

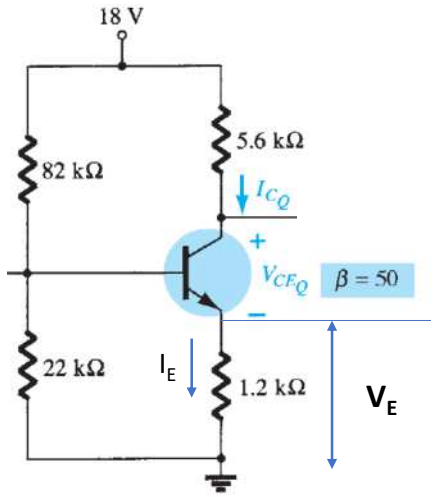
$$V_E = V_B - V_{BE} = 3.81\ \text{V} - 0.7\ \text{V} = 3.11\ \text{V}$$

$$I_{C_Q} \cong I_E = \frac{V_E}{R_E} = \frac{3.11\ \text{V}}{1.2\ \text{k}\Omega} = \mathbf{2.59\ mA}$$

$$V_{CE_Q} = V_{CC} - I_C(R_C + R_E)$$

$$= 18\ \text{V} - (2.59\ \text{mA})(5.6\ \text{k}\Omega + 1.2\ \text{k}\Omega)$$

$$= \mathbf{3.88\ V}$$



Tabulating the results, we have:

Comparing the exact and approximate approaches.

	I_{C_Q} (mA)	V_{CE_Q} (V)
Exact	1.98	4.54
Approximate	2.59	3.88

The results reveal the difference between exact and approximate solutions. I_{C_Q} is about 30% greater with the approximate solution, whereas V_{CE_Q} is about 10% less. The results are notably different in magnitude, but even though βR_E is only about three times larger than R_2, the results are still relatively close to each other. For the future, however, our analysis will be dictated by Eq. (4.33) to ensure a close similarity between exact and approximate solutions.

لیث، عمرو

Transistor Saturation

The output collector–emitter circuit for the voltage-divider configuration has the same appearance as the emitter-biased circuit analyzed in Section 4.4. The resulting equation for the saturation current (when V_{CE} is set to 0 V on the schematic) is therefore the same as obtained for the emitter-biased configuration. That is,

$$I_{C_{sat}} = I_{C_{max}} = \frac{V_{CC}}{R_C + R_E} \tag{4.38}$$

$$V_{CE_Q} = V_{CC} - I_C(R_C + R_E) \tag{4.37}$$

Load-Line Analysis

$$V_{CE_Q} = V_{CC} - I_C(R_C + R_E) \tag{4.37}$$

The similarities with the output circuit of the emitter-biased configuration result in the same intersections for the load line of the voltage-divider configuration. The load line will therefore have the same appearance as that of Fig. 4.25, with

$$I_C = \left.\frac{V_{CC}}{R_C + R_E}\right|_{V_{CE}=0\text{ V}} \tag{4.39}$$

and

$$V_{CE} = V_{CC}|_{I_C=0\text{ mA}} \tag{4.40}$$

The level of I_B is of course determined by a different equation for the voltage-divider bias and the emitter-bias configurations.

لیث، عمرو

EXAMPLE 4.20 Determine V_C and V_B for the network of Fig. 4.55.

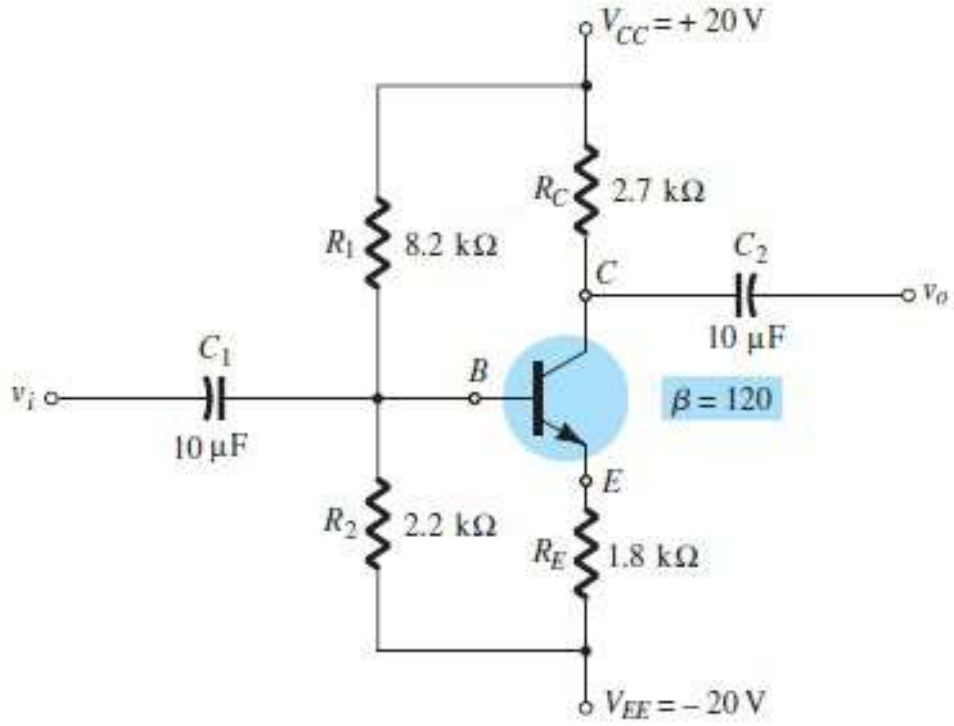


FIG. 4.55
Example 4.20.

Solution: The Thévenin resistance and voltage are determined for the network to the left of the base terminal as shown in Figs. 4.56 and 4.57.

R_{Th}

$$R_{Th} = 8.2\,k\Omega \,\|\, 2.2\,k\Omega = 1.73\,k\Omega$$

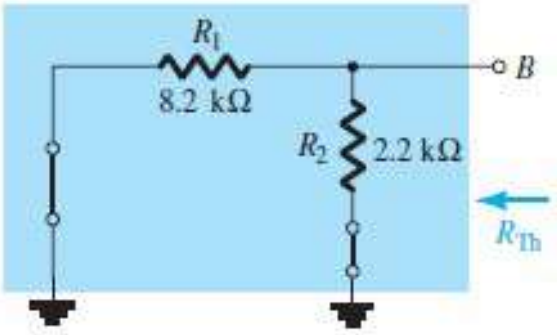


FIG. 4.56
Determining R_{Th}.

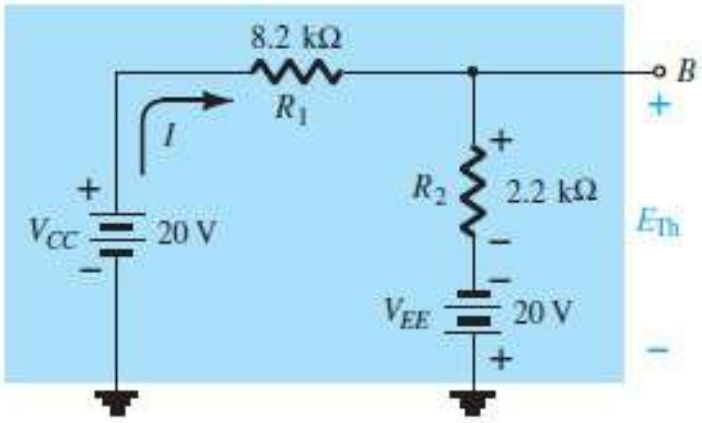


FIG. 4.57
Determining E_{Th}.

E_{Th}

$$I = \frac{V_{CC} + V_{EE}}{R_1 + R_2} = \frac{20\text{ V} + 20\text{ V}}{8.2\text{ k}\Omega + 2.2\text{ k}\Omega} = \frac{40\text{ V}}{10.4\text{ k}\Omega}$$
$$= 3.85\text{ mA}$$
$$E_{Th} = IR_2 - V_{EE}$$
$$= (3.85\text{ mA})(2.2\text{ k}\Omega) - 20\text{ V}$$
$$= -11.53\text{ V}$$

The network can then be redrawn as shown in Fig. 4.58, where the application of Kirchhoff's voltage law results in

$$-E_{Th} - I_BR_{Th} - V_{BE} - I_ER_E + V_{EE} = 0$$

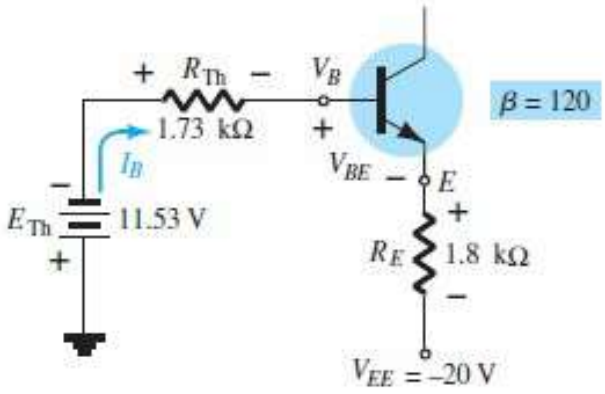


FIG. 4.58
Substituting the Thévenin equivalent circuit.

Substituting $I_E = (\beta + 1)I_B$ gives

$$V_{EE} - E_{Th} - V_{BE} - (\beta + 1)I_BR_E - I_BR_{Th} = 0$$

and

$$I_B = \frac{V_{EE} - E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E}$$
$$= \frac{20\text{ V} - 11.53\text{ V} - 0.7\text{ V}}{1.73\text{ k}\Omega + (121)(1.8\text{ k}\Omega)}$$
$$= \frac{7.77\text{ V}}{219.53\text{ k}\Omega}$$
$$= 35.39\ \mu\text{A}$$
$$I_C = \beta I_B$$
$$= (120)(35.39\ \mu\text{A})$$
$$= 4.25\text{ mA}$$
$$V_C = V_{CC} - I_CR_C$$
$$= 20\text{ V} - (4.25\text{ mA})(2.7\text{ k}\Omega)$$
$$= \mathbf{8.53\text{ V}}$$
$$V_B = -E_{Th} - I_BR_{Th}$$
$$= -(11.53\text{ V}) - (35.39\ \mu\text{A})(1.73\text{ k}\Omega)$$
$$= \mathbf{-11.59\text{ V}}$$

لیث، عمرو

4.6 COLLECTOR FEEDBACK CONFIGURATION

An improved level of stability can also be obtained by introducing a feedback path from collector to base as shown in Fig. 4.38. Although the Q-point is not totally independent of beta (even under approximate conditions), the sensitivity to changes in beta or temperature variations is normally less than encountered for the fixed-bias or emitter-biased configurations. The analysis will again be performed by first analyzing the base–emitter loop, with the results then applied to the collector–emitter loop.

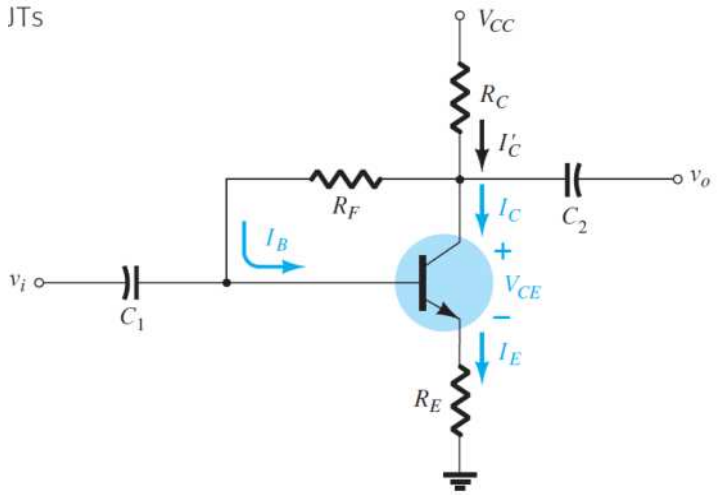


FIG. 4.38
DC bias circuit with voltage feedback.

Base–Emitter Loop

Figure 4.39 shows the base–emitter loop for the voltage feedback configuration. Writing Kirchhoff's voltage law around the indicated loop in the clockwise direction will result in

$$V_{CC} - I'_C R_C - I_B R_F - V_{BE} - I_E R_E = 0$$

It is important to note that the current through R_C is not I_C, but I'_C (where $I'_C = I_C + I_B$). However, the level of I_C and I'_C far exceeds the usual level of I_B, and the approximation $I'_C \cong I_C$ is normally employed. Substituting $I'_C \cong I_C = \beta I_B$ and $I_E \cong I_C$ results in

$$V_{CC} - \beta I_B R_C - I_B R_F - V_{BE} - \beta I_B R_E = 0$$

Gathering terms, we have

$$V_{CC} - V_{BE} - \beta I_B (R_C + R_E) - I_B R_F = 0$$

and solving for I_B yields

$$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta(R_C + R_E)} \qquad (4.41)$$

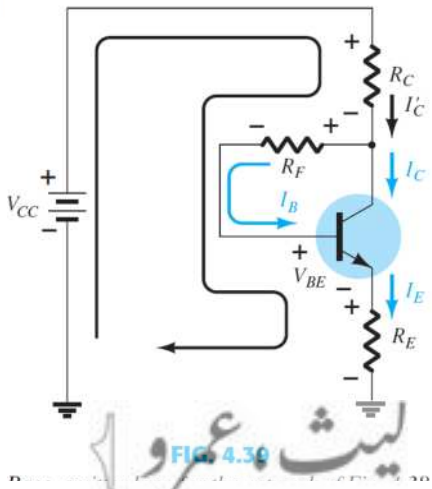


FIG. 4.39

$$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta(R_C + R_E)} \tag{4.41}$$

The result is quite interesting in that the format is very similar to equations for I_B obtained for earlier configurations. The numerator is again the difference of available voltage levels, whereas the denominator is the base resistance plus the collector and emitter resistors reflected by beta. In general, therefore, the feedback path results in a reflection of the resistance R_C back to the input circuit, much like the reflection of R_E.

In general, the equation for I_B has the following format, which can be compared with the result for the fixed-bias and emitter-bias configurations.

$$I_B = \frac{V'}{R_F + \beta R'}$$

For the fixed-bias configuration $\beta R'$ does not exist. For the emitter-bias setup (with $\beta + 1 \cong \beta$), $R' = R_E$.

Because $I_C = \beta I_B$,

$$I_{C_Q} = \frac{\beta V'}{R_F + \beta R'} = \frac{V'}{\frac{R_F}{\beta} + R'}$$

In general, the larger R' is compared with $\frac{R_F}{\beta}$, the more accurate the approximation that

$$I_{C_Q} \cong \frac{V'}{R'}$$

The result is an equation absent of β, which would be very stable for variations in β. Because R' is typically larger for the voltage-feedback configuration than for the emitter-bias configuration, the sensitivity to variations in beta is less. Of course, R' is 0 Ω for the fixed-bias configuration and is therefore quite sensitive to variations in beta.

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Collector–Emitter Loop

The collector–emitter loop for the network of Fig. 4.38 is provided in Fig. 4.40. Applying Kirchhoff's voltage law around the indicated loop in the clockwise direction results in

$$I_E R_E + V_{CE} + I'_C R_C - V_{CC} = 0$$

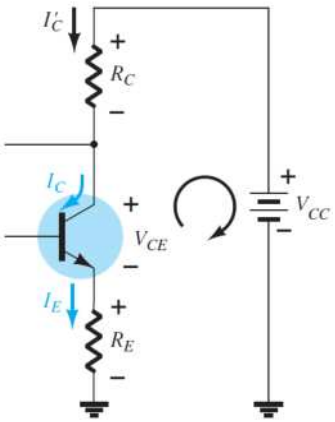
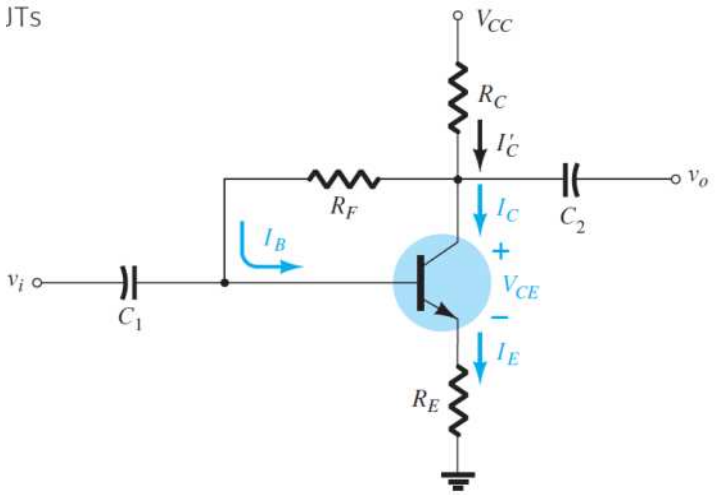


FIG. 4.40
Collector–emitter loop for the network of Fig. 4.38.

JTs



Because $I'_C \cong I_C$ and $I_E \cong I_C$, we have

$$I_C(R_C + R_E) + V_{CE} - V_{CC} = 0$$

and

$$\boxed{V_{CE} = V_{CC} - I_C(R_C + R_E)} \qquad \textbf{(4.42)}$$

which is exactly as obtained for the emitter-bias and voltage-divider bias configurations.

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EXAMPLE 4.12 Determine the quiescent levels of I_{C_Q} and V_{CE_Q} for the network of Fig. 4.41.

Solution: Eq. (4.41):
$$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta(R_C + R_E)}$$
$$= \frac{10\text{ V} - 0.7\text{ V}}{250\text{ k}\Omega + (90)(4.7\text{ k}\Omega + 1.2\text{ k}\Omega)}$$
$$= \frac{9.3\text{ V}}{250\text{ k}\Omega + 531\text{ k}\Omega} = \frac{9.3\text{ V}}{781\text{ k}\Omega}$$
$$= 11.91\ \mu\text{A}$$
$$I_{C_Q} = \beta I_B = (90)(11.91\ \mu\text{A})$$
$$= \mathbf{1.07\text{ mA}}$$
$$V_{CE_Q} = V_{CC} - I_C(R_C + R_E)$$
$$= 10\text{ V} - (1.07\text{ mA})(4.7\text{ k}\Omega + 1.2\text{ k}\Omega)$$
$$= 10\text{ V} - 6.31\text{ V}$$
$$= \mathbf{3.69\text{ V}}$$

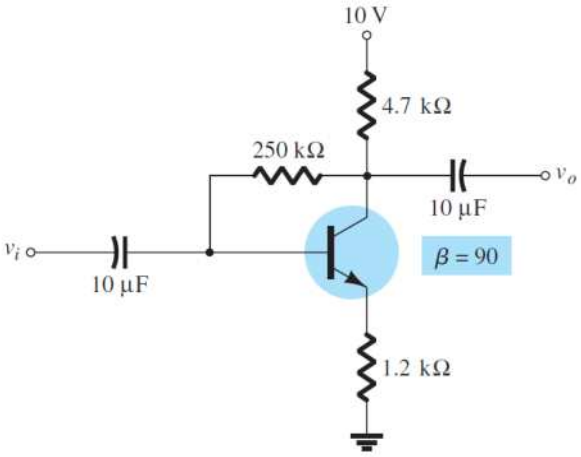


FIG. 4.41
Network for Example 4.12.

EXAMPLE 4.13 Repeat Example 4.12 using a beta of 135 (50% greater than in Example 4.12).

Solution: It is important to note in the solution for I_B in Example 4.12 that the second term in the denominator of the equation is much larger than the first. Recall in a recent discussion that the larger this second term is compared to the first, the less is the sensitivity to changes in beta. In this example, the level of beta is increased by 50%, which will increase the magnitude of this second term even more compared to the first. It is more important to note in these examples, however, that once the second term is relatively large compared to the first, the sensitivity to changes in beta is significantly less.

$$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta(R_C + R_E)} \qquad \textbf{(4.41)}$$

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Solving for I_B gives

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)}$$

$$= \frac{10\text{ V} - 0.7\text{ V}}{250\text{ k}\Omega + (135)(4.7\text{ k}\Omega + 1.2\text{ k}\Omega)}$$

$$= \frac{9.3\text{ V}}{250\text{ k}\Omega + 796.5\text{ k}\Omega} = \frac{9.3\text{ V}}{1046.5\text{ k}\Omega}$$

$$= 8.89\ \mu\text{A}$$

and

$$I_{C_Q} = \beta I_B$$

$$= (135)(8.89\ \mu\text{A})$$

$$= \mathbf{1.2\text{ mA}}$$

and

$$V_{CE_Q} = V_{CC} - I_C(R_C + R_E)$$

$$= 10\text{ V} - (1.2\text{ mA})(4.7\text{ k}\Omega + 1.2\text{ k}\Omega)$$

$$= 10\text{ V} - 7.08\text{ V}$$

$$= \mathbf{2.92\text{ V}}$$

Even though the level of β increased 50%, the level of I_{C_Q} only increased 12.1%, whereas the level of V_{CE_Q} decreased about 20.9%. If the network were a fixed-bias design, a 50% increase in β would have resulted in a 50% increase in I_{C_Q} and a dramatic change in the location of the Q-point.

$\beta = 90$	$\beta = 135$
$I_{C_Q} = \beta I_B = (90)(11.91\ \mu\text{A})$ $= \mathbf{1.07\text{ mA}}$ $V_{CE_Q} = V_{CC} - I_C(R_C + R_E)$ $= 10\text{ V} - (1.07\text{ mA})(4.7\text{ k}\Omega + 1.2\text{ k}\Omega)$ $= 10\text{ V} - 6.31\text{ V}$ $= \mathbf{3.69\text{ V}}$	$I_{C_Q} = \beta I_B$ $= (135)(8.89\ \mu\text{A})$ $= \mathbf{1.2\text{ mA}}$ $V_{CE_Q} = V_{CC} - I_C(R_C + R_E)$ $= 10\text{ V} - (1.2\text{ mA})(4.7\text{ k}\Omega + 1.2\text{ k}\Omega)$ $= 10\text{ V} - 7.08\text{ V}$ $= \mathbf{2.92\text{ V}}$

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EXAMPLE 4.14 Determine the dc level of I_B and V_C for the network of Fig. 4.42.

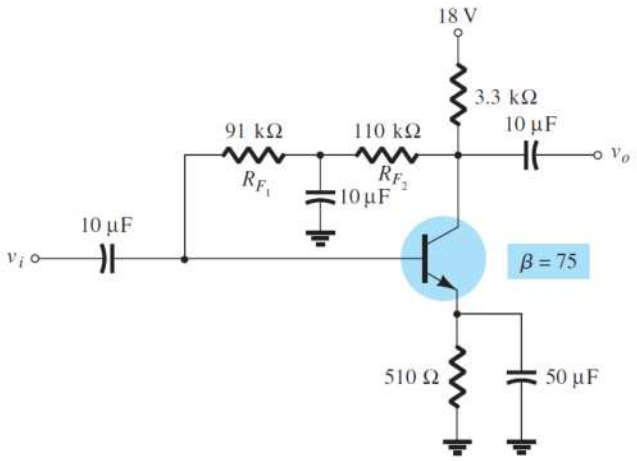


FIG. 4.42
Network for Example 4.14.

Solution: In this case, the base resistance for the dc analysis is composed of two resistors with a capacitor connected from their junction to ground. For the dc mode, the capacitor assumes the open-circuit equivalence, and $R_B = R_{F_1} + R_{F_2}$.

Solving for I_B gives

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)}$$

$$= \frac{18\text{ V} - 0.7\text{ V}}{(91\text{ k}\Omega + 110\text{ k}\Omega) + (75)(3.3\text{ k}\Omega + 0.51\text{ k}\Omega)}$$

$$= \frac{17.3\text{ V}}{201\text{ k}\Omega + 285.75\text{ k}\Omega} = \frac{17.3\text{ V}}{486.75\text{ k}\Omega}$$

$$= \mathbf{35.5\ \mu A}$$

$$I_C = \beta I_B$$

$$= (75)(35.5\ \mu\text{A})$$

$$= 2.66\text{ mA}$$

$$V_C = V_{CC} - I'_C R_C \cong V_{CC} - I_C R_C$$

$$= 18\text{ V} - (2.66\text{ mA})(3.3\text{ k}\Omega)$$

$$= 18\text{ V} - 8.78\text{ V}$$

$$= \mathbf{9.22\text{ V}}$$

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EXAMPLE 4.15 Given the network of Fig. 4.43 and the BJT characteristics of Fig. 4.44.

a. Draw the load line for the network on the characteristics.
b. Determine the dc beta in the center region of the characteristics. Define the choser point as the Q-point.
c. Using the dc beta calculated in part b, find the dc value of I_B.
d. Find I_{C_Q} and I_{CE_Q}.

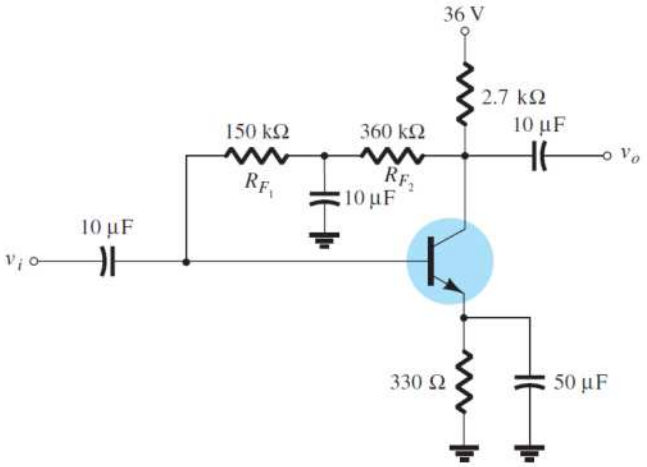


FIG. 4.43

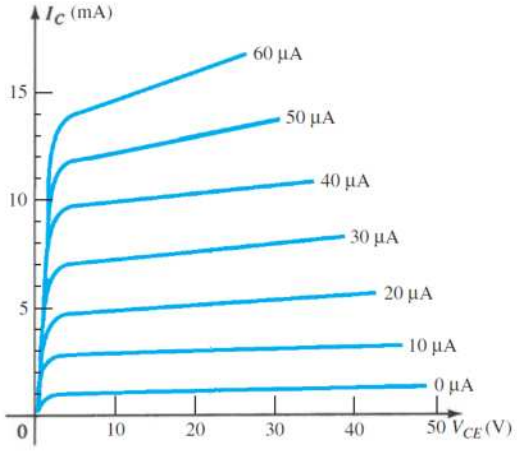


FIG. 4.44

Solution:

a. The load line is drawn on Fig. 4.45 as determined by the following intersections:

$$V_{CE} = 0\text{ V}: I_C = \frac{V_{CC}}{R_C + R_E} = \frac{36\text{ V}}{2.7\text{ k}\Omega + 330\ \Omega} = \mathbf{11.88\text{ mA}}$$

$$I_C = 0\text{ mA}: V_{CE} = V_{CC} = \mathbf{36\text{ V}}$$

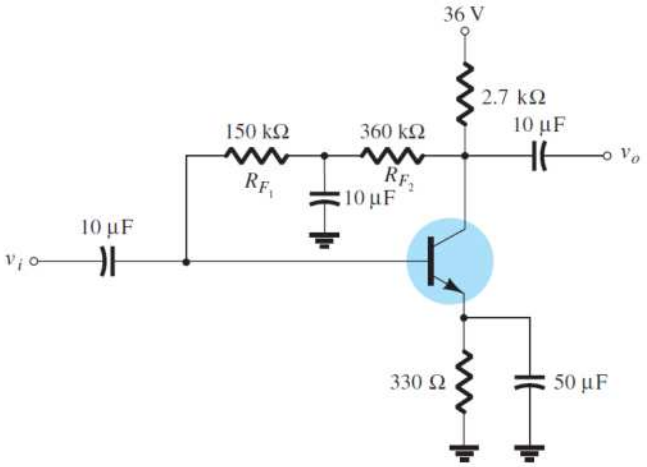


FIG. 4.43

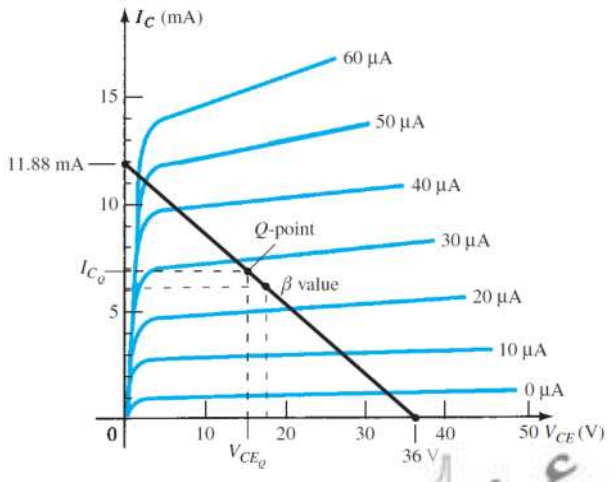


FIG. 4.45

b. The dc beta was determined using $I_B = 25\ \mu A$ and V_{CE} about 17 V.

$$\beta \cong \frac{I_{C_Q}}{I_{B_Q}} = \frac{6.2\ \text{mA}}{25\ \mu\text{A}} = \mathbf{248}$$

c. Using Eq. 4.41:

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{36\ \text{V} - 0.7\ \text{V}}{510\ \text{k}\Omega + 248(2.7\ \text{k}\Omega + 330\ \Omega)}$$

$$= \frac{35.3\ \text{V}}{510\ \text{k}\Omega + 751.44\ \text{k}\Omega}$$

and $$I_B = \frac{35.3\ \text{V}}{1.261\ \text{M}\Omega} = \mathbf{28\ \mu A}$$

d. From Fig. 4.45 the quiescent values are

$I_{C_Q} \cong \mathbf{6.9\ mA}$ and $V_{CE_Q} \cong \mathbf{15\ V}$

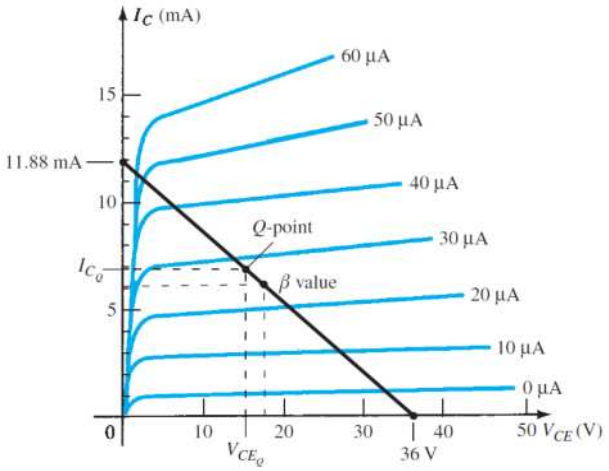


FIG. 4.45

4.7 EMITTER-FOLLOWER CONFIGURATION

The previous sections introduced configurations in which the output voltage is typically taken off the collector terminal of the BJT. This section will examine a configuration where the output is taken off the emitter terminal as shown in Fig. 4.46. The configuration of Fig. 4.46 is not the only one where the output can be taken off the emitter terminal. In fact, any of the configurations just described can be used so long as there is a resistor in the emitter leg.

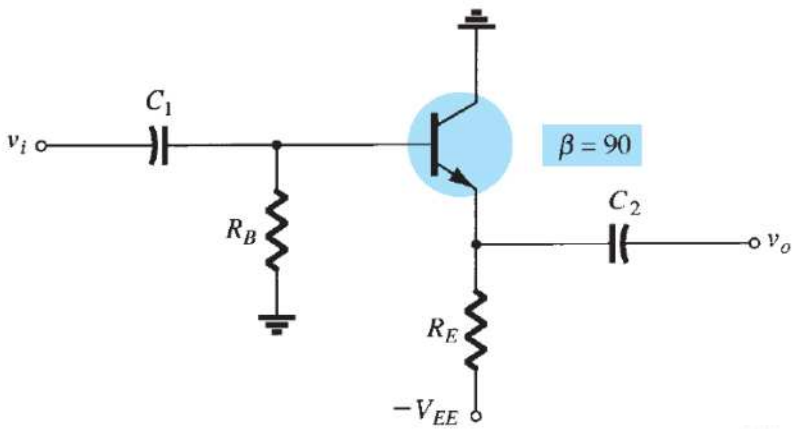


FIG. 4.46

The dc equivalent of the network of Fig. 4.46 appears in Fig. 4.47

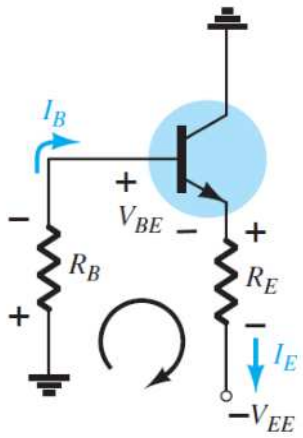


FIG. 4.47

Applying Kirchhoff's voltage rule to the input circuit will result in

$$-I_B R_B - V_{BE} - I_E R_E + V_{EE} = 0$$

and using $I_E = (\beta + 1)I_B$

$$I_B R_B + (\beta + 1)I_B R_E = V_{EE} - V_{BE}$$

so that

$$\boxed{I_B = \frac{V_{EE} - V_{BE}}{R_B + (\beta + 1)R_E}}$$

For the output network, an application of Kirchhoff's voltage law will result in

$$-V_{CE} - I_E R_E + V_{EE} = 0$$

and

$$\boxed{V_{CE} = V_{EE} - I_E R_E}$$

EXAMPLE 4.16 Determine V_{CE_Q} and I_{E_Q} for the network of Fig. 4.48.

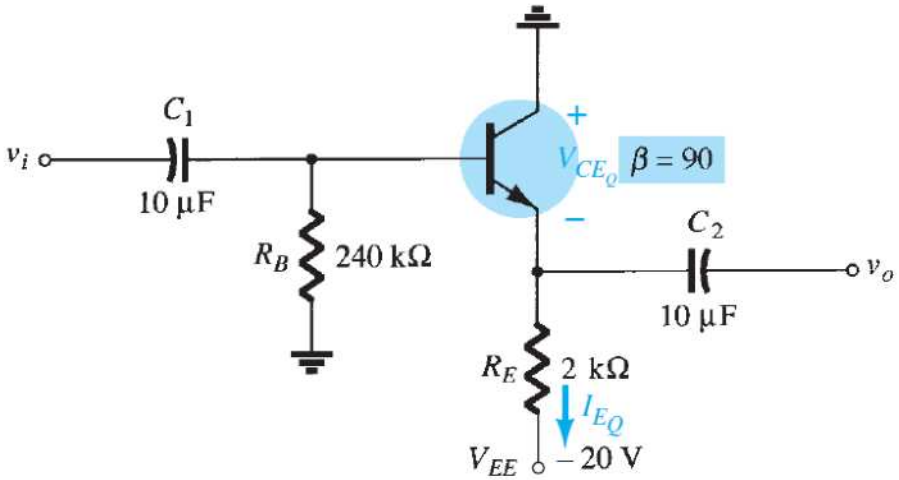


FIG. 4.48

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Solution:

Eq. 4.44:
$$I_B = \frac{V_{EE} - V_{BE}}{R_B + (\beta + 1)R_E}$$
$$= \frac{20\text{ V} - 0.7\text{ V}}{240\text{ k}\Omega + (90 + 1)2\text{ k}\Omega} = \frac{19.3\text{ V}}{240\text{ k}\Omega + 182\text{ k}\Omega}$$
$$= \frac{19.3\text{ V}}{422\text{ k}\Omega} = 45.73\ \mu\text{A}$$

and Eq. 4.45:
$$V_{CE_Q} = V_{EE} - I_E R_E$$
$$= V_{EE} - (\beta + 1)I_B R_E$$
$$= 20\text{ V} - (90 + 1)(45.73\ \mu\text{A})(2\text{ k}\Omega)$$
$$= 20\text{ V} - 8.32\text{ V}$$
$$= \mathbf{11.68\text{ V}}$$
$$I_{E_Q} = (\beta + 1)I_B = (91)(45.73\ \mu\text{A})$$
$$= 4.16\text{ mA}$$

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EXAMPLE 4.19 Determine V_C and V_B for the network of Fig. 4.54.

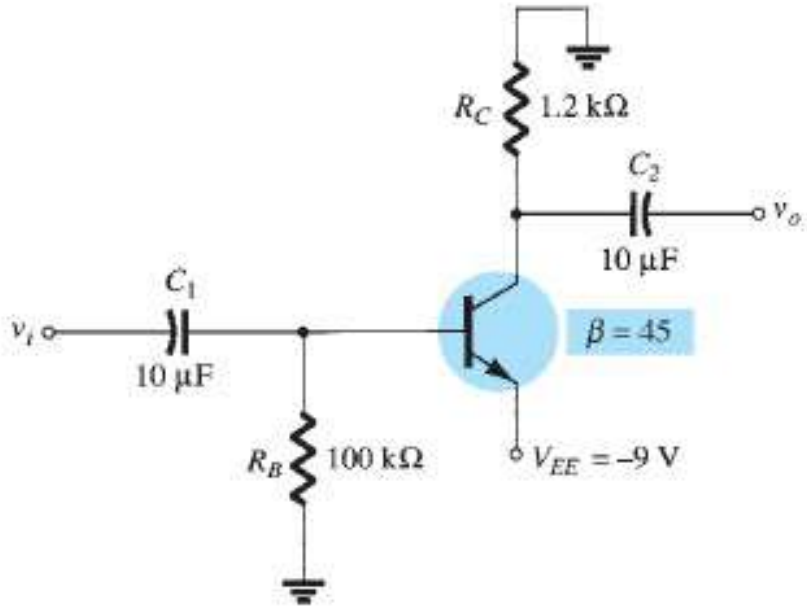


FIG. 4.54
Example 4.19.

Solution: Applying Kirchhoff's voltage law in the clockwise direction for the base–emitter loop results in

$$-I_B R_B - V_{BE} + V_{EE} = 0$$

and

$$I_B = \frac{V_{EE} - V_{BE}}{R_B}$$

Substitution yields

$$I_B = \frac{9\text{ V} - 0.7\text{ V}}{100\text{ k}\Omega}$$

$$= \frac{8.3\text{ V}}{100\text{ k}\Omega}$$

$$= 83\ \mu\text{A}$$

$$I_C = \beta I_B$$

$$= (45)(83\ \mu\text{A})$$

$$= 3.735\text{ mA}$$

$$V_C = -I_C R_C$$

$$= -(3.735\text{ mA})(1.2\text{ k}\Omega)$$

$$= \mathbf{-4.48\text{ V}}$$

$$V_B = -I_B R_B$$

$$= -(83\ \mu\text{A})(100\text{ k}\Omega)$$

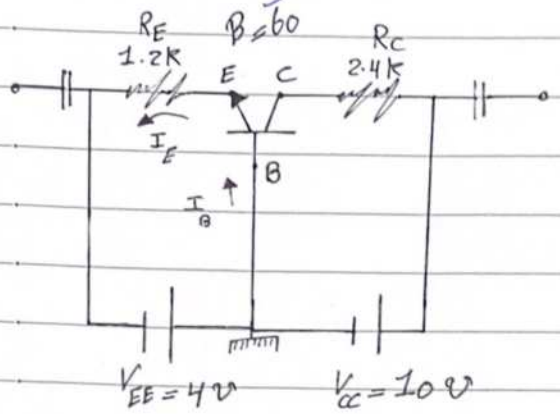
$$= \mathbf{-8.3\text{ V}}$$

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• Final Material

(الوجاهي)

* Common Base configuration :-



Find :- 1) I_E
2) I_B
3) V_{CE}
4) V_{CB}

Sol :- 1) $-V_{EE} + V_{BE} + I_E * R_E = 0 \Rightarrow I_E = \frac{V_{EE} - V_{BE}}{R_E} = \frac{4 - 0.7}{1.2\,k}$

$I_E = 2.75\ mA$

2) $I_E = (\beta + 1) I_B \Rightarrow I_B = \frac{I_E}{\beta + 1} \Rightarrow I_B = \frac{2.75m}{61} = 45.08\ \mu A$

3) $-V_{EE} - V_{cc} + I_c * R_c + V_{CE} + I_E * R_E = 0$ $\quad \therefore I_c \simeq I_E$

$V_{CE} = V_{EE} + V_{cc} - I_E (R_c + R_E) \Rightarrow V_{CE} = 4 + 10 - 2.75m (2.4k + 1.2k)$

$V_{CE} = 4.1\ v$

4) $-V_{cc} + I_c * R_c + V_{CB} = 0 \Rightarrow V_{CB} = V_{cc} - I_c * R_c \Rightarrow V_{CB} = 10 - 2.75m (2.4k)$

$V_{CB} = 3.4\ v$

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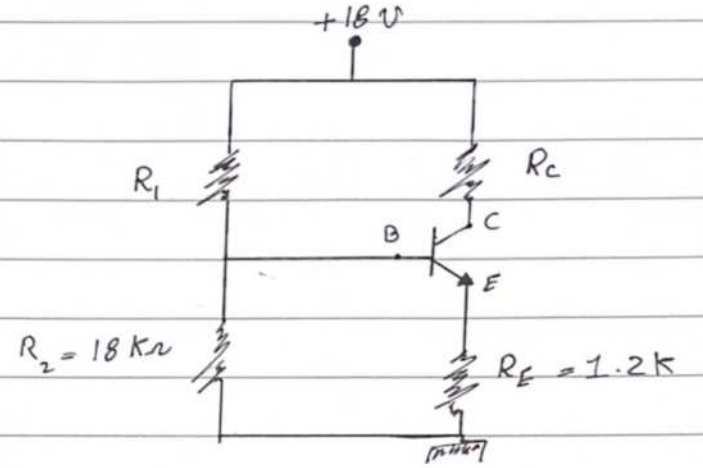
* Design :-

ex1: $I_{CQ} = 2\ mA$

$V_{CEQ} = 10\ V$

Find: 1) R_C

2) R_1



Sol:- 1) $-V_{CC} + I_C * R_C + V_{CE} + I_E * R_E = 0$ $\quad \because I_E \cong I_C$

$V_{CE} = V_{CC} - I_C (R_C + R_E) \Rightarrow 10 = 18 - 2m(R_C + 1.2K)$

$R_C = 2.8\ K$

2) $V_E = I_E * R_E \cong I_C * R_E = (2\ mA)(1.2K) = 2.4\ V$

$V_{BE} = V_B - V_E \rightarrow V_B = V_{BE} + V_E \rightarrow V_B = 0.7 + 2.4 = 3.1\ V$

$V_B = V_{Th} = \frac{V_{CC} R_2}{R_1 + R_2} \Rightarrow R_1 = 86.52\ K\Omega$

• Page 2

ليث، عمرو

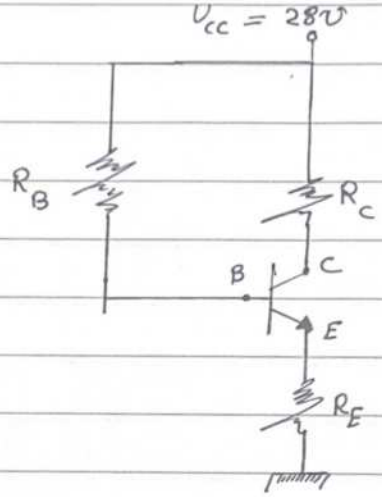
* Emitter-Bias :.

$I_{CQ} = \frac{1}{2} I_{Csat}$

$I_{Csat} = 8\,mA$

• الدائرة غير مُعطاة إنت لازم تعرفها من إسمها .. (Emitter Bias)

Sol :.



$V_C = 18\,V$

$V_{CC} = 28\,V$

$\beta = 110$

Find :. 1] R_C

2] R_E

3] R_B

1) $-V_{CC} + I_C^* R_C + V_C = 0$ $\quad \therefore I_{CQ} = \frac{1}{2}(8\,mA) = 4\,mA$

$R_C = \frac{V_{CC} - V_C}{I_C} \Rightarrow R_C = \frac{28-18}{4\,m} = 2.5\,K\Omega$

2) Saturation Region :. $V_{CE} = 0 \Rightarrow -V_{CC} + I_{Csat} R_C + I_{Csat} R_E = 0$

$R_E \Rightarrow -28 + (8\,mA)(R_E + 2.5K) = 0 \Rightarrow R_E = 1K\Omega$ ($I_E \cong I_{Csat}$)

3) $I_B = \frac{I_C}{\beta} \Rightarrow I_B = \frac{4m}{110} = 36.36\,\mu A$

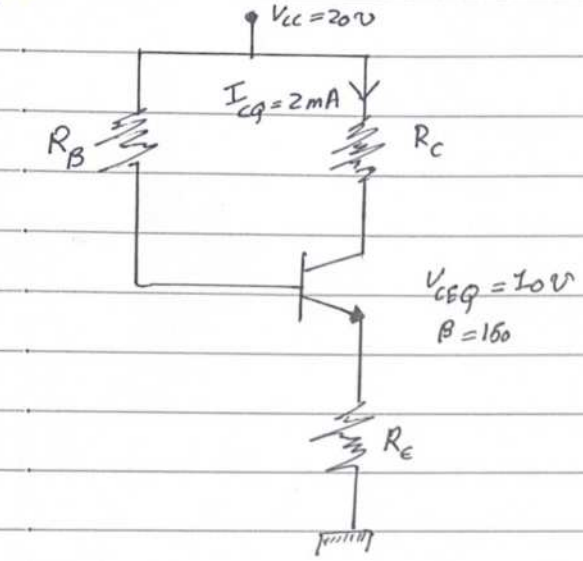
$-V_{CC} + I_B^* R_B + V_{BE} + I_E R_E = 0 \Rightarrow I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta+1)R_E}$

$R_B = 639.8\,K\Omega$

• Page 3

ليث، عمرو

* Design :



Find : R_C , R_B , R_E

Sol :

• في هذا المثال لا يمكن إيجاد المطاليب لأن أي Loop سيكون هناك مجهولان لحل المشكلة نفرض

$$\frac{V}{E} = \frac{1}{10}\left(\frac{V}{cc}\right)$$

• $\frac{V}{E} = \frac{1}{10}(20) = 2\ v$

$$\frac{V}{E} = \frac{I}{E} * \frac{R}{E} \cong \frac{I}{c} * \frac{R}{E} \rightarrow 2 = 2m(R_E) \Rightarrow R_E = 1\ k\Omega$$

• $-\frac{V}{cc} + \frac{I}{c}\frac{R}{c} + \frac{V}{CEQ} + \frac{V}{E} = 0 \Rightarrow R_c = \frac{V_{cc} - V_{CEQ} - V_E}{I_c} \rightarrow R_c = 4k\Omega$

• $-\frac{V}{cc} + \frac{I}{B}\frac{R}{B} + \frac{V}{BE} + \frac{V}{E} = 0 \Rightarrow R_B = \frac{V_{cc} - V_{BE} - V_E}{I_B}$

$\rightarrow I_B = \frac{I_c}{\beta}$

$R_B = 1.298\ M\Omega$

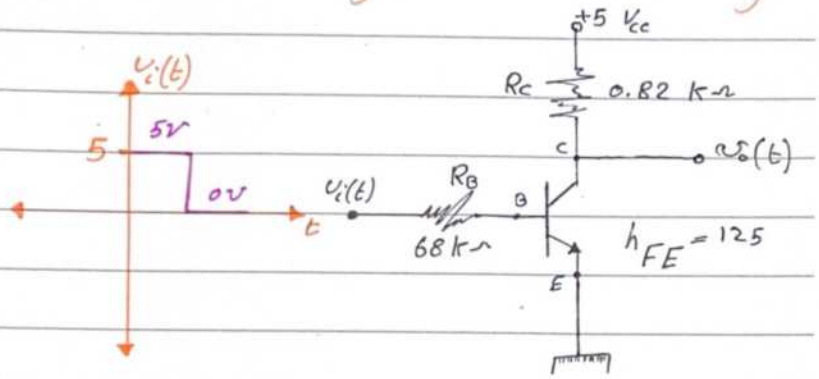
Mega $= 10^6$

ليث، عمرو

* Transistor as switch (BJT)

Transistor as switch in Saturation Region And cutoff Region.

Ex:



Sol:

* Positive half cycle:

$5 \gg V_{BE} \Rightarrow$ Transistor "on"

$$-v_i(t) + I_B R_B + V_{BE} = 0 \Rightarrow I_B = \frac{v_i(t) - V_{BE}}{R_B} \Rightarrow I_B = \frac{5 - 0.7}{68\,k} = 63\,\mu A$$

$$I_{CQ} = \beta I_B \Rightarrow I_{CQ} = (125)(63\,\mu) = 7.8\,mA$$

$$I_{csat} = \frac{V_{CC} - V_{CE}}{R_C} = \frac{5}{0.82\,k} = 6.1\,mA$$

(V_{CE} → 0,(02 ≈ 04))

$I_C = \beta I_B$

بزيد I_B بمقدار 20 μA عشان نضمن فقد العراة ما يخرب الـ switch

$I_{CQ} > I_{csat} \Rightarrow$ sat هذا يعني أننا نعمل بال (switch)

* Negative half cycle:

$I_B = 0\,\mu A$

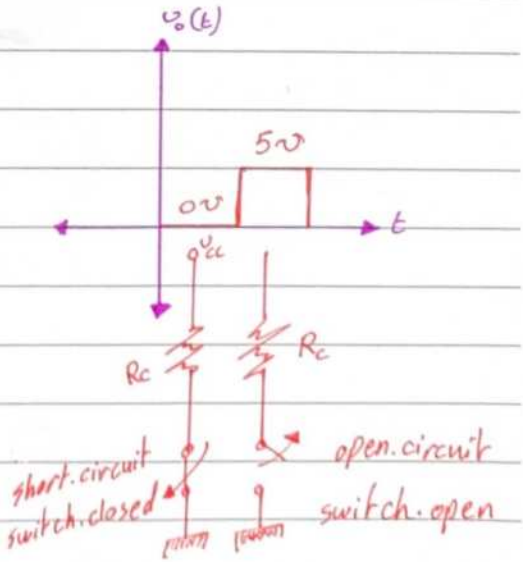
$I_C = 0\,\mu A$

$V_{CE} = V_C = 5V = V_{CC}$

$R_{sat} = \frac{V_{CE\,sat}}{I_{C\,sat}} = 0$

$R_{c.off} = \frac{V_{CE\,cut.off}}{I_{c\,cut.off}} = \infty$

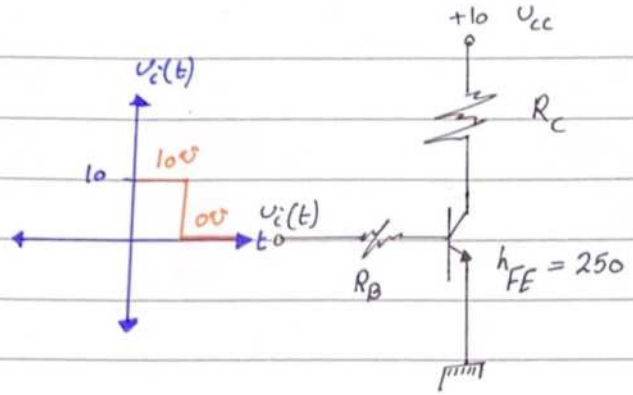
Page 5



ليث، عمرو

Ex:

$I_{c\,sat} = 10\ mA$



Find R_C, R_B :

Sol:

$$R_C = \frac{V_{cc}}{I_{c\,sat}} = \frac{10}{10\ mA} = 1\ k\Omega$$

$$I_c = \beta I_B \Rightarrow I_B = \frac{I_c}{\beta} \Rightarrow I_B = \frac{10\ mA}{250} = 40\ \mu A$$

- $I_B + 20\ \mu A = 60\ \mu A$ ✓

$$I_B = \frac{V_i - V_{BE}}{R_B} \Rightarrow R_B = \frac{V_i - V_{BE}}{I_B} \Rightarrow R_B = \frac{10 - 0.7}{60\ \mu}$$

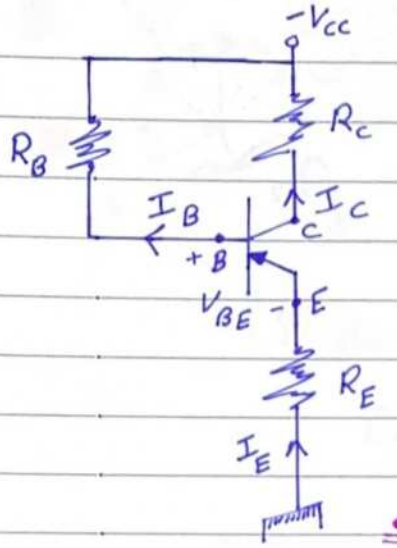
$$R_B = 155\ k\Omega$$

$$I_{CQ} = \beta I_B \Rightarrow I_{CQ} = (250)(60\ \mu) = 15\ mA$$

• Page 6

ليث، عمرو

* PnP Transistor



$V_{BE} = -0.7\ V$

$$-V_{CC} - I_B R_B + V_{BE} - I_E R_E = 0$$

$$I_B = \frac{+V_{CC} + V_{BE}}{R_B + (\beta+1) R_E}$$

$$I_C = \beta I_B$$

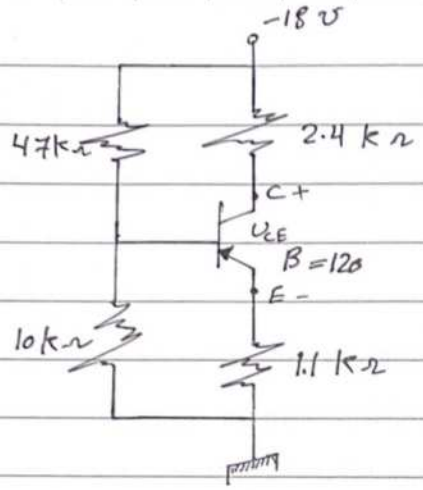
$$-V_{CC} - I_C R_C + V_{CE} - I_E R_E = 0$$

$$V_{CE} = -V_{CC} + I_C (R_C + R_E)$$

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لیث، عمرو

• Find V_{CE} :



Approximate :

$$\beta R_E \geq 10 R_2$$

$$(120)(1.1k) \geq (10)(10k)$$

$$132k \geq 100k \quad \text{(satisfied)}$$

$$V_B = \frac{V_{CC} R_2}{R_1 + R_2} \Rightarrow V_B = \frac{(-18)(10k)}{10k + 47k} = -3.16\ V$$

$$V_{BE} = V_B - V_E$$

$$-0.7 = -3.16 - V_E$$

$$V_E = -2.46\ V$$

$$V_E = -I_E R_E \Rightarrow I_E = 2.24\ mA \cong I_C$$

$$V_{CE} = -V_{CC} + I_C (R_C + R_E)$$

$$V_{CE} = -18 + 2.24m (2.4k + 1.1k)$$

$$V_{CE} = -10.16\ V$$

• Page 8

ليث، عمرو

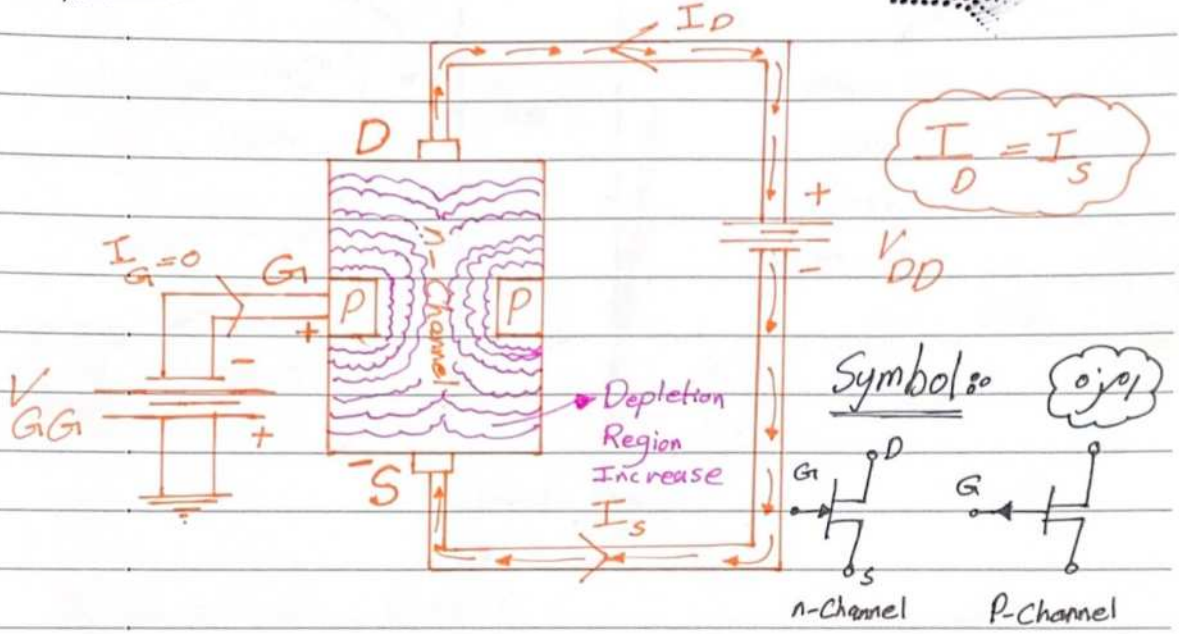
Ch. 6: FET (Field Effect Transistor)

Differences Between BJT & FET

• FET	• BJT
1) Unipolar (أحادي القطبية)	1) Bipolar (ثنائي القطبية)
2) Voltage Control Current V_{GS} control I_D (فولتية تتحكم بتيار)	2) Current Control Current I_B control I_c $[I_c = \beta I_B]$ (تيار يتحكم بتيار)
3) $I_G = 0$ mA	3) $V_{BE} = 0$ V
4) Gate, Source, Drain G, S, D	4) Base, Emitter, Collector B, E, C
5) n-type OR p-type	5) npn OR pnp

ليث، عمرو

1] JFET : Junction Field Effect Transistor



• طريقة العمل : كلما زاد قيمة VGG تكونت Depletion بين الـ P والـ n-Channel وبذلك يقل تيار I_D

Test : ① $V_{GG} = 0V$, $V_{GS} = 0V$, $I_D = I_{DSS}$ ← Saturation ← Maximum Current

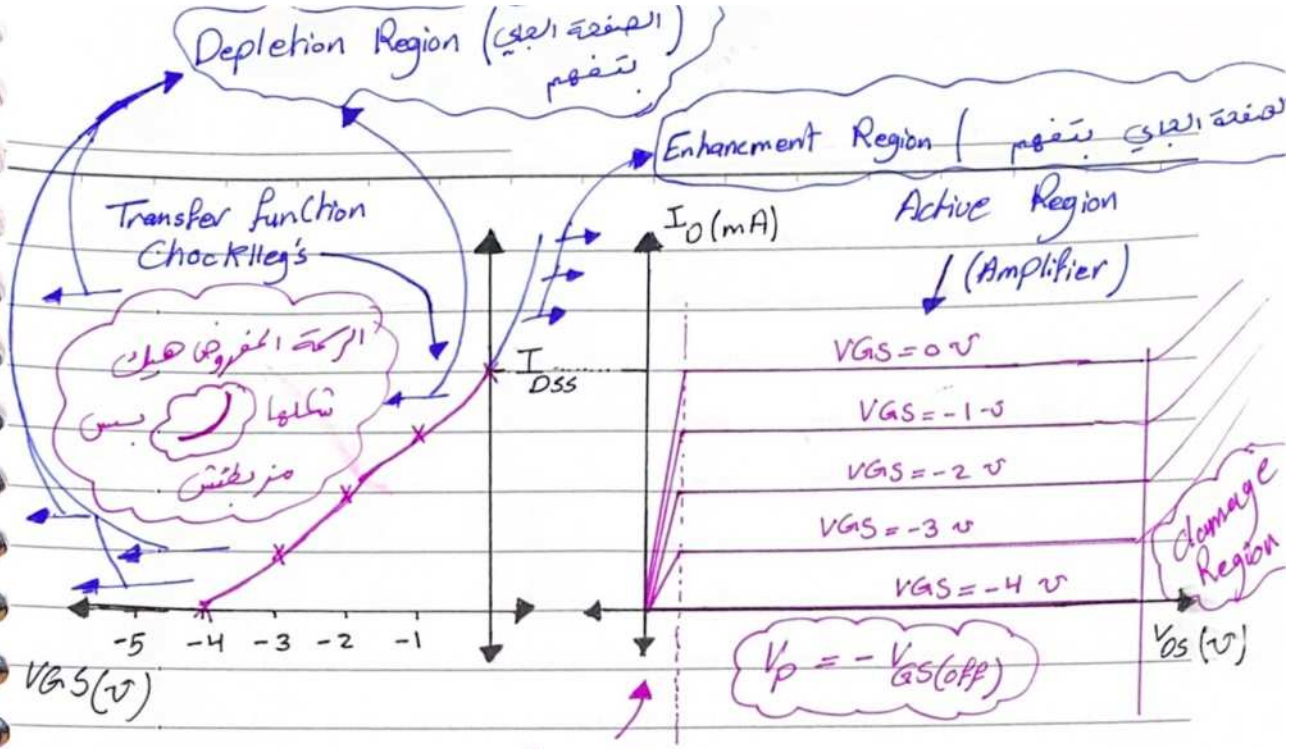
② $V_{GG} = 1V$, $V_{GS} = -1V$, I_D ↓ Decrease

③ $V_{GG} = 2V$, $V_{GS} = -2V$, I_D ↓↓ Decrease

④ $V_{GG} = 3V$, $V_{GS} = -3V$, I_D ↓↓↓ Decrease

⑤ $V_{GG} = 4V$, $V_{GS(off)} = -4V$, $I_D = 0\ mA$

ليث، عمرو



$$I_D = I_{DSS}\left(1 - \frac{VGS}{V_P}\right)^2$$

data sheet

$VGS = 0\ V \Rightarrow I_D = I_{DSS}$ mA

$VGS_{(off)} \Rightarrow I_D = 0$ mA

Four Points ... (حفظ زي إسحاق)

1] $VGS = 0 \Rightarrow I_D = I_{DSS}$ mA

2] $VGS = V_P \Rightarrow I_D = 0$ mA

3] $VGS = V_P/2 \Rightarrow I_D = I_{DSS}/4$ mA

4] $VGS = 0.3\,V_P \Rightarrow I_D = I_{DSS}/2$ mA

• زي إيش ؟؟ زي إسحاق .....

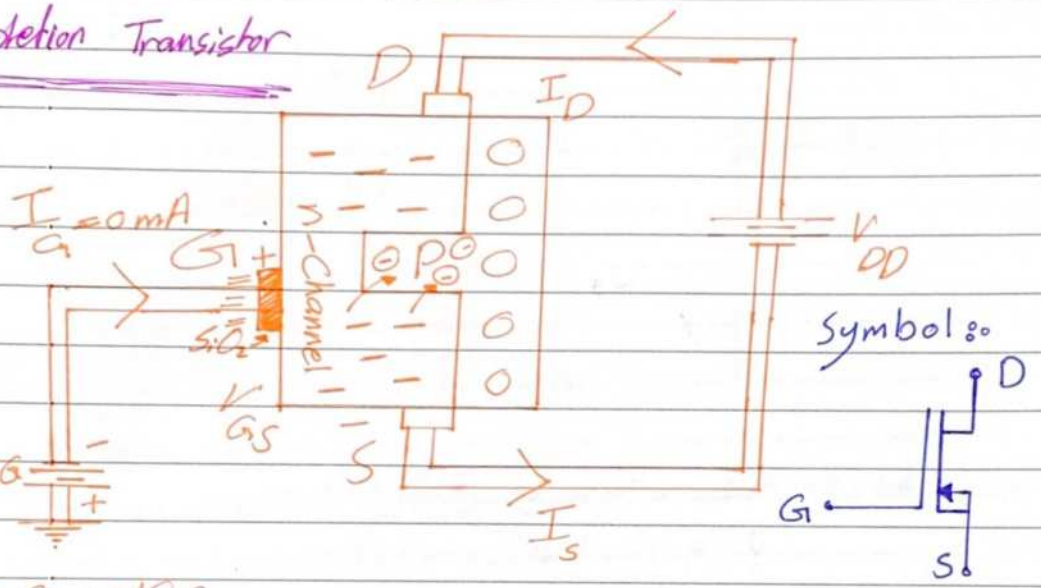
ليث، عمرو

2] MosFET : Metal Oxide Semiconductor Field Effect transistor

SiO_2 (مادة عازلة)

a. Depletion D-MosFET

• Depletion Transistor



$V_{GG} + V_{GS} = 0$

$V_{GS} = -V_{GG}$

• طريقة العمل : عند زيادة فولتية V_{GG} تتكوم الإلكترونات عند الطبقة العازلة SiO_2 فتصبح هنالك قوة تنافر بينها وبين n-Channel فتنتقل إلكترونات n-channel لـ P-type وتستمر بالفجوات

(منطقة عمل بالحالة السابقة على الشمال)

Test :

1) $V_{GG} = 0 \Rightarrow V_{GS} = 0 \Rightarrow I_D = I_{DSS}$ (saturation)

2) $V_{GG} = 1 \Rightarrow V_{GS} = -1 \Rightarrow I_D \downarrow$ Decrease

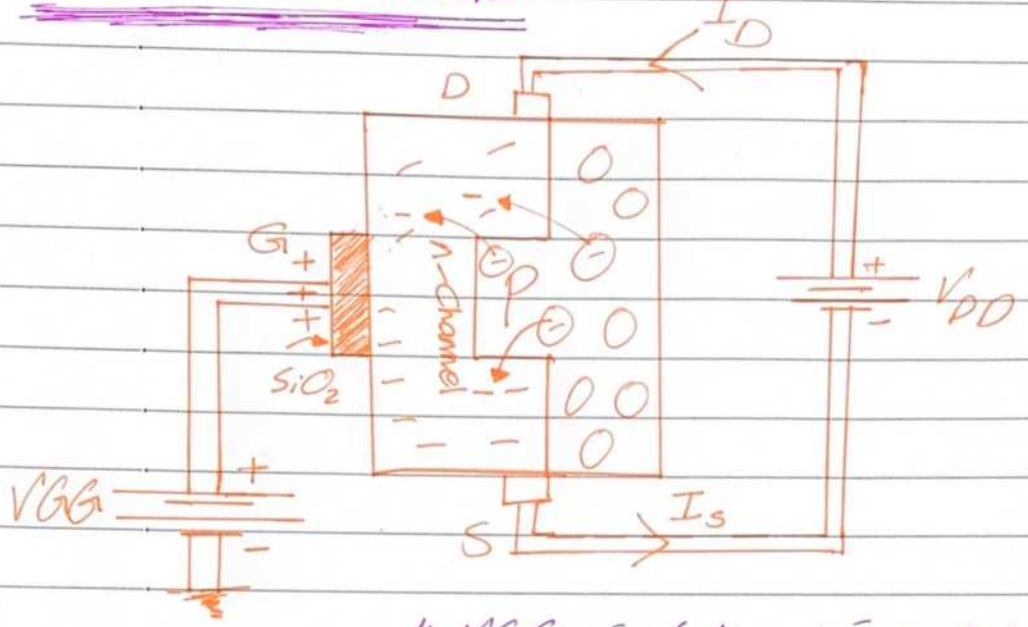
• Page 12

n) $V_{GG} = 4 \Rightarrow V_{GS} = -4 \Rightarrow I_D \downarrow\downarrow\downarrow\downarrow$ Decrease and $I_D = 0$ mA

$V_{GS(off)}$

ليث، عمرو

• Enhancement Transistor



• طريقة العمل هي قطب البطارية VGG الموجب يكوّم شحنات موجبة على حاجز العازل SiO_2 فتتم عملية تجاذب بينها وبين n-Channel فتتحرك الإلكترونات من P إلى n-Chanel ... والتيار يزداد ...
(منطقة عمل بصفة ● !!)

Test so

$$V_{GG} = V_{GS} = 1\,v$$

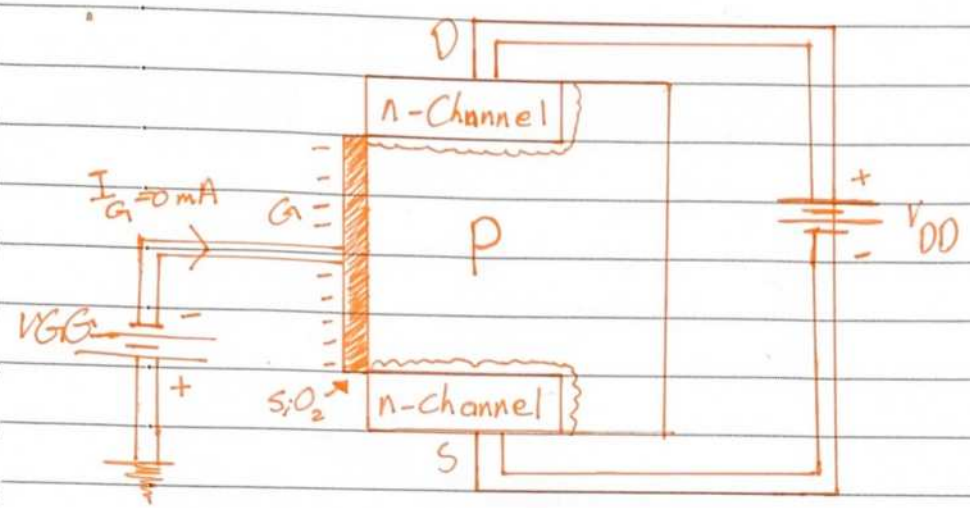
$$I_D > I_{DSS}$$

(شايف الأربع نقاط اللي زي إسمك ؟؟ ضعيف عليهم بالـ MosFet
5] $V_{GS} = 1\,v$ ← (بجد Enhancement

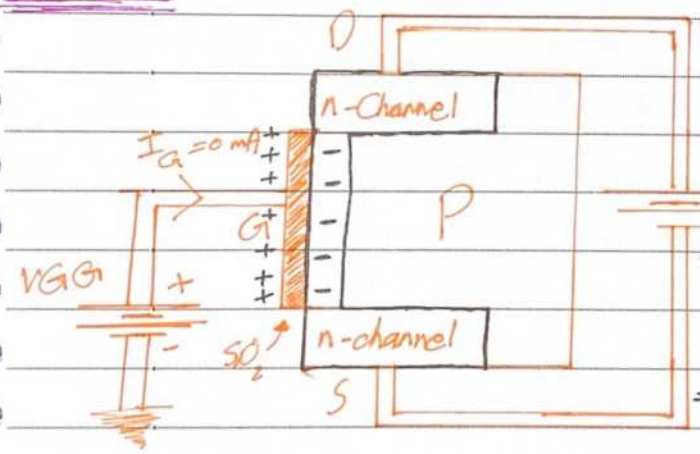
• Page 13 $I_D = I_{DSS}\left[1 - \frac{1}{V_P}\right]^2$

ليث، عمرو

b. Enhancement E-MOSFET



Test so. $V_{GG} = 1$ V , $V_{GS} = -1$ V (Not work)



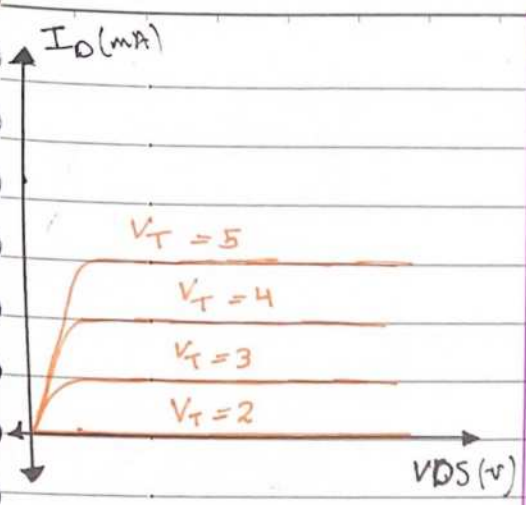
$-V_{GG} + V_{GS} = 0$

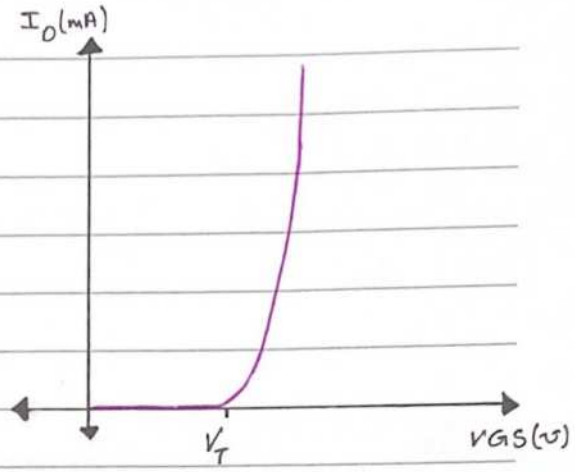
$V_{GS} = V_{GG}$

- حتى يعمل لازم قولتية V_{GS} تكون وصلت V_T
  V_T: on data sheet

- طريقة العمل هو تكوّم الشحنات الموجبة على حاجز العازل فيصبح هناك تجاذب بين الموجب والسالب بالـ n-Channel ويُفتح طريق للإلكترونات فيمر تيار..

ليث، عمرو





V : data sheet
T

Remember 80

FET Transistor 80

1. JFET
2. MOSFET
   - a. D-MosFET → Depletion, Enhancment
   - b. E-MosFET

$$I_D = K(VGS - V_T)^2$$

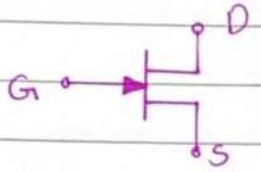
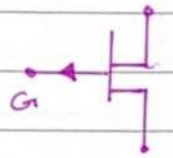
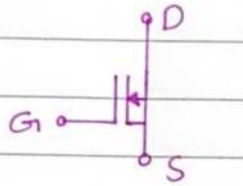
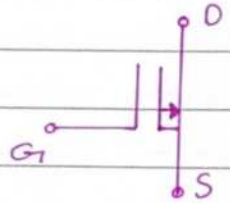
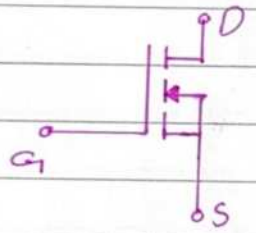
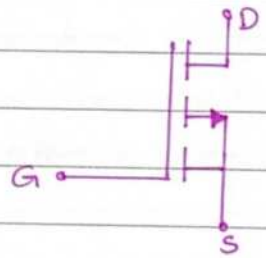
$$K = \frac{I_{D(on)}}{VGS(on) - V_T}$$

For the Graph 80

1. $VGS = V_T \Rightarrow I_D = 0$ mA
2. $VGS = VGS_{(on)} \Rightarrow I_D = I_{D(on)}$
3. $V < VGS < VGS(on) \Rightarrow I_D = ??$
4. $VGS > VGS(on) \Rightarrow I_D = ??$

لیث، عمرو

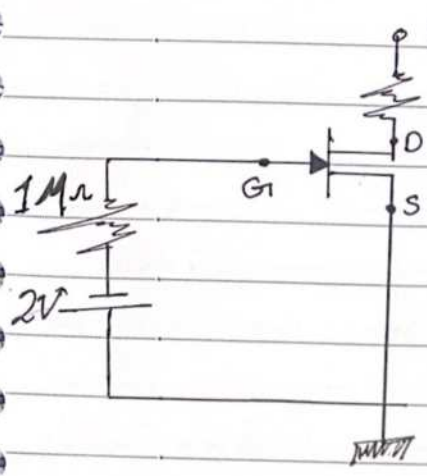
* Symbol For FET Transistor ..

Type	n - Channel	P - Channel
JFET		
D-MosFET		
E-MosFET		

لیث، عمرو

*FET Biasing :-

1 Fixed Bias Configuration :-



JFET n-Channel

$I_{DSS} = 10mA$

$V_P = -8V$

Find:- 1] V_{GSQ}
2] I_{DQ}
3] V_{DS}
4] V_D
5] V_G
6] V_S

Sol:- 1) $2 + V_{GS} = 0 \Rightarrow V_{GS} = -2\ V$

2) $I_{DQ} = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2 \Rightarrow I_{DQ} = 10m\left(1 - \frac{-2}{-8}\right)^2$

$I_{DQ} = 5.625\ mA$

3) $V_{DS} = V_{DD} - I_D R_D \Rightarrow V_{DS} = 16 - (5.625\ m)(2\ k) \Rightarrow V_{DS} = 4.75\ V$

4) $V_D = V_{DS} = 4.75\ V$

5) $V_G = V_{GS} = -2\ V$

6) $V_S = 0\ V$

Cont. in second page ....

ليث، عمرو

Four Points For Graph so

1) $V_{GS} = 0 \Rightarrow I_D = I_{DSS}$ $(0, 10)$

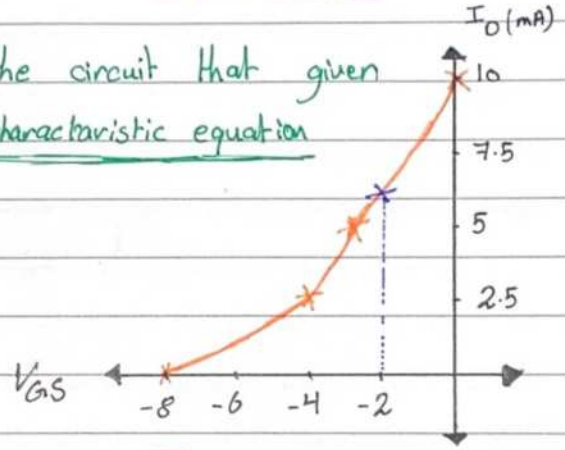
2) $V_{GS} = V_p = -8 \Rightarrow I_D = 0$ $(-8, 0)$

3) $V_{GS} = \frac{V_p}{2} = -4 \Rightarrow I_D = \frac{I_{DSS}}{4} = 2.5$ $(-4, 2.5)$

4) $V_{GS} = 0.3\,V_p = -2.4 \Rightarrow I_D = \frac{I_{DSS}}{2} = 5$ $(-2.4, 5)$

Q-Point • To Find equation from the circuit that given relation between (I_D, V_{GS}) not characteristic equation

$V_{GS} = -2\,V$



1] $V_{GS} = -2\,V$

2] $I_D = 5.6\,mA$

3] $V_{DS} = V_{DD} - I_D R_D$

$V_{DS} = 16 - (5.6m)(2k)$

$V_{DS} = 4.8\,V$

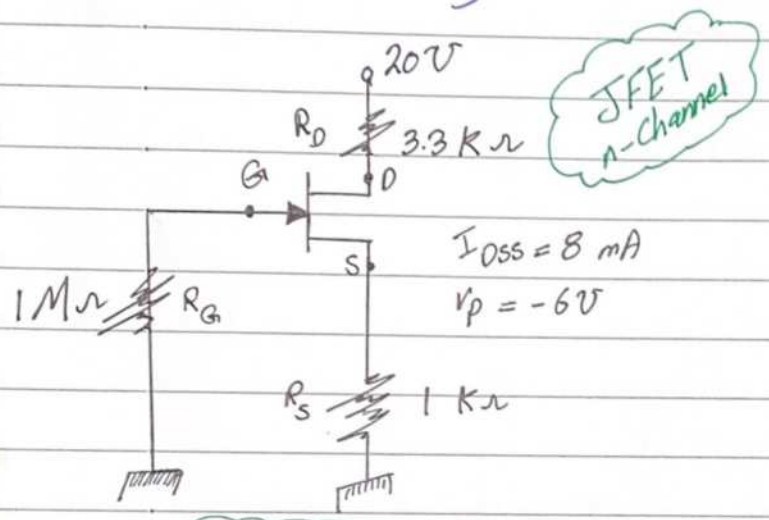
4] $V_D = V_{DS} = 4.8\,V$

5] $V_G = V_{GS} = -2\,V$

6] $V_S = 0\,V$

ليث، عمرو

[2] Self Bias Configuration



JFET n-Channel

Find: 1] V_{GSQ}
2] I_{DQ}
3] V_{DS}
4] V_D
5] V_G
6] V_S

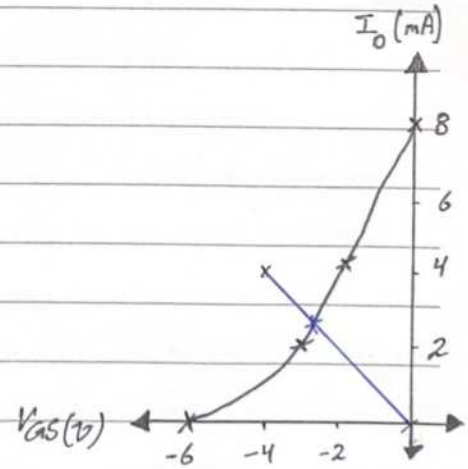
Sol: $V_{GS} + I_D R_S = 0$ ... (1)

$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_p}\right)^2$

Two eq. Two unknowns its easy to find I_D Calc..

$I_D = 8\,mA\left(1 - \frac{V_{GS}}{-6}\right)^2$ ... (2)

Four Point For Graph:
1) (0, 8)
2) (-6, 0)
3) (-3, 2)
4) (-1.8, 4)



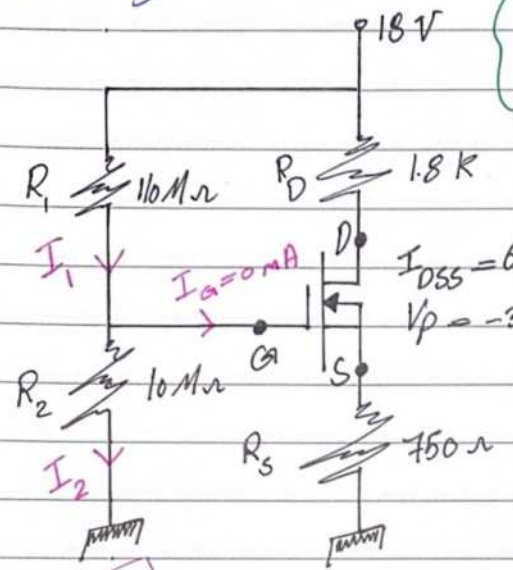
Equation For Load Line:
$V_{GS} = -I_D R_D$ ($R_D = 1$ kΩ)

1] (0,0)
2] (-4,4)

2] $I_{DQ} = 2.6$ mA
1] $V_{GSQ} = -2.6$ V
3] $V_{DS} = V_{DD} - I_D(R_D + R_S) \rightarrow 20 - 2.6(4.3K) = 8.82$ V
6] $V_S = I_D R_S \rightarrow 2.6m(1K) = 2.6$ V
4] $V_D = V_{DS} + V_S \rightarrow 8.82 + 2.6 = 11.42$ V
5] $V_G = 0$ V

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[3] Voltage Divider ∞



Find ∞ 1] I_{DQ}
2] V_{GSQ}
3] V_{DS}

$I_1 = I_2$

Always Approximate ∞ $V_G = \frac{V_{DD} R_2}{R_1 + R_2}$

$V_G = \frac{18 \times 10M}{110M + 10M}$

$V_G = 1.5\ V$

Five Point (Not Four) For Graph ∞

1) (0, 6)
2) (-3, 0)
3) (-1.5, 1.5)
4) (-0.9, 3)
5) $V_{GS} = -1\ V \Rightarrow I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2 \Rightarrow I_D = 6m\left(1 - \frac{1}{-3}\right)^2$
(1, 10.67)

$I_D = 10.67\ mA$

Load Line ∞

$V_{GS} = V_G - V_S$
$V_{GS} = 1.5 - I_D R_S$
$V_{GS} = 1.5 - 750 I_D$

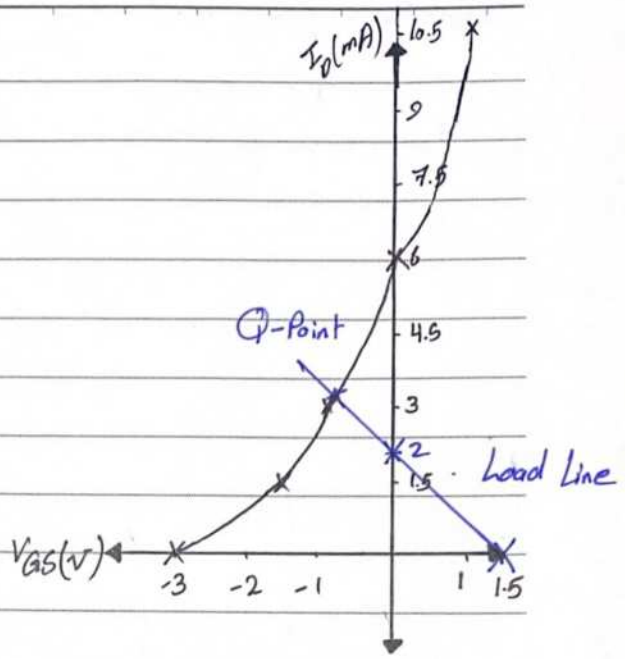
1] (0, 2)
2] (1.5, 0)

Graph on second Page ∞

• Page 20

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Cont. Example :-



1] $I_{DQ} = 3.1\,mA$

2] $V_{GSQ} = -0.8\,V$

3] $V_{DS} = V_{DD} - I_D(R_D + R_S)$

$V_{DS} = 18 - 3.1m(2550)$

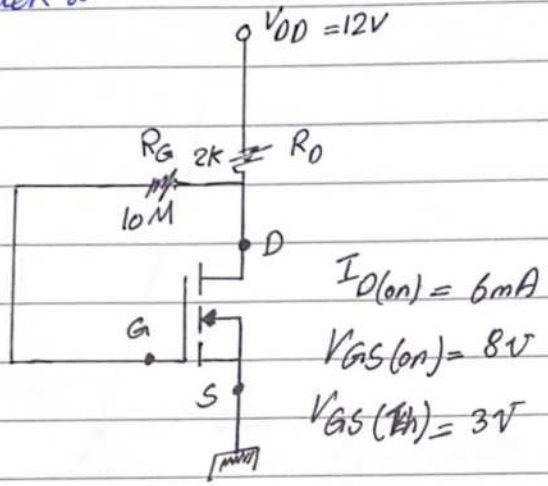
$V_{DS} = 10.1\,V$

"صَلِّ عَلَى أَبَا القَاسِمِ"

• اللهمّ صَلِّ وسلّم عليه ...

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4 Feedback ∞



Find ∞
1] I_{DSQ}
2] V_{GSQ}

Sol ∞

$-V_{DD} + I_D R_D + V_{DS} = 0$

$V_{GS} = V_{DD} - I_D R_D$ ... (1)

$V_{DS} = V_{GS}$

$V_{GS} = 12 - I_D(2000)$

$I_D = K(V_{GS} - V_{Th})^2$

$K = \frac{I_{D(on)}}{(V_{GS(on)} - V_{(Th)})^2} = \frac{6m}{(8-3)^2} = 0.24 * 10^{-3} \ \frac{A}{V^2}$

$I_D = 0.24 * 10^{-3}[V_{GS} - 3]^2$ ... (2)

1) (3, 0)
2) (8, 6)
3) (6, 2.16)
4) (10, 11.76)

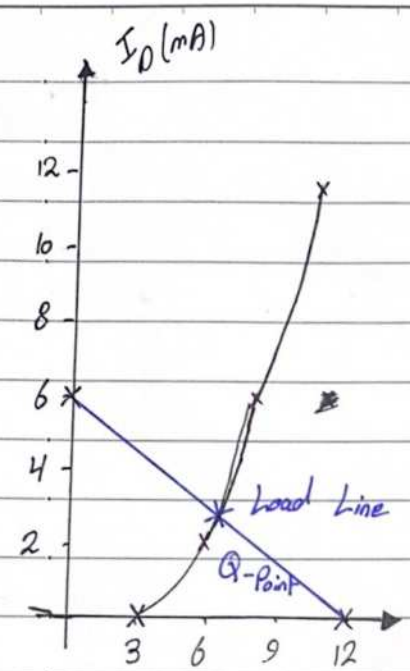
Page (15)

Load Line ∞ $V_{GS} = 12 - I_D(2K)$
1) (12, 0)
2) (0, 6)

Page 22

Graph on second Page

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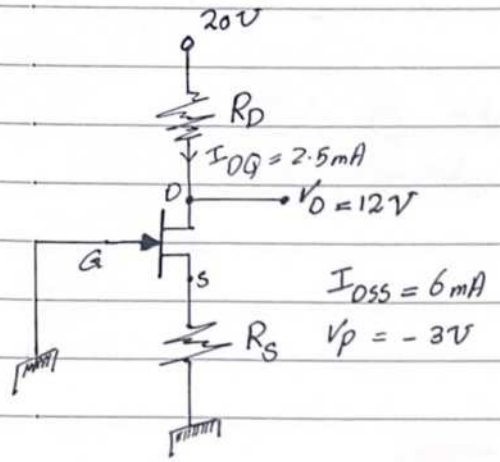


$I_{DQ} = 2.75$ mA

$V_{GSQ} = 6.4$ V $= V_{DS}$

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* Design so



Find so 1] R_D
2] R_S

Sol:

$$-20 + 2.5m(R_D) + 12 = 0$$

$$R_D = \frac{8}{2.5m} \rightarrow \boxed{R_D = 3.2\ K\Omega}$$

$$V_G = 0 \Longrightarrow V_{GS} = -V_S \Longrightarrow V_{GS} = -I_D R_S$$

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_p}\right)^2$$

$$2.5m = 6m\left(1 - \frac{V_{GS}}{-3}\right)^2 \Longrightarrow \boxed{V_{GS} = -1\ V}$$

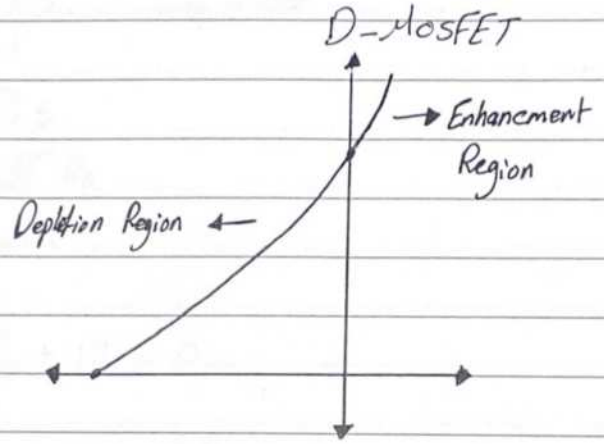
$$-1 = -I_D(R_S)$$

$$R_S = \frac{+1}{+2.5m} = \underline{0.4\ K\Omega}$$

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Extra Question On D-MOSFET

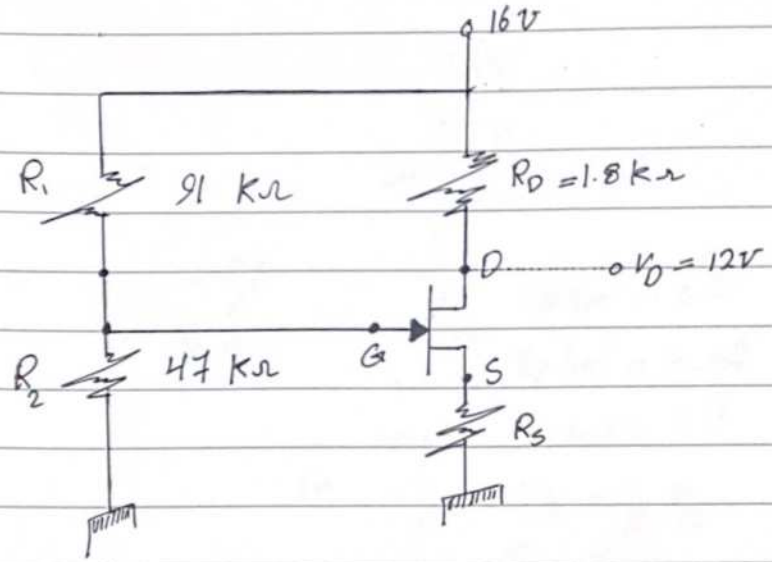
- Find R_D That will Replace the Transistor From Depletion to Enhancment Region..



- Find R_D

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★ Voltage Divider ∞



$V_{GS} = -2\ V$

Find ∞ R_S

Sol ∞ $-16 + I_D R_D + 12 = 0$

$$I_D = 2.22\ mA$$

$V_S = I_D R_S \Longrightarrow V_{GS} = V_G - V_S$

① $-2 = \frac{16 * 47K}{47K + 91K} - V_S$

$V_S = \frac{16 * 47K}{47K + 91K} + 2$

$V_S = 5.44 + 2$

$V_S = 7.44$

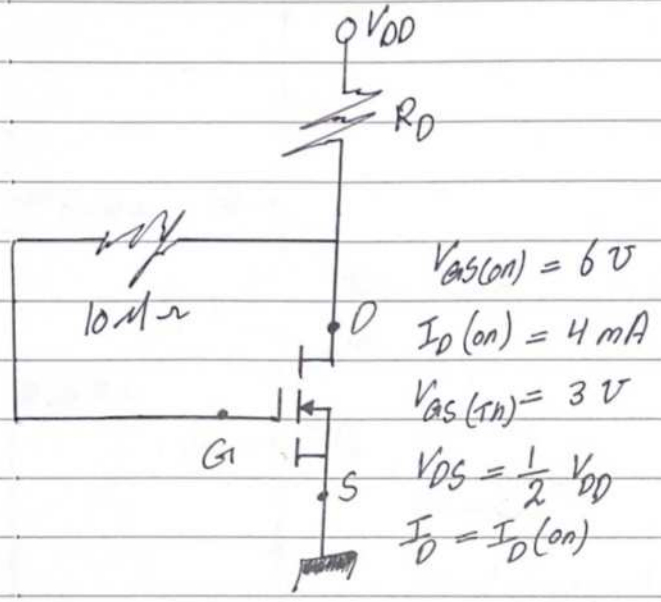
② ⇓

$7.44 = 2.22m(R_S)$

$R_S = 3.35k\Omega$

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* Design



Find :: 1] V_{DD}
2] R_D

Sol:: $V_{DS} = V_{GS(on)} = \frac{1}{2} V_{DD} = 6\ V$ $\left(V_{GS(on)} \rightarrow I_{D(on)}\right)$

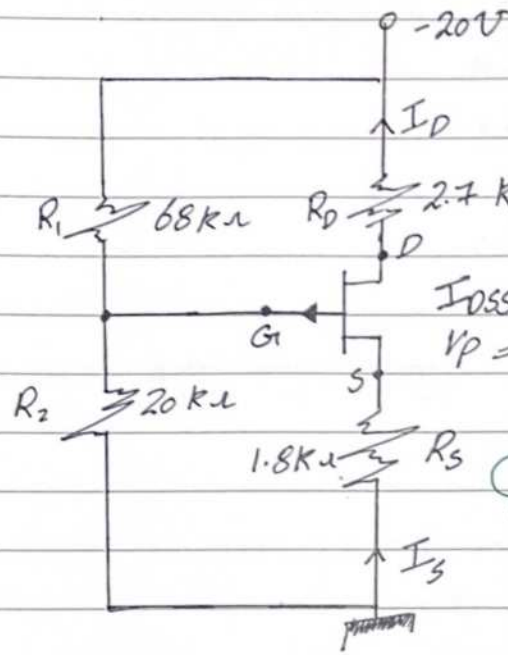
$-V_{DD} + I_{D(on)} R_D + V_{DS} = 0$ $\left(6\ ,\ 4m\right)$

V_{DD}: 12 $I_{D(on)}$: 4m V_{DS}: 6

$R_D = \frac{12 - 6}{4m} \rightarrow R_D = 1.5\ K\Omega$

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* P-Channel :: (Voltage Divider)



Find :: 1] I_{DQ}
2] V_{GSQ}
3] V_{DS}

Sol :: $V_G = \frac{V_{DD} * R_2}{R_1 + R_2}$

$V_G = -4.55$

$V_S = -I_D R_S$

$V_{GS} = V_G - V_S$

$V_{GS} = -4.55 + I_D (1.8k\Omega)$

Load Line :: 1] $V_{GS} = 0 \rightarrow I_D = 2.53\,mA$
2] $I_D = 0 \rightarrow V_{GS} = -4.55V$
(0, 2.53) , (-4.55, 0)

To Sketch The graph ::

1] $V_{GS} = 0 \Rightarrow I_D = 8m$ (0,8)
2] $I_D = 0 \Rightarrow V_{GS} = 4$ (4,0)
3] $V_{GS} = \frac{V_P}{2} = 2 \Rightarrow I_D = \frac{I_{DSS}}{4} = 2m$ (2,2)
4] $V_{GS} = 0.3 V_P = 1.2 \Rightarrow I_D = \frac{I_{DSS}}{2} = 4m$ (1.2, 4)

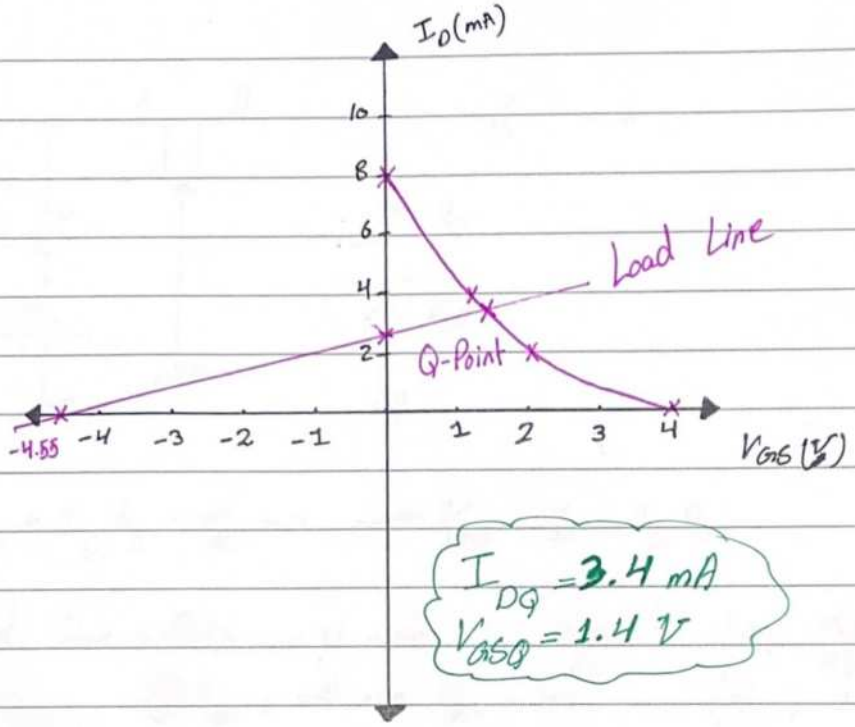
Graph on Second Page

$20 - I_D R_D + V_{DS} - I_D R_S = 0$

$-20 + I_D (2.7K + 1.8K) = V_{DS}$

I_D :: From Graph

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$I_{DQ} = 3.4\ mA$

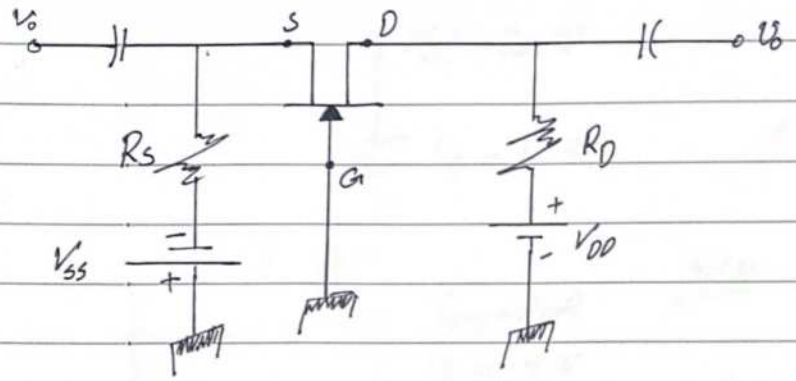
$V_{GSQ} = 1.4\ V$

$V_{DS} = -20 + 3.4m(2.7K + 1.8K)$

$V_{DS} = -4.7\ V$

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* Common Gate 80



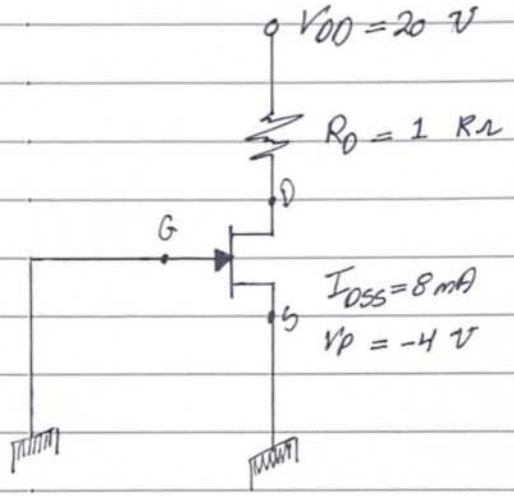
$$V_{GS} + I_D R_S - V_{SS} = 0 \Longrightarrow V_{GS} = V_{SS} - I_D R_S$$

- Load Line :: ① $V_{GS} = 0\ V \Longrightarrow I_D = \frac{V_{SS}}{R_S}$ $\left(0, \frac{V_{SS}}{R_S}\right)$
  ② $I_D = 0\ mA \Longrightarrow V_{GS} = V_{SS}$ $(V_{SS}, 0)$

- To sketch Graph :: ① $(0, I_{DSS})$
  ② $(V_P, 0)$
  ③ $\left(\frac{V_P}{2}, \frac{I_{DSS}}{4}\right)$
  ④ $\left(0.3V_P, \frac{I_{DSS}}{2}\right)$

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* Example 8o



Find 8o • V_{DS}

Sol 8o $V_{GS} = 0$ $I_D = I_{DSS} = 8mA$

$-20 + I_D R_D + V_{DS} = 0$

$V_{DS} = 20 - 8m(1\ K\Omega)$

$V_{DS} = 20 - 8$

$V_{DS} = 12\ V$

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5 BJT AC Analysis

CHAPTER OBJECTIVES

- Become familiar with the r_e, hybrid, and hybrid π models for the BJT transistor.
- Learn to use the equivalent model to find the important ac parameters for an amplifier.
- Understand the effects of a source resistance and load resistor on the overall gain and characteristics of an amplifier.
- Become aware of the general ac characteristics of a variety of important BJT configurations.
- Begin to understand the advantages associated with the two-port systems approach to single- and multistage amplifiers.
- Develop some skill in troubleshooting ac amplifier networks.

5.1 INTRODUCTION

The basic construction, appearance, and characteristics of the transistor were introduced in Chapter 3. The dc biasing of the device was then examined in detail in Chapter 4. We now begin to examine the ac response of the BJT amplifier by reviewing the *models* most frequently used to represent the transistor in the sinusoidal ac domain.

One of our first concerns in the sinusoidal ac analysis of transistor networks is the magnitude of the input signal. It will determine whether *small-signal* or *large-signal* techniques should be applied. There is no set dividing line between the two, but the application—and the magnitude of the variables of interest relative to the scales of the device characteristics—will usually make it quite clear which method is appropriate. The small-signal technique is introduced in this chapter, and large-signal applications are examined in Chapter 12.

There are three models commonly used in the small-signal ac analysis of transistor networks: the r_e model, the hybrid π model, and the hybrid equivalent model. This chapter introduces all three but emphasizes the r_e model.

5.2 AMPLIFICATION IN THE AC DOMAIN

It was demonstrated in Chapter 3 that the transistor can be employed as an amplifying device. That is, the output sinusoidal signal is greater than the input sinusoidal signal, or, stated another way, the output ac power is greater than the input ac power. The question then arises as to how the ac power output can be greater than the input ac power. Conservation of energy dictates that over time the total power output, P_o, of a system cannot be greater than its power

input, P_i, and that the efficiency defined by $\eta = P_o/P_i$ cannot be greater than 1. The factor missing from the discussion above that permits an ac power output greater than the input ac power is the applied dc power. It is the principal contributor to the total output power even though part of it is dissipated by the device and resistive elements. In other words, there is an "exchange" of dc power to the ac domain that permits establishing a higher output ac power. In fact, a *conversion efficiency* is defined by $\eta = P_{o(\text{ac})}/P_{i(\text{dc})}$, where $P_{o(\text{ac})}$ is the ac power to the load and $P_{i(\text{dc})}$ is the dc power supplied.

Perhaps the role of the dc supply can best be described by first considering the simple dc network of Fig. 5.1. The resulting direction of flow is indicated in the figure with a plot of the current i versus time. Let us now insert a control mechanism such as that shown in Fig. 5.2. The control mechanism is such that the application of a relatively small signal to the control mechanism can result in a substantial oscillation in the output circuit.

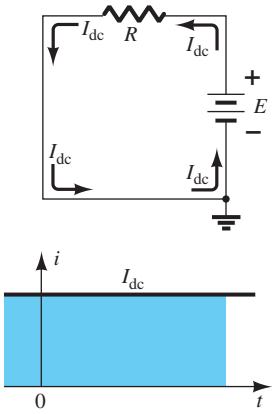


FIG. 5.1
Steady current established by a dc supply.

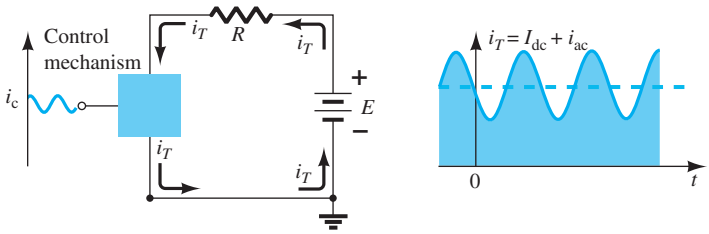


FIG. 5.2
Effect of a control element on the steady-state flow of the electrical system of Fig. 5.1.

That is, for this example,

$$i_{\text{ac(p-p)}} \gg i_{c\text{(p-p)}}$$

and amplification in the ac domain has been established. The peak-to-peak value of the output current far exceeds that of the control current.

For the system of Fig. 5.2, the peak value of the oscillation in the output circuit is controlled by the established dc level. Any attempt to exceed the limit set by the dc level will result in a "clipping" (flattening) of the peak region at the high and low end of the output signal. In general, therefore, proper amplification design requires that the dc and ac components be sensitive to each other's requirements and limitations.

However, it is extremely helpful to realize that:

The superposition theorem is applicable for the analysis and design of the dc and ac components of a BJT network, permitting the separation of the analysis of the dc and ac responses of the system.

In other words, one can make a complete dc analysis of a system before considering the ac response. Once the dc analysis is complete, the ac response can be determined using a completely ac analysis. It happens, however, that one of the components appearing in the ac analysis of BJT networks will be determined by the dc conditions, so there is still an important link between the two types of analysis.

5.3 BJT TRANSISTOR MODELING

The key to transistor small-signal analysis is the use of the equivalent circuits (models) to be introduced in this chapter.

A model is a combination of circuit elements, properly chosen, that best approximates the actual behavior of a semiconductor device under specific operating conditions.

Once the ac equivalent circuit is determined, the schematic symbol for the device can be replaced by this equivalent circuit and the basic methods of circuit analysis applied to determine the desired quantities of the network.

In the formative years of transistor network analysis the *hybrid equivalent network* was employed the most frequently. Specification sheets included the parameters in their listing, and analysis was simply a matter of inserting the equivalent circuit with the listed values.

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The drawback to using this equivalent circuit, however, is that it is *defined for a set of operating conditions that might not match the actual operating conditions.* In most cases, this is not a serious flaw because the actual operating conditions are relatively close to the chosen operating conditions on the data sheets. In addition, there is always a variation in actual resistor values and given transistor beta values, so as an approximate approach it was quite reliable. Manufacturers continue to specify the hybrid parameter values for a particular operating point on their specification sheets. They really have no choice. They want to give the user some idea of the value of each important parameter so comparisons can be made between transistors, but they really do not know the user's actual operating conditions.

In time the use of the r_e *model* became the more desirable approach because an important parameter of the equivalent circuit was determined by the actual operating conditions rather than using a data sheet value that in some cases could be quite different. Unfortunately, however, one must still turn to the data sheets for some of the other parameters of the equivalent circuit. The r_e model also failed to include a feedback term, which in some cases can be important if not simply troublesome.

The r_e model is really a reduced version of the *hybrid* π *model* used almost exclusively for high-frequency analysis. This model also includes a connection between output and input to include the feedback effect of the output voltage and the input quantities. The full hybrid model is introduced in Chapter 9.

Throughout the text the r_e model is the model of choice unless the discussion centers on the description of each model or a region of examination that predetermines the model that should be used. Whenever possible, however, a comparison between models will be discussed to show how closely related they really are. It is also important that once you gain a proficiency with one model it will carry over to an investigation using a different model, so moving from one to another will not be a dramatic undertaking.

In an effort to demonstrate the effect that the ac equivalent circuit will have on the analysis to follow, consider the circuit of Fig. 5.3. Let us assume for the moment that the small-signal ac equivalent circuit for the transistor has already been determined. Because we are interested only in the ac response of the circuit, all the dc supplies can be replaced by a zero-potential equivalent (short circuit) because they determine only the dc (quiescent level) of the output voltage and not the magnitude of the swing of the ac output. This is clearly demonstrated by Fig. 5.4. The dc levels were simply important for determining the proper *Q*-point of operation. Once determined, the dc levels can be ignored in the ac analysis of the network. In addition, the coupling capacitors C_1 and C_2 and bypass capacitor C_3 were chosen to have a very small reactance at the frequency of application. Therefore, they, too, may for all practical purposes be replaced by a low-resistance path or a short circuit. Note that this will result in the "shorting out" of the dc biasing resistor R_E. Recall that capacitors assume an "open-circuit" equivalent under dc steady-state conditions, permitting an isolation between stages for the dc levels and quiescent conditions.

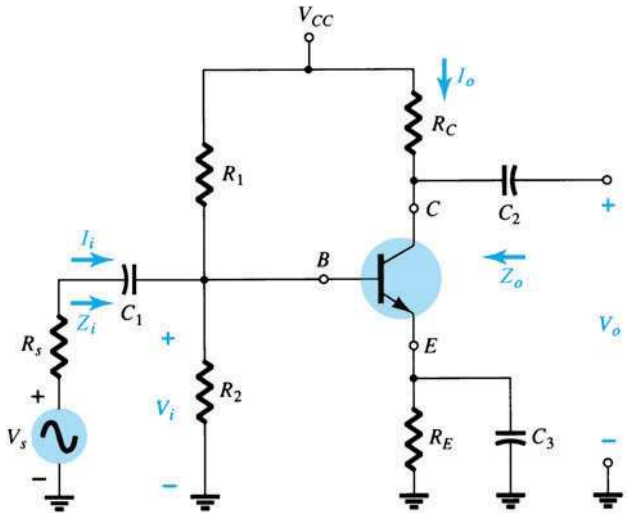


FIG. 5.3
Transistor circuit under examination in this introductory discussion.

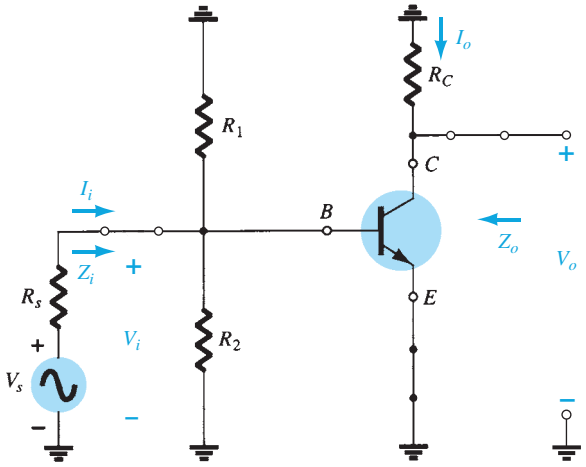


FIG. 5.4
The network of Fig. 5.3 following removal of the dc supply and insertion of the short-circuit equivalent for the capacitors.

It is important as you progress through the modifications of the network to define the ac equivalent that the parameters of interest such as Z_i, Z_o, I_i, and I_o as defined by Fig. 5.5 be carried through properly. Even though the network appearance may change, you want to be sure the quantities you find in the reduced network are the same as defined by the original network. In both networks the input impedance is defined from base to ground, the input current as the base current of the transistor, the output voltage as the voltage from collector to ground, and the output current as the current through the load resistor R_C.

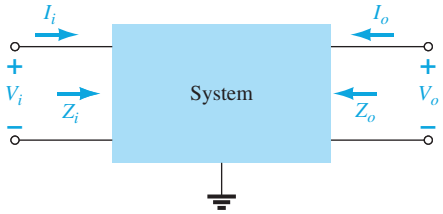


FIG. 5.5
Defining the important parameters of any system.

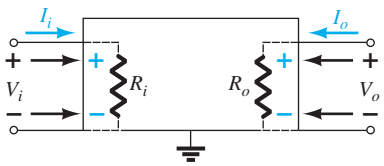


FIG. 5.6
Demonstrating the reason for the defined directions and polarities.

The parameters of Fig. 5.5 can be applied to any system whether it has one or a thousand components. For all the analysis to follow in this text, the directions of the currents, the polarities of the voltages, and the direction of interest for the impedance levels are as appearing in Fig. 5.5. In other words, the input current I_i and output current I_o are, by definition, defined to enter the system. If, in a particular example, the output current is leaving the system rather than entering the system as shown in Fig. 5.5, a minus sign must be applied. The defined polarities for the input and output voltages are also as appearing in Fig. 5.5. If V_o has the opposite polarity, the minus sign must be applied. Note that Z_i is the impedance "looking into" the system, whereas Z_o is the impedance "looking back into" the system from the output side. By choosing the defined directions for the currents and voltages as appearing in Fig. 5.5, both the input impedance and output impedance are defined as having positive values. For example, in Fig. 5.6 the input and output impedances for a particular system are both resistive. For the direction of I_i and I_o the resulting voltage across the resistive elements will have the same polarity as V_i and V_o, respectively. If I_o had been defined as the opposite direction in Fig. 5.5 a minus sign would have to be applied. For each case $Z_i = V_i/I_i$ and $Z_o = V_o/I_o$ with positive results if they all have the defined directions and polarity of Fig. 5.5. If the output current of an actual system has a direction opposite to that

of Fig. 5.5 a minus sign must be applied to the result because V_o must be defined as appearing in Fig. 5.5. Keep Fig. 5.5 in mind as you analyze the BJT networks in this chapter. It is an important introduction to "System Analysis," which is becoming so important with the expanded use of packaged IC systems.

If we establish a common ground and rearrange the elements of Fig. 5.4, R_1 and R_2 will be in parallel, and R_C will appear from collector to emitter as shown in Fig. 5.7. Because the components of the transistor equivalent circuit appearing in Fig. 5.7 employ familiar components such as resistors and independent controlled sources, analysis techniques such as superposition, Thévenin's theorem, and so on, can be applied to determine the desired quantities.

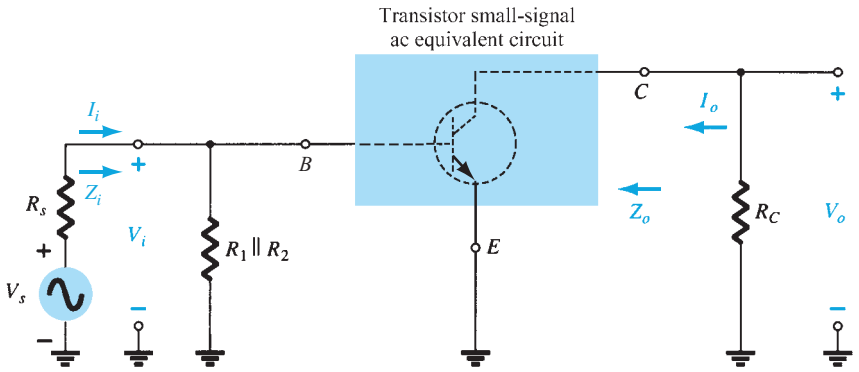


FIG. 5.7
Circuit of Fig. 5.4 redrawn for small-signal ac analysis.

Let us further examine Fig. 5.7 and identify the important quantities to be determined for the system. Because we know that the transistor is an amplifying device, we would expect some indication of how the output voltage V_o is related to the input voltage V_i—the *voltage gain.* Note in Fig. 5.7 for this configuration that the *current gain* is defined by $A_i = I_o/I_i$.

In summary, therefore, the ac equivalent of a transistor network is obtained by:

1. ***Setting all dc sources to zero and replacing them by a short-circuit equivalent***
2. ***Replacing all capacitors by a short-circuit equivalent***
3. ***Removing all elements bypassed by the short-circuit equivalents introduced by steps 1 and 2***
4. ***Redrawing the network in a more convenient and logical form***

In the sections to follow, a transistor equivalent model will be introduced to complete the ac analysis of the network of Fig. 5.7.

5.4 THE r_e TRANSISTOR MODEL

The r_e model for the CE, CB, and CC BJT transistor configurations will now be introduced with a short description of why each is a good approximation to the actual behavior of a BJT transistor.

Common-Emitter Configuration

The equivalent circuit for the common-emitter configuration will be constructed using the device characteristics and a number of approximations. Starting with the input side, we find the applied voltage V_i is equal to the voltage V_{be} with the input current being the base current I_b as shown in Fig. 5.8.

Recall from Chapter 3 that because the current through the forward-biased junction of the transistor is I_E, the characteristics for the input side appear as shown in Fig. 5.9a for various levels of V_{BE}. Taking the average value for the curves of Fig. 5.9a will result in the single curve of Fig. 5.9b, which is simply that of a forward-biased diode.

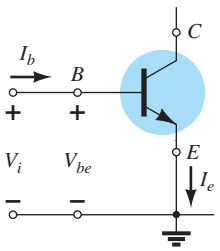


FIG. 5.8
Finding the input equivalent circuit for a BJT transistor.

5.19 THE HYBRID EQUIVALENT MODEL

The hybrid equivalent model was mentioned in the earlier sections of this chapter as one that was used in the early years before the popularity of the r_e model developed. Today there is a mix of usage depending on the level and direction of the investigation.

The r_e model has the advantage that the parameters are defined by the actual operating conditions,

whereas

the parameters of the hybrid equivalent circuit are defined in general terms for any operating conditions.

In other words, the hybrid parameters may not reflect the actual operating conditions but simply provide an indication of the level of each parameter to expect for general use. The r_e model suffers from the fact that parameters such as the output impedance and the feedback elements are not available, whereas the hybrid parameters provide the entire set on the specification sheet. In most cases, if the r_e model is employed, the investigator will simply examine the specification sheet to have some idea of what the additional elements might be. This section will show how one can go from one model to the other and how the parameters are related. Because all specification sheets provide the hybrid parameters and the model is still extensively used, it is important to be aware of both models. The hybrid parameters as shown in Fig. 5.92 are derived from the specification sheet for the 2N4400 transistor described in Chapter 3. The values are provided at a dc collector current of 1 mA and a collector-to-emitter voltage of 10 V. In addition, a range of values is provided for each parameter for guidance in the initial design or analysis of a system. One obvious advantage of the specification sheet listing is the immediate knowledge of typical levels for the parameters of the device as compared to other transistors.

		Min.	Max.	
Input impedance (I_C = 1 mA dc, V_{CE} = 10 V dc, f = 1 kHz)	h_{ie}	0.5	7.5	kΩ
Voltage feedback ratio (I_C = 1 mA dc, V_{CE} = 10 V dc, f = 1 kHz)	h_{re}	0.1	8.0	$\times 10^{-4}$
Small-signal current gain (I_C = 1 mA dc, V_{CE} = 10 V dc, f = 1 kHz)	h_{fe}	20	250	—
Output admittance (I_C = 1 mA dc, V_{CE} = 10 V dc, f = 1 kHz)	h_{oe}	1.0	30	1 μS

FIG. 5.92
Hybrid parameters for the 2N4400 transistor.

The description of the hybrid equivalent model will begin with the general two-port system of Fig. 5.93. The following set of equations (5.131) and (5.132) is only one of a number of ways in which the four variables of Fig. 5.93 can be related. It is the most frequently employed in transistor circuit analysis, however, and therefore is discussed in detail in this chapter.

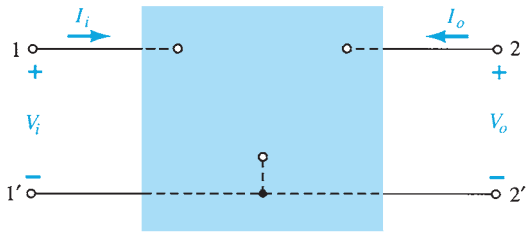


FIG. 5.93
Two-port system.

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$$V_i = h_{11}I_i + h_{12}V_o \tag{5.133}$$

$$I_o = h_{21}I_i + h_{22}V_o \tag{5.134}$$

The parameters relating the four variables are called *h-parameters*, from the word "hybrid." The term *hybrid* was chosen because the mixture of variables (V and I) in each equation results in a "hybrid" set of units of measurement for the h-parameters. A clearer understanding of what the various h-parameters represent and how we can determine their magnitude can be developed by isolating each and examining the resulting relationship.

h_{11} If we arbitrarily set $V_o = 0$ (short circuit the output terminals) and solve for h_{11} in Eq. (5.133), we find

$$h_{11} = \left.\frac{V_i}{I_i}\right|_{V_o=0} \quad \text{ohms} \tag{5.135}$$

The ratio indicates that the parameter h_{11} is an impedance parameter with the units of ohms. Because it is the ratio of the *input* voltage to the *input* current with the output terminals *shorted*, it is called the *short-circuit input-impedance parameter.* The subscript 11 of h_{11} refers to the fact that the parameter is determined by a ratio of quantities measured at the input terminals.

h_{12} If I_i is set equal to zero by opening the input leads, the following results for h_{12}:

$$h_{12} = \left.\frac{V_i}{V_o}\right|_{I_i=0} \quad \text{unitless} \tag{5.136}$$

The parameter h_{12}, therefore, is the ratio of the input voltage to the output voltage with the input current equal to zero. It has no units because it is a ratio of voltage levels and is called the *open-circuit reverse transfer voltage ratio parameter.* The subscript 12 of h_{12} indicates that the parameter is a transfer quantity determined by a ratio of input (1) to output (2) measurements. The first integer of the subscript defines the measured quantity to appear in the numerator; the second integer defines the source of the quantity to appear in the denominator. The term *reverse* is included because the ratio is an input voltage over an output voltage rather than the reverse ratio typically of interest.

h_{21} If in Eq. (5.134) V_o is set equal to zero by again shorting the output terminals, the following results for h_{21}:

$$h_{21} = \left.\frac{I_o}{I_i}\right|_{V_o=0} \quad \text{unitless} \tag{5.137}$$

Note that we now have the ratio of an output quantity to an input quantity. The term *forward* will now be used rather than *reverse* as indicated for h_{12}. The parameter h_{21} is the ratio of the output current to the input current with the output terminals shorted. This parameter, like h_{12}, has no units because it is the ratio of current levels. It is formally called the *short-circuit forward transfer current ratio parameter.* The subscript 21 again indicates that it is a transfer parameter with the output quantity (2) in the numerator and the input quantity (1) in the denominator.

h_{22} The last parameter, h_{22}, can be found by again opening the input leads to set $I_1 = 0$ and solving for h_{22} in Eq. (5.134):

$$h_{22} = \left.\frac{I_o}{V_o}\right|_{I_i=0} \quad \text{siemens} \tag{5.138}$$

Because it is the ratio of the output current to the output voltage, it is the output conductance parameter, and it is measured in siemens (S). It is called the *open-circuit output admittance parameter.* The subscript 22 indicates that it is determined by a ratio of output quantities.

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Because each term of Eq. (5.133) has the unit volt, let us apply Kirchhoff's voltage law "in reverse" to find a circuit that "fits" the equation. Performing this operation results in the circuit of Fig. 5.94. Because the parameter h_{11} has the unit ohm, it is represented by a resistor in Fig. 5.94. The quantity h_{12} is dimensionless and therefore simply appears as a multiplying factor of the "feedback" term in the input circuit.

Because each term of Eq. (5.134) has the units of current, let us now apply Kirchhoff's current law "in reverse" to obtain the circuit of Fig. 5.95. Because h_{22} has the units of admittance, which for the transistor model is conductance, it is represented by the resistor symbol. Keep in mind, however, that the resistance in ohms of this resistor is equal to the reciprocal of conductance ($1/h_{22}$).

The complete "ac" equivalent circuit for the basic three-terminal linear device is indicated in Fig. 5.96 with a new set of subscripts for the *h*-parameters. The notation of Fig. 5.96 is of a more practical nature because it relates the *h*-parameters to the resulting ratio obtained in the last few paragraphs. The choice of letters is obvious from the following listing:

$h_{11} \rightarrow$ *i*nput resistance $\rightarrow h_i$

$h_{12} \rightarrow$ *r*everse transfer voltage ratio $\rightarrow h_r$

$h_{21} \rightarrow$ *f*orward transfer current ratio $\rightarrow h_f$

$h_{22} \rightarrow$ *o*utput conductance $\rightarrow h_o$

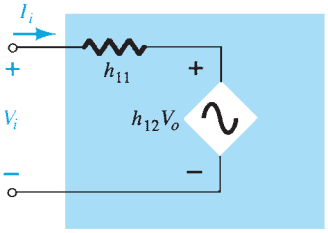


FIG. 5.94
Hybrid input equivalent circuit.

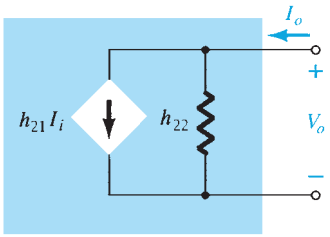


FIG. 5.95
Hybrid output equivalent circuit.

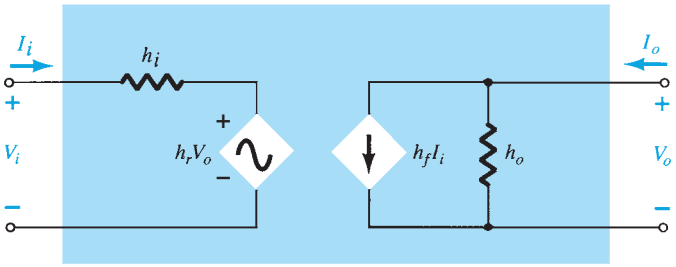


FIG. 5.96
Complete hybrid equivalent circuit.

The circuit of Fig. 5.96 is applicable to any linear three-terminal electronic device or system with no internal independent sources. For the transistor, therefore, even though it has three basic configurations, *they are all three-terminal configurations*, so that the resulting equivalent circuit will have the same format as shown in Fig. 5.96. In each case, the bottom of the input and output sections of the network of Fig. 5.96 can be connected as shown in Fig. 5.97 because the potential level is the same. Essentially, therefore, the transistor model is a three-terminal two-port system. The *h*-parameters, however, will change with each configuration. To distinguish which parameter has been used or which is available, a second

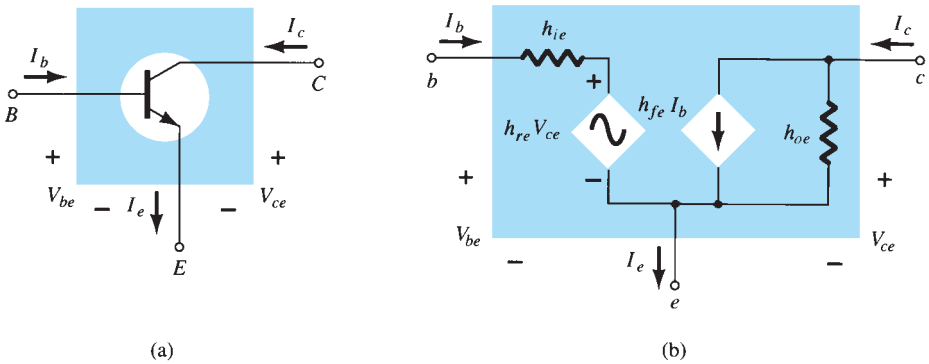


FIG. 5.97
Common-emitter configuration: (a) graphical symbol; (b) hybrid equivalent circuit.

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subscript has been added to the *h*-parameter notation. For the common-base configuration, the lowercase letter *b* was added, whereas for the common-emitter and common-collector configurations, the letters *e* and *c* were added, respectively. The hybrid equivalent network for the common-emitter configuration appears with the standard notation in Fig. 5.97. Note that $I_i = I_b$, $I_o = I_c$, and, through an application of Kirchhoff's current law, $I_e = I_b + I_c$. The input voltage is now V_{be}, with the output voltage V_{ce}. For the common-base configuration of Fig. 5.98, $I_i = I_e$, $I_o = I_c$ with $V_{eb} = V_i$ and $V_{cb} = V_o$. The networks of Figs. 5.97 and 5.98 are applicable for *pnp* or *npn* transistors.

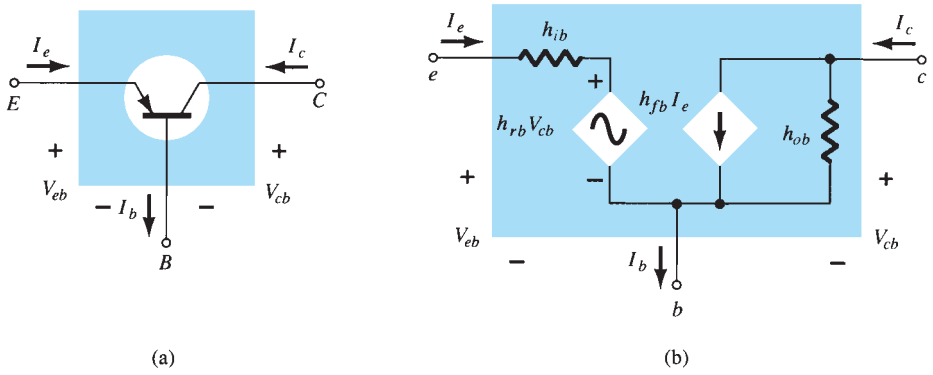


FIG. 5.98
Common-base configuration: (a) graphical symbol; (b) hybrid equivalent circuit.

The fact that both a Thévenin and a Norton circuit appear in the circuit of Fig. 5.96 was further impetus for calling the resultant circuit a *hybrid* equivalent circuit. Two additional transistor equivalent circuits, not to be discussed in this text, called the **z**-parameter and **y**-parameter equivalent circuits, use either the voltage source or the current source, but not both, in the same equivalent circuit. In Appendix A the magnitudes of the various parameters will be found from the transistor characteristics in the region of operation resulting in the desired *small-signal equivalent network* for the transistor.

For the common-emitter and common-base configurations, the magnitude of h_r and h_o is often such that the results obtained for the important parameters such as Z_i, Z_o, A_v, and A_i are only slightly affected if h_r and h_o are not included in the model.

Because h_r is normally a relatively small quantity, its removal is approximated by $h_r \cong 0$ and $h_r V_o = 0$, resulting in a short-circuit equivalent for the feedback element as shown in Fig. 5.99. The resistance determined by $1/h_o$ is often large enough to be ignored in comparison to a parallel load, permitting its replacement by an open-circuit equivalent for the CE and CB models, as shown in Fig. 5.99.

The resulting equivalent of Fig. 5.100 is quite similar to the general structure of the common-base and common-emitter equivalent circuits obtained with the r_e model. In fact,

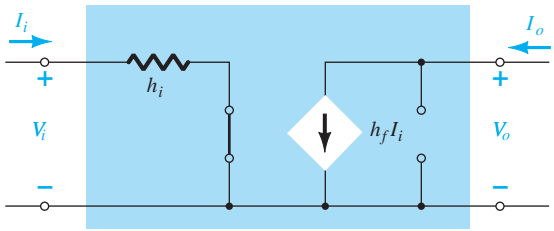


FIG. 5.99
Effect of removing h_{re} and h_{oe} from the hybird equivalent circuit.

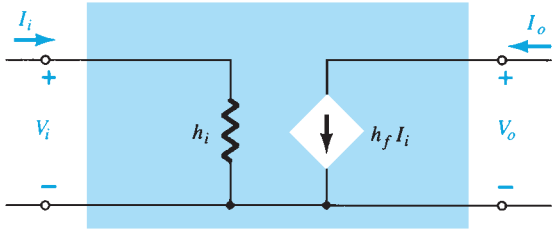


FIG. 5.100
Approximate hybrid equivalent model.

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the hybrid equivalent and the r_e models for each configuration are repeated in Fig. 5.101 for comparison. It should be reasonably clear from Fig. 5.101a that

$$h_{ie} = \beta r_e \tag{5.139}$$

and

$$h_{fe} = \beta_{ac} \tag{5.140}$$

From Fig. 5.101b,

$$h_{ib} = r_e \tag{5.141}$$

and

$$h_{fb} = -\alpha \cong -1 \tag{5.142}$$

In particular, note that the minus sign in Eq. (5.142) accounts for the fact that the current source of the standard hybrid equivalent circuit is pointing down rather than in the actual direction as shown in the r_e model of Fig. 5.101b.

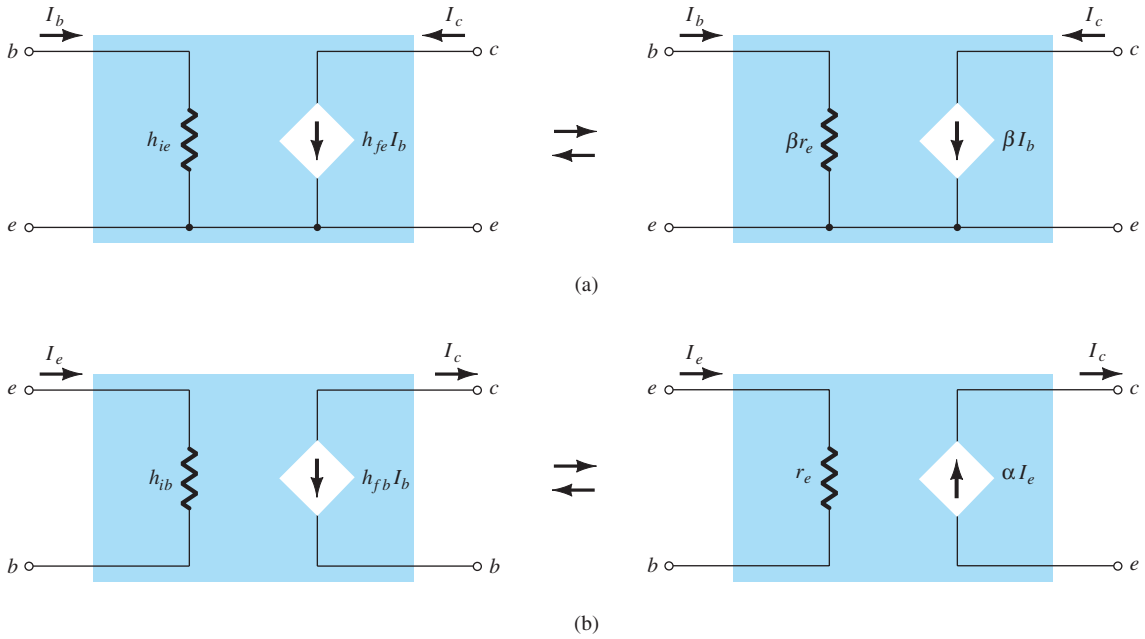


FIG. 5.101
Hybrid versus r_e model: (a) common-emitter configuration; (b) common-base configuration.

EXAMPLE 5.19 Given $I_E = 2.5$ mA, $h_{fe} = 140$, $h_{oe} = 20\ \mu$S (μmho), and $h_{ob} = 0.5\ \mu$S, determine:

a. The common-emitter hybrid equivalent circuit.
b. The common-base r_e model.

Solution:

a. $$r_e = \frac{26\text{ mV}}{I_E} = \frac{26\text{ mV}}{2.5\text{ mA}} = \mathbf{10.4\ \Omega}$$

$$h_{ie} = \beta r_e = (140)(10.4\ \Omega) = \mathbf{1.456\ k\Omega}$$

$$r_o = \frac{1}{h_{oe}} = \frac{1}{20\ \mu\text{S}} = \mathbf{50\ k\Omega}$$

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Note Fig. 5.102.

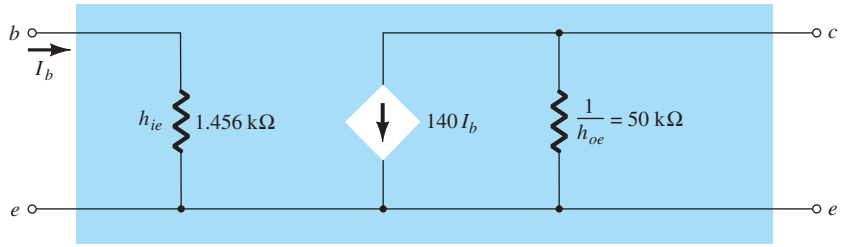


FIG. 5.102
Common-emitter hybrid equivalent circuit for the parameters of Example 5.19.

b. $r_e = \mathbf{10.4\ \Omega}$

$$\alpha \cong 1, \qquad r_o = \frac{1}{h_{ob}} = \frac{1}{0.5\ \mu\text{S}} = \mathbf{2\ M\Omega}$$

Note Fig. 5.103.

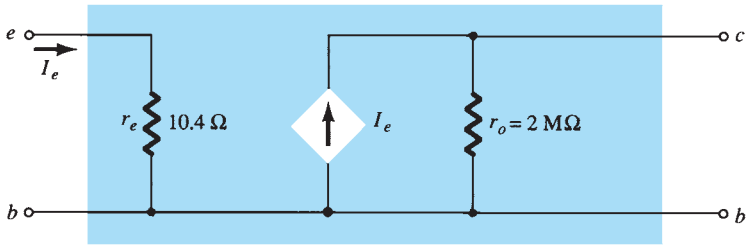


FIG. 5.103
Common-base r_e model for the parameters of Example 5.19.

A series of equations relating the parameters of each configuration for the hybrid equivalent circuit is provided in Appendix B. In Section 5.23 it is demonstrated that the hybrid parameter h_{fe} (β_{ac}) is the least sensitive of the hybrid parameters to a change in collector current. Assuming, therefore, that $h_{fe} = \beta$ is a constant for the range of interest, is a fairly good approximation. It is $h_{ie} = \beta r_e$ that will vary significantly with I_C and should be determined at operating levels because it can have a real effect on the gain levels of a transistor amplifier.

5.20 APPROXIMATE HYBRID EQUIVALENT CIRCUIT

The analysis using the approximate hybrid equivalent circuit of Fig. 5.104 for the common-emitter configuration and of Fig. 5.105 for the common-base configuration is very similar to that just performed using the r_e model. A brief overview of some of the most important configurations will be included in this section to demonstrate the similarities in approach and the resulting equations.

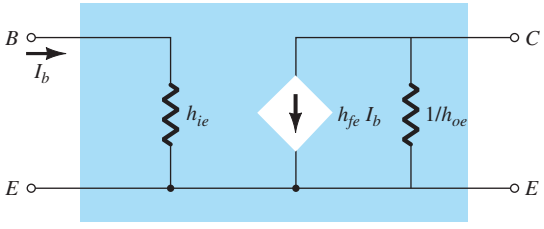


FIG. 5.104
Approximate common-emitter hybrid equivalent circuit.

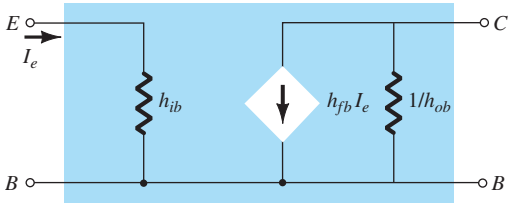


FIG. 5.105
Approximate common-base hybrid equivalent circuit.

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